

# Linear Algebra

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## Vectors

**Exercise 0.1** (Store a data point as one object). A house has a floor area of 120 square meters, 3 bedrooms, and an age of 15 years.

1. Write these three numbers as a single vertical list  $\tilde{x}$ , in the order given, and call its entries  $x_1$ ,  $x_2$  and  $x_3$ . What is  $x_2$ ?
2. Does the list  $(3, 120, 15)$  describe the same house?

### Solution

*Solution 0.1.*

1. Stacking the three numbers in order gives

$$\tilde{x} = \begin{bmatrix} 120 \\ 3 \\ 15 \end{bmatrix},$$

so  $x_2 = 3$ , the number of bedrooms.

2. No. The list  $(3, 120, 15)$  would describe a house with 3 square meters of floor area and 120 bedrooms. An entry's position says what it measures, so the order of the entries matters.

**Definition 0.1** (Column vector). A **column vector** of length  $p$  is an ordered list of  $p$  numbers, written vertically:

$$\tilde{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$$

*Remark 0.1* (Vectors are columns by default). In these notes, as in most statistics textbooks, a vector is a column vector unless stated otherwise. A vector written inline as a list in parentheses is still a column vector: for example,  $\tilde{x} = (2, -1, 5)$  means

$$\tilde{x} = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}.$$

**Definition 0.2** (The set  $\mathbb{R}^p$ ). For a positive integer  $p$ ,  $\mathbb{R}^p$  is the set of all **column vectors** of length  $p$  whose entries are real numbers.

*Remark 0.2* (Reading  $\tilde{x} \in \mathbb{R}^p$ ). Writing  $\tilde{x} \in \mathbb{R}^p$  says, in one symbol, that  $\tilde{x}$  is a column vector with  $p$  real entries. For example,  $(1, -2, 0.5) \in \mathbb{R}^3$ . The vector  $(1, 2)$  is not in  $\mathbb{R}^3$ , because it has only 2 entries; it is in  $\mathbb{R}^2$ . In Exercise [0.1](#),  $\tilde{x} \in \mathbb{R}^3$ .

**Definition 0.3** (Transpose). The **transpose** of a column vector  $\tilde{x}$  is the row vector with the same sequence of entries, written horizontally:

$$\tilde{x}^\top \equiv \tilde{x}' \equiv [x_1, x_2, \dots, x_p]$$

**Example 0.1** (Transposing a column vector). The transpose of the column vector with entries 2, -1, 5 is the row vector with the same entries:

$$\begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}^\top = [2, -1, 5].$$

**Definition 0.4** (Vector addition). The **sum** of two column vectors  $\tilde{x}$  and  $\tilde{y}$  of the same length  $p$  is the column vector  $\tilde{x} + \tilde{y}$  of length  $p$  obtained by adding entry by entry:

$$(\tilde{x} + \tilde{y})_i \stackrel{\text{def}}{=} x_i + y_i, \quad i = 1, \dots, p$$

**Example 0.2** (Adding two vectors).

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1+4 \\ 2+5 \\ 3+6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

**Definition 0.5** (Dot product). For any two real-valued vectors  $\tilde{x} = (x_1, \dots, x_p)$  and  $\tilde{y} = (y_1, \dots, y_p)$  of the same length  $p$ , the **dot product** of  $\tilde{x}$  and  $\tilde{y}$  is:

$$\tilde{x} \cdot \tilde{y} = \tilde{x}^\top \tilde{y} \stackrel{\text{def}}{=} \sum_{i=1}^p x_i y_i \quad (1)$$

See also the definitions in:

- Dobson and Barnett (2018), Section 1.3 (equation 1.1, page 7)
- Kaplan (2022), chapter on vectors
- [wikipedia](#)

The dot product has a different generalization for two matrices; see [wikipedia](#) for more.

**Example 0.3** (A dot product). For  $\tilde{x} = (1, 2, 3)$  and  $\tilde{y} = (4, 5, 6)$ :

$$\begin{aligned} \tilde{x} \cdot \tilde{y} &= 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 && \text{(definition of the dot product)} \\ &= 4 + 10 + 18 && \text{(multiply)} \\ &= 32 && \text{(add)} \end{aligned}$$

**Definition 0.6** (Linear combination). A **linear combination** of the numbers  $a_1, \dots, a_k$  with **coefficients**  $c_1, \dots, c_k$  is the weighted sum

$$c_1 a_1 + \dots + c_k a_k = \sum_{i=1}^k c_i a_i$$

A linear combination of vectors  $\tilde{v}_1, \dots, \tilde{v}_k$  of the same length is defined entry by entry: entry  $j$  of  $\sum_{i=1}^k c_i \tilde{v}_i$  is the linear combination of the  $j$ th entries of  $\tilde{v}_1, \dots, \tilde{v}_k$  with the same coefficients. For example, the linear combination of  $(1, 0)^\top$  and  $(0, 1)^\top$  with coefficients 2 and 3 is  $(2 \cdot 1 + 3 \cdot 0, 2 \cdot 0 + 3 \cdot 1)^\top = (2, 3)^\top$ .

*Remark 0.3* (The dot product is a linear combination). The dot product  $\tilde{x} \cdot \tilde{y}$  is a linear combination (Definition 0.6) of the entries of  $\tilde{y}$ , with coefficients  $x_1, \dots, x_p$ . In Example 0.3,

$\tilde{x} \cdot \tilde{y} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6$  is the linear combination of 4, 5, and 6 with coefficients 1, 2, and 3.

The dot product is also the standard *inner product* on  $\mathbb{R}^p$ ; “inner product” is the general notion, of which the dot product is one example.

**Theorem 0.1** (Dot product is symmetric). *The dot product is symmetric:*

$$\tilde{x} \cdot \tilde{y} = \tilde{y} \cdot \tilde{x}$$

**i** Proof

*Proof.*

$$\begin{aligned} \tilde{x} \cdot \tilde{y} &= \sum_{i=1}^p x_i y_i && \text{(definition of the dot product)} \\ &= \sum_{i=1}^p y_i x_i && \text{(products of numbers are symmetric)} \\ &= \tilde{y} \cdot \tilde{x} && \text{(definition of the dot product)} \end{aligned}$$

□

## Special vectors

**Definition 0.7** (Zero vector). The **zero vector**  $\tilde{0}$  of length  $p$  has all entries equal to zero:

$$\tilde{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

*Remark 0.4* (The zero vector is the additive identity). Adding the zero vector to a vector leaves it unchanged (Definition 0.4):  $\tilde{x} + \tilde{0} = \tilde{x}$  for any vector  $\tilde{x}$  of the same length. For example,  $(2, -1) + (0, 0) = (2 + 0, -1 + 0) = (2, -1)$ .

**Definition 0.8** (Ones vector). The **ones vector**  $\tilde{1}$  of length  $p$  has all entries equal to one:

$$\tilde{\mathbf{1}} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

**Remark 0.5** (The ones vector sums the entries). The dot product (Definition 0.5) of  $\tilde{\mathbf{1}}$  with  $\tilde{x}$  is the sum of the entries of  $\tilde{x}$ :  $\tilde{\mathbf{1}} \cdot \tilde{x} = \sum_{i=1}^p 1 \cdot x_i = \sum_{i=1}^p x_i$ . For example, for  $\tilde{x} = (2, -1, 5)$ ,  $\tilde{\mathbf{1}} \cdot \tilde{x} = 2 + (-1) + 5 = 6$ , and dividing by the length  $p = 3$  gives the mean of the entries,  $6/3 = 2$ .

**Definition 0.9** (Indicator vector / standard basis vector). The  $j$ -th **indicator vector** (or *standard basis vector*)  $\tilde{e}_j$  of length  $p$  has a 1 in position  $j$  and 0s elsewhere:

$$(\tilde{e}_j)_i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \tilde{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow \text{position } j$$

**Remark 0.6** (Indicator vectors build every vector). For  $p = 3$ , the indicator vectors are  $\tilde{e}_1 = (1, 0, 0)$ ,  $\tilde{e}_2 = (0, 1, 0)$ , and  $\tilde{e}_3 = (0, 0, 1)$ . Every vector of length 3 is a weighted sum  $c_1\tilde{e}_1 + c_2\tilde{e}_2 + c_3\tilde{e}_3$  of them, with weights equal to its entries; for example,  $(2, -1, 5) = 2\tilde{e}_1 - 1\tilde{e}_2 + 5\tilde{e}_3$ . No other weights work: entry  $j$  of  $c_1\tilde{e}_1 + c_2\tilde{e}_2 + c_3\tilde{e}_3$  is  $c_j$ , so each weight  $c_j$  must equal entry  $j$  of the vector.

**Theorem 0.2** (Indicator vectors select entries). *For any vector  $\tilde{x}$  of length  $p$  and any  $j \in \{1, \dots, p\}$ :*

$$\tilde{e}_j \cdot \tilde{x} = x_j$$

### **i** Proof

*Proof.* Writing the dot product componentwise:

$$\begin{aligned}
\tilde{e}_j \cdot \tilde{x} &= \sum_{i=1}^p (\tilde{e}_j)_i x_i && \text{(definition of the dot product)} \\
&= \sum_{i=1}^p \begin{cases} 1 \cdot x_i & \text{if } i = j \\ 0 \cdot x_i & \text{if } i \neq j \end{cases} && \text{(definition of } \tilde{e}_j \text{)} \\
&= x_j && \text{(only the } i = j \text{ term is nonzero)}
\end{aligned}$$

□

## Orthogonality

**Definition 0.10** (Orthogonal vectors). Two vectors  $\tilde{x}$  and  $\tilde{y}$  of the same length are **orthogonal** (written  $\tilde{x} \perp \tilde{y}$ ) if their dot product (Definition 0.5) is zero:

$$\tilde{x} \perp \tilde{y} \iff \tilde{x} \cdot \tilde{y} = 0$$

*Remark 0.7* (Orthogonal means perpendicular). Orthogonality extends the geometric idea of perpendicular lines to vectors with any number of entries. In the plane,  $(1, 2)$  and  $(-2, 1)$  are perpendicular, and their dot product is  $1 \cdot (-2) + 2 \cdot 1 = 0$ . The same test works in three dimensions, where pictures are harder to draw:  $(1, 1, 0) \cdot (1, -1, 5) = 1 - 1 + 0 = 0$ , so  $(1, 1, 0) \perp (1, -1, 5)$ .

**Definition 0.11** (Euclidean norm). The **Euclidean norm** (or *length*, also called the  $L_2$  norm and written  $\|\tilde{x}\|_2$ ) of a vector  $\tilde{x}$  of length  $p$  is

$$\|\tilde{x}\| \stackrel{\text{def}}{=} \sqrt{\tilde{x} \cdot \tilde{x}} = \sqrt{\sum_{i=1}^p x_i^2} \quad (2)$$

**Example 0.4** (The length of a vector). For  $\tilde{x} = (3, 4)$ :

$$\begin{aligned}
\|\tilde{x}\| &= \sqrt{3^2 + 4^2} && \text{(definition of the norm)} \\
&= \sqrt{25} && \text{(square and add)} \\
&= 5
\end{aligned}$$

The vector  $(0.6, 0.8)$  has norm  $\sqrt{0.36 + 0.64} = 1$ .

**Exercise 0.2** (How far apart are two points?). Let  $\tilde{x} = (1, 2)$  and  $\tilde{y} = (4, 6)$ .

1. Compute the vector  $\tilde{x} - \tilde{y}$ .
2. Compute the length of  $\tilde{x} - \tilde{y}$ , using the [Euclidean norm](#).
3. Is the result the same if you compute the length of  $\tilde{y} - \tilde{x}$  instead?

Solution

*Solution 0.2.*

1.  $\tilde{x} - \tilde{y} = (1 - 4, 2 - 6) = (-3, -4)$ .
2.  $\|\tilde{x} - \tilde{y}\| = \sqrt{(-3)^2 + (-4)^2} = \sqrt{25} = 5$ .
3.  $\tilde{y} - \tilde{x} = (3, 4)$ , and  $\|\tilde{y} - \tilde{x}\| = \sqrt{3^2 + 4^2} = 5$ . Yes, the result is the same.

**Definition 0.12** (Euclidean distance). The **Euclidean distance** between two vectors  $\tilde{x}$  and  $\tilde{y}$  of length  $p$  is the [Euclidean norm](#) of their difference:

$$d(\tilde{x}, \tilde{y}) \stackrel{\text{def}}{=} \|\tilde{x} - \tilde{y}\| = \sqrt{\sum_{i=1}^p (x_i - y_i)^2} \quad (3)$$

*Remark 0.8* (Distance from the origin). The distance from the origin  $\tilde{0}$  to a vector  $\tilde{x}$  is the norm of  $\tilde{x}$ :  $d(\tilde{x}, \tilde{0}) = \|\tilde{x} - \tilde{0}\| = \|\tilde{x}\|$  (Goodfellow et al. 2016, 39).

For example, the distance from  $\tilde{0}$  to  $(3, 4)$  is  $\|(3, 4)\| = 5$ , as in [Example 0.4](#).

**Definition 0.13** (Orthonormal vectors). A set of vectors  $\{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_k\}$  is **orthonormal** if the vectors are mutually orthogonal ([Definition 0.10](#)) and each has norm 1 ([Definition 0.11](#)), that is, unit length:

$$\tilde{x}_i \cdot \tilde{x}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

**Example 0.5** (Indicator vectors are orthonormal). The indicator vectors  $\tilde{e}_1, \tilde{e}_2, \dots, \tilde{e}_p$  ([Definition 0.9](#)) form an orthonormal set. By [Theorem 0.2](#),  $\tilde{e}_i \cdot \tilde{e}_j$  is entry  $i$  of  $\tilde{e}_j$ , which is 1 if  $i = j$  and 0 if  $i \neq j$ . For  $p = 2$ :  $\tilde{e}_1 \cdot \tilde{e}_1 = 1 \cdot 1 + 0 \cdot 0 = 1$ ,  $\tilde{e}_2 \cdot \tilde{e}_2 = 0 \cdot 0 + 1 \cdot 1 = 1$ , and  $\tilde{e}_1 \cdot \tilde{e}_2 = 1 \cdot 0 + 0 \cdot 1 = 0$ .

**Exercise 0.3** (Compute an inner product and a norm). Let

$$\tilde{a} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \quad \tilde{b} = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$$

Compute  $\tilde{a}^\top \tilde{b}$  and  $\|\tilde{a}\|_2$ .

Solution

*Solution 0.3.* Multiply entry by entry and add:

$$\tilde{a}^\top \tilde{b} = (3)(0) + (-1)(4) + (2)(-2) = 0 - 4 - 4 = -8$$

For the norm, take the inner product of  $\tilde{a}$  with itself first:

$$\tilde{a}^\top \tilde{a} = 3^2 + (-1)^2 + 2^2 = 9 + 1 + 4 = 14$$

so

$$\|\tilde{a}\|_2 = \sqrt{14} \approx 3.742$$

The inner product came out negative while the norm cannot: a norm is a square root of a sum of squares, so it is never below zero.

Python has both operations built in, and R builds them from `sum`.

```
import numpy as np

a = np.array([3.0, -1.0, 2.0])
b = np.array([0.0, 4.0, -2.0])

print(a @ b, round(float(np.linalg.norm(a)), 3))
```

```
a <- c(3, -1, 2)
b <- c(0, 4, -2)

print(c(sum(a * b), round(sqrt(sum(a^2)), 3)))
#> [1] -8.000 3.742
```

Both print  $-8$  and  $3.742$ .

💡 Video lecture

Hutchinson's [Linear Function Basics](#) (24 min) covers dot products and the Euclidean norm, and goes on to hyperplanes and level sets (Hutchinson, n.d.). The login for the video site is posted [on Canvas](#).

## Matrices

**Definition 0.14** (Matrix). A **matrix** of dimensions  $m \times n$  is a rectangular array of  $m \cdot n$  numbers, arranged in  $m$  rows and  $n$  columns:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

*Remark 0.9* (Matrix entries, and vectors as matrices). The entry in row  $i$  and column  $j$  of  $\mathbf{A}$  is written  $a_{ij}$  or  $(\mathbf{A})_{ij}$ . For example, for the  $2 \times 3$  matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix},$$

$a_{12} = 2$  and  $a_{23} = 6$ .

A column vector of length  $p$  is a  $p \times 1$  matrix, and a row vector of length  $p$  is a  $1 \times p$  matrix. For example, the column vector  $(7, 8)$  is a  $2 \times 1$  matrix, and the row vector  $[7, 8]$  is a  $1 \times 2$  matrix.

**Definition 0.15** (Tensor). A **tensor** of order  $k \geq 1$  is an array of numbers with  $k$  indices,

$$\mathcal{A} = (a_{i_1 i_2 \dots i_k}), \quad i_1 = 1, \dots, n_1, \dots, i_k = 1, \dots, n_k,$$

and its dimensions are  $n_1 \times n_2 \times \cdots \times n_k$ .

**Example 0.6** (Vectors, matrices and images as tensors). A vector of length  $p$  is a tensor of order 1, and an  $m \times n$  [matrix](#) is a tensor of order 2. A  $28 \times 28$  grayscale image is a matrix: the entry  $a_{ij}$  is the brightness of the pixel in row  $i$  and column  $j$ . A  $28 \times 28$  color image stores three numbers per pixel (red, green, blue), so it is an order-3 tensor with dimensions  $28 \times 28 \times 3$ , and  $a_{ijc}$  is the value of color channel  $c$  at the pixel in row  $i$  and column  $j$ . It has  $28 \cdot 28 \cdot 3 = 2352$  entries.

## Matrix transpose

**Definition 0.16** (Matrix transpose). The **transpose** of an  $m \times n$  matrix  $\mathbf{A}$  is the  $n \times m$  matrix  $\mathbf{A}^\top$  obtained by swapping the rows and columns of  $\mathbf{A}$ :

$$(\mathbf{A}^\top)_{ij} = a_{ji}$$

## Matrix addition

**Definition 0.17** (Zero matrix). The  $m \times n$  **zero matrix**  $\mathbf{0}_{m \times n}$  (or  $\mathbf{0}$  when dimensions are clear from context) has all entries equal to zero:

$$\mathbf{0}_{m \times n} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

**Definition 0.18** (Matrix addition). Two matrices  $\mathbf{A}$  and  $\mathbf{B}$  of the same dimensions  $m \times n$  can be added element-wise; their **matrix sum** is:

$$(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}$$

**Theorem 0.3** (Matrix addition is commutative). *For  $m \times n$  matrices  $\mathbf{A}$  and  $\mathbf{B}$ :*

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

**Theorem 0.4** (Matrix addition is associative). *For  $m \times n$  matrices  $\mathbf{A}$ ,  $\mathbf{B}$ , and  $\mathbf{C}$ :*

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$$

**Theorem 0.5** (Zero matrix is the additive identity). *For an  $m \times n$  matrix  $\mathbf{A}$ :*

$$\mathbf{A} + \mathbf{0}_{m \times n} = \mathbf{A}$$

**Theorem 0.6** (Additive inverse). *For any  $m \times n$  matrix  $\mathbf{A}$ , the  $m \times n$  matrix  $-\mathbf{A}$  (defined by  $(-\mathbf{A})_{ij} = -a_{ij}$ ) satisfies:*

$$\mathbf{A} + (-\mathbf{A}) = \mathbf{0}_{m \times n}$$

## Scalar multiplication

**Definition 0.19** (Scalar multiplication). The **scalar multiple** of a matrix  $\mathbf{A}$  by a scalar  $c$  is:

$$(c\mathbf{A})_{ij} = c \cdot a_{ij}$$

## Matrix multiplication

**Definition 0.20** (Matrix multiplication). The **product** of an  $m \times k$  matrix  $\mathbf{A}$  and a  $k \times n$  matrix  $\mathbf{B}$  is the  $m \times n$  matrix  $\mathbf{C} = \mathbf{AB}$  with entries:

$$c_{ij} = \sum_{s=1}^k a_{is} b_{sj}$$

*Remark 0.10* (Matrix multiplication is not commutative). When  $\mathbf{AB}$  and  $\mathbf{BA}$  are both defined, they are usually different. For example, with

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

$$\mathbf{AB} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \mathbf{BA}.$$

**Example 0.7** (Dot product as matrix multiplication). The dot product of two column vectors  $\tilde{x}$  and  $\tilde{\beta}$  can be written as a matrix product of the row vector  $\tilde{x}^\top$  with the column vector  $\tilde{\beta}$ :

$$\begin{aligned}
\tilde{x} \cdot \tilde{\beta} &= \tilde{x}^\top \tilde{\beta} \\
&= [x_1, x_2, \dots, x_p] \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} \\
&= x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p
\end{aligned}$$

**Theorem 0.7** (Matrix multiplication is associative). *For an  $m \times k$  matrix  $\mathbf{A}$ , a  $k \times l$  matrix  $\mathbf{B}$ , and an  $l \times n$  matrix  $\mathbf{C}$ :*

$$\underbrace{(\mathbf{AB})\mathbf{C}}_{m \times n} = \underbrace{\mathbf{A}(\mathbf{BC})}_{m \times n}$$

### **i** Proof

*Proof.* Entry  $(i, j)$  of each side, from Definition 0.20:

$$\begin{aligned}
[(\mathbf{AB})\mathbf{C}]_{ij} &= \sum_{t=1}^l (\mathbf{AB})_{it} c_{tj} && \text{(definition of } (\mathbf{AB})\mathbf{C} \text{)} \\
&= \sum_{t=1}^l \left( \sum_{s=1}^k a_{is} b_{st} \right) c_{tj} && \text{(definition of } \mathbf{AB} \text{)} \\
&= \sum_{s=1}^k a_{is} \left( \sum_{t=1}^l b_{st} c_{tj} \right) && \text{(distribute and swap the finite sums)} \\
&= \sum_{s=1}^k a_{is} (\mathbf{BC})_{sj} && \text{(definition of } \mathbf{BC} \text{)} \\
&= [\mathbf{A}(\mathbf{BC})]_{ij} && \text{(definition of } \mathbf{A}(\mathbf{BC}) \text{)}
\end{aligned}$$

□

**Theorem 0.8** (Matrix multiplication is distributive over addition). *For an  $m \times k$  matrix  $\mathbf{A}$  and  $k \times n$  matrices  $\mathbf{B}$  and  $\mathbf{C}$ :*

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

*For  $m \times k$  matrices  $\mathbf{A}$  and  $\mathbf{B}$  and a  $k \times n$  matrix  $\mathbf{C}$ :*

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$$

**Exercise 0.4** (Which of these products are defined?). Let  $\mathbf{A}$  be  $4 \times 3$ , let  $\mathbf{B}$  be  $3 \times 2$ , and let  $\tilde{x} \in \mathbb{R}^3$ . For each expression, say whether it is defined, and if so give the shape of the result:

1.  $\mathbf{AB}$
2.  $\mathbf{BA}$
3.  $\mathbf{A}\tilde{x}$
4.  $\tilde{x}^\top \tilde{x}$
5.  $\tilde{x}\tilde{x}^\top$

#### Solution

*Solution 0.4.* A vector in  $\mathbb{R}^3$  is a  $3 \times 1$  matrix, so the same rule settles every case.

1.  $\mathbf{AB}$ :  $(4 \times 3)(3 \times 2)$ . The inner pair is 3 and 3, so it is defined, and the result is  $4 \times 2$ .
2.  $\mathbf{BA}$ :  $(3 \times 2)(4 \times 3)$ . The inner pair is 2 and 4, which do not match, so it is **not defined**. Matrix multiplication is not commutative, and this is the blunt form of that: swapping the order can leave an expression that means nothing.
3.  $\mathbf{A}\tilde{x}$ :  $(4 \times 3)(3 \times 1)$ , defined, result  $4 \times 1$  — a vector in  $\mathbb{R}^4$ .
4.  $\tilde{x}^\top \tilde{x}$ :  $(1 \times 3)(3 \times 1)$ , defined, result  $1 \times 1$  — a single number. The product is the inner product of Equation 1.
5.  $\tilde{x}\tilde{x}^\top$ :  $(3 \times 1)(1 \times 3)$ , defined, result  $3 \times 3$  — a matrix.

The last two use the same two vectors and differ only in order, and they return objects of different kinds. Reading a transpose as decoration rather than as a shape change is the commonest way to lose track of an expression.

### Matrix-vector multiplication

**Definition 0.21** (Matrix-vector multiplication). The **matrix-vector product** of an  $m \times p$  matrix  $\mathbf{A}$  and a  $p \times 1$  column vector  $\tilde{x}$  is the  $m \times 1$  column vector  $\mathbf{A}\tilde{x}$  with entries:

$$(\mathbf{A}\tilde{x})_i = \sum_{j=1}^p a_{ij} x_j$$

*Remark 0.11* (Each entry is a dot product). Matrix-vector multiplication extends the dot product (Definition 0.5): entry  $i$  of  $\mathbf{A}\tilde{x}$  is the dot product of row  $i$  of  $\mathbf{A}$  with  $\tilde{x}$ . For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot (-1) \\ 3 \cdot 1 + 4 \cdot (-1) \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

## Linear and affine maps

**Definition 0.22** (Linear map). A function  $f : \mathbb{R}^p \rightarrow \mathbb{R}^m$  is a **linear map** (or linear transformation) if it preserves vector addition and scalar multiplication: for all  $\tilde{x}, \tilde{y} \in \mathbb{R}^p$  and  $c \in \mathbb{R}$ ,

1.  $f(\tilde{x} + \tilde{y}) = f(\tilde{x}) + f(\tilde{y})$
2.  $f(c\tilde{x}) = cf(\tilde{x})$

Every such function can be represented as multiplication by an  $m \times p$  matrix  $\mathbf{A}$ :

$$f(\tilde{x}) = \mathbf{A}\tilde{x}$$

A linear map always maps the zero vector to the zero vector:  $f(\tilde{0}) = \tilde{0}$ .

**Definition 0.23** (Affine transformation). An **affine transformation** is a function  $f : \mathbb{R}^p \rightarrow \mathbb{R}^m$  composed of a linear map followed by a translation:

$$f(\tilde{x}) = \mathbf{A}\tilde{x} + \tilde{b}, \quad \mathbf{A} \in \mathbb{R}^{m \times p}, \quad \tilde{b} \in \mathbb{R}^m$$

For a scalar-valued function ( $m = 1$ ), this is an inner product plus an offset:

$$f(\tilde{x}) = \tilde{w}^\top \tilde{x} + b, \quad \tilde{w} \in \mathbb{R}^p, \quad b \in \mathbb{R} \quad (4)$$

When  $\tilde{b} \neq \tilde{0}$  (or  $b \neq 0$ ), an affine transformation does not send  $\tilde{0}$  to  $\tilde{0}$ , so it is not strictly linear. In machine learning and statistics, functions of this form are commonly referred to as **linear models** or linear functions, with  $\tilde{b}$  (or  $b$ ) called the **bias** or **intercept**.

**Definition 0.24** (Hyperplane). A **hyperplane** in  $\mathbb{R}^p$  is an affine subspace of dimension  $p - 1$ . It is defined as the set of points  $\tilde{x} \in \mathbb{R}^p$  satisfying:

$$\tilde{w}^\top \tilde{x} + b = 0$$

where  $\tilde{w} \in \mathbb{R}^p \setminus \{\tilde{0}\}$  is the normal vector orthogonal to the hyperplane and  $b \in \mathbb{R}$  is the offset. When  $b = 0$ , the hyperplane passes through the origin and forms a  $(p - 1)$ -dimensional linear subspace.

Geometrically:

- In  $\mathbb{R}^2$  ( $p = 2$ ), a hyperplane is a 1-dimensional line:  $w_1x_1 + w_2x_2 + b = 0$ .
- In  $\mathbb{R}^3$  ( $p = 3$ ), a hyperplane is a 2-dimensional plane.
- A hyperplane divides the space  $\mathbb{R}^p$  into two open **half-spaces**:  $\{\tilde{x} : \tilde{w}^\top \tilde{x} + b > 0\}$  and  $\{\tilde{x} : \tilde{w}^\top \tilde{x} + b < 0\}$ .

The course website's version of this page has an interactive plot here.

**Exercise 0.5** (Write a rule in matrix form). A clinic scores a patient's risk from three measurements.

| symbol | meaning                 | units             |
|--------|-------------------------|-------------------|
| $x_1$  | age                     | years             |
| $x_2$  | systolic blood pressure | mmHg              |
| $x_3$  | body mass index         | kg/m <sup>2</sup> |

The rule in use is

$$\text{score} = 0.4x_1 + 0.2x_2 + 1.5x_3 - 30$$

Write this rule in the form of Equation 4, naming  $\tilde{w}$ ,  $b$  and  $p$  explicitly, and give the score for a patient aged 50 with blood pressure 130 and BMI 28.

### Solution

*Solution 0.5.* Read the coefficients straight off the rule. There are three inputs, so  $p = 3$ :

$$\tilde{w} = \begin{bmatrix} 0.4 \\ 0.2 \\ 1.5 \end{bmatrix}, \quad b = -30$$

and the rule is  $f(\tilde{x}) = \tilde{w}^\top \tilde{x} + b$ . For the patient,

$$\tilde{x} = \begin{bmatrix} 50 \\ 130 \\ 28 \end{bmatrix}$$

$$\tilde{w}^\top \tilde{x} = (0.4)(50) + (0.2)(130) + (1.5)(28) = 20 + 26 + 42 = 88$$

$$f(\tilde{x}) = 88 - 30 = 58$$

Two things are worth noticing. The coefficients are not comparable as they stand: 1.5 is the largest number but BMI is measured on the smallest scale, so the weight on a feature

says nothing on its own about how much that feature matters. And  $b = -30$  is doing real work — without it the score would be 88, which is a different clinical claim entirely.

**Exercise 0.6** (Which of these are linear in  $\tilde{w}$ ?). Each of the five expressions is a function of the weights  $\tilde{w} \in \mathbb{R}^p$ , with the data  $\tilde{x}$  and  $y$  held fixed. Say which ones have the form Equation 4 — an inner product plus an offset — and which do not.

1.  $\tilde{w}^\top \tilde{x}$
2.  $\tilde{w}^\top \tilde{x} - y$
3.  $(\tilde{w}^\top \tilde{x} - y)^2$
4.  $\|\tilde{w}\|_2^2$
5.  $w_1 x_1 + 3$

### Solution

*Solution 0.6.*

1. **Linear.** It is Equation 4 with  $b = 0$ .
2. **Linear.** Still an inner product plus an offset; here the offset is  $b = -y$ , which is a constant because  $y$  is held fixed.
3. **Not linear.** Squaring is not an inner product plus an offset. This expression is the squared error of one observation, and its curvature in  $\tilde{w}$  is exactly why fitting a model is an optimization problem rather than a lookup.
4. **Not linear.** By Equation 2 this is  $\tilde{w}^\top \tilde{w}$ , which has  $\tilde{w}$  in both slots at once. Hold either slot fixed and it is linear in the other; as a function of the single variable  $\tilde{w}$  it is quadratic.
5. **Linear.** It is Equation 4 with coefficient vector  $\tilde{a} = (x_1, 0, \dots, 0)^\top$  and  $b = 3$ . The data  $\tilde{x}$  is untouched —  $\tilde{a}$  is a new vector built from its first entry.

Items 1, 2 and 5 are linear and items 3 and 4 are not, which is the split that matters: a *model* can be linear in its weights while the *quantity we minimize* is not. Squared error and the norm penalty are both curved in  $\tilde{w}$ , and that curvature is what makes fitting an optimization problem at all: there is always a downhill direction to follow. Curvature alone does not promise a *single* answer. The squared error of one observation is completely flat along a whole line of weight vectors, every one of which fits that observation exactly, and pinning down one of them is part of what the norm penalty is added for in week 3.

## Transposes of sums and products

**Theorem 0.9** (Transpose of a sum). *For  $m \times n$  matrices  $\mathbf{A}$  and  $\mathbf{B}$ :*

$$(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$$

*In particular, for column vectors  $\tilde{x}$  and  $\tilde{y}$  of the same length:*

$$(\tilde{x} + \tilde{y})^\top = \tilde{x}^\top + \tilde{y}^\top$$

### Proof

*Proof.* Entry  $(i, j)$  of each side:

$$\begin{aligned} [(\mathbf{A} + \mathbf{B})^\top]_{ij} &= (\mathbf{A} + \mathbf{B})_{ji} && \text{(definition of the transpose)} \\ &= a_{ji} + b_{ji} && \text{(definition of matrix addition)} \\ &= (\mathbf{A}^\top)_{ij} + (\mathbf{B}^\top)_{ij} && \text{(definition of the transpose)} \\ &= [\mathbf{A}^\top + \mathbf{B}^\top]_{ij} && \text{(definition of matrix addition)} \end{aligned}$$

□

**Theorem 0.10** (Transpose of a product). *For an  $m \times k$  matrix  $\mathbf{A}$  and a  $k \times n$  matrix  $\mathbf{B}$ :*

$$\underbrace{(\mathbf{AB})^\top}_{n \times m} = \underbrace{\mathbf{B}^\top}_{n \times k} \underbrace{\mathbf{A}^\top}_{k \times m}$$

### Proof

*Proof.* Entry  $(i, j)$  of each side:

$$\begin{aligned} [(\mathbf{AB})^\top]_{ij} &= (\mathbf{AB})_{ji} && \text{(definition of the transpose)} \\ &= \sum_{s=1}^k a_{js} b_{si} && \text{(definition of matrix multiplication)} \\ &= \sum_{s=1}^k (\mathbf{B}^\top)_{is} (\mathbf{A}^\top)_{sj} && \text{(definition of the transpose; reorder each product)} \\ &= [\mathbf{B}^\top \mathbf{A}^\top]_{ij} && \text{(definition of matrix multiplication)} \end{aligned}$$

□

*Remark 0.12* (The order of the factors reverses). Transposing a product reverses the order of the factors. Keeping the original order usually gives an undefined product or a different matrix. For example, with  $\mathbf{A} = [1, 2]$  ( $1 \times 2$ ) and  $\mathbf{B} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$  ( $2 \times 1$ ),  $\mathbf{AB} = 1 \cdot 3 + 2 \cdot 4 = 11$ , so  $(\mathbf{AB})^\top = 11$ , and

$$\mathbf{B}^\top \mathbf{A}^\top = [3, 4] \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 11, \quad \mathbf{A}^\top \mathbf{B}^\top = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [3, 4] = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}.$$

If instead  $\mathbf{A}$  is  $2 \times 3$  and  $\mathbf{B}$  is  $3 \times 1$ , then  $\mathbf{B}^\top \mathbf{A}^\top$  is a  $(1 \times 3)(3 \times 2)$  product, but  $\mathbf{A}^\top \mathbf{B}^\top$  is a  $(3 \times 2)(1 \times 3)$  product, which is undefined.

## Rank

**Definition 0.25** (Linearly independent vectors). Vectors  $\tilde{v}_1, \dots, \tilde{v}_k$  of the same length  $p$  are **linearly independent** if the only numbers  $c_1, \dots, c_k$  with

$$c_1 \tilde{v}_1 + \dots + c_k \tilde{v}_k = \tilde{0}$$

are  $c_1 = \dots = c_k = 0$ .

**Example 0.8** (Independent and dependent pairs of vectors).

- $\tilde{v}_1 = (1, 0)$  and  $\tilde{v}_2 = (1, 1)$  are linearly independent:  $c_1 \tilde{v}_1 + c_2 \tilde{v}_2 = (c_1 + c_2, c_2)$ , which is  $\tilde{0}$  only if  $c_2 = 0$  and then  $c_1 = 0$ .
- $\tilde{v}_1 = (1, 2)$  and  $\tilde{v}_2 = (2, 4)$  are not:  $2\tilde{v}_1 - \tilde{v}_2 = \tilde{0}$ .

**Definition 0.26** (Rank). The **rank** of a matrix  $\mathbf{A}$ , written  $\text{rank}(\mathbf{A})$ , is the largest number of columns of  $\mathbf{A}$  that are linearly independent (Definition 0.25).

**Example 0.9** (The rank of two  $3 \times 2$  matrices). The matrix  $\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$  has rank 2: if

$c_1(1, 1, 1) + c_2(1, 2, 3) = \tilde{0}$ , then subtracting the first entry from the second gives  $c_2 = 0$ , and then  $c_1 = 0$ .

The matrix  $\begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$  has rank 1: its second column is twice its first, so the two columns are not linearly independent, but the first column on its own is.

**Definition 0.27** (Full column rank). An  $n \times p$  matrix has **full column rank** if its rank (Definition 0.26) is  $p$ , so that all of its columns are linearly independent.

*Remark 0.13* (Full column rank needs at least as many rows as columns). In Example 0.9, the first matrix has full column rank: its rank is 2, and it has 2 columns. The second does not, because its rank is 1.

An  $n \times p$  matrix can have full column rank only when  $p \leq n$ , because more than  $n$  vectors of length  $n$  are never linearly independent. For example, the  $2 \times 3$  matrix  $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$  has 3 columns of length 2, and  $1 \cdot (1, 0) + 1 \cdot (0, 1) - 1 \cdot (1, 1) = \tilde{0}$ , with coefficients that are not all zero. So its 3 columns are not linearly independent, and its rank is less than 3.

## Subspaces

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**Definition 0.28** (Subspace). A set  $\mathcal{S}$  of vectors in  $\mathbb{R}^p$  is a **subspace** of  $\mathbb{R}^p$  if it is not empty and it is closed under vector addition and scalar multiplication:

1. if  $\tilde{u} \in \mathcal{S}$  and  $\tilde{v} \in \mathcal{S}$ , then  $\tilde{u} + \tilde{v} \in \mathcal{S}$ ;
2. if  $\tilde{u} \in \mathcal{S}$  and  $c \in \mathbb{R}$ , then  $c\tilde{u} \in \mathcal{S}$ .

A subspace is also called a **linear subspace**, a **vector space**, or a **linear space**.

**Example 0.10** (A line through the origin is a subspace). Let  $\mathcal{S} = \{c(1, 2) : c \in \mathbb{R}\}$ , the line in  $\mathbb{R}^2$  through  $(0, 0)$  and  $(1, 2)$ .

- **Addition:**  $a(1, 2) + b(1, 2) = (a + b)(1, 2)$ , which is in  $\mathcal{S}$ .
- **Scalar multiplication:**  $c(a(1, 2)) = (ca)(1, 2)$ , which is in  $\mathcal{S}$ .

So  $\mathcal{S}$  is a subspace of  $\mathbb{R}^2$ . Two other subspaces of  $\mathbb{R}^p$  are the set  $\{\tilde{0}\}$  that holds only the zero vector, and  $\mathbb{R}^p$  itself.

**Example 0.11** (A line that misses the origin is not a subspace). Let  $\mathcal{T} = \{(x, 1) : x \in \mathbb{R}\}$ , the horizontal line in  $\mathbb{R}^2$  at height 1. The vectors  $(0, 1)$  and  $(2, 1)$  are in  $\mathcal{T}$ , but their sum  $(2, 2)$  is not, because its second entry is 2, not 1. So  $\mathcal{T}$  is not closed under addition, and it is not a subspace.

**Theorem 0.11** (Every subspace contains the zero vector). *If  $\mathcal{S}$  is a subspace of  $\mathbb{R}^p$  (Definition 0.28), then  $\tilde{0} \in \mathcal{S}$ .*

### **i** Proof

*Proof.* A subspace is not empty, so it contains some vector  $\tilde{u}$ . Closure under scalar multiplication with  $c = 0$  puts  $0 \cdot \tilde{u}$  in  $\mathcal{S}$ , and  $0 \cdot \tilde{u} = \tilde{0}$ . □

**Example 0.12** (Using the zero vector to rule out a subspace). The plane  $\{(x, y, z) : x + y + z = 1\}$  in  $\mathbb{R}^3$  does not contain  $\tilde{0} = (0, 0, 0)$ , because  $0 + 0 + 0 = 0 \neq 1$ . So, by Theorem 0.11, it is not a subspace. The line  $\mathcal{T}$  in Example 0.11 fails the same test:  $(0, 0)$  is not in  $\mathcal{T}$ , because its second entry is not 1. Containing  $\tilde{0}$  is necessary but not sufficient. The union of the two coordinate axes in  $\mathbb{R}^2$ ,  $\{(x, 0) : x \in \mathbb{R}\} \cup \{(0, y) : y \in \mathbb{R}\}$ , contains  $\tilde{0}$ , but  $(1, 0) + (0, 1) = (1, 1)$  lies on neither axis, so the union is not closed under addition and is not a subspace.

**Definition 0.29** (Span). The **span** of vectors  $\tilde{v}_1, \dots, \tilde{v}_k \in \mathbb{R}^p$  is the set of all their linear combinations:

$$\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\} \stackrel{\text{def}}{=} \{c_1 \tilde{v}_1 + \dots + c_k \tilde{v}_k : c_1, \dots, c_k \in \mathbb{R}\}.$$

The vectors  $\tilde{v}_1, \dots, \tilde{v}_k$  **span** a set  $\mathcal{S}$  if their span equals  $\mathcal{S}$ .

By convention, the span of the empty list ( $k = 0$ ) is  $\{\tilde{0}\}$ , and the empty list counts as linearly independent (Definition 0.25), since it has no coefficients that could fail to be zero.

**Example 0.13** (Two spans in  $\mathbb{R}^3$ ).

- $\text{span}\{(1, 0, 0), (0, 1, 0)\}$  is the set of vectors  $(c_1, c_2, 0)$ , the plane  $z = 0$  in  $\mathbb{R}^3$ .
- $\text{span}\{(1, 2, 3), (2, 4, 6)\}$  is only the line  $\{c(1, 2, 3) : c \in \mathbb{R}\}$ , because  $(2, 4, 6) = 2(1, 2, 3)$ , so  $c_1(1, 2, 3) + c_2(2, 4, 6) = (c_1 + 2c_2)(1, 2, 3)$ . Adding a vector to a list does not always enlarge its span.

**Theorem 0.12** (A span is a subspace). *For any vectors  $\tilde{v}_1, \dots, \tilde{v}_k \in \mathbb{R}^p$ ,  $\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\}$  (Definition 0.29) is a subspace of  $\mathbb{R}^p$  (Definition 0.28).*

### **i** Proof

*Proof.* The span is not empty: it contains  $0 \cdot \tilde{v}_1 + \dots + 0 \cdot \tilde{v}_k = \tilde{0}$ .

**Addition.** Take two vectors in the span,  $\tilde{u} = a_1 \tilde{v}_1 + \dots + a_k \tilde{v}_k$  and  $\tilde{w} = b_1 \tilde{v}_1 + \dots + b_k \tilde{v}_k$ . Then

$$\begin{aligned} \tilde{u} + \tilde{w} &= (a_1 \tilde{v}_1 + \dots + a_k \tilde{v}_k) + (b_1 \tilde{v}_1 + \dots + b_k \tilde{v}_k) \quad (\text{substitute the two combinations}) \\ &= (a_1 + b_1) \tilde{v}_1 + \dots + (a_k + b_k) \tilde{v}_k \quad (\text{regroup, and distribute each } \tilde{v}_i) \end{aligned}$$

which is a linear combination of  $\tilde{v}_1, \dots, \tilde{v}_k$ , so it is in the span.

**Scalar multiplication.** For  $c \in \mathbb{R}$ ,

$$\begin{aligned} c \tilde{u} &= c(a_1 \tilde{v}_1 + \dots + a_k \tilde{v}_k) \quad (\text{substitute the combination}) \\ &= (ca_1) \tilde{v}_1 + \dots + (ca_k) \tilde{v}_k \quad (\text{distribute } c \text{ over the sum}) \end{aligned}$$

which is again in the span. □

**Example 0.14** (The line in Example 0.10 is a span). The line  $\mathcal{S} = \{c(1, 2) : c \in \mathbb{R}\}$  in Example 0.10 is  $\text{span}\{(1, 2)\}$ , so Theorem 0.12 gives a second proof that it is a subspace. Likewise, the plane  $z = 0$  in Example 0.13 is a subspace of  $\mathbb{R}^3$ , because it is

$\text{span}\{(1, 0, 0), (0, 1, 0)\}$ .

**Theorem 0.13** (A matrix-vector product combines the columns). *If  $\mathbf{A}$  is an  $m \times n$  matrix with columns  $\tilde{a}_1, \dots, \tilde{a}_n$  and  $\tilde{x} \in \mathbb{R}^n$ , then*

$$\mathbf{A}\tilde{x} = x_1\tilde{a}_1 + \dots + x_n\tilde{a}_n.$$

**i** Proof

*Proof.* Compare entry  $i$  of the two sides, for each  $i = 1, \dots, m$ . Entry  $i$  of  $\tilde{a}_j$  is  $a_{ij}$ , so

$$\begin{aligned} (\mathbf{A}\tilde{x})_i &= a_{i1}x_1 + \dots + a_{in}x_n && (\text{@def-matvec-mult}) \\ &= x_1a_{i1} + \dots + x_na_{in} && (\text{multiplication of numbers is commutative}) \\ &= (x_1\tilde{a}_1 + \dots + x_n\tilde{a}_n)_i && (\text{entry } i \text{ of each } x_j\tilde{a}_j \text{ is } x_ja_{ij}) \end{aligned}$$

□

**Example 0.15** (A product as a combination of columns).

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= 1 \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 2 \\ 4 \end{bmatrix} && (\text{@thm-matvec-columns}) \\ &= \begin{bmatrix} -1 \\ -1 \end{bmatrix}, && (\text{arithmetic}) \end{aligned}$$

the same answer as computing each entry as a row-times-vector dot product (Remark 0.11).

**Theorem 0.14** (A matrix maps a combination to the same combination of images). *If  $\mathbf{A}$  is an  $m \times n$  matrix,  $\tilde{z}_1, \dots, \tilde{z}_k \in \mathbb{R}^n$  and  $c_1, \dots, c_k \in \mathbb{R}$ , then*

$$\mathbf{A}(c_1\tilde{z}_1 + \dots + c_k\tilde{z}_k) = c_1\mathbf{A}\tilde{z}_1 + \dots + c_k\mathbf{A}\tilde{z}_k.$$

**i** Proof

*Proof.* Write  $z_{ij}$  for entry  $j$  of  $\tilde{z}_i$ , so entry  $j$  of  $\sum_i c_i\tilde{z}_i$  is  $\sum_i c_iz_{ij}$ . Compare entry  $r$  of the two sides, for each  $r = 1, \dots, m$ :

$$\begin{aligned}
\left( \mathbf{A} \sum_{i=1}^k c_i \tilde{z}_i \right)_r &= \sum_{j=1}^n a_{rj} \sum_{i=1}^k c_i z_{ij} \quad (\text{@def-matvec-mult}) \\
&= \sum_{j=1}^n \sum_{i=1}^k a_{rj} c_i z_{ij} \quad (\text{distribute } a_{rj} \text{ over the inner sum}) \\
&= \sum_{i=1}^k \sum_{j=1}^n a_{rj} c_i z_{ij} \quad (\text{reorder the finite double sum}) \\
&= \sum_{i=1}^k \sum_{j=1}^n c_i a_{rj} z_{ij} \quad (\text{multiplication of numbers is commutative}) \\
&= \sum_{i=1}^k c_i \sum_{j=1}^n a_{rj} z_{ij} \quad (\text{factor } c_i \text{ out of the inner sum}) \\
&= \sum_{i=1}^k c_i (\mathbf{A} \tilde{z}_i)_r \quad (\text{@def-matvec-mult})
\end{aligned}$$

which is entry  $r$  of  $\sum_i c_i \mathbf{A} \tilde{z}_i$ . □

**Example 0.16** (Applying a matrix to a combination). Let  $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ ,  $\tilde{z}_1 = (1, 0)$  and  $\tilde{z}_2 = (0, 1)$ . Then  $\mathbf{A} \tilde{z}_1 = (1, 3)$  and  $\mathbf{A} \tilde{z}_2 = (2, 4)$ , and for the combination  $2\tilde{z}_1 - \tilde{z}_2 = (2, -1)$ :

$$\begin{aligned}
\mathbf{A} (2, -1) &= (1 \cdot 2 + 2 \cdot (-1), 3 \cdot 2 + 4 \cdot (-1)) \quad (\text{@def-matvec-mult}) \\
&= (0, 2) \quad (\text{arithmetic}) \\
&= 2(1, 3) - (2, 4), \quad (\text{arithmetic})
\end{aligned}$$

which is  $2\mathbf{A} \tilde{z}_1 - \mathbf{A} \tilde{z}_2$ , as Theorem 0.14 says.

**Theorem 0.15** (More than  $p$  vectors in  $\mathbb{R}^p$  are linearly dependent). *If  $k > p$ , then any  $k$  vectors  $\tilde{v}_1, \dots, \tilde{v}_k \in \mathbb{R}^p$  are not linearly independent (Definition 0.25): some numbers  $c_1, \dots, c_k$ , not all zero, satisfy  $c_1 \tilde{v}_1 + \dots + c_k \tilde{v}_k = \tilde{0}_p$ .*

### **i** Proof

*Proof.* Induction on  $p$ .

**Base case**,  $p = 1$ . Each  $\tilde{v}_i$  is a single number  $v_i$ , and  $k \geq 2$ . If  $v_1 = 0$ , take  $c_1 = 1$  and every other  $c_i = 0$ . Otherwise take  $c_1 = v_2$ ,  $c_2 = -v_1$  and every other  $c_i = 0$ ; then  $c_2 \neq 0$ , and  $v_2 v_1 - v_1 v_2 = 0$ .

**Inductive step.** Let  $p \geq 2$ , assume the theorem holds for vectors in  $\mathbb{R}^{p-1}$ , and take

$k > p$  vectors in  $\mathbb{R}^p$ . Write  $v_{i,p}$  for the last entry of  $\tilde{v}_i$ .

*Case 1: every  $v_{i,p} = 0$ .* Deleting the last entry gives  $k > p - 1$  vectors in  $\mathbb{R}^{p-1}$ , which by the inductive hypothesis have a combination with coefficients not all zero that equals  $\tilde{0}_{p-1}$ . The same coefficients combine the original vectors to  $\tilde{0}_p$ , since every last entry is 0.

*Case 2: some  $v_{i,p} \neq 0$ .* Renumber the vectors so that  $v_{k,p} \neq 0$ , and for  $i = 1, \dots, k-1$  let

$$\tilde{w}_i \stackrel{\text{def}}{=} \tilde{v}_i - \frac{v_{i,p}}{v_{k,p}} \tilde{v}_k.$$

The last entry of  $\tilde{w}_i$  is  $v_{i,p} - \frac{v_{i,p}}{v_{k,p}} v_{k,p} = 0$ . Deleting that zero last entry gives  $k-1 > p-1$  vectors in  $\mathbb{R}^{p-1}$ , so by the inductive hypothesis there are numbers  $c_1, \dots, c_{k-1}$ , not all zero, whose combination of the shortened vectors is  $\tilde{0}_{p-1}$ . Putting the zero last entries back gives  $c_1 \tilde{w}_1 + \dots + c_{k-1} \tilde{w}_{k-1} = \tilde{0}_p$ . Then

$$\begin{aligned} \tilde{0}_p &= c_1 \tilde{w}_1 + \dots + c_{k-1} \tilde{w}_{k-1} && \text{(choice of } c_1, \dots, c_{k-1}) \\ &= \sum_{i=1}^{k-1} c_i \left( \tilde{v}_i - \frac{v_{i,p}}{v_{k,p}} \tilde{v}_k \right) && \text{(definition of } \tilde{w}_i) \\ &= \sum_{i=1}^{k-1} c_i \tilde{v}_i - \sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \tilde{v}_k && \text{(distribute each } c_i, \text{ and split the sum)} \\ &= c_1 \tilde{v}_1 + \dots + c_{k-1} \tilde{v}_{k-1} - \left( \sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \right) \tilde{v}_k, && \text{(factor } \tilde{v}_k \text{ out of the second sum)} \end{aligned}$$

a combination of  $\tilde{v}_1, \dots, \tilde{v}_k$  equal to  $\tilde{0}_p$  whose first  $k-1$  coefficients are not all zero.  $\square$

**Example 0.17** (Three vectors in  $\mathbb{R}^2$ ). Theorem 0.15 says the three vectors  $(1, 0)$ ,  $(0, 1)$  and  $(1, 1)$  in  $\mathbb{R}^2$  are linearly dependent, and indeed  $1 \cdot (1, 0) + 1 \cdot (0, 1) - 1 \cdot (1, 1) = (0, 0)$ . The theorem says nothing about  $k \leq p$  vectors:  $(1, 0)$  and  $(0, 1)$  are linearly independent, while  $(1, 2)$  and  $(2, 4)$  are not (Example 0.8).

**Definition 0.30** (Basis). A **basis** of a subspace  $\mathcal{S}$  (Definition 0.28) is a list of vectors that are linearly independent (Definition 0.25) and span  $\mathcal{S}$  (Definition 0.29).

**Example 0.18** (Two bases of  $\mathbb{R}^2$ ).

- $(1, 0)$  and  $(0, 1)$  form a basis of  $\mathbb{R}^2$ : they are linearly independent, and any  $(x, y) \in \mathbb{R}^2$  equals  $x(1, 0) + y(0, 1)$ .
- $(1, 0)$  and  $(1, 1)$  also form a basis of  $\mathbb{R}^2$ : Example 0.8 shows they are linearly independent, and any  $(x, y)$  equals  $(x - y)(1, 0) + y(1, 1)$ , because  $(x - y) + y = x$ .

in the first entry and  $0 + y = y$  in the second.

**Example 0.19** (Lists that are not bases of  $\mathbb{R}^2$ ).

- $(1, 0)$  and  $(2, 0)$  are not a basis of  $\mathbb{R}^2$ . They are not linearly independent, since  $2(1, 0) - (2, 0) = \tilde{0}$ , and they do not span  $\mathbb{R}^2$ : every combination  $c_1(1, 0) + c_2(2, 0)$  has second entry 0, so  $(0, 1)$  is not in their span.
- $(1, 0)$ ,  $(0, 1)$  and  $(1, 1)$  span  $\mathbb{R}^2$ , but they are not a basis:  $(1, 0) + (0, 1) - (1, 1) = \tilde{0}$ , so they are not linearly independent.

**Theorem 0.16** (Coordinates in a basis are unique). *If  $\tilde{a}_1, \dots, \tilde{a}_k$  is a basis of a subspace  $\mathcal{S}$  (Definition 0.30), then every  $\tilde{x} \in \mathcal{S}$  is a linear combination  $\tilde{x} = c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k$  for exactly one list of numbers  $c_1, \dots, c_k$ .*

### Proof

*Proof.* At least one such list exists, because the basis spans  $\mathcal{S}$  (Definition 0.29). Suppose two lists both work:

$$c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k = \tilde{x} = d_1\tilde{a}_1 + \dots + d_k\tilde{a}_k.$$

Subtracting the right side from the left side gives

$$\begin{aligned} \tilde{0} &= (c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k) - (d_1\tilde{a}_1 + \dots + d_k\tilde{a}_k) \quad (\text{both combinations equal } \tilde{x}) \\ &= (c_1 - d_1)\tilde{a}_1 + \dots + (c_k - d_k)\tilde{a}_k \quad (\text{regroup, and distribute each } \tilde{a}_i) \end{aligned}$$

Because  $\tilde{a}_1, \dots, \tilde{a}_k$  are linearly independent (Definition 0.25), every coefficient  $c_i - d_i$  is 0, so  $c_i = d_i$  for each  $i$ .  $\square$

**Example 0.20** (Coordinates of  $(3, 5)$  in two bases). In the basis  $(1, 0), (0, 1)$  of Example 0.18,  $(3, 5) = 3(1, 0) + 5(0, 1)$ , and no other coefficients work. In the basis  $(1, 0), (1, 1)$ , the formula in Example 0.18 gives  $(3, 5) = (3 - 5)(1, 0) + 5(1, 1) = -2(1, 0) + 5(1, 1)$ . The same vector has different coordinates in different bases, but within one basis its coordinates are unique.

The list  $(1, 0), (0, 1), (1, 1)$  in Example 0.19 spans  $\mathbb{R}^2$  but is not linearly independent, and its coordinates are not unique:  $(3, 5) = 3(1, 0) + 5(0, 1) + 0(1, 1) = 0(1, 0) + 2(0, 1) + 3(1, 1)$ .

**Theorem 0.17** (All bases of a subspace have the same number of vectors). *If  $\tilde{a}_1, \dots, \tilde{a}_k$  and  $\tilde{b}_1, \dots, \tilde{b}_l$  are both bases of a subspace  $\mathcal{S}$  of  $\mathbb{R}^p$  (Definition 0.30), then  $k = l$ .*

**i** Proof

*Proof.* Let  $\mathbf{A}$  be the  $p \times k$  matrix with columns  $\tilde{a}_1, \dots, \tilde{a}_k$ , and  $\mathbf{B}$  the  $p \times l$  matrix with columns  $\tilde{b}_1, \dots, \tilde{b}_l$ .

**Write  $\mathbf{A}$  in terms of  $\mathbf{B}$ .** Each  $\tilde{a}_j$  is in  $\mathcal{S}$ , and the  $\tilde{b}$ 's span  $\mathcal{S}$  (Definition 0.29), so  $\tilde{a}_j = \mathbf{B}\tilde{c}_j$  for some  $\tilde{c}_j \in \mathbb{R}^l$  (Definition 0.21). Let  $\mathbf{C}$  be the  $l \times k$  matrix with columns  $\tilde{c}_1, \dots, \tilde{c}_k$ . Then  $\underset{p \times k}{\mathbf{A}} = \underset{p \times l}{\mathbf{B}} \underset{l \times k}{\mathbf{C}}$ , because column  $j$  of  $\mathbf{BC}$  is  $\mathbf{B}\tilde{c}_j$ .

**The columns of  $\mathbf{C}$  are linearly independent.** Suppose  $\mathbf{C}\tilde{x} = \tilde{0}$  for some  $\tilde{x} \in \mathbb{R}^k$ . Then

$$\begin{aligned} \mathbf{A}\tilde{x} &= (\mathbf{BC})\tilde{x} \quad (\text{substitute } \mathbf{A} = \mathbf{BC}) \\ &= \mathbf{B}(\mathbf{C}\tilde{x}) \quad (\text{@thm-matmul-assoc}) \\ &= \mathbf{B}\tilde{0} \quad (\text{substitute } \mathbf{C}\tilde{x} = \tilde{0}) \\ &= \tilde{0} \quad (\text{a matrix times the zero vector is } \tilde{0}) \end{aligned}$$

$\mathbf{A}\tilde{x}$  is the combination  $x_1\tilde{a}_1 + \dots + x_k\tilde{a}_k$  (Theorem 0.13), and the  $\tilde{a}$ 's are linearly independent (Definition 0.25), so  $\tilde{x} = \tilde{0}$ . So the only combination of the columns of  $\mathbf{C}$  that equals  $\tilde{0}$  has all coefficients 0.

**Count.**  $\mathbf{C}$  has  $k$  linearly independent columns of length  $l$ . More than  $l$  vectors in  $\mathbb{R}^l$  are never linearly independent (Theorem 0.15), so  $k \leq l$ . Exchanging the roles of the two bases gives  $l \leq k$ , so  $k = l$ .  $\square$

**Example 0.21** (Both bases of  $\mathbb{R}^2$  in Example 0.18 have two vectors). The bases  $(1, 0), (0, 1)$  and  $(1, 0), (1, 1)$  of  $\mathbb{R}^2$  in Example 0.18 both have 2 vectors, as Theorem 0.17 requires. So no list of 3 vectors is a basis of  $\mathbb{R}^2$ , which agrees with Example 0.19: the list  $(1, 0), (0, 1), (1, 1)$  spans  $\mathbb{R}^2$  but is not linearly independent.

**Definition 0.31** (Dimension). The **dimension** of a subspace  $\mathcal{S}$ , written  $\dim(\mathcal{S})$ , is the number of vectors in any basis of  $\mathcal{S}$  (Definition 0.30). Theorem 0.17 makes this number well defined. By convention, the empty list is the basis of  $\{\tilde{0}\}$ , so  $\dim(\{\tilde{0}\}) = 0$ .

**Example 0.22** (The dimensions of some subspaces).

- $\dim(\{\tilde{0}\}) = 0$ , by the convention in Definition 0.31.
- $\dim(\mathbb{R}^2) = 2$ , by Example 0.18. More generally,  $\dim(\mathbb{R}^p) = p$ : the indicator vectors  $\tilde{e}_1, \dots, \tilde{e}_p$  (Definition 0.9) are linearly independent, since  $c_1\tilde{e}_1 + \dots + c_p\tilde{e}_p = (c_1, \dots, c_p)$ , which is  $\tilde{0}_p$  only if every  $c_i = 0$ ; and they span  $\mathbb{R}^p$ , since any  $\tilde{x} = x_1\tilde{e}_1 + \dots + x_p\tilde{e}_p$ .
- The line in Example 0.10 has dimension 1: the single vector  $(1, 2)$  is linearly independent and spans the line.

- The plane  $z = 0$  in Example 0.13 has dimension 2, with basis  $(1, 0, 0), (0, 1, 0)$ .

**Theorem 0.18** (Adding a vector outside the span keeps a list independent). *If  $\tilde{v}_1, \dots, \tilde{v}_k$  are linearly independent (Definition 0.25) and  $\tilde{w}$  is not in  $\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\}$  (Definition 0.29), then  $\tilde{v}_1, \dots, \tilde{v}_k, \tilde{w}$  are linearly independent.*

**i** Proof

*Proof.* Suppose  $c_1\tilde{v}_1 + \dots + c_k\tilde{v}_k + d\tilde{w} = \tilde{0}$ .

**The coefficient  $d$  is 0.** If  $d \neq 0$ , then

$$\begin{aligned}\tilde{w} &= \frac{1}{d}(d\tilde{w}) && (d \neq 0) \\ &= \frac{1}{d}(-c_1\tilde{v}_1 - \dots - c_k\tilde{v}_k) && (\text{solve the supposed equation for } d\tilde{w}) \\ &= \left(-\frac{c_1}{d}\right)\tilde{v}_1 + \dots + \left(-\frac{c_k}{d}\right)\tilde{v}_k, && (\text{distribute } \frac{1}{d})\end{aligned}$$

which puts  $\tilde{w}$  in  $\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\}$ , contrary to the assumption.

**The other coefficients are 0.** With  $d = 0$ , the equation reads  $c_1\tilde{v}_1 + \dots + c_k\tilde{v}_k = \tilde{0}$ , so every  $c_i = 0$ , because  $\tilde{v}_1, \dots, \tilde{v}_k$  are linearly independent.  $\square$

**Example 0.23** (Adding  $(0, 0, 1)$  to two independent vectors in  $\mathbb{R}^3$ ). The vectors  $(1, 0, 0)$  and  $(0, 1, 0)$  are linearly independent, and their span is the plane  $z = 0$  (Example 0.13). The vector  $(0, 0, 1)$  has third entry 1, so it is not in that plane, and by Theorem 0.18 the three vectors  $(1, 0, 0), (0, 1, 0), (0, 0, 1)$  are linearly independent. Adding  $(1, 1, 0)$  instead would not work:  $(1, 1, 0)$  is in the span, and  $(1, 0, 0) + (0, 1, 0) - (1, 1, 0) = \tilde{0}$ .

**Theorem 0.19** (An independent list extends to a basis). *If  $\mathcal{S}$  is a subspace of  $\mathbb{R}^p$  (Definition 0.28) and  $\tilde{v}_1, \dots, \tilde{v}_k \in \mathcal{S}$  are linearly independent (Definition 0.25), then adding vectors of  $\mathcal{S}$  to the list gives a basis of  $\mathcal{S}$  (Definition 0.30). Starting from the empty list shows that every subspace has a basis.*

**i** Proof

*Proof.* Repeat the following step. If the current list spans  $\mathcal{S}$ , stop: it is linearly independent and spans  $\mathcal{S}$ , so it is a basis. Otherwise, some  $\tilde{w} \in \mathcal{S}$  is not in the span of the current list; add  $\tilde{w}$  to the list, which stays linearly independent by Theorem 0.18.

Every list produced is a linearly independent list of vectors in  $\mathbb{R}^p$ , so it has at most  $p$  vectors (Theorem 0.15). Each step adds one vector, so the process stops after at most

$p - k$  steps, and when it stops the list is a basis of  $\mathcal{S}$ .

For the empty list, which is linearly independent and spans  $\{\tilde{0}\}$  by the conventions in Definition 0.29, the first step asks whether  $\mathcal{S} = \{\tilde{0}\}$ : if so, the empty list is already a basis; if not, the process adds vectors as described.  $\square$

**Example 0.24** (Extending  $(1, 1, 0)$  to a basis of  $\mathbb{R}^3$ ). Start with the single vector  $(1, 1, 0)$ .

1.  $(1, 0, 0)$  is not in  $\text{span}\{(1, 1, 0)\}$ , because every multiple  $c(1, 1, 0)$  has equal first and second entries. Add it.
2.  $(0, 0, 1)$  is not in  $\text{span}\{(1, 1, 0), (1, 0, 0)\}$ , because every combination of those two vectors has third entry 0. Add it.
3. The list  $(1, 1, 0), (1, 0, 0), (0, 0, 1)$  spans  $\mathbb{R}^3$ :  $(x, y, z) = y(1, 1, 0) + (x - y)(1, 0, 0) + z(0, 0, 1)$ . Stop.

The result is a basis of  $\mathbb{R}^3$  that contains the starting vector.

**Theorem 0.20** (An independent list in a subspace has at most  $\dim$  vectors). *Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^p$  with  $\dim(\mathcal{S}) = d$  (Definition 0.31).*

1. Any linearly independent list of vectors in  $\mathcal{S}$  has at most  $d$  vectors.
2. If  $\mathcal{T}$  is a subspace with  $\mathcal{T} \subseteq \mathcal{S}$ , then  $\dim(\mathcal{T}) \leq \dim(\mathcal{S})$ .

### **i** Proof

*Proof. Part 1.* If  $d = 0$ , then  $\mathcal{S} = \{\tilde{0}\}$ , so every vector in a nonempty list from  $\mathcal{S}$  is  $\tilde{0}$ , and  $1 \cdot \tilde{0} = \tilde{0}$  shows the list is not linearly independent; only the empty list, with  $0 = d$  vectors, is. Now let  $d \geq 1$ , and let  $\tilde{a}_1, \dots, \tilde{a}_d$  be a basis of  $\mathcal{S}$ , and let  $\tilde{u}_1, \dots, \tilde{u}_k \in \mathcal{S}$  be linearly independent. Each  $\tilde{u}_j$  is in  $\mathcal{S}$ , so  $\tilde{u}_j = c_{1j}\tilde{a}_1 + \dots + c_{dj}\tilde{a}_d$  for some coordinate vector  $\tilde{c}_j = (c_{1j}, \dots, c_{dj}) \in \mathbb{R}^d$ . Suppose  $k > d$ . Then  $\tilde{c}_1, \dots, \tilde{c}_k$  are  $k > d$  vectors in  $\mathbb{R}^d$ , so some  $x_1, \dots, x_k$ , not all zero, satisfy  $x_1\tilde{c}_1 + \dots + x_k\tilde{c}_k = \tilde{0}_d$  (Theorem 0.15); entry  $i$  of that equation says  $\sum_{j=1}^k x_j c_{ij} = 0$ . Then

$$\begin{aligned}
\sum_{j=1}^k x_j \tilde{u}_j &= \sum_{j=1}^k x_j \sum_{i=1}^d c_{ij} \tilde{a}_i && \text{(substitute each } \tilde{u}_j \text{)} \\
&= \sum_{j=1}^k \sum_{i=1}^d x_j c_{ij} \tilde{a}_i && \text{(distribute each } x_j \text{ over the inner sum)} \\
&= \sum_{i=1}^d \sum_{j=1}^k x_j c_{ij} \tilde{a}_i && \text{(swap the order of the finite sums)} \\
&= \sum_{i=1}^d \left( \sum_{j=1}^k x_j c_{ij} \right) \tilde{a}_i && \text{(factor } \tilde{a}_i \text{ out of the inner sum)} \\
&= \sum_{i=1}^d 0 \cdot \tilde{a}_i && \text{(entry } i \text{ of } \sum_j x_j \tilde{c}_j = \tilde{0}_d \text{)} \\
&= \tilde{0}_p, && \text{(a zero multiple of a vector is } \tilde{0} \text{)}
\end{aligned}$$

a combination of  $\tilde{u}_1, \dots, \tilde{u}_k$  with coefficients not all zero that equals  $\tilde{0}_p$ . That contradicts their linear independence, so  $k \leq d$ .

**Part 2.**  $\mathcal{T}$  has a basis (Theorem 0.19) with  $\dim(\mathcal{T})$  vectors. Those vectors are linearly independent and lie in  $\mathcal{S}$ , so by Part 1 there are at most  $\dim(\mathcal{S})$  of them.  $\square$

**Example 0.25** (Subspaces of a plane have dimension at most two). The plane  $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$  in  $\mathbb{R}^3$  has dimension 2 (Example 0.22). By Theorem 0.20, no three vectors with third entry 0 are linearly independent; for instance,  $(1, 0, 0) + (0, 1, 0) - (1, 1, 0) = \tilde{0}$ . The line  $\mathcal{T} = \text{span}\{(1, 1, 0)\}$  lies in  $\mathcal{S}$ , and  $\dim(\mathcal{T}) = 1 \leq 2 = \dim(\mathcal{S})$ .

**Theorem 0.21** (A subspace inside another of the same dimension equals it). *If  $\mathcal{T}$  and  $\mathcal{S}$  are subspaces of  $\mathbb{R}^p$  with  $\mathcal{T} \subseteq \mathcal{S}$  and  $\dim(\mathcal{T}) = \dim(\mathcal{S})$  (Definition 0.31), then  $\mathcal{T} = \mathcal{S}$ .*

### **i** Proof

*Proof.* Let  $d = \dim(\mathcal{S})$ , and let  $\tilde{t}_1, \dots, \tilde{t}_d$  be a basis of  $\mathcal{T}$  (Theorem 0.19); it has  $d$  vectors because  $\dim(\mathcal{T}) = d$ . (When  $d = 0$  it is the empty list, whose span is  $\{\tilde{0}\}$  by the convention in Definition 0.29, and the argument is unchanged.) Suppose some  $\tilde{s} \in \mathcal{S}$  is not in  $\text{span}\{\tilde{t}_1, \dots, \tilde{t}_d\} = \mathcal{T}$ . Then  $\tilde{t}_1, \dots, \tilde{t}_d, \tilde{s}$  are linearly independent (Theorem 0.18), and they are  $d+1$  vectors of  $\mathcal{S}$ , which contradicts part 1 of Theorem 0.20. So every vector of  $\mathcal{S}$  is in  $\mathcal{T}$ , that is,  $\mathcal{S} \subseteq \mathcal{T}$ . Together with  $\mathcal{T} \subseteq \mathcal{S}$ , this inclusion gives  $\mathcal{T} = \mathcal{S}$ .  $\square$

**Example 0.26** (Two independent vectors in a plane span it). Let  $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$ , the plane  $z = 0$ , which has dimension 2 (Example 0.22), and let  $\mathcal{T} =$

$\text{span}\{(1, 0, 0), (1, 1, 0)\}$ . Both vectors have third entry 0, so  $\mathcal{T} \subseteq \mathcal{S}$ . They are linearly independent:  $c_1(1, 0, 0) + c_2(1, 1, 0) = (c_1 + c_2, c_2, 0)$ , which is  $\tilde{0}$  only if  $c_2 = 0$  and then  $c_1 = 0$ . So  $\dim(\mathcal{T}) = 2$ , and Theorem 0.21 gives  $\mathcal{T} = \mathcal{S}$  without solving for any coefficients. As a check,  $(x, y, 0) = (x - y)(1, 0, 0) + y(1, 1, 0)$ .

**Example 0.27** (Equal dimensions without containment). The lines  $\text{span}\{(1, 0, 0)\}$  and  $\text{span}\{(0, 1, 0)\}$  in  $\mathbb{R}^3$  both have dimension 1, but they are not equal:  $(1, 0, 0)$  is in the first and not the second. Theorem 0.21 does not apply, because neither line contains the other.

## Column space and null space

**Definition 0.32** (Column space). The **column space** of an  $m \times n$  matrix  $\mathbf{A}$  is

$$\mathcal{C}(\mathbf{A}) \stackrel{\text{def}}{=} \{\mathbf{A}\tilde{x} : \tilde{x} \in \mathbb{R}^n\},$$

the set of all vectors  $\mathbf{A}\tilde{x}$  in  $\mathbb{R}^m$ . It is also called the **range** or **image** of  $\mathbf{A}$ . The **row space** of  $\mathbf{A}$  is  $\mathcal{C}(\mathbf{A}^\top)$ , a set of vectors in  $\mathbb{R}^n$ .

**Example 0.28** (The column space of a rank-one matrix). Let

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix}.$$

For any  $\tilde{x} \in \mathbb{R}^3$ ,  $\mathbf{A}\tilde{x} = (x_1 - 2x_2 - 2x_3, 3(x_1 - 2x_2 - 2x_3)) = (x_1 - 2x_2 - 2x_3)(1, 3)$  (Definition 0.21). The number  $x_1 - 2x_2 - 2x_3$  can be any real number (take  $x_2 = x_3 = 0$ ), so  $\mathcal{C}(\mathbf{A}) = \{c(1, 3) : c \in \mathbb{R}\}$ , a line in  $\mathbb{R}^2$ . The vector  $(1, 0)$  is not in  $\mathcal{C}(\mathbf{A})$ :  $c(1, 3)$  has second entry  $3c$ , which is 0 only if  $c = 0$ , and then the first entry is 0, not 1.

**Theorem 0.22** (The column space is the span of the columns). *If  $\mathbf{A}$  is an  $m \times n$  matrix with columns  $\tilde{a}_1, \dots, \tilde{a}_n$ , then*

$$\mathcal{C}(\mathbf{A}) = \text{span}\{\tilde{a}_1, \dots, \tilde{a}_n\},$$

*so  $\mathcal{C}(\mathbf{A})$  is a subspace of  $\mathbb{R}^m$ , and the row space  $\mathcal{C}(\mathbf{A}^\top)$  is a subspace of  $\mathbb{R}^n$ .*

### **i** Proof

*Proof.* By Theorem 0.13,  $\mathbf{A}\tilde{x} = x_1\tilde{a}_1 + \dots + x_n\tilde{a}_n$ , and as  $\tilde{x}$  ranges over  $\mathbb{R}^n$ , the coefficients  $x_1, \dots, x_n$  range over all lists of  $n$  numbers. So the set of all  $\mathbf{A}\tilde{x}$  (Definition 0.32) is the

set of all linear combinations of the columns, which is their span (Definition 0.29). A span is a subspace (Theorem 0.12). Applying the same argument to  $\mathbf{A}^\top$ , whose columns are the rows of  $\mathbf{A}$ , shows the row space is a subspace of  $\mathbb{R}^n$ .  $\square$

**Example 0.29** (Reading off the column space and row space from the columns and rows). For  $\mathbf{A}$  in Example 0.28, the columns are  $(1, 3)$ ,  $(-2, -6) = -2(1, 3)$  and  $(-2, -6) = -2(1, 3)$ , so by Theorem 0.22  $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$ , the same line found in Example 0.28. The rows are  $(1, -2, -2)$  and  $(3, -6, -6) = 3(1, -2, -2)$ , so the row space is  $\text{span}\{(1, -2, -2)\}$ , a line in  $\mathbb{R}^3$ .

**Definition 0.33** (Null space). The **null space** of an  $m \times n$  matrix  $\mathbf{A}$  is

$$\mathcal{N}(\mathbf{A}) \stackrel{\text{def}}{=} \{\tilde{x} \in \mathbb{R}^n : \mathbf{A}\tilde{x} = \tilde{0}_m\},$$

the set of solutions of  $\mathbf{A}\tilde{x} = \tilde{0}_m$ . It is also called the **kernel** of  $\mathbf{A}$ . The **left null space** of  $\mathbf{A}$  is  $\mathcal{N}(\mathbf{A}^\top)$ , a set of vectors in  $\mathbb{R}^m$ .

**Example 0.30** (The null space and left null space of a rank-one matrix). For  $\mathbf{A}$  in Example 0.28,  $\mathbf{A}\tilde{x} = (x_1 - 2x_2 - 2x_3, 3(x_1 - 2x_2 - 2x_3))$ , so  $\mathbf{A}\tilde{x} = \tilde{0}_2$  exactly when  $x_1 = 2x_2 + 2x_3$ . Writing  $\tilde{x} = (2x_2 + 2x_3, x_2, x_3) = x_2(2, 1, 0) + x_3(2, 0, 1)$  shows  $\mathcal{N}(\mathbf{A}) = \text{span}\{(2, 1, 0), (2, 0, 1)\}$ , a plane in  $\mathbb{R}^3$ . For the left null space,  $\mathbf{A}^\top \tilde{y} = (y_1 + 3y_2, -2(y_1 + 3y_2), -2(y_1 + 3y_2))$ , which is  $\tilde{0}_3$  exactly when  $y_1 = -3y_2$ , so  $\mathcal{N}(\mathbf{A}^\top) = \text{span}\{(-3, 1)\}$ , a line in  $\mathbb{R}^2$ .

**Theorem 0.23** (A null space is a subspace). *For any  $m \times n$  matrix  $\mathbf{A}$ ,  $\mathcal{N}(\mathbf{A})$  (Definition 0.33) is a subspace of  $\mathbb{R}^n$  (Definition 0.28).*

### **i** Proof

*Proof.*  $\mathbf{A}\tilde{0}_n = \tilde{0}_m$ , so  $\tilde{0}_n \in \mathcal{N}(\mathbf{A})$ , and the null space is not empty. If  $\tilde{u}, \tilde{w} \in \mathcal{N}(\mathbf{A})$  and  $c \in \mathbb{R}$ , then

$$\begin{aligned} \mathbf{A}(\tilde{u} + \tilde{w}) &= \mathbf{A}\tilde{u} + \mathbf{A}\tilde{w} && \text{(@thm-matmul-distrib)} \\ &= \tilde{0}_m + \tilde{0}_m && (\tilde{u} \text{ and } \tilde{w} \text{ are in the null space)} \\ &= \tilde{0}_m && (\text{adding } \tilde{0}_m \text{ changes nothing}) \end{aligned}$$

and, for each entry  $i = 1, \dots, m$ ,

$$\begin{aligned}
(\mathbf{A}(c\tilde{u}))_i &= a_{i1}(cu_1) + \cdots + a_{in}(cu_n) && \text{(@def-matvec-mult)} \\
&= c(a_{i1}u_1 + \cdots + a_{in}u_n) && \text{(factor } c \text{ out of each term)} \\
&= c(\mathbf{A}\tilde{u})_i && \text{(@def-matvec-mult)} \\
&= c \cdot 0 && (\tilde{u} \text{ is in the null space)} \\
&= 0, && \text{(arithmetic)}
\end{aligned}$$

so  $\mathbf{A}(c\tilde{u}) = \tilde{0}_m$ . Therefore  $\tilde{u} + \tilde{w}$  and  $c\tilde{u}$  are in  $\mathcal{N}(\mathbf{A})$ .  $\square$

**Example 0.31** (The solutions of  $\mathbf{A}\tilde{x} = \tilde{b}$  with  $\tilde{b} \neq \tilde{0}$  are not a subspace). In Example 0.30,  $(2, 1, 0)$  and  $(2, 0, 1)$  are in  $\mathcal{N}(\mathbf{A})$ , and so is their sum  $(4, 1, 1)$ , since  $4 - 2 \cdot 1 - 2 \cdot 1 = 0$ . By contrast, the solutions of  $\mathbf{A}\tilde{x} = (1, 3)$  for the same  $\mathbf{A}$  are the vectors with  $x_1 - 2x_2 - 2x_3 = 1$ . That set does not contain  $\tilde{0}_3$ , so by Theorem 0.11 it is not a subspace.

**Theorem 0.24** (The row space and null space share only the zero vector). *For any  $m \times n$  matrix  $\mathbf{A}$ ,*

$$\mathcal{C}(\mathbf{A}^\top) \cap \mathcal{N}(\mathbf{A}) = \{\tilde{0}_n\}.$$

### Proof

*Proof.* Both sets are subspaces, by Theorem 0.22 and Theorem 0.23, so both contain  $\tilde{0}_n$  (Theorem 0.11).

Now take any  $\tilde{x}$  in both sets. Being in the row space means  $\tilde{x} = \mathbf{A}^\top \tilde{u}$  for some  $\tilde{u} \in \mathbb{R}^m$  (Definition 0.32), and being in the null space means  $\mathbf{A}\tilde{x} = \tilde{0}_m$  (Definition 0.33). Then

$$\begin{aligned}
\tilde{x}^\top \tilde{x} &= (\mathbf{A}^\top \tilde{u})^\top \tilde{x} && \text{(substitute } \tilde{x} = \mathbf{A}^\top \tilde{u} \text{ in the first factor)} \\
&= (\tilde{u}^\top (\mathbf{A}^\top)^\top) \tilde{x} && \text{(@thm-transpose-product)} \\
&= (\tilde{u}^\top \mathbf{A}) \tilde{x} && \text{(transposing twice returns the original matrix)} \\
&= \tilde{u}^\top (\mathbf{A}\tilde{x}) && \text{(@thm-matmul-assoc)} \\
&= \tilde{u}^\top \tilde{0}_m && \text{(substitute } \mathbf{A}\tilde{x} = \tilde{0}_m \text{)} \\
&= 0. && \text{(every term of the inner product is 0)}
\end{aligned}$$

But  $\tilde{x}^\top \tilde{x} = x_1^2 + \cdots + x_n^2$ , which is 0 only when every  $x_i = 0$ . So  $\tilde{x} = \tilde{0}_n$ .  $\square$

**Example 0.32** (The row space and null space in Example 0.30). For  $\mathbf{A}$  in Example 0.28, the row space is  $\text{span}\{(1, -2, -2)\}$  and the null space is  $\text{span}\{(2, 1, 0), (2, 0, 1)\}$  (Example 0.30). A vector  $c(1, -2, -2)$  of the row space is in the null space only if

$$\begin{aligned}\mathbf{A}(c(1, -2, -2)) &= c(1 - 2 \cdot (-2) - 2 \cdot (-2), 3(1 - 2 \cdot (-2) - 2 \cdot (-2))) \quad (\text{@exm-null-space's formula for } \mathbf{A}\tilde{x}) \\ &= c(9, 27) \quad (\text{arithmetic})\end{aligned}$$

is  $\tilde{0}_2$ , that is, only if  $c = 0$ . So the two subspaces share only  $\tilde{0}_3$ .

**Theorem 0.25** ( $\mathbf{A}^\top \mathbf{A}$  has the same null space as  $\mathbf{A}$ ). For any  $m \times n$  matrix  $\mathbf{A}$ ,

$$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}).$$

### **i** Proof

*Proof.*  $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}^\top \mathbf{A})$ . If  $\mathbf{A}\tilde{x} = \tilde{0}_m$ , then  $\mathbf{A}^\top \mathbf{A}\tilde{x} = \mathbf{A}^\top \tilde{0}_m = \tilde{0}_n$ .  
 $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$ . If  $\mathbf{A}^\top \mathbf{A}\tilde{x} = \tilde{0}_n$ , then

$$\begin{aligned}0 &= \tilde{x}^\top \tilde{0}_n \quad (\text{every term of the inner product is } 0) \\ &= \tilde{x}^\top (\mathbf{A}^\top \mathbf{A}\tilde{x}) \quad (\text{substitute } \tilde{0}_n = \mathbf{A}^\top \mathbf{A}\tilde{x}) \\ &= (\tilde{x}^\top \mathbf{A}^\top) (\mathbf{A}\tilde{x}) \quad (\text{@thm-matmul-assoc}) \\ &= (\mathbf{A}\tilde{x})^\top (\mathbf{A}\tilde{x}) \quad (\text{@thm-transpose-product}) \\ &= (\mathbf{A}\tilde{x}) \cdot (\mathbf{A}\tilde{x}) \quad (\text{@exm-dot-product-matmul}) \\ &= \|\mathbf{A}\tilde{x}\|^2. \quad (\text{square both sides of @eq-l2-norm})\end{aligned}$$

A vector whose Euclidean norm is 0 has every entry 0, so  $\mathbf{A}\tilde{x} = \tilde{0}_m$ . □

**Example 0.33** (The null space of a Gram matrix). For  $\mathbf{A}$  in Example 0.28,

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix} = \begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix}.$$

Every row is a multiple of  $(1, -2, -2)$ , so  $\mathbf{A}^\top \mathbf{A}\tilde{x} = \tilde{0}_3$  exactly when  $x_1 - 2x_2 - 2x_3 = 0$ , the same condition as for  $\mathcal{N}(\mathbf{A})$  in Example 0.30.

## Rank-nullity and rank factorization

### i Source

This section is adapted from Zhou (2024e), used under the MIT License (see the license text in Section ).

**Theorem 0.26** (The rank is the dimension of the column space). *For any  $m \times n$  matrix  $\mathbf{A}$ ,*

$$\text{rank}(\mathbf{A}) = \dim(\mathcal{C}(\mathbf{A})).$$

### i Proof

*Proof.* Let  $r = \text{rank}(\mathbf{A})$  (Definition 0.26), and choose  $r$  linearly independent columns of  $\mathbf{A}$ ; by Definition 0.26, no larger set of columns is linearly independent. We show the chosen columns are a basis of  $\mathcal{C}(\mathbf{A})$  (Definition 0.30).

**Every column is in the span of the chosen columns.** A chosen column is in that span trivially. If some unchosen column  $\tilde{a}_j$  were not in the span, then adding it to the chosen columns would give  $r + 1$  linearly independent columns (Theorem 0.18), more than  $r$ . So every column of  $\mathbf{A}$  is in the span of the chosen columns.

**The chosen columns span  $\mathcal{C}(\mathbf{A})$ .** The span of the chosen columns is a subspace (Theorem 0.12), so it is closed under addition and scalar multiplication, and it contains every linear combination of the columns of  $\mathbf{A}$ . By Theorem 0.22, those combinations make up  $\mathcal{C}(\mathbf{A})$ , so  $\mathcal{C}(\mathbf{A})$  is contained in the span of the chosen columns. The chosen columns are themselves columns of  $\mathbf{A}$ , so their span is contained in  $\mathcal{C}(\mathbf{A})$ , and the two sets are equal.

The chosen columns are linearly independent and span  $\mathcal{C}(\mathbf{A})$ , so they are a basis of it, and  $\dim(\mathcal{C}(\mathbf{A})) = r$  (Definition 0.31). If  $r = 0$ , every column is  $\tilde{0}_m$ ,  $\mathcal{C}(\mathbf{A}) = \{\tilde{0}_m\}$ , and both sides are 0.  $\square$

**Example 0.34** (The rank of the matrix in Example 0.28). For  $\mathbf{A}$  in Example 0.28,  $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$  (1), and  $(1, 3)$  is linearly independent because it is nonzero, so it is a basis and  $\mathcal{C}(\mathbf{A})$  has dimension 1. So  $\text{rank}(\mathbf{A}) = 1$ , by Theorem 0.26. Directly: the first column  $(1, 3)$  is linearly independent on its own, and any two columns are dependent, because each column is a multiple of  $(1, 3)$ .

**Definition 0.34** (Nullity). The **nullity** of a matrix  $\mathbf{A}$  is the dimension of its null space (Definition 0.33):

$$\text{nullity}(\mathbf{A}) \stackrel{\text{def}}{=} \dim(\mathcal{N}(\mathbf{A})).$$

**Example 0.35** (The nullity of the matrix in Example 0.28). For  $\mathbf{A}$  in Example 0.28,  $\mathcal{N}(\mathbf{A}) = \text{span}\{(2, 1, 0), (2, 0, 1)\}$  (Example 0.30). Those two vectors are linearly independent:  $c_1(2, 1, 0) + c_2(2, 0, 1) = (2c_1 + 2c_2, c_1, c_2)$ , which is  $\tilde{\mathbf{0}}_3$  only if  $c_1 = c_2 = 0$ . So they are a basis of  $\mathcal{N}(\mathbf{A})$ , and  $\text{nullity}(\mathbf{A}) = 2$ . The  $2 \times 2$  identity matrix has nullity 0, because  $\mathbf{I}\tilde{x} = \tilde{x}$  is  $\tilde{\mathbf{0}}_2$  only when  $\tilde{x} = \tilde{\mathbf{0}}_2$ .

**Theorem 0.27** (Rank-nullity theorem). *For any  $m \times n$  matrix  $\mathbf{A}$ ,*

$$\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n.$$

### **i** Proof

*Proof.* Let  $\nu = \text{nullity}(\mathbf{A})$ . Choose a basis  $\tilde{x}_1, \dots, \tilde{x}_\nu$  of  $\mathcal{N}(\mathbf{A})$ , and extend it to a basis  $\tilde{x}_1, \dots, \tilde{x}_\nu, \tilde{y}_1, \dots, \tilde{y}_{n-\nu}$  of  $\mathbb{R}^n$  (Theorem 0.19); it has  $n$  vectors, because every basis of  $\mathbb{R}^n$  has  $\dim(\mathbb{R}^n) = n$  vectors (Theorem 0.17, Example 0.22). We show that  $\mathbf{A}\tilde{y}_1, \dots, \mathbf{A}\tilde{y}_{n-\nu}$  are a basis of  $\mathcal{C}(\mathbf{A})$ , using Theorem 0.14 to apply  $\mathbf{A}$  to linear combinations.

**They are linearly independent.** Suppose  $\sum_{i=1}^{n-\nu} v_i \mathbf{A}\tilde{y}_i = \tilde{\mathbf{0}}_m$ . Then  $\mathbf{A}(\sum_{i=1}^{n-\nu} v_i \tilde{y}_i) = \tilde{\mathbf{0}}_m$  (Theorem 0.14), so  $\sum_{i=1}^{n-\nu} v_i \tilde{y}_i \in \mathcal{N}(\mathbf{A})$ , and it equals  $\sum_{j=1}^\nu u_j \tilde{x}_j$  for some numbers  $u_j$ , since the  $\tilde{x}_j$  span  $\mathcal{N}(\mathbf{A})$ . Moving everything to one side,

$$\sum_{j=1}^\nu (-u_j) \tilde{x}_j + \sum_{i=1}^{n-\nu} v_i \tilde{y}_i = \tilde{\mathbf{0}}_n,$$

and because the  $\tilde{x}$ 's and  $\tilde{y}$ 's together form a basis of  $\mathbb{R}^n$ , which is linearly independent, every coefficient is 0; in particular every  $v_i = 0$ .

**They span  $\mathcal{C}(\mathbf{A})$ .** Take any  $\tilde{w} \in \mathcal{C}(\mathbf{A})$ , so  $\tilde{w} = \mathbf{A}\tilde{z}$  for some  $\tilde{z} \in \mathbb{R}^n$  (Definition 0.32). Write  $\tilde{z} = \sum_j a_j \tilde{x}_j + \sum_i b_i \tilde{y}_i$  in the basis of  $\mathbb{R}^n$ . Then

$$\begin{aligned} \tilde{w} &= \mathbf{A} \left( \sum_{j=1}^\nu a_j \tilde{x}_j + \sum_{i=1}^{n-\nu} b_i \tilde{y}_i \right) && \text{(substitute } \tilde{z} \text{)} \\ &= \sum_{j=1}^\nu a_j \mathbf{A}\tilde{x}_j + \sum_{i=1}^{n-\nu} b_i \mathbf{A}\tilde{y}_i && \text{(@thm-matvec-linear)} \\ &= \sum_{j=1}^\nu a_j \tilde{\mathbf{0}}_m + \sum_{i=1}^{n-\nu} b_i \mathbf{A}\tilde{y}_i && \text{(each } \tilde{x}_j \in \mathcal{N}(\mathbf{A}) \text{)} \\ &= \sum_{i=1}^{n-\nu} b_i \mathbf{A}\tilde{y}_i, && \text{(drop the zero terms)} \end{aligned}$$

a linear combination of  $\mathbf{A}\tilde{\mathbf{y}}_1, \dots, \mathbf{A}\tilde{\mathbf{y}}_{n-\nu}$ . These vectors are in  $\mathcal{C}(\mathbf{A})$ , so their span is exactly  $\mathcal{C}(\mathbf{A})$ .

So  $\dim(\mathcal{C}(\mathbf{A})) = n - \nu$ , and  $\text{rank}(\mathbf{A}) = n - \nu$  by Theorem 0.26.  $\square$

**Example 0.36** (Checking rank-nullity on the matrix in Example 0.28). The matrix  $\mathbf{A}$  in Example 0.28 has  $n = 3$  columns, rank 1 (Example 0.34) and nullity 2 (Example 0.35), and  $1 + 2 = 3$ . The  $2 \times 2$  identity matrix has rank 2 and nullity 0 (Example 0.35), and  $2 + 0 = 2$ .

**Theorem 0.28** (Multiplying on the right cannot increase the rank). *If  $\mathbf{A}$  is  $m \times n$  and  $\mathbf{B}$  is  $n \times k$ , then*

$$\text{rank}(\mathbf{AB}) \leq \text{rank}(\mathbf{A}).$$

### **i** Proof

*Proof.*  $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$ . Any vector in  $\mathcal{C}(\mathbf{AB})$  is  $(\mathbf{AB})\tilde{\mathbf{x}}$  for some  $\tilde{\mathbf{x}} \in \mathbb{R}^k$  (Definition 0.32), and

$$(\mathbf{AB})\tilde{\mathbf{x}} = \mathbf{A}(\mathbf{B}\tilde{\mathbf{x}}) \quad (\text{@thm-matmul-assoc})$$

is  $\mathbf{A}$  times the vector  $\mathbf{B}\tilde{\mathbf{x}} \in \mathbb{R}^n$ , so it is in  $\mathcal{C}(\mathbf{A})$ .

**Compare dimensions.** Both column spaces are subspaces (Theorem 0.22), so

$$\begin{aligned} \text{rank}(\mathbf{AB}) &= \dim(\mathcal{C}(\mathbf{AB})) && (\text{@thm-rank-dim}) \\ &\leq \dim(\mathcal{C}(\mathbf{A})) && (\text{@thm-dim-bound, part 2}) \\ &= \text{rank}(\mathbf{A}). && (\text{@thm-rank-dim}) \end{aligned}$$

$\square$

**Example 0.37** (A product can lose rank). Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{AB} = \begin{bmatrix} 1 \cdot 0 + 0 \cdot 1 & 1 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 0 + 0 \cdot 1 & 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$\mathbf{A}$  has rank 1, but  $\mathbf{AB}$  has rank 0, so the inequality in Theorem 0.28 can be strict. With the  $2 \times 2$  identity in place of  $\mathbf{B}$ ,  $\mathbf{AI} = \mathbf{A}$  keeps rank 1, and the inequality is an equality.

**Theorem 0.29** (Fundamental theorem of ranks). *For any  $m \times n$  matrix  $\mathbf{A}$ ,*

$$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A} \mathbf{A}^\top).$$

*In particular, the largest number of linearly independent rows of  $\mathbf{A}$  equals the largest number of linearly independent columns.*

### **i** Proof

*Proof.*  $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A})$ . Both matrices have  $n$  columns, and  $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$  (Theorem 0.25), so they have the same nullity (Definition 0.34). Then

$$\begin{aligned} \text{rank}(\mathbf{A}^\top \mathbf{A}) &= n - \text{nullity}(\mathbf{A}^\top \mathbf{A}) \quad (\text{@thm-rank-nullity}) \\ &= n - \text{nullity}(\mathbf{A}) \quad (\text{equal null spaces}) \\ &= \text{rank}(\mathbf{A}). \quad (\text{@thm-rank-nullity}) \end{aligned}$$

$\text{rank}(\mathbf{A} \mathbf{A}^\top) = \text{rank}(\mathbf{A}^\top)$ . Apply the previous step to the  $n \times m$  matrix  $\mathbf{A}^\top$ , using  $(\mathbf{A}^\top)^\top = \mathbf{A}$ .

**Chain the inequalities.**

$$\begin{aligned} \text{rank}(\mathbf{A}) &= \text{rank}(\mathbf{A}^\top \mathbf{A}) \quad (\text{first step}) \\ &\leq \text{rank}(\mathbf{A}^\top) \quad (\text{@thm-rank-product, with } \mathbf{A}^\top \text{ on the left}) \\ &= \text{rank}(\mathbf{A} \mathbf{A}^\top) \quad (\text{second step}) \\ &\leq \text{rank}(\mathbf{A}). \quad (\text{@thm-rank-product, with } \mathbf{A} \text{ on the left}) \end{aligned}$$

The chain starts and ends at  $\text{rank}(\mathbf{A})$ , so every quantity in it equals  $\text{rank}(\mathbf{A})$ . The columns of  $\mathbf{A}^\top$  are the rows of  $\mathbf{A}$ , which gives the statement about rows.  $\square$

**Example 0.38** (Four equal ranks). For  $\mathbf{A}$  in Example 0.28,  $\text{rank}(\mathbf{A}) = 1$  (Example 0.34).

Its transpose  $\mathbf{A}^\top = \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}$  has second column 3 times its first, so its rank is 1.  $\mathbf{A}^\top \mathbf{A}$  in Example 0.33 has every row a multiple of  $(1, -2, -2)$ , so its rank is 1 too, and

$$\mathbf{A} \mathbf{A}^\top = \begin{bmatrix} 1 \cdot 1 + (-2)(-2) + (-2)(-2) & 1 \cdot 3 + (-2)(-6) + (-2)(-6) \\ 3 \cdot 1 + (-6)(-2) + (-6)(-2) & 3 \cdot 3 + (-6)(-6) + (-6)(-6) \end{bmatrix} = \begin{bmatrix} 9 & 27 \\ 27 & 81 \end{bmatrix},$$

whose second column is 3 times its first, so its rank is 1.

**Corollary 0.1** (The rank is at most the smaller dimension). *For any  $m \times n$  matrix  $\mathbf{A}$ ,  $\text{rank}(\mathbf{A}) \leq \min\{m, n\}$ .*

**i** Proof

*Proof.*  $\mathbf{A}$  has  $n$  columns, so at most  $n$  of them are linearly independent, and  $\text{rank}(\mathbf{A}) \leq n$  (Definition 0.26). Likewise  $\mathbf{A}^\top$  has  $m$  columns, so  $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top) \leq m$  (Theorem 0.29).  $\square$

**Example 0.39** (A  $2 \times 3$  matrix has rank at most 2). By Corollary 0.1, every  $2 \times 3$  matrix has rank at most  $\min\{2, 3\} = 2$ , even though it has 3 columns. The matrix  $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$  of Remark 0.13 reaches that bound: its first two columns  $(1, 0)$  and  $(0, 1)$  are linearly independent, so its rank is 2.

**Definition 0.35** (Rank factorization). Let  $\mathbf{A}$  be an  $m \times n$  matrix with  $\text{rank}(\mathbf{A}) = r \geq 1$ . A **rank factorization** of  $\mathbf{A}$  is a product

$$\underset{m \times n}{\mathbf{A}} = \underset{m \times r}{\mathbf{C}} \underset{r \times n}{\mathbf{R}}.$$

**Example 0.40** (A rank factorization of a rank-one matrix). The matrix  $\mathbf{A}$  in Example 0.28 has rank 1 (Example 0.34), and

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 & 1 \cdot (-2) & 1 \cdot (-2) \\ 3 \cdot 1 & 3 \cdot (-2) & 3 \cdot (-2) \end{bmatrix} = \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix},$$

so  $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$  ( $2 \times 1$ ) and  $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$  ( $1 \times 3$ ) are a rank factorization. It is not unique:  $\mathbf{C} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$  and  $\mathbf{R} = \begin{bmatrix} \frac{1}{2} & -1 & -1 \end{bmatrix}$  give the same product. A product  $\mathbf{C}\mathbf{R}$  with  $\mathbf{C}$  of size  $2 \times 2$  is not a rank factorization of this  $\mathbf{A}$ , because the inner dimension must equal  $\text{rank}(\mathbf{A}) = 1$ .

**Theorem 0.30** (Every nonzero matrix has a rank factorization). *Every  $m \times n$  matrix  $\mathbf{A}$  with  $\text{rank}(\mathbf{A}) = r \geq 1$  has a rank factorization (Definition 0.35). One is given by taking the columns of  $\mathbf{C}$  to be any basis of  $\mathcal{C}(\mathbf{A})$ .*

### i Proof

*Proof.*  $\mathcal{C}(\mathbf{A})$  has dimension  $r$  (Theorem 0.26), so it has a basis  $\tilde{c}_1, \dots, \tilde{c}_r$  (Theorem 0.19); let  $\mathbf{C}$  be the  $m \times r$  matrix with these columns. Each column  $\tilde{a}_j$  of  $\mathbf{A}$  is in  $\mathcal{C}(\mathbf{A})$ , so  $\tilde{a}_j = r_{1j}\tilde{c}_1 + \dots + r_{rj}\tilde{c}_r$  for some numbers  $r_{ij}$ , because the basis spans  $\mathcal{C}(\mathbf{A})$ . Let  $\mathbf{R}$  be the  $r \times n$  matrix with entries  $r_{ij}$ , whose column  $j$  is  $\tilde{r}_j = (r_{1j}, \dots, r_{rj})$ . Then column  $j$  of  $\mathbf{CR}$  is

$$\begin{aligned} \mathbf{C}\tilde{r}_j &= r_{1j}\tilde{c}_1 + \dots + r_{rj}\tilde{c}_r && (\text{@thm-matvec-columns}) \\ &= \tilde{a}_j, && (\text{choice of the } r_{ij}) \end{aligned}$$

so  $\mathbf{CR} = \mathbf{A}$ . □

**Example 0.41** (Building a rank factorization from a basis). For  $\mathbf{A}$  in Example 0.28,  $(1, 3)$  spans  $\mathcal{C}(\mathbf{A})$  (1), and it is linearly independent because it is nonzero ( $c(1, 3) = \tilde{0}_2$  forces  $c = 0$ ), so it is a basis of  $\mathcal{C}(\mathbf{A})$ . The columns of  $\mathbf{A}$  are  $(1, 3) = 1 \cdot (1, 3)$ ,  $(-2, -6) = -2 \cdot (1, 3)$  and  $(-2, -6) = -2 \cdot (1, 3)$ , so the construction in Theorem 0.30 gives  $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$  and  $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$ , the factorization in Example 0.40.

## Sums and direct sums

### i Source

This section, Section and Section are adapted from Zhou (2024d), used under the MIT License (see the license text in Section ). The source proves that a subspace and its orthogonal complement together make up  $\mathbb{R}^p$  by extending an orthonormal basis; the proof here uses the rank-nullity theorem (Theorem 0.27) instead.

**Definition 0.36** (Sum of subspaces). The **sum** of two subspaces  $\mathcal{S}_1$  and  $\mathcal{S}_2$  of  $\mathbb{R}^p$  (Definition 0.28) is the set of all sums of a vector from  $\mathcal{S}_1$  and a vector from  $\mathcal{S}_2$ :

$$\mathcal{S}_1 + \mathcal{S}_2 \stackrel{\text{def}}{=} \{\tilde{x}_1 + \tilde{x}_2 : \tilde{x}_1 \in \mathcal{S}_1, \tilde{x}_2 \in \mathcal{S}_2\}.$$

**Example 0.42** (Two lines sum to a plane). Let  $\mathcal{S}_1 = \text{span}\{(1, 0, 0)\}$  and  $\mathcal{S}_2 = \text{span}\{(0, 1, 0)\}$ , two lines in  $\mathbb{R}^3$ . A vector of  $\mathcal{S}_1 + \mathcal{S}_2$  has the form  $a(1, 0, 0) + b(0, 1, 0) = (a, b, 0)$ , and  $a$  and  $b$  can be any numbers, so  $\mathcal{S}_1 + \mathcal{S}_2 = \{(a, b, 0) : a, b \in \mathbb{R}\}$ , the plane  $z = 0$ .

**Example 0.43** (The sum is not the union). For the two lines in Example 0.42, the union  $\mathcal{S}_1 \cup \mathcal{S}_2$  contains  $(1, 0, 0)$  and  $(0, 1, 0)$  but not their sum  $(1, 1, 0)$ : every multiple of  $(1, 0, 0)$  has second entry 0, and every multiple of  $(0, 1, 0)$  has first entry 0. So the union is not closed under addition and is not a subspace, while the sum, the whole plane  $z = 0$ , is.

**Theorem 0.31** (Sums and intersections of subspaces are subspaces). *If  $\mathcal{S}_1$  and  $\mathcal{S}_2$  are subspaces of  $\mathbb{R}^p$  (Definition 0.28), then  $\mathcal{S}_1 + \mathcal{S}_2$  (Definition 0.36) and  $\mathcal{S}_1 \cap \mathcal{S}_2$  are subspaces of  $\mathbb{R}^p$ .*

### **i** Proof

*Proof.* Both  $\mathcal{S}_1$  and  $\mathcal{S}_2$  contain  $\tilde{0}$  (Theorem 0.11).

**The sum.** It is not empty: it contains  $\tilde{0} + \tilde{0} = \tilde{0}$ . Take two of its vectors,  $\tilde{u} = \tilde{u}_1 + \tilde{u}_2$  and  $\tilde{w} = \tilde{w}_1 + \tilde{w}_2$ , with  $\tilde{u}_1, \tilde{w}_1 \in \mathcal{S}_1$  and  $\tilde{u}_2, \tilde{w}_2 \in \mathcal{S}_2$ , and a number  $c$ . Then

$$\begin{aligned}\tilde{u} + \tilde{w} &= (\tilde{u}_1 + \tilde{u}_2) + (\tilde{w}_1 + \tilde{w}_2) \quad (\text{substitute}) \\ &= (\tilde{u}_1 + \tilde{w}_1) + (\tilde{u}_2 + \tilde{w}_2), \quad (\text{regroup; vector addition is entrywise})\end{aligned}$$

where  $\tilde{u}_1 + \tilde{w}_1 \in \mathcal{S}_1$  and  $\tilde{u}_2 + \tilde{w}_2 \in \mathcal{S}_2$ , because each subspace is closed under addition. Likewise

$$\begin{aligned}c\tilde{u} &= c(\tilde{u}_1 + \tilde{u}_2) \quad (\text{substitute}) \\ &= c\tilde{u}_1 + c\tilde{u}_2, \quad (\text{distribute } c \text{ entrywise})\end{aligned}$$

where  $c\tilde{u}_1 \in \mathcal{S}_1$  and  $c\tilde{u}_2 \in \mathcal{S}_2$ , because each subspace is closed under scalar multiplication. So  $\tilde{u} + \tilde{w}$  and  $c\tilde{u}$  are in  $\mathcal{S}_1 + \mathcal{S}_2$ .

**The intersection.** It is not empty: it contains  $\tilde{0}$ . If  $\tilde{u}$  and  $\tilde{w}$  are in both  $\mathcal{S}_1$  and  $\mathcal{S}_2$ , then  $\tilde{u} + \tilde{w}$  and  $c\tilde{u}$  are in  $\mathcal{S}_1$ , because  $\mathcal{S}_1$  is a subspace, and in  $\mathcal{S}_2$  for the same reason, so they are in  $\mathcal{S}_1 \cap \mathcal{S}_2$ .  $\square$

**Example 0.44** (Two planes in  $\mathbb{R}^3$ ). Let  $\mathcal{S}_1 = \{(a, b, 0) : a, b \in \mathbb{R}\}$ , the plane  $z = 0$ , and  $\mathcal{S}_2 = \{(0, b, c) : b, c \in \mathbb{R}\}$ , the plane  $x = 0$ .

- **Intersection:** a vector in both has third entry 0 and first entry 0, so  $\mathcal{S}_1 \cap \mathcal{S}_2 = \{(0, b, 0) : b \in \mathbb{R}\}$ , the  $y$ -axis, which is a line through the origin.
- **Sum:**  $(a, b, 0) + (0, b', c) = (a, b + b', c)$ , and any  $(x, y, z)$  arises this way, with  $a = x$ ,  $b = y$ ,  $b' = 0$  and  $c = z$ . So  $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$ .

**Theorem 0.32** (The dimension of a sum of subspaces). *If  $\mathcal{S}_1$  and  $\mathcal{S}_2$  are subspaces of  $\mathbb{R}^p$ , then*

$$\dim(\mathcal{S}_1 + \mathcal{S}_2) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2) - \dim(\mathcal{S}_1 \cap \mathcal{S}_2).$$

### **i** Proof

*Proof.* All three sets in the formula are subspaces (Theorem 0.31), so each has a dimension (Definition 0.31). Let  $k = \dim(\mathcal{S}_1 \cap \mathcal{S}_2)$ ,  $d_1 = \dim(\mathcal{S}_1)$  and  $d_2 = \dim(\mathcal{S}_2)$ .

**Build a list.** Choose a basis  $\tilde{z}_1, \dots, \tilde{z}_k$  of  $\mathcal{S}_1 \cap \mathcal{S}_2$  (Theorem 0.19). These vectors are linearly independent and lie in  $\mathcal{S}_1$ , so Theorem 0.19 extends them to a basis  $\tilde{z}_1, \dots, \tilde{z}_k, \tilde{x}_1, \dots, \tilde{x}_{d_1-k}$  of  $\mathcal{S}_1$ , which has  $d_1$  vectors (Theorem 0.17). In the same way, extend them to a basis  $\tilde{z}_1, \dots, \tilde{z}_k, \tilde{y}_1, \dots, \tilde{y}_{d_2-k}$  of  $\mathcal{S}_2$ . We show that the  $\tilde{z}$ 's,  $\tilde{x}$ 's and  $\tilde{y}$ 's together,  $k + (d_1 - k) + (d_2 - k) = d_1 + d_2 - k$  vectors, are a basis of  $\mathcal{S}_1 + \mathcal{S}_2$  (Definition 0.30).

**They span  $\mathcal{S}_1 + \mathcal{S}_2$ .** Any  $\tilde{v} \in \mathcal{S}_1 + \mathcal{S}_2$  is  $\tilde{v}_1 + \tilde{v}_2$  with  $\tilde{v}_1 \in \mathcal{S}_1$  and  $\tilde{v}_2 \in \mathcal{S}_2$ .  $\tilde{v}_1$  is a linear combination of the  $\tilde{z}$ 's and  $\tilde{x}$ 's, and  $\tilde{v}_2$  is a linear combination of the  $\tilde{z}$ 's and  $\tilde{y}$ 's, so their sum is a linear combination of all three groups. Conversely, each vector in the list is in  $\mathcal{S}_1 + \mathcal{S}_2$  (for example,  $\tilde{x}_1 = \tilde{x}_1 + \tilde{0}$ ), and  $\mathcal{S}_1 + \mathcal{S}_2$  is a subspace, so it contains every linear combination of them.

**They are linearly independent.** Suppose

$$\sum_{i=1}^k a_i \tilde{z}_i + \sum_{j=1}^{d_1-k} b_j \tilde{x}_j + \sum_{l=1}^{d_2-k} c_l \tilde{y}_l = \tilde{0}.$$

Let  $\tilde{w} \stackrel{\text{def}}{=} \sum_l c_l \tilde{y}_l$ . Then  $\tilde{w} \in \mathcal{S}_2$ , because it is a linear combination of vectors of  $\mathcal{S}_2$ . Subtracting  $\sum_i a_i \tilde{z}_i + \sum_j b_j \tilde{x}_j$  from both sides of the supposed equation gives

$$\tilde{w} = \sum_{i=1}^k (-a_i) \tilde{z}_i + \sum_{j=1}^{d_1-k} (-b_j) \tilde{x}_j,$$

a linear combination of vectors of  $\mathcal{S}_1$ , so  $\tilde{w} \in \mathcal{S}_1$  too. So  $\tilde{w} \in \mathcal{S}_1 \cap \mathcal{S}_2$ , and  $\tilde{w} = \sum_i e_i \tilde{z}_i$  for some numbers  $e_i$ , since the  $\tilde{z}$ 's span the intersection. Then

$$\begin{aligned} \tilde{0} &= \tilde{w} - \tilde{w} && \text{(a vector minus itself)} \\ &= \sum_{i=1}^k e_i \tilde{z}_i - \sum_{l=1}^{d_2-k} c_l \tilde{y}_l && \text{(substitute the two expressions for } \tilde{w}) \\ &= \sum_{i=1}^k e_i \tilde{z}_i + \sum_{l=1}^{d_2-k} (-c_l) \tilde{y}_l, && \text{(write each subtraction as adding } -1 \text{ times)} \end{aligned}$$

and because the  $\tilde{z}$ 's and  $\tilde{y}$ 's are a basis of  $\mathcal{S}_2$ , every  $c_l = 0$ . The supposed equation then reads  $\sum_i a_i \tilde{z}_i + \sum_j b_j \tilde{x}_j = \tilde{0}$ , and because the  $\tilde{z}$ 's and  $\tilde{x}$ 's are a basis of  $\mathcal{S}_1$ , every  $a_i$  and  $b_j$  is 0.

So  $\dim(\mathcal{S}_1 + \mathcal{S}_2) = d_1 + d_2 - k$ . When some of the groups are empty (for example  $k = 0$ , or  $\mathcal{S}_1 = \mathcal{S}_1 \cap \mathcal{S}_2$ ), the sums over them are  $\tilde{0}$  and the argument goes through unchanged.  $\square$

**Example 0.45** (Two planes in  $\mathbb{R}^3$ , by dimension). For the planes  $z = 0$  and  $x = 0$  in Example 0.44, each has dimension 2: the plane  $z = 0$  has basis  $(1, 0, 0), (0, 1, 0)$  (Example 0.22), and the plane  $x = 0$  has basis  $(0, 1, 0), (0, 0, 1)$  by the same argument. Their intersection, the  $y$ -axis  $\text{span}\{(0, 1, 0)\}$ , has dimension 1, because the single nonzero vector  $(0, 1, 0)$  is linearly independent ( $c(0, 1, 0) = (0, c, 0)$  is  $\tilde{0}$  only if  $c = 0$ ) and so is a basis of it. Theorem 0.32 gives

$$\dim(\mathcal{S}_1 + \mathcal{S}_2) = 2 + 2 - 1 = 3,$$

which agrees with  $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$  (Example 0.44) and  $\dim(\mathbb{R}^3) = 3$  (Example 0.22).

**Corollary 0.2** (The dimension of a sum is at most the sum of the dimensions). *If  $\mathcal{S}_1$  and  $\mathcal{S}_2$  are subspaces of  $\mathbb{R}^p$ , then*

$$\dim(\mathcal{S}_1 + \mathcal{S}_2) \leq \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2),$$

*with equality exactly when  $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$ .*

#### **i** Proof

*Proof.* By Theorem 0.32, the two sides differ by  $\dim(\mathcal{S}_1 \cap \mathcal{S}_2) \geq 0$ . That dimension is 0 exactly when a basis of the intersection is the empty list, that is, when the intersection is the span of the empty list,  $\{\tilde{0}\}$  (Definition 0.29, Example 0.22).  $\square$

**Example 0.46** (Equality and strict inequality).

- For the lines  $\mathcal{S}_1 = \text{span}\{(1, 0, 0)\}$  and  $\mathcal{S}_2 = \text{span}\{(0, 1, 0)\}$  in Example 0.42, a common vector satisfies  $a(1, 0, 0) = b(0, 1, 0)$ , so  $(a, -b, 0) = \tilde{0}$  and  $a = b = 0$ . The intersection is  $\{\tilde{0}\}$ , and indeed  $\dim(\mathcal{S}_1 + \mathcal{S}_2) = 2 = 1 + 1$ : the sum is the plane  $z = 0$  (Example 0.42), which has dimension 2 (Example 0.22), and each line has dimension 1, with its spanning vector as a basis.
- For  $\mathcal{S}_1 = \mathcal{S}_2 = \text{span}\{(1, 0, 0)\}$ , the sum is the same line, because  $a(1, 0, 0) + b(1, 0, 0) = (a + b)(1, 0, 0)$ , so  $\dim(\mathcal{S}_1 + \mathcal{S}_2) = 1 < 1 + 1$ .

**Theorem 0.33** (Rank is subadditive). *For  $m \times n$  matrices  $\mathbf{A}$  and  $\mathbf{B}$ ,*

$$\text{rank}(\mathbf{A} + \mathbf{B}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}).$$

**i** Proof

*Proof.*  $\mathcal{C}(\mathbf{A} + \mathbf{B}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})$ . Any vector of  $\mathcal{C}(\mathbf{A} + \mathbf{B})$  is  $(\mathbf{A} + \mathbf{B})\tilde{x}$  for some  $\tilde{x} \in \mathbb{R}^n$  (Definition 0.32), and  $(\mathbf{A} + \mathbf{B})\tilde{x} = \mathbf{A}\tilde{x} + \mathbf{B}\tilde{x}$  (Theorem 0.8, with  $\tilde{x}$  as an  $n \times 1$  matrix), a vector of  $\mathcal{C}(\mathbf{A})$  plus a vector of  $\mathcal{C}(\mathbf{B})$ .

**Compare dimensions.** All the sets involved are subspaces of  $\mathbb{R}^m$  (Theorem 0.22, Theorem 0.31), so

$$\begin{aligned} \text{rank}(\mathbf{A} + \mathbf{B}) &= \dim(\mathcal{C}(\mathbf{A} + \mathbf{B})) && (\text{@thm-rank-dim}) \\ &\leq \dim(\mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})) && (\text{the containment, and part 2 of @thm-dim-bound}) \\ &\leq \dim(\mathcal{C}(\mathbf{A})) + \dim(\mathcal{C}(\mathbf{B})) && (\text{@cor-dim-subadditive}) \\ &= \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}). && (\text{@thm-rank-dim}) \end{aligned}$$

□

**Example 0.47** (Subadditivity with equality and without). Let  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ . Each has one nonzero column and one zero column. The nonzero column on its own is linearly independent ( $c\tilde{v} = \tilde{0}_2$  with  $\tilde{v} \neq \tilde{0}_2$  forces  $c = 0$ ), and the two columns together are not, since 1 times the zero column is  $\tilde{0}_2$ ; so each matrix has rank 1 (Definition 0.26). By the same argument,  $-\mathbf{A}$  has rank 1.

- $\mathbf{A} + \mathbf{B}$  is the  $2 \times 2$  identity matrix, which has rank 2 (Example 0.36), so the bound  $2 \leq 1 + 1$  holds with equality.
- $\mathbf{A} + (-\mathbf{A})$  is the  $2 \times 2$  zero matrix, which has rank 0: any nonempty list of its columns contains  $\tilde{0}_2$ , and  $1 \cdot \tilde{0}_2 = \tilde{0}_2$ . So the bound  $0 \leq \text{rank}(\mathbf{A}) + \text{rank}(-\mathbf{A}) = 1 + 1$  is strict.

**Definition 0.37** (Direct sum). Let  $\mathcal{S}_1$ ,  $\mathcal{S}_2$  and  $\mathcal{V}$  be subspaces of  $\mathbb{R}^p$ .  $\mathcal{S}_1$  and  $\mathcal{S}_2$  are **complementary** in  $\mathcal{V}$  if

$$\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2 \quad \text{and} \quad \mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$$

(Definition 0.36). Then  $\mathcal{V}$  is the **direct sum** of  $\mathcal{S}_1$  and  $\mathcal{S}_2$ , written  $\mathcal{V} = \mathcal{S}_1 \oplus \mathcal{S}_2$ .

**Example 0.48** ( $\mathbb{R}^3$  is a plane plus a line). Let  $\mathcal{S}_1 = \{(a, b, 0) : a, b \in \mathbb{R}\}$ , the plane  $z = 0$ , and  $\mathcal{S}_2 = \{(0, 0, c) : c \in \mathbb{R}\}$ , the  $z$ -axis.

- **Sum:** any  $(x, y, z) \in \mathbb{R}^3$  is  $(x, y, 0) + (0, 0, z)$ , so  $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$ .

- **Intersection:** a vector in both has third entry 0 and first and second entries 0, so  $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$ .

So  $\mathbb{R}^3 = \mathcal{S}_1 \oplus \mathcal{S}_2$ .

**Example 0.49** (Two planes that sum to  $\mathbb{R}^3$  but not directly). The planes  $z = 0$  and  $x = 0$  sum to  $\mathbb{R}^3$  (Example 0.44), but they share the  $y$ -axis, so they are not complementary, and  $\mathbb{R}^3$  is not their direct sum. The shared line shows up as more than one way to split a vector:  $(0, 1, 0) = (0, 1, 0) + \tilde{0} = \tilde{0} + (0, 1, 0)$ , with the first term in the plane  $z = 0$  and the second in the plane  $x = 0$  both times.

**Theorem 0.34** (Three ways to recognize a direct sum). *Let  $\mathcal{S}_1$  and  $\mathcal{S}_2$  be subspaces of  $\mathbb{R}^p$ , and let  $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$ . The following statements are equivalent:*

1.  $\mathcal{V} = \mathcal{S}_1 \oplus \mathcal{S}_2$  (Definition 0.37).
2.  $\dim(\mathcal{V}) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2)$ .
3. Every  $\tilde{x} \in \mathcal{V}$  can be written as  $\tilde{x} = \tilde{x}_1 + \tilde{x}_2$  with  $\tilde{x}_1 \in \mathcal{S}_1$  and  $\tilde{x}_2 \in \mathcal{S}_2$  in only one way.

### Proof

*Proof.* Since  $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$  is given, statement 1 says exactly that  $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$ .

**1 and 2 are equivalent.** By Corollary 0.2,  $\dim(\mathcal{S}_1 + \mathcal{S}_2) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2)$  exactly when  $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$ .

**1 implies 3.** Every  $\tilde{x} \in \mathcal{V}$  has at least one such expression, because  $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$ . Suppose  $\tilde{x} = \tilde{u}_1 + \tilde{u}_2 = \tilde{v}_1 + \tilde{v}_2$ , with  $\tilde{u}_1, \tilde{v}_1 \in \mathcal{S}_1$  and  $\tilde{u}_2, \tilde{v}_2 \in \mathcal{S}_2$ . Then

$$\begin{aligned} \tilde{u}_1 - \tilde{v}_1 &= (\tilde{u}_1 + \tilde{u}_2) - \tilde{u}_2 - \tilde{v}_1 && \text{(add and subtract } \tilde{u}_2) \\ &= (\tilde{v}_1 + \tilde{v}_2) - \tilde{u}_2 - \tilde{v}_1 && \text{(substitute } \tilde{u}_1 + \tilde{u}_2 = \tilde{v}_1 + \tilde{v}_2) \\ &= \tilde{v}_2 - \tilde{u}_2. && \text{(cancel } \tilde{v}_1) \end{aligned}$$

The left side is in  $\mathcal{S}_1$  and the right side is in  $\mathcal{S}_2$ , because each subspace is closed under addition and scalar multiplication (the difference is  $\tilde{u}_1 + (-1)\tilde{v}_1$ ). So both sides are in  $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$ , which gives  $\tilde{u}_1 = \tilde{v}_1$  and  $\tilde{u}_2 = \tilde{v}_2$ .

**3 implies 1.** Take any  $\tilde{x} \in \mathcal{S}_1 \cap \mathcal{S}_2$ . Then  $\tilde{x} = \tilde{x} + \tilde{0}$ , with  $\tilde{x} \in \mathcal{S}_1$  and  $\tilde{0} \in \mathcal{S}_2$ , and  $\tilde{x} = \tilde{0} + \tilde{x}$ , with  $\tilde{0} \in \mathcal{S}_1$  and  $\tilde{x} \in \mathcal{S}_2$  (Theorem 0.11). By statement 3 these two expressions are the same, so  $\tilde{x} = \tilde{0}$ , and  $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\tilde{0}\}$ .  $\square$

**Example 0.50** (Checking the three statements). For the plane  $z = 0$  and the  $z$ -axis in Example 0.48:

- the dimensions add up:  $2 + 1 = 3 = \dim(\mathbb{R}^3)$ , since the plane has dimension 2 (Example 0.22), the  $z$ -axis has the single nonzero vector  $(0, 0, 1)$  as a basis, and  $\dim(\mathbb{R}^3) = 3$  (Example 0.22);
- the split is unique: if  $(x, y, z) = (a, b, 0) + (0, 0, c) = (a, b, c)$ , then  $a = x$ ,  $b = y$  and  $c = z$ .

For the planes  $z = 0$  and  $x = 0$  in Example 0.49, both statements fail:  $2 + 2 = 4 \neq 3 = \dim(\mathcal{S}_1 + \mathcal{S}_2)$  (Example 0.45), and  $(0, 1, 0)$  splits in two ways.

## Orthogonal complements

**Theorem 0.35** (The dot product is linear in each slot). *For vectors  $\tilde{x}, \tilde{u}, \tilde{w} \in \mathbb{R}^p$  and numbers  $a, b$ ,*

$$\tilde{x} \cdot (a \tilde{u} + b \tilde{w}) = a (\tilde{x} \cdot \tilde{u}) + b (\tilde{x} \cdot \tilde{w})$$

*and*

$$(a \tilde{u} + b \tilde{w}) \cdot \tilde{x} = a (\tilde{u} \cdot \tilde{x}) + b (\tilde{w} \cdot \tilde{x}).$$

### **i** Proof

*Proof.* **Second slot.**

$$\begin{aligned}
\tilde{x} \cdot (a\tilde{u} + b\tilde{w}) &= \sum_{i=1}^p x_i (a\tilde{u} + b\tilde{w})_i && (\text{@def-dot-product}) \\
&= \sum_{i=1}^p x_i (a u_i + b w_i) && (\text{@def-scalar-mult, @def-vector-addition}) \\
&= \sum_{i=1}^p (x_i a u_i + x_i b w_i) && (\text{distribute each } x_i) \\
&= \sum_{i=1}^p (a x_i u_i + b x_i w_i) && (\text{commute the factors in each product}) \\
&= \sum_{i=1}^p a x_i u_i + \sum_{i=1}^p b x_i w_i && (\text{split the finite sum}) \\
&= a \sum_{i=1}^p x_i u_i + b \sum_{i=1}^p x_i w_i && (\text{factor } a \text{ and } b \text{ out of the sums}) \\
&= a(\tilde{x} \cdot \tilde{u}) + b(\tilde{x} \cdot \tilde{w}). && (\text{@def-dot-product})
\end{aligned}$$

**First slot.**

$$\begin{aligned}
(a\tilde{u} + b\tilde{w}) \cdot \tilde{x} &= \tilde{x} \cdot (a\tilde{u} + b\tilde{w}) && (\text{@thm-lincom-symmetric}) \\
&= a(\tilde{x} \cdot \tilde{u}) + b(\tilde{x} \cdot \tilde{w}) && (\text{second slot}) \\
&= a(\tilde{u} \cdot \tilde{x}) + b(\tilde{w} \cdot \tilde{x}). && (\text{@thm-lincom-symmetric, twice})
\end{aligned}$$

Two special cases are used often. With  $a = b = 1$ , and  $1\tilde{u} = \tilde{u}$ , the theorem gives  $\tilde{x} \cdot (\tilde{u} + \tilde{w}) = \tilde{x} \cdot \tilde{u} + \tilde{x} \cdot \tilde{w}$ . With  $\tilde{w} = \tilde{u}$  and  $b = 0$ , and  $0\tilde{u} = \tilde{0}$ , it gives  $\tilde{x} \cdot (a\tilde{u}) = a(\tilde{x} \cdot \tilde{u})$ , and likewise in the first slot.  $\square$

**Example 0.51** (Splitting a dot product). Let  $\tilde{x} = (1, 2)$ ,  $\tilde{u} = (3, 0)$ ,  $\tilde{w} = (0, 1)$ ,  $a = 2$  and  $b = -1$ . Directly,  $a\tilde{u} + b\tilde{w} = (6, -1)$  and  $\tilde{x} \cdot (6, -1) = 6 - 2 = 4$ . By Theorem 0.35,  $a(\tilde{x} \cdot \tilde{u}) + b(\tilde{x} \cdot \tilde{w}) = 2 \cdot 3 + (-1) \cdot 2 = 4$ , the same number.

**Definition 0.38** (Orthogonal complement). The **orthogonal complement** of a set  $\mathcal{X}$  of vectors in  $\mathbb{R}^p$  is the set of vectors orthogonal (Definition 0.10) to every vector of  $\mathcal{X}$ :

$$\mathcal{X}^\perp \stackrel{\text{def}}{=} \{\tilde{u} \in \mathbb{R}^p : \tilde{x} \cdot \tilde{u} = 0 \text{ for all } \tilde{x} \in \mathcal{X}\}.$$

$\mathcal{X}^\perp$  is read “ $\mathcal{X}$  perp”.  $\mathcal{X}$  need not be a subspace.

**Example 0.52** (Orthogonal complements in  $\mathbb{R}^3$ ).

- **A plane.** Let  $\mathcal{X} = \{(a, b, 0) : a, b \in \mathbb{R}\}$ , the plane  $z = 0$ . A vector  $\tilde{u}$  is in  $\mathcal{X}^\perp$  when  $(a, b, 0) \cdot \tilde{u} = au_1 + bu_2 = 0$  for all  $a$  and  $b$ . Taking  $(a, b) = (1, 0)$  forces  $u_1 = 0$ , and taking  $(a, b) = (0, 1)$  forces  $u_2 = 0$ ; conversely, if  $u_1 = u_2 = 0$ , then  $au_1 + bu_2 = 0$  for all  $a$  and  $b$ . So  $\mathcal{X}^\perp = \{(0, 0, c) : c \in \mathbb{R}\}$ , the  $z$ -axis.
- **Two vectors.** The two-element set  $\{(1, 0, 0), (0, 1, 0)\}$  is not a subspace, and its orthogonal complement is the  $z$ -axis too: the two conditions  $u_1 = 0$  and  $u_2 = 0$  are the same as for the plane.
- **The extremes.**  $\{\tilde{0}\}^\perp = \mathbb{R}^3$ , because  $\tilde{0} \cdot \tilde{u} = 0$  for every  $\tilde{u}$ .  $(\mathbb{R}^3)^\perp = \{\tilde{0}\}$ , because a  $\tilde{u}$  in it is orthogonal to itself, and  $\tilde{u} \cdot \tilde{u} = u_1^2 + u_2^2 + u_3^2 = 0$  forces  $\tilde{u} = \tilde{0}$ .

**Theorem 0.36** (An orthogonal complement is a subspace). *For any set  $\mathcal{X}$  of vectors in  $\mathbb{R}^p$ ,  $\mathcal{X}^\perp$  (Definition 0.38) is a subspace of  $\mathbb{R}^p$  (Definition 0.28).*

### i Proof

*Proof.*  $\mathcal{X}^\perp$  is not empty:  $\tilde{x} \cdot \tilde{0} = \sum_i x_i \cdot 0 = 0$  for every  $\tilde{x}$ , so  $\tilde{0} \in \mathcal{X}^\perp$ . Let  $\tilde{u}, \tilde{w} \in \mathcal{X}^\perp$ , let  $c \in \mathbb{R}$ , and let  $\tilde{x} \in \mathcal{X}$ .

**Addition.**

$$\begin{aligned} \tilde{x} \cdot (\tilde{u} + \tilde{w}) &= \tilde{x} \cdot \tilde{u} + \tilde{x} \cdot \tilde{w} && (\text{@thm-dot-linear, special case } a = b = 1) \\ &= 0 + 0 && (\tilde{u}, \tilde{w} \in \mathcal{X}^\perp) \\ &= 0. && (\text{arithmetic}) \end{aligned}$$

**Scalar multiplication.**

$$\begin{aligned} \tilde{x} \cdot (c\tilde{u}) &= c(\tilde{x} \cdot \tilde{u}) && (\text{@thm-dot-linear, special case } b = 0) \\ &= c \cdot 0 && (\tilde{u} \in \mathcal{X}^\perp) \\ &= 0. && (\text{arithmetic}) \end{aligned}$$

These two equations hold for every  $\tilde{x} \in \mathcal{X}$ , so  $\tilde{u} + \tilde{w}$  and  $c\tilde{u}$  are in  $\mathcal{X}^\perp$ . □

**Example 0.53** (The complement of a single vector is a line). The one-element set  $\mathcal{X} = \{(1, 2)\}$  in  $\mathbb{R}^2$  is not a subspace:  $2(1, 2) = (2, 4)$  is not in it. Its orthogonal complement is

$$\mathcal{X}^\perp = \{\tilde{u} \in \mathbb{R}^2 : u_1 + 2u_2 = 0\} = \{c(-2, 1) : c \in \mathbb{R}\},$$

since  $u_1 + 2u_2 = 0$  means  $\tilde{u} = (-2u_2, u_2) = u_2(-2, 1)$ . That set is  $\text{span}\{(-2, 1)\}$ , a subspace by Theorem 0.12, as Theorem 0.36 says it must be. It contains  $(-2, 1)$ , the vector Remark 0.7 found perpendicular to  $(1, 2)$ .

**Theorem 0.37** (The orthogonal complement of a column space is a null space). *For any  $m \times n$  matrix  $\mathbf{A}$ ,*

$$\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top).$$

### **i** Proof

*Proof.* Let  $\tilde{a}_1, \dots, \tilde{a}_n$  be the columns of  $\mathbf{A}$ , so  $\tilde{a}_j = (a_{1j}, \dots, a_{mj})$ . For  $\tilde{x} \in \mathbb{R}^m$ , entry  $j$  of  $\mathbf{A}^\top \tilde{x}$  is

$$\begin{aligned} (\mathbf{A}^\top \tilde{x})_j &= \sum_{i=1}^m (\mathbf{A}^\top)_{ji} x_i && (\text{@def-matvec-mult}) \\ &= \sum_{i=1}^m a_{ij} x_i && (\text{@def-matrix-transpose}) \\ &= \tilde{a}_j \cdot \tilde{x}. && (\text{@def-dot-product}) \end{aligned}$$

So  $\tilde{x} \in \mathcal{N}(\mathbf{A}^\top)$  (Definition 0.33) exactly when  $\tilde{a}_j \cdot \tilde{x} = 0$  for every column  $\tilde{a}_j$ .

$\mathcal{C}(\mathbf{A})^\perp \subseteq \mathcal{N}(\mathbf{A}^\top)$ . Each column  $\tilde{a}_j$  is in  $\mathcal{C}(\mathbf{A})$  (Theorem 0.22), so a vector orthogonal to all of  $\mathcal{C}(\mathbf{A})$  is orthogonal to every column.

$\mathcal{N}(\mathbf{A}^\top) \subseteq \mathcal{C}(\mathbf{A})^\perp$ . Suppose  $\tilde{a}_j \cdot \tilde{x} = 0$  for every  $j$ , and take any  $\tilde{y} \in \mathcal{C}(\mathbf{A})$ , so  $\tilde{y} = \mathbf{A}\tilde{z}$  for some  $\tilde{z} \in \mathbb{R}^n$  (Definition 0.32). Then

$$\begin{aligned} \tilde{y} \cdot \tilde{x} &= \sum_{i=1}^m (\mathbf{A}\tilde{z})_i x_i && (\text{@def-dot-product}) \\ &= \sum_{i=1}^m \left( \sum_{j=1}^n a_{ij} z_j \right) x_i && (\text{@def-matvec-mult}) \\ &= \sum_{i=1}^m \sum_{j=1}^n a_{ij} z_j x_i && (\text{distribute each } x_i \text{ over the inner sum}) \\ &= \sum_{j=1}^n \sum_{i=1}^m a_{ij} z_j x_i && (\text{swap the order of the finite sums}) \\ &= \sum_{j=1}^n z_j \sum_{i=1}^m a_{ij} x_i && (\text{commute, then factor } z_j \text{ out of the inner sum}) \\ &= \sum_{j=1}^n z_j (\tilde{a}_j \cdot \tilde{x}) && (\text{@def-dot-product}) \\ &= \sum_{j=1}^n z_j \cdot 0 && (\text{each } \tilde{a}_j \cdot \tilde{x} = 0) \\ &= 0. && (\text{arithmetic}) \end{aligned}$$

So  $\tilde{x}$  is orthogonal to every vector of  $\mathcal{C}(\mathbf{A})$ . □

**Example 0.54** (The complement of the column space in Example 0.28). For  $\mathbf{A}$  in Example 0.28,  $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$  (1). Example 0.30 found  $\mathcal{N}(\mathbf{A}^\top) = \text{span}\{(-3, 1)\}$ , so by Theorem 0.37 this line is  $\mathcal{C}(\mathbf{A})^\perp$ . Directly, for any  $c$ ,

$$\begin{aligned} (c(1, 3)) \cdot (-3, 1) &= c((1, 3) \cdot (-3, 1)) && \text{(@thm-dot-linear, special case } b = 0, \text{ first slot)} \\ &= c(-3 + 3) && \text{(@def-dot-product)} \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

**Theorem 0.38** (A subspace and its orthogonal complement make up the whole space). *Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^p$ . Then*

1.  $\mathcal{S} \cap \mathcal{S}^\perp = \{\tilde{0}\}$ ;
2.  $\dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) = p$ ;
3.  $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{S}^\perp$  (Definition 0.37).

*So every  $\tilde{y} \in \mathbb{R}^p$  can be written as  $\tilde{y} = \tilde{u} + \tilde{v}$  with  $\tilde{u} \in \mathcal{S}$  and  $\tilde{v} \in \mathcal{S}^\perp$  in only one way (Theorem 0.34).*

### **i** Proof

*Proof.*  $\mathcal{S}^\perp$  is a subspace (Theorem 0.36), so it has a dimension (Definition 0.31).

**Part 1.** Both  $\mathcal{S}$  and  $\mathcal{S}^\perp$  contain  $\tilde{0}$  (Theorem 0.11). If  $\tilde{x}$  is in both, then  $\tilde{x}$  is orthogonal to every vector of  $\mathcal{S}$ ,  $\tilde{x}$  itself included, so  $\tilde{x} \cdot \tilde{x} = x_1^2 + \cdots + x_p^2 = 0$ , which forces every  $x_i = 0$ .

**Part 2.** Let  $d = \dim(\mathcal{S})$ . If  $d = 0$ , then  $\mathcal{S}$  has a basis (Theorem 0.19) with 0 vectors (Definition 0.31), the empty list, whose span is  $\{\tilde{0}\}$  (Definition 0.29); so  $\mathcal{S} = \{\tilde{0}\}$ ,  $\mathcal{S}^\perp = \mathbb{R}^p$  (every  $\tilde{u}$  satisfies  $\tilde{0} \cdot \tilde{u} = 0$ ), and  $0 + p = p$  (Example 0.22). If  $d \geq 1$ , choose a basis  $\tilde{a}_1, \dots, \tilde{a}_d$  of  $\mathcal{S}$  (Theorem 0.19), and let  $\mathbf{A}$  be the  $p \times d$  matrix with these columns. The basis spans  $\mathcal{S}$ , so  $\mathcal{S} = \mathcal{C}(\mathbf{A})$  (Theorem 0.22), and  $\mathcal{S}^\perp = \mathcal{N}(\mathbf{A}^\top)$  (Theorem 0.37).  $\mathbf{A}^\top$  is  $d \times p$ , so

$$\begin{aligned} \dim(\mathcal{S}^\perp) &= \dim(\mathcal{N}(\mathbf{A}^\top)) && \text{(@thm-complement-null-space)} \\ &= \text{nullity}(\mathbf{A}^\top) && \text{(@def-nullity)} \\ &= p - \text{rank}(\mathbf{A}^\top) && \text{(@thm-rank-nullity, for the } p \text{ columns of } \mathbf{A}^\top) \\ &= p - \text{rank}(\mathbf{A}) && \text{(@thm-rank-transpose)} \\ &= p - \dim(\mathcal{C}(\mathbf{A})) && \text{(@thm-rank-dim)} \\ &= p - d. && (\mathcal{C}(\mathbf{A}) = \mathcal{S}) \end{aligned}$$

**Part 3.**  $\mathcal{S} + \mathcal{S}^\perp$  is a subspace of  $\mathbb{R}^p$  (Theorem 0.31), and

$$\begin{aligned}
\dim(\mathcal{S} + \mathcal{S}^\perp) &= \dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) - \dim(\mathcal{S} \cap \mathcal{S}^\perp) && \text{(@thm-dim-sum)} \\
&= p - \dim(\mathcal{S} \cap \mathcal{S}^\perp) && \text{(part 2)} \\
&= p - 0 && \text{(part 1, and } \dim(\{\tilde{0}\}) = 0 \text{ by @exm-dimension)} \\
&= p. && \text{(arithmetic)}
\end{aligned}$$

Since  $\mathcal{S} + \mathcal{S}^\perp \subseteq \mathbb{R}^p$ ,  $\mathbb{R}^p$  is a subspace (Example 0.10) and  $\dim(\mathbb{R}^p) = p$  (Example 0.22), Theorem 0.21 gives  $\mathcal{S} + \mathcal{S}^\perp = \mathbb{R}^p$ . Together with part 1,  $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{S}^\perp$ . The uniqueness of the split then follows from Theorem 0.34 (statement 1 implies statement 3).  $\square$

**Example 0.55** (Splitting a vector of  $\mathbb{R}^3$  along a line and its complement). Let  $\mathcal{S} = \text{span}\{(1, 1, 0)\}$  in  $\mathbb{R}^3$ . A single nonzero vector is linearly independent ( $c\tilde{v} = \tilde{0}$  with  $\tilde{v} \neq \tilde{0}$  forces  $c = 0$ ), so  $(1, 1, 0)$  is a basis of  $\mathcal{S}$  and  $\dim(\mathcal{S}) = 1$ . A vector  $\tilde{u}$  is in  $\mathcal{S}^\perp$  exactly when  $u_1 + u_2 = 0$ : that condition says  $(1, 1, 0) \cdot \tilde{u} = 0$ , and then  $c(1, 1, 0) \cdot \tilde{u} = c(u_1 + u_2) = 0$  for every  $c$ . So  $\mathcal{S}^\perp = \{(a, -a, c) : a, c \in \mathbb{R}\} = \text{span}\{(1, -1, 0), (0, 0, 1)\}$ . Those two vectors are linearly independent, because  $a(1, -1, 0) + c(0, 0, 1) = (a, -a, c)$  is  $\tilde{0}$  only if  $a = c = 0$ , so  $\dim(\mathcal{S}^\perp) = 2$ , and  $1 + 2 = 3$ , as part 2 says.

To split  $\tilde{y} = (3, 1, 2)$ , look for  $\tilde{u} = t(1, 1, 0)$  with  $\tilde{v} = \tilde{y} - \tilde{u} = (3 - t, 1 - t, 2)$  in  $\mathcal{S}^\perp$ : that membership needs  $(3 - t) + (1 - t) = 0$ , so  $t = 2$ . Then  $\tilde{u} = (2, 2, 0)$  and  $\tilde{v} = (1, -1, 2)$ , with  $\tilde{u} + \tilde{v} = (3, 1, 2)$  and  $(1, 1, 0) \cdot (1, -1, 2) = 1 - 1 + 0 = 0$ .

**Theorem 0.39** (The orthogonal complement of the orthogonal complement). *For any subspace  $\mathcal{S}$  of  $\mathbb{R}^p$ ,*

$$(\mathcal{S}^\perp)^\perp = \mathcal{S}.$$

### **i** Proof

*Proof.*  $\mathcal{S} \subseteq (\mathcal{S}^\perp)^\perp$ . Take  $\tilde{s} \in \mathcal{S}$  and any  $\tilde{v} \in \mathcal{S}^\perp$ . By Definition 0.38,  $\tilde{s} \cdot \tilde{v} = 0$ , so  $\tilde{v} \cdot \tilde{s} = 0$  too (Theorem 0.1). So  $\tilde{s}$  is orthogonal to every vector of  $\mathcal{S}^\perp$ , that is,  $\tilde{s} \in (\mathcal{S}^\perp)^\perp$ .

**Equal dimensions.**  $\mathcal{S}^\perp$  is a subspace (Theorem 0.36), so part 2 of Theorem 0.38 applies to it as well as to  $\mathcal{S}$ :

$$\begin{aligned}
\dim((\mathcal{S}^\perp)^\perp) &= p - \dim(\mathcal{S}^\perp) && \text{(@thm-orthogonal-direct-sum, applied to } \mathcal{S}^\perp\text{)} \\
&= p - (p - \dim(\mathcal{S})) && \text{(@thm-orthogonal-direct-sum, applied to } \mathcal{S}\text{)} \\
&= \dim(\mathcal{S}). && \text{(arithmetic)}
\end{aligned}$$

$(\mathcal{S}^\perp)^\perp$  is a subspace (Theorem 0.36) that contains  $\mathcal{S}$  and has the same dimension, so it equals  $\mathcal{S}$  (Theorem 0.21).  $\square$

**Example 0.56** (Back to the line). In Example 0.55,  $\mathcal{S} = \text{span}\{(1, 1, 0)\}$  and  $\mathcal{S}^\perp = \text{span}\{(1, -1, 0), (0, 0, 1)\}$ . Put those two spanning vectors in the columns of the  $3 \times 2$  matrix  $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \end{bmatrix}$ , so  $\mathcal{S}^\perp = \mathcal{C}(\mathbf{B})$  (Theorem 0.22) and  $(\mathcal{S}^\perp)^\perp = \mathcal{N}(\mathbf{B}^\top)$  (Theorem 0.37). Since  $\mathbf{B}^\top \tilde{w} = (w_1 - w_2, w_3)$  (Definition 0.21),  $\tilde{w} \in (\mathcal{S}^\perp)^\perp$  exactly when  $w_1 - w_2 = 0$  and  $w_3 = 0$ . So  $(\mathcal{S}^\perp)^\perp = \{(a, a, 0) : a \in \mathbb{R}\} = \mathcal{S}$ .

**Example 0.57** (A set that is not a subspace does not come back). The one-element set  $\mathcal{X} = \{(1, 1, 0)\}$  is not a subspace. Its orthogonal complement is the same as that of  $\mathcal{S}$  in Example 0.56, since both are defined by the single condition  $u_1 + u_2 = 0$ , so  $(\mathcal{X}^\perp)^\perp = \text{span}\{(1, 1, 0)\}$ , which contains  $(2, 2, 0) \notin \mathcal{X}$ . Theorem 0.39 needs  $\mathcal{S}$  to be a subspace.

## The fundamental theorem of linear algebra

**Theorem 0.40** (Fundamental theorem of linear algebra). *Let  $\mathbf{A}$  be an  $m \times n$  matrix with  $\text{rank}(\mathbf{A}) = r$ . Then*

1.  $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$ , and  $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top)$ ;
2.  $\mathcal{C}(\mathbf{A}) = \mathcal{N}(\mathbf{A}^\top)^\perp$ ;
3.  $\mathcal{N}(\mathbf{A})^\perp = \mathcal{C}(\mathbf{A}^\top)$ , and  $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$ ;
4.  $\dim(\mathcal{C}(\mathbf{A})) = \dim(\mathcal{C}(\mathbf{A}^\top)) = r$ ,  $\dim(\mathcal{N}(\mathbf{A})) = n - r$ , and  $\dim(\mathcal{N}(\mathbf{A}^\top)) = m - r$ .

### Proof

*Proof.* **Part 1.** The first equation is Theorem 0.37.  $\mathcal{C}(\mathbf{A})$  is a subspace of  $\mathbb{R}^m$  (Theorem 0.22), so

$$\begin{aligned} \mathbb{R}^m &= \mathcal{C}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})^\perp && (\text{@thm-orthogonal-direct-sum}) \\ &= \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top). && (\text{first equation}) \end{aligned}$$

**Part 2.**

$$\begin{aligned} \mathcal{C}(\mathbf{A}) &= (\mathcal{C}(\mathbf{A})^\perp)^\perp && (\text{@thm-double-complement}) \\ &= \mathcal{N}(\mathbf{A}^\top)^\perp. && (\text{part 1}) \end{aligned}$$

**Part 3.** Apply parts 2 and 1 to the  $n \times m$  matrix  $\mathbf{A}^\top$ , using  $(\mathbf{A}^\top)^\top = \mathbf{A}$ : part 2 gives  $\mathcal{C}(\mathbf{A}^\top) = \mathcal{N}(\mathbf{A})^\perp$ , and part 1 gives  $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$ .

**Part 4.**  $\dim(\mathcal{C}(\mathbf{A})) = r$  by Theorem 0.26.  $\dim(\mathcal{C}(\mathbf{A}^\top)) = \text{rank}(\mathbf{A}^\top)$  by Theorem 0.26, and that is  $r$  by Theorem 0.29.  $\mathbf{A}$  has  $n$  columns, so  $\dim(\mathcal{N}(\mathbf{A})) = n - r$  (Definition 0.34, Theorem 0.27).  $\mathbf{A}^\top$  has  $m$  columns and rank  $r$ , so  $\dim(\mathcal{N}(\mathbf{A}^\top)) = m - r$  in the same way.  $\square$

**Example 0.58** (The four subspaces of the matrix in Example 0.28). For  $\mathbf{A}$  in Example 0.28,  $m = 2$ ,  $n = 3$  and  $r = 1$  (Example 0.34).

- **In  $\mathbb{R}^3$ :** the row space  $\mathcal{C}(\mathbf{A}^\top) = \text{span}\{(1, -2, -2)\}$ 
  - (1) has dimension  $1 = r$ , and  $\mathcal{N}(\mathbf{A}) = \text{span}\{(2, 1, 0), (2, 0, 1)\}$  has dimension  $2 = n - r$  (Example 0.35). The spanning vectors are orthogonal:  $(1, -2, -2) \cdot (2, 1, 0) = 2 - 2 + 0 = 0$  and  $(1, -2, -2) \cdot (2, 0, 1) = 2 + 0 - 2 = 0$ .
- **In  $\mathbb{R}^2$ :**  $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$  has dimension  $1 = r$ , and  $\mathcal{N}(\mathbf{A}^\top) = \text{span}\{(-3, 1)\}$  has dimension  $1 = m - r$  (Example 0.54).

By part 3,  $\tilde{y} = (1, 0, 0)$  splits uniquely into a row-space part  $t(1, -2, -2)$  and a null-space part. The null-space part  $\tilde{y} - t(1, -2, -2)$  lies in  $\mathcal{N}(\mathbf{A})$  (part 3), and  $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp$  (Theorem 0.37 applied to  $\mathbf{A}^\top$ , with  $(\mathbf{A}^\top)^\top = \mathbf{A}$ ), so it must be orthogonal to  $(1, -2, -2)$ :

$$\begin{aligned}
0 &= (1, -2, -2) \cdot ((1, 0, 0) - t(1, -2, -2)) && \text{(orthogonality)} \\
&= (1, -2, -2) \cdot (1 \cdot (1, 0, 0) + (-t)(1, -2, -2)) && \text{(write the difference as a linear combination)} \\
&= 1 \cdot ((1, -2, -2) \cdot (1, 0, 0)) + (-t)((1, -2, -2) \cdot (1, -2, -2)) && \text{(@thm-dot-linear)} \\
&= 1 \cdot 1 + (-t) \cdot 9 && \text{(@def-dot-product)} \\
&= 1 - 9t, && \text{(arithmetic)}
\end{aligned}$$

so  $t = \frac{1}{9}$ . The row-space part is  $(\frac{1}{9}, -\frac{2}{9}, -\frac{2}{9})$  and the null-space part is  $(\frac{8}{9}, \frac{2}{9}, \frac{2}{9})$ ; as a check, row 1 of  $\mathbf{A}$  gives  $\frac{8}{9} - 2 \cdot \frac{2}{9} - 2 \cdot \frac{2}{9} = 0$ , and row 2 is 3 times row 1, so  $\mathbf{A}$  sends the null-space part to  $\tilde{0}_2$ .

## Cauchy-Schwarz and the triangle inequality

### Source

This section is adapted from Zhou (2024h), used under the MIT License (see the license text in Section ). The source leaves the proof of the triangle inequality to class; it is written out here.

**Theorem 0.41** (Basic properties of the Euclidean norm). *For any  $\tilde{x} \in \mathbb{R}^p$  and any number  $c$ :*

1.  $\|\tilde{x}\| \geq 0$ , and  $\|\tilde{x}\| = 0$  if and only if  $\tilde{x} = \tilde{0}$ ;
2.  $\|c\tilde{x}\| = |c| \|\tilde{x}\|$ .

**i** Proof

*Proof. Part 1.*  $\|\tilde{x}\| = \sqrt{x_1^2 + \cdots + x_p^2}$  (Equation 2) is the square root of a sum of squares, so it is at least 0. It is 0 exactly when  $x_1^2 + \cdots + x_p^2 = 0$ , and a sum of squares is 0 exactly when every term is 0, that is, when every  $x_i = 0$ : if some  $x_i \neq 0$ , its square is positive and the sum is positive, and if every  $x_i = 0$ , the sum is 0.

**Part 2.**

$$\begin{aligned}
 \|c\tilde{x}\| &= \sqrt{\sum_{i=1}^p (cx_i)^2} && (\text{@eq-l2-norm, and @def-scalar-mult}) \\
 &= \sqrt{\sum_{i=1}^p c^2 x_i^2} && ((cx_i)^2 = c^2 x_i^2) \\
 &= \sqrt{c^2 \sum_{i=1}^p x_i^2} && (\text{factor } c^2 \text{ out of the sum}) \\
 &= \sqrt{c^2} \sqrt{\sum_{i=1}^p x_i^2} && (\text{the square root of a product of nonnegative numbers}) \\
 &= |c| \|\tilde{x}\|. && (\sqrt{c^2} = |c|, \text{ and @eq-l2-norm})
 \end{aligned}$$

□

**Example 0.59** (Scaling a vector scales its length). For  $\tilde{x} = (3, 4)$ ,  $\|\tilde{x}\| = 5$  (Example 0.4). Then  $-2\tilde{x} = (-6, -8)$  has norm  $\sqrt{36 + 64} = 10 = |-2| \cdot 5$ , as part 2 says. Dividing by the norm gives a vector of length 1:  $\|\frac{1}{5}\tilde{x}\| = \frac{1}{5} \cdot 5 = 1$ , and  $\frac{1}{5}\tilde{x} = (0.6, 0.8)$  is the unit vector in Example 0.4.

**Theorem 0.42** (The squared norm of a sum). *For any  $\tilde{x}, \tilde{y} \in \mathbb{R}^p$ ,*

$$\|\tilde{x} + \tilde{y}\|^2 = \|\tilde{x}\|^2 + 2(\tilde{x} \cdot \tilde{y}) + \|\tilde{y}\|^2.$$

*In particular, if  $\tilde{x} \perp \tilde{y}$  (Definition 0.10), then  $\|\tilde{x} + \tilde{y}\|^2 = \|\tilde{x}\|^2 + \|\tilde{y}\|^2$  (the **Pythagorean theorem**).*

### **i** Proof

*Proof.*

$$\begin{aligned}
\|\tilde{x} + \tilde{y}\|^2 &= (\tilde{x} + \tilde{y}) \cdot (\tilde{x} + \tilde{y}) && \text{(square both sides of @eq-l2-norm)} \\
&= \tilde{x} \cdot (\tilde{x} + \tilde{y}) + \tilde{y} \cdot (\tilde{x} + \tilde{y}) && \text{(@thm-dot-linear, first slot, } a = b = 1) \\
&= \tilde{x} \cdot \tilde{x} + \tilde{x} \cdot \tilde{y} + \tilde{y} \cdot \tilde{x} + \tilde{y} \cdot \tilde{y} && \text{(@thm-dot-linear, second slot, } a = b = 1, \text{ twice)} \\
&= \tilde{x} \cdot \tilde{x} + \tilde{x} \cdot \tilde{y} + \tilde{x} \cdot \tilde{y} + \tilde{y} \cdot \tilde{y} && \text{(@thm-lincom-symmetric)} \\
&= \tilde{x} \cdot \tilde{x} + 2(\tilde{x} \cdot \tilde{y}) + \tilde{y} \cdot \tilde{y} && \text{(combine like terms)} \\
&= \|\tilde{x}\|^2 + 2(\tilde{x} \cdot \tilde{y}) + \|\tilde{y}\|^2. && \text{(@eq-l2-norm, squared)}
\end{aligned}$$

If  $\tilde{x} \perp \tilde{y}$ , then  $\tilde{x} \cdot \tilde{y} = 0$ , and the middle term drops out. □

**Example 0.60** (Checking the expansion). For  $\tilde{x} = (3, 0)$  and  $\tilde{y} = (1, 4)$ :  $\tilde{x} + \tilde{y} = (4, 4)$ , so  $\|\tilde{x} + \tilde{y}\|^2 = 16 + 16 = 32$ , and  $\|\tilde{x}\|^2 + 2(\tilde{x} \cdot \tilde{y}) + \|\tilde{y}\|^2 = 9 + 2 \cdot 3 + 17 = 32$ . For the orthogonal vectors  $(3, 0)$  and  $(0, 4)$ ,  $\|(3, 4)\|^2 = 25 = 9 + 16$ , the 3-4-5 right triangle.

**Theorem 0.43** (Cauchy-Schwarz inequality). *For any  $\tilde{x}, \tilde{y} \in \mathbb{R}^p$ ,*

$$|\tilde{x} \cdot \tilde{y}| \leq \|\tilde{x}\| \|\tilde{y}\|,$$

*with equality exactly when  $\tilde{x}$  and  $\tilde{y}$  are linearly dependent (Definition 0.25).*

### **i** Proof

*Proof.* **If  $\tilde{y} = \tilde{0}$ .** Then  $\tilde{x} \cdot \tilde{0} = \sum_i x_i \cdot 0 = 0$  (Definition 0.5) and  $\|\tilde{0}\| = 0$  (Theorem 0.41, part 1), so both sides are 0 and equality holds; and  $\tilde{x}, \tilde{y}$  are linearly dependent, because  $0\tilde{x} + 1\tilde{y} = \tilde{0}$ .

**If  $\tilde{y} \neq \tilde{0}$ .** Then  $\|\tilde{y}\| > 0$  (Theorem 0.41, part 1). Let  $t \stackrel{\text{def}}{=} (\tilde{x} \cdot \tilde{y}) / \|\tilde{y}\|^2$ . Then

$$\begin{aligned}
\|\tilde{x} - t\tilde{y}\|^2 &= \|\tilde{x} + (-1)(t\tilde{y})\|^2 && (\tilde{x} - \tilde{v} = \tilde{x} + (-1)\tilde{v}) \\
&= \|\tilde{x} + (-t)\tilde{y}\|^2 && ((-1)(t\tilde{y}) = (-t)\tilde{y}, \text{ entrywise}) \\
&= \|\tilde{x}\|^2 + 2(\tilde{x} \cdot (-t)\tilde{y}) + \|(-t)\tilde{y}\|^2 && (@\text{thm-norm-sum-square}) \\
&= \|\tilde{x}\|^2 + 2(-t)(\tilde{x} \cdot \tilde{y}) + \|(-t)\tilde{y}\|^2 && (@\text{thm-dot-linear}, \text{ special case } b = 0) \\
&= \|\tilde{x}\|^2 + 2(-t)(\tilde{x} \cdot \tilde{y}) + (|-t| \|\tilde{y}\|)^2 && (@\text{thm-norm-properties}, \text{ part 2}) \\
&= \|\tilde{x}\|^2 + 2(-t)(\tilde{x} \cdot \tilde{y}) + |-t|^2 \|\tilde{y}\|^2 && ((ab)^2 = a^2b^2) \\
&= \|\tilde{x}\|^2 - 2t(\tilde{x} \cdot \tilde{y}) + t^2 \|\tilde{y}\|^2 && (2(-t) = -2t \text{ and } |-t|^2 = t^2) \\
&= \|\tilde{x}\|^2 - 2\frac{(\tilde{x} \cdot \tilde{y})^2}{\|\tilde{y}\|^2} + \frac{(\tilde{x} \cdot \tilde{y})^2}{\|\tilde{y}\|^4} \|\tilde{y}\|^2 && (\text{substitute } t) \\
&= \|\tilde{x}\|^2 - 2\frac{(\tilde{x} \cdot \tilde{y})^2}{\|\tilde{y}\|^2} + \frac{(\tilde{x} \cdot \tilde{y})^2}{\|\tilde{y}\|^2} && (\text{cancel one factor } \|\tilde{y}\|^2) \\
&= \|\tilde{x}\|^2 - \frac{(\tilde{x} \cdot \tilde{y})^2}{\|\tilde{y}\|^2}. && (\text{combine the last two terms})
\end{aligned} \tag{5}$$

The left side is at least 0 (Theorem 0.41, part 1), so

$$\begin{aligned}
0 &\leq \|\tilde{x}\|^2 - \frac{(\tilde{x} \cdot \tilde{y})^2}{\|\tilde{y}\|^2} && (@\text{eq-cauchy-schwarz-chain}) \\
\frac{(\tilde{x} \cdot \tilde{y})^2}{\|\tilde{y}\|^2} &\leq \|\tilde{x}\|^2 && (\text{add } (\tilde{x} \cdot \tilde{y})^2/\|\tilde{y}\|^2 \text{ to both sides}) \\
(\tilde{x} \cdot \tilde{y})^2 &\leq \|\tilde{x}\|^2 \|\tilde{y}\|^2, && (\text{multiply both sides by } \|\tilde{y}\|^2 > 0)
\end{aligned}$$

and taking nonnegative square roots of both sides, which preserves the inequality, gives  $|\tilde{x} \cdot \tilde{y}| \leq \|\tilde{x}\| \|\tilde{y}\|$ .

**When equality holds** ( $\tilde{y} \neq \tilde{0}$ ). Both sides of the inequality are nonnegative, so equality holds exactly when  $(\tilde{x} \cdot \tilde{y})^2 = \|\tilde{x}\|^2 \|\tilde{y}\|^2$ , that is (dividing by  $\|\tilde{y}\|^2 > 0$  and rearranging), when  $\|\tilde{x}\|^2 - (\tilde{x} \cdot \tilde{y})^2/\|\tilde{y}\|^2 = 0$ . By Equation 5, that quantity is  $\|\tilde{x} - t\tilde{y}\|^2$ , so equality holds exactly when  $\tilde{x} - t\tilde{y} = \tilde{0}$  (Theorem 0.41, part 1), that is, when  $\tilde{x} = t\tilde{y}$ .

- If  $\tilde{x} = t\tilde{y}$ , then  $1\tilde{x} + (-t)\tilde{y} = \tilde{0}$ , so the two vectors are linearly dependent.
- Conversely, suppose  $a\tilde{x} + b\tilde{y} = \tilde{0}$  with  $a, b$  not both 0. Then  $a \neq 0$ : otherwise  $b \neq 0$  and  $b\tilde{y} = \tilde{0}$ , so  $\tilde{y} = \frac{1}{b}(b\tilde{y}) = \tilde{0}$ , contrary to assumption. Subtracting  $b\tilde{y}$  from both sides gives  $a\tilde{x} = -b\tilde{y}$ , and multiplying by  $\frac{1}{a}$  gives  $\tilde{x} = s\tilde{y}$  with  $s = -b/a$ . Then

$$\begin{aligned}
|\tilde{x} \cdot \tilde{y}| &= |(s \tilde{y}) \cdot \tilde{y}| && \text{(substitute } \tilde{x} = s \tilde{y}) \\
&= |s (\tilde{y} \cdot \tilde{y})| && \text{(@thm-dot-linear, special case } b = 0, \text{ first slot)} \\
&= |s| |\tilde{y} \cdot \tilde{y}| && (|ab| = |a| |b|) \\
&= |s| \|\tilde{y}\|^2 && (\tilde{y} \cdot \tilde{y} = \|\tilde{y}\|^2 \geq 0, \text{ @eq-l2-norm}) \\
&= (|s| \|\tilde{y}\|) \|\tilde{y}\| && \text{(regroup)} \\
&= \|s \tilde{y}\| \|\tilde{y}\| && \text{(@thm-norm-properties, part 2)} \\
&= \|\tilde{x}\| \|\tilde{y}\|, && (s \tilde{y} = \tilde{x})
\end{aligned}$$

so equality holds. □

**Example 0.61** (A strict case and an equality case).

- For  $\tilde{x} = (1, 2)$  and  $\tilde{y} = (3, 4)$ :  $\tilde{x} \cdot \tilde{y} = 3 + 8 = 11$ , while  $\|\tilde{x}\| \|\tilde{y}\| = \sqrt{5} \cdot 5 \approx 11.18$ . The inequality is strict, and the vectors are linearly independent:  $(3, 4)$  is not a multiple of  $(1, 2)$ , since  $3 \cdot 2 \neq 4$ .
- For  $\tilde{x} = (1, 2)$  and  $\tilde{y} = (2, 4) = 2\tilde{x}$ :  $\tilde{x} \cdot \tilde{y} = 2 + 8 = 10$  and  $\|\tilde{x}\| \|\tilde{y}\| = \sqrt{5} \sqrt{20} = \sqrt{100} = 10$ , so equality holds.

**Theorem 0.44** (Triangle inequality). *For any  $\tilde{x}, \tilde{y} \in \mathbb{R}^p$ ,*

$$\|\tilde{x} + \tilde{y}\| \leq \|\tilde{x}\| + \|\tilde{y}\|.$$

### **i** Proof

*Proof.*

$$\begin{aligned}
\|\tilde{x} + \tilde{y}\|^2 &= \|\tilde{x}\|^2 + 2(\tilde{x} \cdot \tilde{y}) + \|\tilde{y}\|^2 && \text{(@thm-norm-sum-square)} \\
&\leq \|\tilde{x}\|^2 + 2|\tilde{x} \cdot \tilde{y}| + \|\tilde{y}\|^2 && \text{(a number is at most its absolute value)} \\
&\leq \|\tilde{x}\|^2 + 2\|\tilde{x}\| \|\tilde{y}\| + \|\tilde{y}\|^2 && \text{(@thm-cauchy-schwarz)} \\
&= (\|\tilde{x}\| + \|\tilde{y}\|)^2. && \text{(expand the square)}
\end{aligned}$$

Both  $\|\tilde{x} + \tilde{y}\|$  and  $\|\tilde{x}\| + \|\tilde{y}\|$  are nonnegative (Theorem 0.41), and for nonnegative numbers  $a^2 \leq b^2$  implies  $a \leq b$ . □

**Example 0.62** (A strict case and an equality case).

- For  $\tilde{x} = (3, 0)$  and  $\tilde{y} = (0, 4)$ :  $\|\tilde{x} + \tilde{y}\| = \|(3, 4)\| = 5$ , while  $\|\tilde{x}\| + \|\tilde{y}\| = 3 + 4 = 7$ .

- For  $\tilde{x} = (1, 0)$  and  $\tilde{y} = (2, 0)$ , which point the same way:  $\|(3, 0)\| = 3 = 1 + 2$ , so equality holds.

**Definition 0.39** (Angle between two vectors). The **angle** between two nonzero vectors  $\tilde{x}, \tilde{y} \in \mathbb{R}^p$  is the unique number  $\theta \in [0, \pi]$  such that

$$\cos \theta = \frac{\tilde{x} \cdot \tilde{y}}{\|\tilde{x}\| \|\tilde{y}\|}.$$

*Remark 0.14* (Why the angle is well defined). The denominator is positive, because both vectors are nonzero (Theorem 0.41), and by Theorem 0.43 the ratio lies between  $-1$  and  $1$ . On  $[0, \pi]$ ,  $\cos$  is decreasing from  $1$  to  $-1$  and takes each value in  $[-1, 1]$  exactly once, so there is exactly one such  $\theta$ . The angle is  $\pi/2$  exactly when  $\tilde{x} \cdot \tilde{y} = 0$ , so for nonzero vectors, orthogonal (Definition 0.10) means at a right angle.

**Example 0.63** (Some angles in the plane).

- $(1, 0)$  and  $(1, 1)$ :  $\cos \theta = \frac{1}{1 \cdot \sqrt{2}}$ , so  $\theta = \pi/4$ .
- $(1, 2)$  and  $(-2, 1)$ :  $\cos \theta = \frac{-2+2}{\sqrt{5}\sqrt{5}} = 0$ , so  $\theta = \pi/2$ .
- $(1, 0)$  and  $(-3, 0)$ :  $\cos \theta = \frac{-3}{1 \cdot 3} = -1$ , so  $\theta = \pi$ .
- $(1, 0)$  and  $\tilde{0}$ : there is no angle, because the ratio would divide by  $\|\tilde{0}\| = 0$ .

## Orthonormal bases and Gram-Schmidt

### Source

Like Section , this section is adapted from Zhou (2024h), used under the MIT License (see the license text in Section ). The source leaves the proof that orthonormal vectors are linearly independent to class; it is written out here.

**Theorem 0.45** (The dot product of a vector with a linear combination). *For vectors  $\tilde{x}, \tilde{u}_1, \dots, \tilde{u}_k \in \mathbb{R}^p$  and numbers  $c_1, \dots, c_k$ , with  $k \geq 1$ ,*

$$\tilde{x} \cdot \left( \sum_{j=1}^k c_j \tilde{u}_j \right) = \sum_{j=1}^k c_j (\tilde{x} \cdot \tilde{u}_j).$$

**i** Proof

*Proof.* Write  $u_{ji}$  for entry  $i$  of  $\tilde{u}_j$ . Then

$$\begin{aligned}
 \tilde{x} \cdot \left( \sum_{j=1}^k c_j \tilde{u}_j \right) &= \sum_{i=1}^p x_i \sum_{j=1}^k c_j u_{ji} \quad (\text{@def-dot-product, @def-linear-combination}) \\
 &= \sum_{i=1}^p \sum_{j=1}^k x_i c_j u_{ji} \quad (\text{distribute each } x_i \text{ over the inner sum}) \\
 &= \sum_{j=1}^k \sum_{i=1}^p x_i c_j u_{ji} \quad (\text{swap the order of the finite sums}) \\
 &= \sum_{j=1}^k \sum_{i=1}^p c_j x_i u_{ji} \quad (\text{commute the factors in each product}) \\
 &= \sum_{j=1}^k c_j \sum_{i=1}^p x_i u_{ji} \quad (\text{factor } c_j \text{ out of the inner sum}) \\
 &= \sum_{j=1}^k c_j (\tilde{x} \cdot \tilde{u}_j). \quad (\text{@def-dot-product})
 \end{aligned}$$

□

**Example 0.64** (Dotting with a combination of three vectors). Let  $\tilde{x} = (1, 2, 3)$  and take the combination  $2(1, 0, 0) - (0, 1, 0) + 4(0, 0, 1) = (2, -1, 4)$ . Directly,  $\tilde{x} \cdot (2, -1, 4) = 2 - 2 + 12 = 12$ . By Theorem 0.45,  $2 \cdot 1 - 1 \cdot 2 + 4 \cdot 3 = 12$ , the same number.

**Theorem 0.46** (Orthonormal vectors are linearly independent). *If  $\tilde{q}_1, \dots, \tilde{q}_k$  are orthonormal (Definition 0.13), then they are linearly independent (Definition 0.25).*

**i** Proof

*Proof.* Suppose  $\sum_{j=1}^k c_j \tilde{q}_j = \tilde{0}$ , and fix any  $i$  between 1 and  $k$ . Then

$$\begin{aligned}
0 &= \tilde{q}_i \cdot \tilde{0} && \text{(every term of the dot product is 0)} \\
&= \tilde{q}_i \cdot \left( \sum_{j=1}^k c_j \tilde{q}_j \right) && \text{(substitute the supposed equation)} \\
&= \sum_{j=1}^k c_j (\tilde{q}_i \cdot \tilde{q}_j) && \text{(@thm-dot-linear-sum)} \\
&= c_i. && (\tilde{q}_i \cdot \tilde{q}_j \text{ is 1 if } j = i \text{ and 0 otherwise})
\end{aligned}$$

So every  $c_i = 0$ . □

**Example 0.65** (Three orthonormal vectors in  $\mathbb{R}^3$ ). Let

$$\tilde{q}_1 = (0, 0, 1), \quad \tilde{q}_2 = \frac{1}{\sqrt{2}} (1, 1, 0), \quad \tilde{q}_3 = \frac{1}{\sqrt{2}} (1, -1, 0).$$

Each has norm 1:  $\|\tilde{q}_2\|^2 = \frac{1}{2} (1 + 1 + 0) = 1$ , and likewise for the others. They are mutually orthogonal:  $\tilde{q}_1 \cdot \tilde{q}_2 = 0$  and  $\tilde{q}_1 \cdot \tilde{q}_3 = 0$ , because  $\tilde{q}_1$  has its only nonzero entry where the others have 0, and  $\tilde{q}_2 \cdot \tilde{q}_3 = \frac{1}{2} (1 - 1 + 0) = 0$ . So by Theorem 0.46 they are linearly independent, with no need to solve  $c_1 \tilde{q}_1 + c_2 \tilde{q}_2 + c_3 \tilde{q}_3 = \tilde{0}$ .

The converse fails:  $(1, 0)$  and  $(1, 1)$  are linearly independent, since  $c_1(1, 0) + c_2(1, 1) = (c_1 + c_2, c_2)$  is  $\tilde{0}$  only if  $c_2 = 0$  and then  $c_1 = 0$ , but they are not orthonormal, since  $(1, 0) \cdot (1, 1) = 1 \neq 0$ .

**Definition 0.40** (Orthonormal basis). An **orthonormal basis** of a subspace  $\mathcal{S}$  of  $\mathbb{R}^p$  is a basis of  $\mathcal{S}$  (Definition 0.30) whose vectors are orthonormal (Definition 0.13).

**Example 0.66** (Orthonormal bases of  $\mathbb{R}^3$ ).

- The indicator vectors  $\tilde{e}_1, \tilde{e}_2, \tilde{e}_3$  are orthonormal (2) and a basis of  $\mathbb{R}^3$  (Example 0.22).
- The vectors  $\tilde{q}_1, \tilde{q}_2, \tilde{q}_3$  of Example 0.65 are orthonormal and linearly independent, so they are a basis of their span (Definition 0.30), which therefore has dimension 3 (Definition 0.31) and is all of  $\mathbb{R}^3$  (Theorem 0.21, Example 0.22). So they are an orthonormal basis of  $\mathbb{R}^3$  too.

**Example 0.67** (Bases that are not orthonormal, and orthonormal lists that are not bases).

- $(1, 0), (1, 1)$  is a basis of  $\mathbb{R}^2$  (Example 0.18). It is not an orthonormal basis, since

$$(1, 0) \cdot (1, 1) = 1.$$

- $(1, 0, 0), (0, 1, 0)$  is orthonormal but not a basis of  $\mathbb{R}^3$ : its span is the plane  $z = 0$  (Example 0.13), which does not contain  $(0, 0, 1)$ .

**Theorem 0.47** (Coordinates in an orthonormal basis are dot products). *If  $\tilde{q}_1, \dots, \tilde{q}_k$  is an orthonormal basis of a subspace  $\mathcal{S}$  (Definition 0.40), then every  $\tilde{x} \in \mathcal{S}$  satisfies*

$$\tilde{x} = \sum_{i=1}^k (\tilde{q}_i \cdot \tilde{x}) \tilde{q}_i.$$

### i Proof

*Proof.* The basis spans  $\mathcal{S}$ , so  $\tilde{x} = \sum_{j=1}^k c_j \tilde{q}_j$  for some numbers  $c_j$ . For each  $i$ ,

$$\begin{aligned} \tilde{q}_i \cdot \tilde{x} &= \tilde{q}_i \cdot \left( \sum_{j=1}^k c_j \tilde{q}_j \right) \quad (\text{substitute}) \\ &= \sum_{j=1}^k c_j (\tilde{q}_i \cdot \tilde{q}_j) \quad (\text{@thm-dot-linear-sum}) \\ &= c_i, \quad (\text{@def-orthonormal-vectors}) \end{aligned}$$

so each coefficient  $c_i$  is  $\tilde{q}_i \cdot \tilde{x}$ . □

**Example 0.68** (Expanding a vector in an orthonormal basis). With the orthonormal basis  $\tilde{q}_1, \tilde{q}_2, \tilde{q}_3$  of  $\mathbb{R}^3$  from Example 0.66 and  $\tilde{x} = (1, 2, 3)$ :  $\tilde{q}_1 \cdot \tilde{x} = 3$ ,  $\tilde{q}_2 \cdot \tilde{x} = \frac{1}{\sqrt{2}}(1+2) = \frac{3}{\sqrt{2}}$  and  $\tilde{q}_3 \cdot \tilde{x} = \frac{1}{\sqrt{2}}(1-2) = -\frac{1}{\sqrt{2}}$ . Then

$$\begin{aligned} \sum_{i=1}^3 (\tilde{q}_i \cdot \tilde{x}) \tilde{q}_i &= 3(0, 0, 1) + \frac{3}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1, 1, 0) - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1, -1, 0) \quad (\text{substitute}) \\ &= (0, 0, 3) + \left(\frac{3}{2}, \frac{3}{2}, 0\right) + \left(-\frac{1}{2}, \frac{1}{2}, 0\right) \quad (\text{multiply out each term}) \\ &= (1, 2, 3), \quad (\text{add entrywise}) \end{aligned}$$

which is  $\tilde{x}$ , as Theorem 0.47 says.

**Definition 0.41** (Gram-Schmidt process). The **Gram-Schmidt process** takes vectors  $\tilde{a}_1, \dots, \tilde{a}_k \in \mathbb{R}^p$  and, for  $i = 1, 2, \dots, k$  in turn:

1. **Orthogonalize:**  $\tilde{q}_i \stackrel{\text{def}}{=} \tilde{a}_i - \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j$  (for  $i = 1$  the sum is empty, so  $\tilde{q}_1 = \tilde{a}_1$ );

2. **Test:** if  $\tilde{q}_i = \tilde{0}$ , stop;
3. **Normalize:**  $\tilde{q}_i \stackrel{\text{def}}{=} \tilde{q}_i / \|\tilde{q}_i\|$ .

The output is the list  $\tilde{q}_1, \tilde{q}_2, \dots$  produced before the process stops, or all  $k$  of them if it never stops.

**Example 0.69** (Two steps of Gram-Schmidt in  $\mathbb{R}^3$ ). Let  $\tilde{a}_1 = (1, 1, 0)$  and  $\tilde{a}_2 = (1, 0, 1)$ .

- **Step 1.**  $\tilde{q}_1 = (1, 1, 0)$ , with norm  $\sqrt{2}$ , so  $\tilde{q}_1 = \frac{1}{\sqrt{2}} (1, 1, 0)$ .
- **Step 2.**  $\tilde{q}_1 \cdot \tilde{a}_2 = \frac{1}{\sqrt{2}} (1 + 0 + 0) = \frac{1}{\sqrt{2}}$ , so

$$\begin{aligned} \tilde{q}_2 &= (1, 0, 1) - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (1, 1, 0) && \text{(orthogonalize)} \\ &= (1, 0, 1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) && \text{(multiply out)} \\ &= \left(\frac{1}{2}, -\frac{1}{2}, 1\right), && \text{(subtract entrywise)} \end{aligned}$$

with norm  $\sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$ , so  $\tilde{q}_2 = \sqrt{\frac{2}{3}} \left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \frac{1}{\sqrt{6}} (1, -1, 2)$ .

As a check,  $\tilde{q}_1 \cdot \tilde{q}_2 = \frac{1}{\sqrt{12}} (1 - 1 + 0) = 0$ .

**Example 0.70** (Gram-Schmidt stops on dependent vectors). Let  $\tilde{a}_1 = (1, 2)$  and  $\tilde{a}_2 = (2, 4)$ . Step 1 gives  $\tilde{q}_1 = \frac{1}{\sqrt{5}} (1, 2)$ . In step 2,  $\tilde{q}_1 \cdot \tilde{a}_2 = \frac{1}{\sqrt{5}} (2 + 8) = \frac{10}{\sqrt{5}}$ , and

$$\tilde{q}_2 = (2, 4) - \frac{10}{\sqrt{5}} \cdot \frac{1}{\sqrt{5}} (1, 2) = (2, 4) - 2(1, 2) = \tilde{0},$$

so the process stops:  $\tilde{a}_2$  is a multiple of  $\tilde{a}_1$ .

**Theorem 0.48** (What Gram-Schmidt produces). *Run the Gram-Schmidt process (Definition 0.41) on  $\tilde{a}_1, \dots, \tilde{a}_k \in \mathbb{R}^p$ .*

1. *If it has produced  $\tilde{q}_1, \dots, \tilde{q}_i$  without stopping, then  $\tilde{q}_1, \dots, \tilde{q}_i$  are orthonormal and  $\text{span}\{\tilde{q}_1, \dots, \tilde{q}_i\} = \text{span}\{\tilde{a}_1, \dots, \tilde{a}_i\}$ .*
2. *If it reaches step  $i$  (that is, it did not stop at steps  $1, \dots, i-1$ ), it stops there exactly when  $\tilde{a}_i \in \text{span}\{\tilde{a}_1, \dots, \tilde{a}_{i-1}\}$ .*
3. *It completes all  $k$  steps exactly when  $\tilde{a}_1, \dots, \tilde{a}_k$  are linearly independent; then  $\tilde{q}_1, \dots, \tilde{q}_k$  is an orthonormal basis of  $\text{span}\{\tilde{a}_1, \dots, \tilde{a}_k\}$  (Definition 0.40).*

## i Proof

*Proof. Parts 1 and 2, by induction on  $i$ .* Before step 1, the list of  $\tilde{q}$ 's is empty, which is orthonormal, and both spans are  $\{\tilde{0}\}$  (Definition 0.29). Suppose part 1 holds after step  $i - 1$ , and write  $\mathcal{S}_{i-1} = \text{span}\{\tilde{a}_1, \dots, \tilde{a}_{i-1}\} = \text{span}\{\tilde{q}_1, \dots, \tilde{q}_{i-1}\}$ . The orthonormal vectors  $\tilde{q}_1, \dots, \tilde{q}_{i-1}$  are linearly independent (Theorem 0.46) and span  $\mathcal{S}_{i-1}$ , so they are an orthonormal basis of it.

$\tilde{q}_i$  is orthogonal to the earlier  $\tilde{q}$ 's. For each  $l < i$ ,

$$\begin{aligned}
 \tilde{q}_l \cdot \tilde{q}_i &= \tilde{q}_l \cdot \left( 1 \tilde{a}_i + \sum_{j=1}^{i-1} (-\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j \right) && \text{(@def-gram-schmidt, as a linear combination)} \\
 &= 1 (\tilde{q}_l \cdot \tilde{a}_i) + \sum_{j=1}^{i-1} (-\tilde{q}_j \cdot \tilde{a}_i) (\tilde{q}_l \cdot \tilde{q}_j) && \text{(@thm-dot-linear-sum)} \\
 &= \tilde{q}_l \cdot \tilde{a}_i - \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) (\tilde{q}_l \cdot \tilde{q}_j) && (1z = z, \text{ and pull the minus signs out of the sum}) \\
 &= \tilde{q}_l \cdot \tilde{a}_i - \tilde{q}_l \cdot \tilde{a}_i && \text{(only the } j = l \text{ term survives, @def-orthonormal-vectors)} \\
 &= 0. && \text{(arithmetic)}
 \end{aligned}$$

*Part 2 at step  $i$ .* If  $\tilde{q}_i = \tilde{0}$ , then  $\tilde{a}_i = \sum_{j < i} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j \in \mathcal{S}_{i-1}$ . Conversely, if  $\tilde{a}_i \in \mathcal{S}_{i-1}$ , then  $\tilde{a}_i = \sum_{j < i} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j$  (Theorem 0.47), so  $\tilde{q}_i = \tilde{0}$ .

*Part 1 at step  $i$ , if the process does not stop.* Then  $\tilde{q}_i \neq \tilde{0}$ , so  $\|\tilde{q}_i\| > 0$ , and  $\tilde{q}_i = \tilde{q}_i / \|\tilde{q}_i\|$  has norm  $\|\tilde{q}_i\| / \|\tilde{q}_i\| = 1$  (Theorem 0.41) and for  $l < i$

$$\begin{aligned}
 \tilde{q}_l \cdot \tilde{q}_i &= \frac{1}{\|\tilde{q}_i\|} (\tilde{q}_l \cdot \tilde{q}_i) && \text{(@thm-dot-linear, special case } b = 0) \\
 &= 0, && (\tilde{q}_i \text{ is orthogonal to } \tilde{q}_l)
 \end{aligned}$$

and  $\tilde{q}_i \cdot \tilde{q}_l = 0$  too (Theorem 0.1). So  $\tilde{q}_1, \dots, \tilde{q}_i$  are orthonormal. For the spans:

- $\tilde{a}_i = \|\tilde{q}_i\| \tilde{q}_i + \sum_{j < i} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j$  is in  $\text{span}\{\tilde{q}_1, \dots, \tilde{q}_i\}$ , and so is each of  $\tilde{a}_1, \dots, \tilde{a}_{i-1}$ , which lie in  $\mathcal{S}_{i-1} = \text{span}\{\tilde{q}_1, \dots, \tilde{q}_{i-1}\} \subseteq \text{span}\{\tilde{q}_1, \dots, \tilde{q}_i\}$ .
- $\tilde{q}_i = \left( \tilde{a}_i - \sum_{j < i} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j \right) / \|\tilde{q}_i\|$  is in  $\text{span}\{\tilde{a}_1, \dots, \tilde{a}_i\}$ , because each  $\tilde{q}_j$  with  $j < i$  lies in  $\mathcal{S}_{i-1}$ ; and so is each of  $\tilde{q}_1, \dots, \tilde{q}_{i-1}$ .

A span is a subspace (Theorem 0.12), so it contains every linear combination of vectors in it; each span therefore contains the other, and they are equal.

**Part 3.** If the process completes, then  $\tilde{a}_1 \notin \text{span}$  of the empty list, and for each  $i \geq 2$ ,  $\tilde{a}_i \notin \text{span}\{\tilde{a}_1, \dots, \tilde{a}_{i-1}\}$  (part 2). Applying Theorem 0.18  $k$  times, starting from

the empty list, shows that  $\tilde{a}_1, \dots, \tilde{a}_k$  are linearly independent. Conversely, if they are linearly independent, no  $\tilde{a}_i$  is a linear combination  $\sum_{j<i} c_j \tilde{a}_j$  of the earlier ones, because then  $\sum_{j<i} c_j \tilde{a}_j - \tilde{a}_i = \tilde{0}$  would be a combination with a nonzero coefficient; so by part 2 the process never stops. When it completes,  $\tilde{q}_1, \dots, \tilde{q}_k$  are orthonormal, hence linearly independent (Theorem 0.46), and span  $\text{span}\{\tilde{a}_1, \dots, \tilde{a}_k\}$  (part 1), so they are an orthonormal basis of it.  $\square$

**Example 0.71** (Finishing an orthonormal basis of  $\mathbb{R}^3$ ). Continue Example 0.69 with  $\tilde{a}_3 = (0, 1, 1)$ .  $\tilde{q}_1 \cdot \tilde{a}_3 = \frac{1}{\sqrt{2}}$  and  $\tilde{q}_2 \cdot \tilde{a}_3 = \frac{1}{\sqrt{6}}(0 - 1 + 2) = \frac{1}{\sqrt{6}}$ , so

$$\begin{aligned} \tilde{q}_3 &= (0, 1, 1) - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (1, 1, 0) - \frac{1}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}} (1, -1, 2) && \text{(orthogonalize)} \\ &= (0, 1, 1) - \frac{1}{2} (1, 1, 0) - \frac{1}{6} (1, -1, 2) && \text{(multiply the scalars)} \\ &= \left(0 - \frac{1}{2} - \frac{1}{6}, 1 - \frac{1}{2} + \frac{1}{6}, 1 - 0 - \frac{2}{6}\right) && \text{(subtract entrywise)} \\ &= \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right), && \text{(arithmetic)} \end{aligned}$$

with norm  $\sqrt{3 \cdot \frac{4}{9}} = \frac{2}{\sqrt{3}}$ , so  $\tilde{q}_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)$ . The process did not stop, so by Theorem 0.48  $\tilde{a}_1, \tilde{a}_2, \tilde{a}_3$  are linearly independent and  $\tilde{q}_1, \tilde{q}_2, \tilde{q}_3$  is an orthonormal basis of their span. That span therefore has dimension 3 (Definition 0.31), so it is  $\mathbb{R}^3$  (Theorem 0.21, Example 0.22). As a check,  $\tilde{q}_1 \cdot \tilde{q}_3 = \frac{1}{\sqrt{6}}(-1 + 1 + 0) = 0$  and  $\tilde{q}_2 \cdot \tilde{q}_3 = \frac{1}{\sqrt{18}}(-1 - 1 + 2) = 0$ .

**Corollary 0.3** (Every subspace has an orthonormal basis). *Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^p$ . Every orthonormal list of vectors in  $\mathcal{S}$  extends to an orthonormal basis of  $\mathcal{S}$  (Definition 0.40). Starting from the empty list shows that  $\mathcal{S}$  has an orthonormal basis.*

### **i** Proof

*Proof.* Let  $\tilde{u}_1, \dots, \tilde{u}_r \in \mathcal{S}$  be orthonormal. They are linearly independent (Theorem 0.46), so they extend to a basis  $\tilde{u}_1, \dots, \tilde{u}_r, \tilde{a}_{r+1}, \dots, \tilde{a}_d$  of  $\mathcal{S}$  (Theorem 0.19). Run the Gram-Schmidt process on this basis. It is linearly independent, so the process completes, and the output is an orthonormal basis of its span, which is  $\mathcal{S}$  (Theorem 0.48, part 3).

The first  $r$  outputs are  $\tilde{u}_1, \dots, \tilde{u}_r$  themselves, by induction on  $i \leq r$ . Suppose the outputs before step  $i$  are  $\tilde{u}_1, \dots, \tilde{u}_{i-1}$  (for  $i = 1$  there are none, and the sum below is empty). Then  $\tilde{q}_i = \tilde{u}_i - \sum_{j<i} (\tilde{u}_j \cdot \tilde{u}_i) \tilde{u}_j = \tilde{u}_i$ , because each  $\tilde{u}_j \cdot \tilde{u}_i = 0$ ;  $\tilde{q}_i = \tilde{u}_i \neq \tilde{0}$ , because  $\|\tilde{u}_i\| = 1$ , so the process does not stop; and  $\tilde{q}_i = \tilde{u}_i / \|\tilde{u}_i\| = \tilde{u}_i$ . So the orthonormal basis contains the starting list.  $\square$

**Example 0.72** (Extending one unit vector to an orthonormal basis of  $\mathbb{R}^3$ ). Start from the unit vector  $\tilde{u}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ . Extend it to a basis of  $\mathbb{R}^3$  as in Example 0.24, starting from the empty list:  $\tilde{u}_1$  is not in the span of the empty list,  $\{\tilde{0}\}$ , because  $\|\tilde{u}_1\| = 1$ ;  $(1, 0, 0)$  is not in  $\text{span}\{\tilde{u}_1\}$ , because every multiple of  $\tilde{u}_1$  has equal first and second entries, and  $(0, 0, 1)$  is not in  $\text{span}\{\tilde{u}_1, (1, 0, 0)\}$ , because every combination of those two vectors has third entry 0. So  $\tilde{u}_1, (1, 0, 0), (0, 0, 1)$  are linearly independent (Theorem 0.18), and three linearly independent vectors in  $\mathbb{R}^3$  are a basis of it (they are a basis of their span, which has dimension 3 (Definition 0.31) and so is  $\mathbb{R}^3$  (Theorem 0.21, Example 0.22)). Gram-Schmidt on  $\tilde{u}_1, (1, 0, 0), (0, 0, 1)$ :

1.  $\tilde{q}_1 = \tilde{u}_1$ .
2.  $\tilde{q}_1 \cdot (1, 0, 0) = \frac{1}{\sqrt{2}}$ , so  $\tilde{q}_2 = (1, 0, 0) - \frac{1}{2}(1, 1, 0) = (\frac{1}{2}, -\frac{1}{2}, 0)$ , with norm  $\frac{1}{\sqrt{2}}$ , and  $\tilde{q}_2 = \frac{1}{\sqrt{2}}(1, -1, 0)$ .
3.  $(0, 0, 1)$  is orthogonal to  $\tilde{q}_1$  and  $\tilde{q}_2$ , so  $\tilde{q}_3 = (0, 0, 1)$ , which already has norm 1, and  $\tilde{q}_3 = (0, 0, 1)$ .

The result is the orthonormal basis of Example 0.66, in a different order.

## Outer product

**Definition 0.42** (Outer product). The **outer product** of a vector  $\tilde{u}$  of length  $m$  and a vector  $\tilde{v}$  of length  $n$  is the  $m \times n$  matrix product (Definition 0.20)  $\tilde{u} \tilde{v}^\top$  of the column vector  $\tilde{u}$  with the row vector  $\tilde{v}^\top$ . Its entries are

$$\left( \begin{array}{c} \tilde{u} \\ \hline \tilde{v}^\top \end{array} \right)_{ij} = u_i v_j \quad (\text{definition of matrix multiplication; the sum has one term})$$

**Example 0.73** (An outer product). For  $\tilde{u} = (1, 2, 3)$  and  $\tilde{v} = (4, 5)$ :

$$\begin{aligned}
\tilde{u} \tilde{v}^\top &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 5 \end{bmatrix} && \text{(definition of the outer product)} \\
&= \begin{bmatrix} 1 \cdot 4 & 1 \cdot 5 \\ 2 \cdot 4 & 2 \cdot 5 \\ 3 \cdot 4 & 3 \cdot 5 \end{bmatrix} && \text{(entry } (i, j) \text{ is } u_i v_j) \\
&= \begin{bmatrix} 4 & 5 \\ 8 & 10 \\ 12 & 15 \end{bmatrix} && \text{(multiply)}
\end{aligned}$$

*Remark 0.15* (Outer product and dot product). The dot product  $\tilde{u}^\top \tilde{v}$  (Example 0.7) needs two vectors of the same length and gives a number. The outer product  $\tilde{u} \tilde{v}^\top$  takes vectors of any two lengths and gives a matrix (Banerjee and Roy 2014, chap. 1, p. 11). For example,  $\tilde{u} = (1, 2, 3)$  and  $\tilde{v} = (4, 5)$  in Example 0.73 have different lengths, so they have no dot product, but their outer product is a  $3 \times 2$  matrix. For  $\tilde{a} = (1, 2)$  and  $\tilde{b} = (3, 4)$ , which have the same length, both products exist:

$$\tilde{a}^\top \tilde{b} = 1 \cdot 3 + 2 \cdot 4 = 11 \quad \tilde{a} \tilde{b}^\top = \begin{bmatrix} 1 \cdot 3 & 1 \cdot 4 \\ 2 \cdot 3 & 2 \cdot 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

**Theorem 0.49** (An outer product has rank one). *If  $\tilde{u}$  is a nonzero vector of length  $m$  and  $\tilde{v}$  is a nonzero vector of length  $n$ , then  $\text{rank}(\tilde{u} \tilde{v}^\top) = 1$  (Definition 0.26).*

### **i** Proof

*Proof.* Column  $j$  of  $\tilde{u} \tilde{v}^\top$  has entries  $u_1 v_j, \dots, u_m v_j$ , so it is the vector  $v_j \tilde{u}$ .

**The rank is at least one.** Because  $\tilde{v} \neq \tilde{0}$ , some entry  $v_j$  is nonzero. Then column  $j$ ,  $v_j \tilde{u}$ , is a nonzero vector, and a single nonzero vector is linearly independent (Definition 0.25):  $c(v_j \tilde{u}) = \tilde{0}$  forces  $c = 0$ .

**The rank is at most one.** Take any two columns  $j \neq k$ . If  $v_j = v_k = 0$ , both columns are  $\tilde{0}$ , and  $1 \cdot (v_j \tilde{u}) + 1 \cdot (v_k \tilde{u}) = \tilde{0}$ . Otherwise, use the coefficients  $c_j = v_k$  and  $c_k = -v_j$ , which are not both zero:

$$\begin{aligned}
v_k(v_j \tilde{u}) - v_j(v_k \tilde{u}) &= (v_k v_j - v_j v_k) \tilde{u} && \text{(collect the scalar multiples of } \tilde{u}) \\
&= 0 \cdot \tilde{u} && \text{(multiplication of numbers is commutative)} \\
&= \tilde{0} && \text{(a zero multiple of a vector is } \tilde{0})
\end{aligned}$$

Either way, every pair of columns has a combination with coefficients that are not all

zero and that equals  $\tilde{0}$ . Now take any set of two or more columns. It contains a pair  $j \neq k$ . Use that pair's coefficients for columns  $j$  and  $k$ , and the coefficient 0 for every other column in the set. This combination equals  $\tilde{0}$ , and its coefficients are not all zero, so the set is not linearly independent (Definition 0.25). The largest linearly independent set of columns therefore has one column.  $\square$

**Example 0.74** (The rank of an outer product). In Example 0.73, the second column  $(5, 10, 15)$  of  $\tilde{u} \tilde{v}^\top$  is  $\frac{5}{4}$  times the first column  $(4, 8, 12)$ , because the columns are  $v_1 \tilde{u} = 4\tilde{u}$  and  $v_2 \tilde{u} = 5\tilde{u}$ . So the two columns are not linearly independent, and the  $3 \times 2$  matrix has rank 1.

## Special Matrices

One special matrix, the zero matrix (Definition 0.17), appeared earlier, in the section on matrix addition.

**Definition 0.43** (Square matrix). A matrix is **square** if it has the same number of rows as columns.

**Definition 0.44** (Order of a square matrix). The **order** of a square matrix (Definition 0.43) is its number of rows, which equals its number of columns.

**Definition 0.45** (Matrix power). For a square matrix  $\mathbf{A}$  of order  $p$  and a positive integer  $k$ , the  $k$ -th **power** of  $\mathbf{A}$  is:

$$\mathbf{A}^k = \underbrace{\mathbf{A} \mathbf{A} \cdots \mathbf{A}}_{k \text{ copies}}$$

In particular,  $\mathbf{A}^2 = \mathbf{A} \mathbf{A}$ .

**Definition 0.46** (Identity matrix). The  $p \times p$  **identity matrix**  $\mathbf{I}_p$  (or  $\mathbf{I}$  when the size is clear from context) has ones on the main diagonal and zeros elsewhere:

$$(\mathbf{I}_p)_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \mathbf{I}_p = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Equivalently, the entries of the identity matrix are given by the **Kronecker delta**:  $(\mathbf{I}_p)_{ij} = \delta_{ij}$ .

**Theorem 0.50** (Identity matrix is a multiplicative identity). For any  $m \times p$  matrix  $\mathbf{A}$ :

$$\mathbf{A} \mathbf{I}_p = \mathbf{A}$$

$$\mathbf{I}_m \mathbf{A} = \mathbf{A}$$

**Definition 0.47** (Symmetric matrix). A square matrix  $\mathbf{A}$  is **symmetric** if  $\mathbf{A}^\top = \mathbf{A}$ , i.e.,  $a_{ij} = a_{ji}$  for all  $i$  and  $j$ .

*Remark 0.16* (Symmetric matrices in statistics). Covariance matrices are symmetric: entry  $(i, j)$  of the covariance matrix of a random vector  $\tilde{Y}$  is  $\text{Cov}(Y_i, Y_j)$ , and  $\text{Cov}(Y_i, Y_j) = \text{Cov}(Y_j, Y_i)$ . For example, if  $\text{Var}(Y_1) = 4$ ,  $\text{Var}(Y_2) = 9$ , and  $\text{Cov}(Y_1, Y_2) = 1$ , the covariance matrix of  $(Y_1, Y_2)$  is

$$\begin{bmatrix} 4 & 1 \\ 1 & 9 \end{bmatrix}.$$

**Definition 0.48** (Diagonal matrix). A square matrix  $\mathbf{D}$  is a **diagonal matrix** if all off-diagonal entries are zero:  $d_{ij} = 0$  whenever  $i \neq j$ :

$$\mathbf{D} = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_p \end{bmatrix}$$

*Remark 0.17* (diag notation). For numbers  $d_1, \dots, d_p$ ,  $\text{diag}(d_1, \dots, d_p)$  is the  $p \times p$  diagonal matrix with diagonal entries  $d_1, \dots, d_p$ . For example,

$$\text{diag}(2, 5) = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix},$$

and the  $p \times p$  identity matrix is  $\mathbf{I}_p = \text{diag}(1, \dots, 1)$ .

**Definition 0.49** (Matrix inverse). For a square  $p \times p$  matrix  $\mathbf{A}$ , the **inverse**  $\mathbf{A}^{-1}$  (if it exists) is the unique matrix satisfying:

$$\mathbf{A} \mathbf{A}^{-1} = \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}_p$$

**Definition 0.50** (Invertible matrix). A  $p \times p$  matrix  $\mathbf{A}$  is **invertible** (or *non-singular*) if some  $p \times p$  matrix  $\mathbf{B}$  satisfies

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}_p,$$

where  $\mathbf{I}_p$  is the identity matrix (Definition 0.46). A square matrix that is not invertible is *singular*.

**Example 0.75** (An invertible matrix and a singular one). The matrix  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$  is invertible, with  $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$ : multiplying out gives  $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_2$ . The matrix  $\mathbf{M} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  is singular: for any  $2 \times 2$  matrix  $\mathbf{C}$ , the two rows of  $\mathbf{MC}$  are equal, so  $\mathbf{MC}$  can never be  $\mathbf{I}_2$ , whose two rows differ.

*Remark 0.18* (Invertible matrices and inverses). Let  $\mathbf{A}$  be a  $p \times p$  matrix, and suppose that two  $p \times p$  matrices  $\mathbf{B}$  and  $\mathbf{C}$  satisfy  $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_p$  and  $\mathbf{AC} = \mathbf{CA} = \mathbf{I}_p$ . Then

$$\begin{aligned} \mathbf{B} &= \mathbf{BI}_p && \text{(identity matrix)} \\ &= \mathbf{B}(\mathbf{AC}) && (\mathbf{AC} = \mathbf{I}_p) \\ &= (\mathbf{BA})\mathbf{C} && \text{(matrix multiplication is associative)} \\ &= \mathbf{I}_p \mathbf{C} && (\mathbf{BA} = \mathbf{I}_p) \\ &= \mathbf{C} && \text{(identity matrix)} \end{aligned}$$

The steps use Theorem 0.7 and Theorem 0.50. Since  $\mathbf{B} = \mathbf{C}$ , at most one matrix satisfies these equations, which is the uniqueness that Definition 0.49 asserts. When  $\mathbf{A}$  is invertible, the matrix  $\mathbf{B}$  in Definition 0.50 is therefore the inverse  $\mathbf{A}^{-1}$  of  $\mathbf{A}$ , and  $\mathbf{A}$  is invertible exactly when it has an inverse. For example, for  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$  in Example 0.75,  $\begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$  is the only  $2 \times 2$  matrix  $\mathbf{B}$  with  $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_2$ , so it is  $\mathbf{A}^{-1}$ .

**Theorem 0.51** (Inverse of a product). If  $\mathbf{A}$  and  $\mathbf{B}$  are invertible  $p \times p$  matrices, then  $\mathbf{AB}$  is invertible, and

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

**i** Proof

*Proof.* Multiply  $\mathbf{AB}$  by the candidate inverse on the right:

$$\begin{aligned}
 (\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) &= \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1} && \text{(matrix multiplication is associative)} \\
 &= \mathbf{A}\mathbf{I}_p\mathbf{A}^{-1} && \text{(definition of } \mathbf{B}^{-1} \text{)} \\
 &= \mathbf{AA}^{-1} && \text{(identity matrix)} \\
 &= \mathbf{I}_p && \text{(definition of } \mathbf{A}^{-1} \text{)}
 \end{aligned}$$

and on the left:

$$\begin{aligned}
 (\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) &= \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} && \text{(matrix multiplication is associative)} \\
 &= \mathbf{B}^{-1}\mathbf{I}_p\mathbf{B} && \text{(definition of } \mathbf{A}^{-1} \text{)} \\
 &= \mathbf{B}^{-1}\mathbf{B} && \text{(identity matrix)} \\
 &= \mathbf{I}_p && \text{(definition of } \mathbf{B}^{-1} \text{)}
 \end{aligned}$$

So  $\mathbf{B}^{-1}\mathbf{A}^{-1}$  satisfies Definition 0.49 for  $\mathbf{AB}$ . The steps use Theorem 0.7 and Theorem 0.50.  $\square$

**Theorem 0.52** (Inverse of a transpose). *If  $\mathbf{A}$  is an invertible  $p \times p$  matrix, then  $\mathbf{A}^\top$  is invertible, and*

$$(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$$

**i** Proof

*Proof.* Multiply  $\mathbf{A}^\top$  by the candidate inverse on each side:

$$\begin{aligned}
 \mathbf{A}^\top (\mathbf{A}^{-1})^\top &= (\mathbf{A}^{-1}\mathbf{A})^\top && \text{(transpose of a product)} \\
 &= \mathbf{I}_p^\top && \text{(definition of } \mathbf{A}^{-1} \text{)} \\
 &= \mathbf{I}_p && \text{(\mathbf{I}_p \text{ is symmetric)} } \\
 (\mathbf{A}^{-1})^\top \mathbf{A}^\top &= (\mathbf{AA}^{-1})^\top && \text{(transpose of a product)} \\
 &= \mathbf{I}_p^\top && \text{(definition of } \mathbf{A}^{-1} \text{)} \\
 &= \mathbf{I}_p && \text{(\mathbf{I}_p \text{ is symmetric)} }
 \end{aligned}$$

So  $(\mathbf{A}^{-1})^\top$  satisfies Definition 0.49 for  $\mathbf{A}^\top$ . The first step of each display is Theorem 0.10.  $\square$

**Example 0.76** (Inverting a transpose). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ , Example 0.75 gives  $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$ , so Theorem 0.52 says

$$(\mathbf{A}^\top)^{-1} = \left( \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \right)^{-1} = (\mathbf{A}^{-1})^\top = \begin{bmatrix} 0.5 & 0 \\ -0.5 & 1 \end{bmatrix},$$

and multiplying  $\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 \\ -0.5 & 1 \end{bmatrix}$  out does give  $\mathbf{I}_2$ .

**Corollary 0.4** (The inverse of a symmetric matrix is symmetric). *If  $\mathbf{A}$  is symmetric and invertible, then  $\mathbf{A}^{-1}$  is symmetric.*

**i** Proof

*Proof.*

$$\begin{aligned} (\mathbf{A}^{-1})^\top &= (\mathbf{A}^\top)^{-1} && \text{(inverse of a transpose)} \\ &= \mathbf{A}^{-1} && (\mathbf{A} \text{ is symmetric}) \end{aligned}$$

The first step is Theorem 0.52. □

**Example 0.77** (Inverting a symmetric matrix).  $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$  is symmetric, and  $\mathbf{S}^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$  is symmetric too, as Corollary 0.4 says.

**Definition 0.51** (Idempotent matrix). A square matrix  $\mathbf{A}$  is **idempotent** if

$$\mathbf{A}^2 = \mathbf{A}$$

**Definition 0.52** (Orthogonal projection matrix). A square matrix  $\mathbf{P}$  is an **orthogonal projection matrix** if it is both symmetric (Definition 0.47) and idempotent (Definition 0.51):

$$\mathbf{P}^\top = \mathbf{P} \quad \text{and} \quad \mathbf{P}^2 = \mathbf{P}$$

**Definition 0.53** (Oblique projection). A square matrix is an **oblique projection** if it is idempotent (Definition 0.51) but not symmetric (Definition 0.47).

**Example 0.78** (An orthogonal projection and an oblique one).  $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  is symmetric, and  $\mathbf{P}^2 = \mathbf{P}$ , so  $\mathbf{P}$  is an orthogonal projection matrix; it maps  $(v_1, v_2)$  to  $(v_1, 0)$ .

$\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$  is idempotent:

$$\mathbf{Q}^2 = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 1 + 1 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 & 0 \cdot 1 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \mathbf{Q},$$

but  $\mathbf{Q}^\top \neq \mathbf{Q}$ , so  $\mathbf{Q}$  is an oblique projection, not an orthogonal projection matrix.

*Remark 0.19* (What “projection matrix” means in these notes). Some texts call any idempotent matrix a projection matrix, so that both  $\mathbf{P}$  and  $\mathbf{Q}$  in Example 0.78 would count as one. Regression texts often say “projection matrix” when they mean an orthogonal one. In these notes, “projection matrix” always means an orthogonal projection matrix in the sense of Definition 0.52, such as  $\mathbf{P}$  in Example 0.78, and never an oblique projection (Definition 0.53) such as  $\mathbf{Q}$ .

**Theorem 0.53** (Complement of a projection matrix). *If  $\mathbf{P}$  is a  $p \times p$  orthogonal projection matrix (Definition 0.52), then  $\mathbf{I}_p - \mathbf{P}$  is also an orthogonal projection matrix.*

#### **i** Proof

*Proof.* We verify symmetry and idempotency.

**Symmetry:**

$$(\mathbf{I} - \mathbf{P})^\top = \mathbf{I}^\top - \mathbf{P}^\top = \mathbf{I} - \mathbf{P}$$

**Idempotency:**

$$\begin{aligned} (\mathbf{I} - \mathbf{P})^2 &= (\mathbf{I} - \mathbf{P})(\mathbf{I} - \mathbf{P}) \\ &= \mathbf{I} - \mathbf{P} - \mathbf{P} + \mathbf{P}^2 \\ &= \mathbf{I} - \mathbf{P} - \mathbf{P} + \mathbf{P} \\ &= \mathbf{I} - \mathbf{P} \end{aligned}$$

□

**Theorem 0.54** (Projection matrices produce orthogonal decompositions). *If  $\mathbf{P}$  is a  $p \times p$  orthogonal projection matrix (Definition 0.52) and  $\tilde{v}$  is any vector of length  $p$ , then the two components of the decomposition*

$$\tilde{v} = \underbrace{\mathbf{P}\tilde{v}}_{\text{projected}} + \underbrace{(\mathbf{I}_p - \mathbf{P})\tilde{v}}_{\text{residual}}$$

are orthogonal:

$$\mathbf{P}\tilde{v} \perp (\mathbf{I}_p - \mathbf{P})\tilde{v}$$

### **i** Proof

*Proof.*

$$\begin{aligned} (\mathbf{P}\tilde{v})^\top (\mathbf{I} - \mathbf{P})\tilde{v} &= \tilde{v}^\top \mathbf{P}^\top (\mathbf{I} - \mathbf{P})\tilde{v} \\ &= \tilde{v}^\top \mathbf{P} (\mathbf{I} - \mathbf{P})\tilde{v} \\ &= \tilde{v}^\top (\mathbf{P} - \mathbf{P}^2)\tilde{v} \\ &= \tilde{v}^\top (\mathbf{P} - \mathbf{P})\tilde{v} \\ &= \tilde{v}^\top \mathbf{0} \tilde{v} \\ &= 0 \end{aligned}$$

where the second line uses symmetry ( $\mathbf{P}^\top = \mathbf{P}$ ) and the fourth line uses idempotency ( $\mathbf{P}^2 = \mathbf{P}$ ).  $\square$

**Definition 0.54** (Orthogonal matrix). A  $p \times p$  matrix  $\mathbf{Q}$  is **orthogonal** if

$$\underbrace{\mathbf{Q}^\top}_{p \times p} \underbrace{\mathbf{Q}}_{p \times p} = \mathbf{I}_p$$

**Example 0.79** (A rotation matrix). The matrix

$$\mathbf{Q} = \begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix}$$

rotates each vector in the plane counterclockwise by the angle  $\theta$  with  $\cos \theta = 0.6$  and  $\sin \theta = 0.8$  (about 53 degrees) (Banerjee and Roy 2014, chap. 8, Example 8.1, p. 207). It is orthogonal:

$$\begin{aligned} \mathbf{Q}^\top \mathbf{Q} &= \begin{bmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{bmatrix} \begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix} && \text{(definition of the transpose)} \\ &= \begin{bmatrix} 0.36 + 0.64 & -0.48 + 0.48 \\ -0.48 + 0.48 & 0.64 + 0.36 \end{bmatrix} && \text{(definition of matrix multiplication)} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} && \text{(add)} \end{aligned}$$

*Remark 0.20* (The columns of an orthogonal matrix are orthonormal). Entry  $(i, j)$  of  $\mathbf{Q}^\top \mathbf{Q}$  is the dot product of column  $i$  and column  $j$  of  $\mathbf{Q}$ , so  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$  says that the columns of  $\mathbf{Q}$  are orthonormal (Definition 0.13). For a square matrix,  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$  also implies  $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_p$ , so  $\mathbf{Q}^{-1} = \mathbf{Q}^\top$  (Definition 0.49) (Banerjee and Roy 2014, chap. 8, Theorem 8.1 and Definition 8.1, p. 209).

For example, the columns of  $\mathbf{Q}$  in Example 0.79 are  $(0.6, 0.8)$  and  $(-0.8, 0.6)$ . The diagonal entries  $0.36 + 0.64 = 1$  of  $\mathbf{Q}^\top \mathbf{Q}$  are their squared norms, and the off-diagonal entries  $-0.48 + 0.48 = 0$  are their dot product. Multiplying in the other order also gives  $\mathbf{I}_2$ :

$$\begin{aligned}\mathbf{Q}\mathbf{Q}^\top &= \begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix} \begin{bmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{bmatrix} && \text{(definition of the transpose)} \\ &= \begin{bmatrix} 0.36 + 0.64 & 0.48 - 0.48 \\ 0.48 - 0.48 & 0.64 + 0.36 \end{bmatrix} && \text{(definition of matrix multiplication)} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} && \text{(add)}\end{aligned}$$

So  $\mathbf{Q}^{-1} = \mathbf{Q}^\top$ .

**Theorem 0.55** (Orthogonal matrices preserve length). *If  $\mathbf{Q}$  is a  $p \times p$  orthogonal matrix (Definition 0.54) and  $\tilde{x}$  is a vector of length  $p$ , then*

$$\|\mathbf{Q}\tilde{x}\| = \|\tilde{x}\|$$

### Proof

*Proof.*

$$\begin{aligned}\|\mathbf{Q}\tilde{x}\|^2 &= (\mathbf{Q}\tilde{x}) \cdot (\mathbf{Q}\tilde{x}) && \text{(definition of the Euclidean norm)} \\ &= (\mathbf{Q}\tilde{x})^\top (\mathbf{Q}\tilde{x}) && \text{(dot product as a matrix product)} \\ &= \tilde{x}^\top \mathbf{Q}^\top (\mathbf{Q}\tilde{x}) && \text{(transpose of a product)} \\ &= \tilde{x}^\top (\mathbf{Q}^\top \mathbf{Q}) \tilde{x} && \text{(regroup; matrix multiplication is associative)} \\ &= \tilde{x}^\top \mathbf{I}_p \tilde{x} && \text{(definition of an orthogonal matrix)} \\ &= \tilde{x}^\top \tilde{x} && \text{(identity matrix)} \\ &= \|\tilde{x}\|^2 && \text{(definition of the Euclidean norm)}\end{aligned}$$

Both norms are nonnegative square roots, so equal squares give  $\|\mathbf{Q}\tilde{x}\| = \|\tilde{x}\|$ . The second step is Example 0.7, the third is Theorem 0.10, and the fourth is Theorem 0.7.  $\square$

**Example 0.80** (Rotating a vector keeps its length). With  $\mathbf{Q}$  from Example 0.79 and  $\tilde{x} = (3, 4)$  from Example 0.4:

$$\begin{aligned}
 \mathbf{Q}\tilde{x} &= \begin{bmatrix} 0.6 \cdot 3 - 0.8 \cdot 4 \\ 0.8 \cdot 3 + 0.6 \cdot 4 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
 &= \begin{bmatrix} 1.8 - 3.2 \\ 2.4 + 2.4 \end{bmatrix} && \text{(multiply)} \\
 &= \begin{bmatrix} -1.4 \\ 4.8 \end{bmatrix} && \text{(add)} \\
 \|\mathbf{Q}\tilde{x}\| &= \sqrt{(-1.4)^2 + 4.8^2} && \text{(definition of the norm)} \\
 &= \sqrt{1.96 + 23.04} && \text{(square)} \\
 &= \sqrt{25} && \text{(add)} \\
 &= 5 && \text{(take the square root)}
 \end{aligned}$$

which is  $\|\tilde{x}\| = 5$ .

## Quadratic Forms

**Definition 0.55** (Quadratic form). A **quadratic form** is a mathematical expression of the structure

$$\tilde{x}^\top \mathbf{S} \tilde{x}$$

where  $\tilde{x}$  is a  $p \times 1$  vector and  $\mathbf{S}$  is a  $p \times p$  matrix.

*Remark 0.21* (Quadratic forms in statistics). A quadratic form extends the scalar expression  $cx^2$  to vectors: with  $p = 1$  and  $\mathbf{S} = [c]$ ,  $\tilde{x}^\top \mathbf{S} \tilde{x} = cx^2$ . With  $p = 2$ ,

$$\tilde{x}^\top \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \tilde{x} = x_1^2 + 2x_1x_2 + 3x_2^2,$$

which is  $1 + 4 + 12 = 17$  at  $\tilde{x} = (1, 2)$ .

Quadratic forms occur often in statistics:

- The residual sum of squares in linear regression (see [Vector Calculus](#)) is a quadratic form.
- The variance of a linear combination of estimates (see [Inference about Gaussian Linear Regression Models](#)) is a quadratic form:  $\text{Var}\left(\tilde{x}^\top \hat{\beta}\right) = \tilde{x}^\top \text{Var}\left(\hat{\beta}\right) \tilde{x}$ .

**Definition 0.56** (Symmetric part of a square matrix). The **symmetric part** of a  $p \times p$  matrix  $\mathbf{S}$  is

$$\frac{1}{2}(\mathbf{S} + \mathbf{S}^\top)$$

**Example 0.81** (The symmetric part of a  $2 \times 2$  matrix). For  $\mathbf{S} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ , the symmetric part is

$$\frac{1}{2} \left( \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix},$$

which is symmetric.

**Theorem 0.56** (A quadratic form depends only on the symmetric part). *If  $\mathbf{S}$  is a  $p \times p$  matrix and  $\tilde{x}$  is a vector of length  $p$ , then*

$$\tilde{x}^\top \mathbf{S} \tilde{x} = \tilde{x}^\top \left( \frac{1}{2}(\mathbf{S} + \mathbf{S}^\top) \right) \tilde{x}.$$

*So the value of a quadratic form depends only on the symmetric part (Definition 0.56) of  $\mathbf{S}$ .*

### **i** Proof

*Proof.* The quadratic form  $\tilde{x}^\top \mathbf{S} \tilde{x}$  is a  $1 \times 1$  matrix, so it equals its own transpose:

$$\begin{aligned} \tilde{x}^\top \mathbf{S} \tilde{x} &= (\tilde{x}^\top \mathbf{S} \tilde{x})^\top && \text{(a } 1 \times 1 \text{ matrix is symmetric)} \\ &= \tilde{x}^\top \mathbf{S}^\top (\tilde{x}^\top)^\top && \text{(transpose of a product, twice)} \\ &= \tilde{x}^\top \mathbf{S}^\top \tilde{x} && \text{(transposing twice changes nothing)} \end{aligned}$$

Averaging the two expressions for  $\tilde{x}^\top \mathbf{S} \tilde{x}$ :

$$\begin{aligned} \tilde{x}^\top \mathbf{S} \tilde{x} &= \frac{1}{2}(\tilde{x}^\top \mathbf{S} \tilde{x} + \tilde{x}^\top \mathbf{S}^\top \tilde{x}) && \text{(average of two equal numbers)} \\ &= \frac{1}{2} \tilde{x}^\top (\mathbf{S} + \mathbf{S}^\top) \tilde{x} && \text{(distributive law)} \\ &= \tilde{x}^\top \left( \frac{1}{2}(\mathbf{S} + \mathbf{S}^\top) \right) \tilde{x} && \text{(move the scalar } \frac{1}{2} \text{ inside)} \end{aligned}$$

The distributive step is Theorem 0.8, and the transpose step is Theorem 0.10. □

**Example 0.82** (Replacing a matrix by its symmetric part). With  $\mathbf{S} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$  from Example 0.81 and  $\tilde{x} = (1, 1)$ :

$$\tilde{x}^\top \mathbf{S} \tilde{x} = 1 + 2 + 0 + 3 = 6 \quad \tilde{x}^\top \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} \tilde{x} = 1 + 1 + 1 + 3 = 6$$

Both quadratic forms equal the sum of their matrix's entries, because every entry of  $\tilde{x}$  is 1.

## Trace and Matrix Inner Product

**Definition 0.57** (Trace). The **trace** of a  $p \times p$  matrix  $\mathbf{M}$  is the sum of its diagonal entries:

$$\text{tr}(\mathbf{M}) \stackrel{\text{def}}{=} \sum_{i=1}^p M_{ii}$$

**Example 0.83** (The trace of a  $2 \times 2$  matrix).

$$\text{tr}\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\right) = 1 + 4 = 5$$

**Theorem 0.57** (The trace of a product does not depend on the order). *For an  $m \times n$  matrix  $\mathbf{A}$  and an  $n \times m$  matrix  $\mathbf{B}$ :*

$$\text{tr}\left(\underbrace{\mathbf{AB}}_{m \times m}\right) = \text{tr}\left(\underbrace{\mathbf{BA}}_{n \times n}\right)$$

**i** Proof

*Proof.*

$$\begin{aligned}
 \text{tr}(\mathbf{AB}) &= \sum_{i=1}^m (\mathbf{AB})_{ii} && \text{(definition of the trace)} \\
 &= \sum_{i=1}^m \sum_{s=1}^n a_{is} b_{si} && \text{(definition of matrix multiplication)} \\
 &= \sum_{s=1}^n \sum_{i=1}^m a_{is} b_{si} && \text{(swap the order of two finite sums)} \\
 &= \sum_{s=1}^n \sum_{i=1}^m b_{si} a_{is} && \text{(multiplication of numbers is commutative)} \\
 &= \sum_{s=1}^n (\mathbf{BA})_{ss} && \text{(definition of matrix multiplication)} \\
 &= \text{tr}(\mathbf{BA}) && \text{(definition of the trace)}
 \end{aligned}$$

□

**Example 0.84** (Traces of the two products of a  $2 \times 3$  and a  $3 \times 2$  matrix). Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$$

The products are

$$\mathbf{AB} = \begin{bmatrix} 5 & 2 \\ 2 & 4 \end{bmatrix} \quad \mathbf{BA} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 1 & 3 \end{bmatrix}$$

The trace needs only the diagonal entries. For  $\mathbf{AB}$ , which is  $2 \times 2$ :

$$\begin{aligned}
 \text{tr}(\mathbf{AB}) &= (\mathbf{AB})_{11} + (\mathbf{AB})_{22} && \text{(definition of the trace)} \\
 &= (1 \cdot 1 + 2 \cdot 2 + 0 \cdot 0) + (0 \cdot 0 + 1 \cdot 1 + 3 \cdot 1) && \text{(definition of matrix multiplication)} \\
 &= (1 + 4 + 0) + (0 + 1 + 3) && \text{(multiply)} \\
 &= 5 + 4 && \text{(add within each entry)} \\
 &= 9 && \text{(add)}
 \end{aligned}$$

For  $\mathbf{BA}$ , which is  $3 \times 3$ :

$$\begin{aligned}
\text{tr}(\mathbf{BA}) &= (\mathbf{BA})_{11} + (\mathbf{BA})_{22} + (\mathbf{BA})_{33} && \text{(definition of the trace)} \\
&= (1 \cdot 1 + 0 \cdot 0) + (2 \cdot 2 + 1 \cdot 1) + (0 \cdot 0 + 1 \cdot 3) && \text{(definition of matrix multiplication)} \\
&= (1 + 0) + (4 + 1) + (0 + 3) && \text{(multiply)} \\
&= 1 + 5 + 3 && \text{(add within each entry)} \\
&= 9 && \text{(add)}
\end{aligned}$$

The products have different sizes, but their traces agree.

*Remark 0.22* (Moving a factor of a triple product). Let  $\mathbf{A}$  be  $m \times n$ ,  $\mathbf{B}$  be  $n \times k$ , and  $\mathbf{C}$  be  $k \times m$ , so that  $\mathbf{AB}$  is  $m \times k$  and  $\mathbf{ABC}$  is square. Applying Theorem 0.57 to the  $m \times k$  matrix  $\mathbf{AB}$  and the  $k \times m$  matrix  $\mathbf{C}$  moves the last factor of the triple product to the front (Banerjee and Roy 2014, chap. 1, Theorem 1.5 and eq. 1.17, p. 19):

$$\begin{aligned}
\text{tr}(\mathbf{ABC}) &= \text{tr}((\mathbf{AB})\mathbf{C}) && \text{(matrix multiplication is associative)} \\
&= \text{tr}(\mathbf{C}(\mathbf{AB})) && \text{(the theorem, applied to the two factors } \mathbf{AB} \text{ and } \mathbf{C}) \\
&= \text{tr}(\mathbf{CAB}) && \text{(matrix multiplication is associative)}
\end{aligned}$$

For example, let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Then

$$\mathbf{AB} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 1 & 1 \cdot 0 + 2 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 0 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \quad \mathbf{CA} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 0 & 0 \cdot 2 + 1 \cdot 1 \\ 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 2 + 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$$

so

$$(\mathbf{AB})\mathbf{C} = \begin{bmatrix} 3 \cdot 0 + 2 \cdot 1 & 3 \cdot 1 + 2 \cdot 0 \\ 1 \cdot 0 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} \quad (\mathbf{CA})\mathbf{B} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 0 + 1 \cdot 1 \\ 1 \cdot 1 + 2 \cdot 1 & 1 \cdot 0 + 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$$

and both traces are the same:  $\text{tr}(\mathbf{ABC}) = 2 + 1 = 3$  and  $\text{tr}(\mathbf{CAB}) = 1 + 2 = 3$ .

**Definition 0.58** (Matrix inner product). The **inner product** of two  $n \times p$  matrices  $\mathbf{A}$  and  $\mathbf{B}$  is

$$\langle \mathbf{A}, \mathbf{B} \rangle \stackrel{\text{def}}{=} \text{tr} \left( \underbrace{\mathbf{A}^\top \mathbf{B}}_{p \times p} \right)$$

Banerjee and Roy (2014, chap. 15, eq. 15.7, p. 491) defines the same inner product on  $n \times p$  matrices with the trace.

**Theorem 0.58** (The matrix inner product multiplies matching entries). *For two  $n \times p$  matrices  $\mathbf{A}$  and  $\mathbf{B}$ :*

$$\langle \mathbf{A}, \mathbf{B} \rangle = \sum_{i=1}^n \sum_{j=1}^p a_{ij} b_{ij}$$

### **i** Proof

*Proof.*

$$\begin{aligned} \langle \mathbf{A}, \mathbf{B} \rangle &= \text{tr}(\mathbf{A}^\top \mathbf{B}) && \text{(definition of the inner product)} \\ &= \sum_{j=1}^p (\mathbf{A}^\top \mathbf{B})_{jj} && \text{(definition of the trace)} \\ &= \sum_{j=1}^p \sum_{i=1}^n (\mathbf{A}^\top)_{ji} b_{ij} && \text{(definition of matrix multiplication)} \\ &= \sum_{j=1}^p \sum_{i=1}^n a_{ij} b_{ij} && \text{(definition of the transpose)} \\ &= \sum_{i=1}^n \sum_{j=1}^p a_{ij} b_{ij} && \text{(swap the order of two finite sums)} \end{aligned}$$

□

**Example 0.85** (The inner product of two  $2 \times 2$  matrices). Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

From Definition 0.58:

$$\begin{aligned}
\langle \mathbf{A}, \mathbf{B} \rangle &= \text{tr}(\mathbf{A}^\top \mathbf{B}) && \text{(definition of the inner product)} \\
&= \text{tr} \left( \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^\top \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \right) && \text{(substitute)} \\
&= \text{tr} \left( \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \right) && \text{(definition of the transpose)} \\
&= \text{tr} \left( \begin{bmatrix} -3 & 7 \\ -4 & 10 \end{bmatrix} \right) && \text{(multiply)} \\
&= -3 + 10 && \text{(definition of the trace)} \\
&= 7 && \text{(add)}
\end{aligned}$$

From Theorem 0.58:

$$\begin{aligned}
\langle \mathbf{A}, \mathbf{B} \rangle &= 1 \cdot 0 + 2 \cdot 1 + 3 \cdot (-1) + 4 \cdot 2 && \text{(multiply matching entries and add)} \\
&= 0 + 2 - 3 + 8 && \text{(multiply)} \\
&= 7 && \text{(add)}
\end{aligned}$$

*Remark 0.23* (The matrix inner product as a dot product). Theorem 0.58 says that the matrix inner product is the dot product (Definition 0.5) of the two matrices' entries, each listed as one vector of length  $np$ , with both matrices' entries listed in the same order. For example, listing the entries of  $\mathbf{A}$  and  $\mathbf{B}$  in Example 0.85 row by row gives  $(1, 2, 3, 4)$  and  $(0, 1, -1, 2)$ , and  $(1, 2, 3, 4) \cdot (0, 1, -1, 2) = 7 = \langle \mathbf{A}, \mathbf{B} \rangle$ .

**Definition 0.59** (Frobenius norm). The **Frobenius norm** of an  $n \times p$  matrix  $\mathbf{A}$  is

$$\|\mathbf{A}\|_F \stackrel{\text{def}}{=} \sqrt{\langle \mathbf{A}, \mathbf{A} \rangle}$$

where  $\langle \cdot, \cdot \rangle$  is the matrix inner product (Definition 0.58).

**Example 0.86** (The Frobenius norm of a  $2 \times 2$  matrix). For  $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ :

$$\begin{aligned}
\|\mathbf{A}\|_F &= \sqrt{\langle \mathbf{A}, \mathbf{A} \rangle} && \text{(definition of the Frobenius norm)} \\
&= \sqrt{1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 + 4 \cdot 4} && \text{(sum of products of matching entries)} \\
&= \sqrt{1 + 4 + 9 + 16} && \text{(multiply)} \\
&= \sqrt{30} && \text{(add)}
\end{aligned}$$

The second step is Theorem 0.58.

*Remark 0.24* (The Frobenius norm is the Euclidean norm of the entries). By Theorem 0.58,  $\|\mathbf{A}\|_F^2 = \sum_{i=1}^n \sum_{j=1}^p a_{ij}^2$ , a sum of squares, so the square root is always defined.  $\|\mathbf{A}\|_F$  is the Euclidean norm (Definition 0.11) of the entries of  $\mathbf{A}$ , listed as one vector of length  $np$  (Banerjee and Roy 2014, chap. 15, Definition 15.4, p. 492).

For example, listing the entries of  $\mathbf{A}$  in Example 0.86 row by row gives the vector  $(1, 2, 3, 4)$  of length  $2 \cdot 2 = 4$ , and  $\|(1, 2, 3, 4)\| = \sqrt{1 + 4 + 9 + 16} = \sqrt{30} = \|\mathbf{A}\|_F$ .

## Matrix Decompositions

**Definition 0.60** (Eigenvalue and eigenvector). Let  $\mathbf{A}$  be a  $p \times p$  matrix. A real number  $\lambda$  is an **eigenvalue** of  $\mathbf{A}$  if some real vector  $\tilde{v} \neq \tilde{0}$  of length  $p$  satisfies

$$\underset{p \times p}{\mathbf{A}} \underset{p \times 1}{\tilde{v}} = \lambda \underset{p \times 1}{\tilde{v}}$$

Any such  $\tilde{v}$  is an **eigenvector** of  $\mathbf{A}$  for  $\lambda$ .

**Example 0.87** (Eigenvectors of a  $2 \times 2$  matrix). Let  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ .

$(1, 1)$  is an eigenvector for the eigenvalue 3:

$$\begin{aligned} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} 2+1 \\ 1+2 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\ &= 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} && \text{(factor out 3)} \end{aligned}$$

$(1, -1)$  is an eigenvector for the eigenvalue 1:

$$\begin{aligned} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= \begin{bmatrix} 2-1 \\ 1-2 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\ &= 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} && \text{(factor out 1)} \end{aligned}$$

*Remark 0.25* (Multiples of an eigenvector). Multiplying by  $\mathbf{A}$  rescales an eigenvector by  $\lambda$  and does not change the line it lies on. Any nonzero multiple  $c\tilde{v}$  of an eigenvector is also an eigenvector for the same  $\lambda$ :

$$\begin{aligned}
\mathbf{A}(c\tilde{v}) &= c\mathbf{A}\tilde{v} && \text{(move the scalar } c \text{ to the front)} \\
&= c\lambda\tilde{v} && (\tilde{v} \text{ is an eigenvector for } \lambda) \\
&= \lambda(c\tilde{v}) && \text{(multiplication of numbers is commutative)}
\end{aligned}$$

For example, in Example 0.87,  $2(1, 1) = (2, 2)$  is also an eigenvector for the eigenvalue 3:

$$\begin{aligned}
\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} &= \begin{bmatrix} 2 \cdot 2 + 1 \cdot 2 \\ 1 \cdot 2 + 2 \cdot 2 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
&= \begin{bmatrix} 4 + 2 \\ 2 + 4 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 6 \\ 6 \end{bmatrix} && \text{(add)} \\
&= 3 \begin{bmatrix} 2 \\ 2 \end{bmatrix} && \text{(factor out 3)}
\end{aligned}$$

**Example 0.88** (A vector that is not an eigenvector). For the same  $\mathbf{A}$ ,  $(1, 0)$  is not an eigenvector:

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

and no number  $\lambda$  gives  $(2, 1) = \lambda(1, 0)$ , because the second entries would need  $1 = \lambda \cdot 0$ .

*Remark 0.26* (Real and complex eigenvalues). These notes take  $\lambda$  and  $\tilde{v}$  to be real. Banerjee and Roy (2014, chap. 11, Definition 11.1, pp. 312-313) also allows complex eigenvalues and eigenvectors, because some real matrices, such as a rotation by 90 degrees, have no real eigenvalues (Banerjee and Roy 2014, chap. 11, Example 11.2, p. 312).

For example,  $\mathbf{R} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$  rotates each vector in the plane counterclockwise by 90 degrees:  $\mathbf{R}(v_1, v_2) = (-v_2, v_1)$ . Suppose  $\mathbf{R}\tilde{v} = \lambda\tilde{v}$  for a real number  $\lambda$ . Matching entries gives  $-v_2 = \lambda v_1$  and  $v_1 = \lambda v_2$ . Substituting the second equation into the first gives  $-v_2 = \lambda^2 v_2$ , so  $(1 + \lambda^2)v_2 = 0$ . Since  $1 + \lambda^2 > 0$ ,  $v_2 = 0$ , and then  $v_1 = \lambda v_2 = 0$ . So the only solution is  $\tilde{v} = \tilde{0}$ , and  $\mathbf{R}$  has no real eigenvalue.

**Theorem 0.59** (Spectral theorem for symmetric matrices). *If  $\mathbf{A}$  is a  $p \times p$  symmetric matrix (Definition 0.47) with real entries, then there are a  $p \times p$  orthogonal matrix  $\mathbf{Q}$  (Definition 0.54) and a  $p \times p$  diagonal matrix (Definition 0.48) with real diagonal entries  $\lambda_1, \dots, \lambda_p$  such that*

$$\underbrace{\mathbf{A}}_{p \times p} = \underbrace{\mathbf{Q}}_{p \times p} \underbrace{\underbrace{\mathbf{Q}^\top}_{p \times p}}_{p \times p}$$

Each  $\lambda_i$  is an eigenvalue of  $\mathbf{A}$  (Definition 0.60), and column  $i$  of  $\mathbf{Q}$  is an eigenvector of  $\mathbf{A}$  for  $\lambda_i$ .

*Remark 0.27* (An equivalent form of the spectral theorem). The proof, by induction on  $p$ , is outside the scope of these notes; see Banerjee and Roy (2014, chap. 11, Theorem 11.27, p. 349), which states the result in the equivalent form  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{D}$ . The two forms are equivalent because  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q} \mathbf{Q}^\top = \mathbf{I}_p$  (Remark 0.20). Multiplying  $\mathbf{A} = \mathbf{Q} \mathbf{Q}^\top \mathbf{D}$  by  $\mathbf{Q}^\top$  on the left and by  $\mathbf{Q}$  on the right gives

$$\begin{aligned} \mathbf{Q}^\top \mathbf{A} \mathbf{Q} &= \mathbf{Q}^\top (\mathbf{Q} \mathbf{Q}^\top) \mathbf{Q} && \text{(substitute)} \\ &= (\mathbf{Q}^\top \mathbf{Q}) (\mathbf{Q}^\top \mathbf{Q}) && \text{(regroup; matrix multiplication is associative)} \\ &= \mathbf{I}_p \mathbf{I}_p && (\mathbf{Q} \text{ is orthogonal}) \\ &= \mathbf{D} && \text{(multiplying by } \mathbf{I}_p \text{ changes nothing)} \end{aligned}$$

Multiplying  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{D}$  by  $\mathbf{Q}$  on the left and by  $\mathbf{Q}^\top$  on the right gives back the first form:

$$\begin{aligned} \mathbf{Q} \mathbf{Q}^\top &= \mathbf{Q} (\mathbf{Q}^\top \mathbf{A} \mathbf{Q}) \mathbf{Q}^\top && \text{(substitute)} \\ &= (\mathbf{Q} \mathbf{Q}^\top) \mathbf{A} (\mathbf{Q} \mathbf{Q}^\top) && \text{(regroup; matrix multiplication is associative)} \\ &= \mathbf{I}_p \mathbf{A} \mathbf{I}_p && (\mathbf{Q} \mathbf{Q}^\top = \mathbf{I}_p) \\ &= \mathbf{A} && \text{(multiplying by } \mathbf{I}_p \text{ changes nothing)} \end{aligned}$$

For example, let  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  and  $\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ , which is symmetric, so  $\mathbf{Q}^\top = \mathbf{Q}$ . Then

$$\begin{aligned}
\mathbf{Q}^\top \mathbf{A} \mathbf{Q} &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute)} \\
&= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalars to the front)} \\
&= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply the scalars)} \\
&= \frac{1}{2} \begin{bmatrix} 1 \cdot 2 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot 2 \\ 1 \cdot 2 + (-1) \cdot 1 & 1 \cdot 1 + (-1) \cdot 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(definition of matrix multiplication, first two matrices)} \\
&= \frac{1}{2} \begin{bmatrix} 2+1 & 1+2 \\ 2-1 & 1-2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 3 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(add)} \\
&= \frac{1}{2} \begin{bmatrix} 3 \cdot 1 + 3 \cdot 1 & 3 \cdot 1 + 3 \cdot (-1) \\ 1 \cdot 1 + (-1) \cdot 1 & 1 \cdot 1 + (-1) \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{2} \begin{bmatrix} 3+3 & 3-3 \\ 1-1 & 1+1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} && \text{(multiply by } \frac{1}{2} \text{)}
\end{aligned}$$

a diagonal matrix with the eigenvalues 3 and 1 of  $\mathbf{A}$  from Example 0.87 on its diagonal.

**Definition 0.61** (Eigendecomposition). Let  $\mathbf{A}$  be a  $p \times p$  symmetric matrix with real entries. An **eigendecomposition**, or **spectral decomposition**, of  $\mathbf{A}$  is a factorization

$$\mathbf{A} = \mathbf{Q} \mathbf{Q}^\top$$

with  $\mathbf{Q}$  a  $p \times p$  orthogonal matrix (Definition 0.54) and a  $p \times p$  diagonal matrix (Definition 0.48). Theorem 0.59 says that every such  $\mathbf{A}$  has one.

**Example 0.89** (An eigendecomposition of a  $2 \times 2$  symmetric matrix). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ , Example 0.87 found the eigenvectors  $(1, 1)$  for 3 and  $(1, -1)$  for 1. They are orthogonal (Definition 0.10):

$$\begin{aligned}
(1, 1) \cdot (1, -1) &= 1 \cdot 1 + 1 \cdot (-1) && \text{(definition of the dot product)} \\
&= 1 - 1 && \text{(multiply)} \\
&= 0 && \text{(add)}
\end{aligned}$$

Each has norm  $\sqrt{1^2 + 1^2} = \sqrt{2}$  (Definition 0.11), so dividing each by  $\sqrt{2}$  gives two orthonormal eigenvectors (Definition 0.13). Put them in the columns of  $\mathbf{Q}$ , and the matching eigenvalues on the diagonal of :

$$\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

This  $\mathbf{Q}$  is symmetric, so  $\mathbf{Q}^\top = \mathbf{Q}$ . Then

$$\begin{aligned}
\mathbf{Q}\mathbf{Q}^\top &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} && \text{(substitute)} \\
&= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} && \text{(move the scalars to the front)} \\
&= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} && \text{(multiply the scalars)} \\
&= \frac{1}{2} \begin{bmatrix} 1 \cdot \frac{1}{\sqrt{2}} + 1 \cdot \frac{1}{\sqrt{2}} & 1 \cdot \frac{1}{\sqrt{2}} + 1 \cdot (-\frac{1}{\sqrt{2}}) \\ 1 \cdot \frac{1}{\sqrt{2}} + (-1) \cdot \frac{1}{\sqrt{2}} & 1 \cdot \frac{1}{\sqrt{2}} + (-1) \cdot (-\frac{1}{\sqrt{2}}) \end{bmatrix} && \text{(definition of matrix multiplication, first two matrices)} \\
&= \frac{1}{2} \begin{bmatrix} \frac{2}{\sqrt{2}} & 0 \\ 0 & \frac{2}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} \frac{2}{\sqrt{2}} & 0 \\ 0 & \frac{2}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} && \text{(add)} \\
&= \frac{1}{2} \begin{bmatrix} \frac{2}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 \cdot \frac{1}{\sqrt{2}} & \frac{2}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 \cdot (-\frac{1}{\sqrt{2}}) \\ 0 \cdot \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} & 0 \cdot \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}} \cdot (-\frac{1}{\sqrt{2}}) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(add)} \\
&= \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} && \text{(multiply by } \frac{1}{2} \text{)} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}
\end{aligned}$$

which is  $\mathbf{A}$ .

*Remark 0.28* (An eigendecomposition is not unique). A symmetric matrix can have more than one eigendecomposition. Reordering the eigenvalues on the diagonal of , and the

columns of  $\mathbf{Q}$  with them, gives another one, and so does multiplying a column of  $\mathbf{Q}$  by  $-1$ .

For example, take  $\mathbf{A}$  from Example 0.89. Swapping the two eigenvalues and the two columns gives

$$\mathbf{Q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \quad \mathbf{Q}_1^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

and multiplying the second column of  $\mathbf{Q}$  by  $-1$  gives

$$\mathbf{Q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \quad \mathbf{Q}_2^{-1} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

The columns of  $\mathbf{Q}_1$  and  $\mathbf{Q}_2$  are the columns of  $\mathbf{Q}$ , reordered or multiplied by  $-1$ , so they are still orthonormal, and  $\mathbf{Q}_1$  and  $\mathbf{Q}_2$  are orthogonal. Both products give back  $\mathbf{A}$ :

$$\begin{aligned} \mathbf{Q}_1 \mathbf{Q}_1^\top &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \left( \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \right)^\top && \text{(substitute)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(definition of the transpose)} \\ &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(move the scalars to the front)} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(multiply the scalars)} \\ &= \frac{1}{2} \begin{bmatrix} 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 0 + 1 \cdot 3 \\ -1 \cdot 1 + 1 \cdot 0 & -1 \cdot 0 + 1 \cdot 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(definition of matrix multiplication, first two matrices)} \\ &= \frac{1}{2} \begin{bmatrix} 1 + 0 & 0 + 3 \\ -1 + 0 & 0 + 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(multiply)} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 3 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(add)} \\ &= \frac{1}{2} \begin{bmatrix} 1 \cdot 1 + 3 \cdot 1 & 1 \cdot (-1) + 3 \cdot 1 \\ -1 \cdot 1 + 3 \cdot 1 & -1 \cdot (-1) + 3 \cdot 1 \end{bmatrix} && \text{(definition of matrix multiplication)} \\ &= \frac{1}{2} \begin{bmatrix} 1 + 3 & -1 + 3 \\ -1 + 3 & 1 + 3 \end{bmatrix} && \text{(multiply)} \\ &= \frac{1}{2} \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} && \text{(add)} \\ &= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} && \text{(multiply by } \frac{1}{2} \text{)} \end{aligned}$$

$$\begin{aligned}
\mathbf{Q}_{22}\mathbf{Q}_2^\top &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \left( \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \right)^\top && \text{(substitute)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(definition of the transpose)} \\
&= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(move the scalars to the front)} \\
&= \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(multiply the scalars)} \\
&= \frac{1}{2} \begin{bmatrix} 1 \cdot 3 + (-1) \cdot 0 & 1 \cdot 0 + (-1) \cdot 1 \\ 1 \cdot 3 + 1 \cdot 0 & 1 \cdot 0 + 1 \cdot 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(definition of matrix multiplication, first two matrices)} \\
&= \frac{1}{2} \begin{bmatrix} 3 + 0 & 0 - 1 \\ 3 + 0 & 0 + 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 3 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(add)} \\
&= \frac{1}{2} \begin{bmatrix} 3 \cdot 1 + (-1) \cdot (-1) & 3 \cdot 1 + (-1) \cdot 1 \\ 3 \cdot 1 + 1 \cdot (-1) & 3 \cdot 1 + 1 \cdot 1 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{2} \begin{bmatrix} 3 + 1 & 3 - 1 \\ 3 - 1 & 3 + 1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} && \text{(multiply by } \frac{1}{2} \text{)}
\end{aligned}$$

**Theorem 0.60** (Singular value decomposition). *Let  $\mathbf{A}$  be an  $n \times p$  matrix with real entries and  $\text{rank}(\mathbf{A}) = r$  (Definition 0.26). Then there are an  $n \times n$  orthogonal matrix  $\mathbf{U}$ , a  $p \times p$  orthogonal matrix  $\mathbf{V}$  (Definition 0.54), and numbers  $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$  such that*

$$\underbrace{\mathbf{A}}_{n \times p} = \underbrace{\mathbf{U}}_{n \times n} \underbrace{\mathbf{D}}_{n \times p} \underbrace{\mathbf{V}^\top}_{p \times p}$$

where  $\mathbf{D}$  has entries  $d_{ii} = \sigma_i$  for  $i = 1, \dots, r$  and every other entry 0.

*Remark 0.29* (The SVD applies to every real matrix). The proof is outside the scope of these notes; see Banerjee and Roy (2014, chap. 12, Theorem 12.1, p. 373). Unlike the spectral theorem (Theorem 0.59), Theorem 0.60 applies to every real matrix, including

one that is not square or not symmetric.

For example, the  $1 \times 2$  matrix  $\mathbf{A} = \begin{bmatrix} 1 & 1 \end{bmatrix}$  is not square, so Theorem 0.59 does not apply to it. It has rank 1, and Theorem 0.60 holds with  $\mathbf{U} = [1]$ ,  $\mathbf{D} = \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix}$ , and  $\mathbf{V}$  the orthogonal matrix  $\mathbf{Q}$  from Example 0.89, which is symmetric, so  $\mathbf{V}^\top = \mathbf{V}$ :

$$\begin{aligned}
 \mathbf{UDV}^\top &= [1] \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute)} \\
 &= \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiplying by the } 1 \times 1 \text{ identity changes nothing)} \\
 &= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalar to the front)} \\
 &= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} \cdot 1 + 0 \cdot 1 & \sqrt{2} \cdot 1 + 0 \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
 &= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} + 0 & \sqrt{2} + 0 \end{bmatrix} && \text{(multiply)} \\
 &= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & \sqrt{2} \end{bmatrix} && \text{(add)} \\
 &= \begin{bmatrix} 1 & 1 \end{bmatrix} && \text{(divide by } \sqrt{2})
 \end{aligned}$$

$\mathbf{U}$  is orthogonal because  $\mathbf{U}^\top \mathbf{U} = [1] = \mathbf{I}_1$ , and the single singular value is  $\sigma_1 = \sqrt{2}$ .

**Definition 0.62** (Singular value decomposition and singular values). Let  $\mathbf{A}$  be an  $n \times p$  matrix with real entries and  $\text{rank}(\mathbf{A}) = r$ . A **singular value decomposition** (SVD) of  $\mathbf{A}$  is a factorization  $\mathbf{A} = \mathbf{UDV}^\top$  with  $\mathbf{U}$ ,  $\mathbf{D}$  and  $\mathbf{V}$  as in Theorem 0.60. The numbers  $\sigma_1 \geq \dots \geq \sigma_r > 0$  on the diagonal of  $\mathbf{D}$  are the **singular values** of  $\mathbf{A}$ .

**Example 0.90** (An SVD of a  $3 \times 2$  matrix). Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \quad \mathbf{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$\mathbf{V}$  is the orthogonal matrix  $\mathbf{Q}$  from Example 0.89.  $\mathbf{U}$  is orthogonal too. Entry  $(i, j)$  of  $\mathbf{U}^\top \mathbf{U}$  is the dot product of columns  $i$  and  $j$  of  $\mathbf{U}$ , and those columns are  $\tilde{u}_1 = (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$ ,  $\tilde{u}_2 = (0, 1, 0)$  and  $\tilde{u}_3 = (\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}})$ :

$$\begin{aligned}
\tilde{u}_1 \cdot \tilde{u}_1 &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 \cdot 0 + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} && \text{(definition of the dot product)} \\
&= \frac{1}{2} + 0 + \frac{1}{2} && \text{(multiply)} \\
&= 1 && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\tilde{u}_2 \cdot \tilde{u}_2 &= 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 && \text{(definition of the dot product)} \\
&= 0 + 1 + 0 && \text{(multiply)} \\
&= 1 && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\tilde{u}_3 \cdot \tilde{u}_3 &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 \cdot 0 + \left(-\frac{1}{\sqrt{2}}\right) \cdot \left(-\frac{1}{\sqrt{2}}\right) && \text{(definition of the dot product)} \\
&= \frac{1}{2} + 0 + \frac{1}{2} && \text{(multiply)} \\
&= 1 && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\tilde{u}_1 \cdot \tilde{u}_2 &= \frac{1}{\sqrt{2}} \cdot 0 + 0 \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 && \text{(definition of the dot product)} \\
&= 0 + 0 + 0 && \text{(multiply)} \\
&= 0 && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\tilde{u}_1 \cdot \tilde{u}_3 &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 \cdot 0 + \frac{1}{\sqrt{2}} \cdot \left(-\frac{1}{\sqrt{2}}\right) && \text{(definition of the dot product)} \\
&= \frac{1}{2} + 0 - \frac{1}{2} && \text{(multiply)} \\
&= 0 && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\tilde{u}_2 \cdot \tilde{u}_3 &= 0 \cdot \frac{1}{\sqrt{2}} + 1 \cdot 0 + 0 \cdot \left(-\frac{1}{\sqrt{2}}\right) && \text{(definition of the dot product)} \\
&= 0 + 0 + 0 && \text{(multiply)} \\
&= 0 && \text{(add)}
\end{aligned}$$

The dot product is symmetric, so these six values fill in all nine entries, and  $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_3$ .  $\mathbf{V}$  is symmetric, so  $\mathbf{V}^\top = \mathbf{V}$ . Then

$$\begin{aligned}
\mathbf{D}\mathbf{V}^\top &= \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalar to the front)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \cdot 1 + 0 \cdot 1 & 2 \cdot 1 + 0 \cdot (-1) \\ 0 \cdot 1 + \sqrt{2} \cdot 1 & 0 \cdot 1 + \sqrt{2} \cdot (-1) \\ 0 \cdot 1 + 0 \cdot 1 & 0 \cdot 1 + 0 \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 + 0 & 2 + 0 \\ 0 + \sqrt{2} & 0 - \sqrt{2} \\ 0 + 0 & 0 + 0 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 2 \\ \sqrt{2} & -\sqrt{2} \\ 0 & 0 \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(divide by } \sqrt{2})
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{U}\mathbf{D}\mathbf{V}^\top &= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(substitute)} \\
&= \begin{bmatrix} \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 & \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot (-1) + \frac{1}{\sqrt{2}} \cdot 0 \\ 0 \cdot \sqrt{2} + 1 \cdot 1 + 0 \cdot 0 & 0 \cdot \sqrt{2} + 1 \cdot (-1) + 0 \cdot 0 \\ \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot 1 - \frac{1}{\sqrt{2}} \cdot 0 & \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot (-1) - \frac{1}{\sqrt{2}} \cdot 0 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \begin{bmatrix} 1 + 0 + 0 & 1 + 0 + 0 \\ 0 + 1 + 0 & 0 - 1 + 0 \\ 1 + 0 - 0 & 1 + 0 - 0 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(add)}
\end{aligned}$$

which is  $\mathbf{A}$ .  $\mathbf{A}$  has rank 2, because its two columns are linearly independent, and its singular values are  $\sigma_1 = 2$  and  $\sigma_2 = \sqrt{2}$ .

*Remark 0.30* (An SVD is not unique). An SVD is not unique (Banerjee and Roy 2014, chap. 12, Examples 12.2 and 12.3, p. 378). For any  $i \leq r$ , multiplying column  $i$  of both  $\mathbf{U}$  and  $\mathbf{V}$  by  $-1$  gives another one. The singular values do not depend on which SVD is chosen (Banerjee and Roy 2014, chap. 12, pp. 371 and 378).

For example, in Example 0.90, multiplying the first columns of  $\mathbf{U}$  and  $\mathbf{V}$  by  $-1$  gives

$$\mathbf{U}_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \quad \mathbf{V}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}$$

Their columns are the columns of  $\mathbf{U}$  and  $\mathbf{V}$ , some multiplied by  $-1$ , so they are still orthonormal. With the same  $\mathbf{D}$ :

$$\begin{aligned} \mathbf{D}\mathbf{V}_1^\top &= \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \left( \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \right)^\top && \text{(substitute)} \\ &= \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} && \text{(definition of the transpose)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalar to the front)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \cdot (-1) + 0 \cdot 1 & 2 \cdot (-1) + 0 \cdot (-1) \\ 0 \cdot (-1) + \sqrt{2} \cdot 1 & 0 \cdot (-1) + \sqrt{2} \cdot (-1) \\ 0 \cdot (-1) + 0 \cdot 1 & 0 \cdot (-1) + 0 \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} -2 + 0 & -2 + 0 \\ 0 + \sqrt{2} & 0 - \sqrt{2} \\ 0 + 0 & 0 + 0 \end{bmatrix} && \text{(multiply)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} -2 & -2 \\ \sqrt{2} & -\sqrt{2} \\ 0 & 0 \end{bmatrix} && \text{(add)} \\ &= \begin{bmatrix} -\sqrt{2} & -\sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(divide by } \sqrt{2}) \end{aligned}$$

and

$$\begin{aligned}
\mathbf{U}_1 \mathbf{D} \mathbf{V}_1^\top &= \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} -\sqrt{2} & -\sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(substitute)} \\
&= \begin{bmatrix} -\frac{1}{\sqrt{2}} \cdot (-\sqrt{2}) + 0 \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 & -\frac{1}{\sqrt{2}} \cdot (-\sqrt{2}) + 0 \cdot (-1) + \frac{1}{\sqrt{2}} \cdot 0 \\ 0 \cdot (-\sqrt{2}) + 1 \cdot 1 + 0 \cdot 0 & 0 \cdot (-\sqrt{2}) + 1 \cdot (-1) + 0 \cdot 0 \\ -\frac{1}{\sqrt{2}} \cdot (-\sqrt{2}) + 0 \cdot 1 - \frac{1}{\sqrt{2}} \cdot 0 & -\frac{1}{\sqrt{2}} \cdot (-\sqrt{2}) + 0 \cdot (-1) - \frac{1}{\sqrt{2}} \cdot 0 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \begin{bmatrix} 1 + 0 + 0 & 1 + 0 + 0 \\ 0 + 1 + 0 & 0 - 1 + 0 \\ 1 + 0 - 0 & 1 + 0 - 0 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(add)}
\end{aligned}$$

which is  $\mathbf{A}$  again. Both SVDs have the singular values  $\sigma_1 = 2$  and  $\sigma_2 = \sqrt{2}$ .

*Remark 0.31* (Counting zero singular values). Some texts, and R's `svd()`, also count  $\min(n, p) - r$  singular values equal to 0, so that every  $n \times p$  matrix has  $\min(n, p)$  singular values.

For example,  $\mathbf{B} = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$  has rank  $r = 1$ . Taking  $\mathbf{U} = \mathbf{V} = \mathbf{I}_2$  and  $\mathbf{D} = \mathbf{B}$  gives an SVD  $\mathbf{B} = \mathbf{I}_2 \mathbf{B} \mathbf{I}_2^\top$ , so by Definition 0.62,  $\mathbf{B}$  has one singular value,  $\sigma_1 = 3$ . R's `svd()` reports  $\min(2, 2) = 2$  singular values for  $\mathbf{B}$ : 3 and 0.

**Theorem 0.61** (An SVD gives an eigendecomposition of  $\mathbf{A}^\top \mathbf{A}$ ). *Let  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{V}^\top$  be a singular value decomposition (Definition 0.62) of an  $n \times p$  matrix  $\mathbf{A}$  with singular values  $\sigma_1, \dots, \sigma_r$ . Then  $\mathbf{A}^\top \mathbf{A}$  is symmetric (Definition 0.47), and*

$$\underbrace{\mathbf{A}^\top \mathbf{A}}_{p \times p} = \underbrace{\mathbf{V}}_{p \times p} \underbrace{\mathbf{D}^\top \mathbf{D}}_{p \times p} \underbrace{\mathbf{V}^\top}_{p \times p}$$

where  $\stackrel{\text{def}}{=} \mathbf{D}^\top \mathbf{D}$  is the  $p \times p$  diagonal matrix whose  $i$ -th diagonal entry  $\lambda_i$  is  $\sigma_i^2$  for  $i \leq r$  and 0 for  $i > r$ . So  $\mathbf{V} \mathbf{V}^\top$  is an eigendecomposition (Definition 0.61) of  $\mathbf{A}^\top \mathbf{A}$ : column  $i$  of  $\mathbf{V}$  is an eigenvector of  $\mathbf{A}^\top \mathbf{A}$  for the eigenvalue  $\lambda_i$ .

## i Proof

*Proof. Symmetry.*

$$\begin{aligned} (\mathbf{A}^\top \mathbf{A})^\top &= \mathbf{A}^\top (\mathbf{A}^\top)^\top && \text{(transpose of a product)} \\ &= \mathbf{A}^\top \mathbf{A} && \text{(transposing twice changes nothing)} \end{aligned}$$

The first step is Theorem 0.10, and the second follows from Definition 0.16, which swaps rows and columns, so swapping them again restores  $\mathbf{A}$ .

**The factorization.**

$$\begin{aligned} \mathbf{A}^\top \mathbf{A} &= (\mathbf{U} \mathbf{D} \mathbf{V}^\top)^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top && \text{(substitute the SVD)} \\ &= (\mathbf{D} \mathbf{V}^\top)^\top \mathbf{U}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top && \text{(transpose of the product of } \mathbf{U} \text{ and } \mathbf{D} \mathbf{V}^\top) \\ &= (\mathbf{V}^\top)^\top \mathbf{D}^\top \mathbf{U}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top && \text{(transpose of the product of } \mathbf{D} \text{ and } \mathbf{V}^\top) \\ &= \mathbf{V} \mathbf{D}^\top \mathbf{U}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top && \text{(transposing twice changes nothing)} \\ &= \mathbf{V} \mathbf{D}^\top (\mathbf{U}^\top \mathbf{U}) \mathbf{D} \mathbf{V}^\top && \text{(regroup; matrix multiplication is associative)} \\ &= \mathbf{V} \mathbf{D}^\top \mathbf{I}_n \mathbf{D} \mathbf{V}^\top && \text{(} \mathbf{U} \text{ is orthogonal)} \\ &= \mathbf{V} (\mathbf{D}^\top \mathbf{D}) \mathbf{V}^\top && \text{(identity matrix)} \\ &= \mathbf{V} \mathbf{V}^\top && \text{(definition of )} \end{aligned}$$

**The entries of  $\lambda$ .** Write  $\lambda_{ij}$  for entry  $(i, j)$  of  $\lambda$ , so  $\lambda_i = \lambda_{ii}$ . Only the entries  $d_{kk}$  with  $k \leq r$  of  $\mathbf{D}$  can be nonzero.

$$\begin{aligned} \lambda_{ij} &= \sum_{k=1}^n (\mathbf{D}^\top)_{ik} d_{kj} && \text{(definition of matrix multiplication)} \\ &= \sum_{k=1}^n d_{ki} d_{kj} && \text{(definition of the transpose)} \end{aligned}$$

The term for  $k$  is nonzero only if  $k = i \leq r$  and  $k = j$ . So  $\lambda_{ij} = 0$  when  $i \neq j$ . On the diagonal, only the term  $k = i$  can be nonzero, and for  $i \leq r$

$$\begin{aligned} \lambda_i &= \lambda_{ii} && \text{(notation)} \\ &= d_{ii} d_{ii} && \text{(the only term that can be nonzero)} \\ &= \sigma_i \cdot \sigma_i && \text{(definition of } \mathbf{D}) \\ &= \sigma_i^2 && \text{(notation for a square)} \end{aligned}$$

For  $i > r$ , every term is 0, so  $\lambda_i = 0$ .

**The eigenvectors.** Write  $\tilde{v}_i$  for column  $i$  of  $\mathbf{V}$ , and  $\tilde{e}_i$  for the indicator vector (Definition 0.9) of length  $p$ . Column  $i$  of  $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_p$  says  $\mathbf{V}^\top \tilde{v}_i = \tilde{e}_i$ . Then

$$\begin{aligned}
\mathbf{A}^\top \mathbf{A} \tilde{v}_i &= \mathbf{V} \mathbf{V}^\top \tilde{v}_i && \text{(the factorization)} \\
&= \mathbf{V} \tilde{e}_i && (\mathbf{V}^\top \tilde{v}_i = \tilde{e}_i) \\
&= \mathbf{V} (\lambda_i \tilde{e}_i) && \text{(column } i \text{ of the diagonal matrix )} \\
&= \lambda_i \mathbf{V} \tilde{e}_i && \text{(move the scalar } \lambda_i \text{ to the front)} \\
&= \lambda_i \tilde{v}_i && (\mathbf{V} \tilde{e}_i \text{ is column } i \text{ of } \mathbf{V})
\end{aligned}$$

and  $\tilde{v}_i \neq \tilde{0}$  because it has norm 1. □

**Example 0.91** ( $\mathbf{A}^\top \mathbf{A}$  for the matrix of the SVD example). For  $\mathbf{A}$  in Example 0.90:

$$\begin{aligned}
\mathbf{A}^\top \mathbf{A} &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(definition of the transpose)} \\
&= \begin{bmatrix} 1+1+1 & 1-1+1 \\ 1-1+1 & 1+1+1 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} && \text{(add)}
\end{aligned}$$

Theorem 0.61 says its eigenvalues are  $\sigma_1^2 = 4$  and  $\sigma_2^2 = 2$ , with the columns of  $\mathbf{V}$  as eigenvectors. Checking the first column, up to its factor  $\frac{1}{\sqrt{2}}$ :

$$\begin{aligned}
\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} 3 \cdot 1 + 1 \cdot 1 \\ 1 \cdot 1 + 3 \cdot 1 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
&= \begin{bmatrix} 3+1 \\ 1+3 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 4 \\ 4 \end{bmatrix} && \text{(add)} \\
&= 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} && \text{(factor out 4)}
\end{aligned}$$

and the second:

$$\begin{aligned}
\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= \begin{bmatrix} 3 \cdot 1 + 1 \cdot (-1) \\ 1 \cdot 1 + 3 \cdot (-1) \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
&= \begin{bmatrix} 3-1 \\ 1-3 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 2 \\ -2 \end{bmatrix} && \text{(add)} \\
&= 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} && \text{(factor out 2)}
\end{aligned}$$

*Remark 0.32* (Building an SVD from an eigendecomposition). Theorem 0.61 matches Banerjee and Roy (2014, chap. 12, pp. 372-373), which works in the other direction: it constructs an SVD from a spectral decomposition of  $\mathbf{A}^\top \mathbf{A}$  and sets  $\sigma_i = \sqrt{\lambda_i}$ . For example, in Example 0.91 the eigenvalues of  $\mathbf{A}^\top \mathbf{A}$  are 4 and 2, and  $\sqrt{4} = 2$  and  $\sqrt{2}$  are the singular values  $\sigma_1$  and  $\sigma_2$  of  $\mathbf{A}$  from Example 0.90.

## Definite Matrices

**Definition 0.63** (Positive semidefinite matrix). A  $p \times p$  matrix  $\mathbf{A}$  is **positive semidefinite** if it satisfies both conditions:

- $\mathbf{A}$  is symmetric (Definition 0.47).
- Every quadratic form (Definition 0.55) in  $\mathbf{A}$  is non-negative:  $\tilde{x}^\top \mathbf{A} \tilde{x} \geq 0$  for every vector  $\tilde{x}$  of length  $p$ .

**Example 0.92** (A positive semidefinite matrix). Let  $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ , which is symmetric. For any  $\tilde{x} = (x_1, x_2)$ :

$$\begin{aligned} \tilde{x}^\top \mathbf{B} \tilde{x} &= x_1^2 + x_1 x_2 + x_2 x_1 + x_2^2 && \text{(multiply out the quadratic form)} \\ &= (x_1 + x_2)^2 && \text{(complete the square)} \\ &\geq 0 && \text{(a square is non-negative)} \end{aligned}$$

So  $\mathbf{B}$  is positive semidefinite.

**Definition 0.64** (Positive definite matrix). A  $p \times p$  matrix  $\mathbf{A}$  is **positive definite** if it satisfies both conditions:

- $\mathbf{A}$  is symmetric (Definition 0.47).
- Every quadratic form (Definition 0.55) in  $\mathbf{A}$  at a nonzero vector is positive:  $\tilde{x}^\top \mathbf{A} \tilde{x} > 0$  for every vector  $\tilde{x} \neq \tilde{0}$  of length  $p$ .

**Example 0.93** (Positive definite, semidefinite, and neither).

- The identity matrix  $\mathbf{I}_p$  (Definition 0.46) is positive definite:  $\tilde{x}^\top \mathbf{I}_p \tilde{x} = \sum_{i=1}^p x_i^2$ , which is positive unless every  $x_i$  is 0.
- $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  from Example 0.92 is positive semidefinite but not positive definite: at  $\tilde{x} = (1, -1) \neq \tilde{0}$ ,  $\tilde{x}^\top \mathbf{B} \tilde{x} = (1 - 1)^2 = 0$ .

- $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$  is symmetric but not positive semidefinite: at  $\tilde{x} = (1, -1)$ ,  $\tilde{x}^\top \mathbf{D} \tilde{x} = 1 - 2 - 2 + 1 = -2 < 0$ .

*Remark 0.33* (Why the definition requires symmetry). A positive definite matrix is positive semidefinite (Definition 0.63):  $\tilde{x}^\top \mathbf{A} \tilde{x} > 0$  for every  $\tilde{x} \neq \tilde{0}$ , and  $\tilde{0}^\top \mathbf{A} \tilde{0} = 0$ . For example,  $\mathbf{I}_p$  in Example 0.93 is both.

Some sources drop the symmetry condition from both definitions. These notes keep it, because without it a matrix can pass the quadratic-form condition and still have no real eigenvalues (Definition 0.60). For example,  $\mathbf{C} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$  is not symmetric, and for any  $\tilde{x} = (x_1, x_2)$ :

$$\begin{aligned} \tilde{x}^\top \mathbf{C} \tilde{x} &= x_1(x_1 + x_2) + x_2(-x_1 + x_2) && \text{(multiply out the quadratic form)} \\ &= x_1^2 + x_1x_2 - x_2x_1 + x_2^2 && \text{(distribute)} \\ &= x_1^2 + x_2^2 && \text{(the middle terms cancel)} \end{aligned}$$

which is positive for every  $\tilde{x} \neq \tilde{0}$ . But suppose  $\mathbf{C}\tilde{v} = \lambda\tilde{v}$  for a real number  $\lambda$ . Matching entries gives  $v_1 + v_2 = \lambda v_1$  and  $-v_1 + v_2 = \lambda v_2$ , so  $v_2 = (\lambda - 1)v_1$  and  $-v_1 = (\lambda - 1)v_2$ . Substituting the first into the second gives  $-v_1 = (\lambda - 1)^2 v_1$ , so  $(1 + (\lambda - 1)^2)v_1 = 0$ . Since  $1 + (\lambda - 1)^2 > 0$ ,  $v_1 = 0$ , and then  $v_2 = (\lambda - 1)v_1 = 0$ . So no real  $\lambda$  has an eigenvector  $\tilde{v} \neq \tilde{0}$ .

See also [https://en.wikipedia.org/wiki/Definite\\_matrix](https://en.wikipedia.org/wiki/Definite_matrix).

**Theorem 0.62** (Definiteness and eigenvalues). *Let  $\mathbf{A}$  be a  $p \times p$  symmetric matrix with real entries, with eigendecomposition  $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$  (Definition 0.61) and eigenvalues  $\lambda_1, \dots, \lambda_p$  on the diagonal of  $\mathbf{Q}$ . Then:*

- $\mathbf{A}$  is positive semidefinite (Definition 0.63) if and only if every  $\lambda_i \geq 0$ .
- $\mathbf{A}$  is positive definite (Definition 0.64) if and only if every  $\lambda_i > 0$ .

#### **i** Proof

*Proof.* For any vector  $\tilde{x}$  of length  $p$ , let  $\tilde{y} = \mathbf{Q}^\top \tilde{x}$ . Then:

$$\begin{aligned}
\tilde{x}^\top \mathbf{A} \tilde{x} &= \tilde{x}^\top \mathbf{Q} \mathbf{Q}^\top \tilde{x} && \text{(substitute the eigendecomposition)} \\
&= (\mathbf{Q}^\top \tilde{x})^\top (\mathbf{Q}^\top \tilde{x}) && \text{(transpose of a product)} \\
&= \tilde{y}^\top \tilde{y} && \text{(definition of } \tilde{y}) \\
&= \sum_{i=1}^p \lambda_i y_i^2 && \text{(is diagonal)}
\end{aligned}$$

The second step is Theorem 0.10. Also,  $\tilde{x} = \tilde{0}$  exactly when  $\tilde{y} = \tilde{0}$ :  $\mathbf{Q} \mathbf{Q}^\top = \mathbf{I}_p$  for an orthogonal matrix (Definition 0.54), so  $\tilde{x} = \mathbf{Q} \tilde{y}$ .

If every  $\lambda_i \geq 0$ , then every term  $\lambda_i y_i^2 \geq 0$ , so  $\tilde{x}^\top \mathbf{A} \tilde{x} \geq 0$ . If every  $\lambda_i > 0$  and  $\tilde{x} \neq \tilde{0}$ , then some  $y_i \neq 0$ , so at least one term is positive and the rest are non-negative, and  $\tilde{x}^\top \mathbf{A} \tilde{x} > 0$ . Conversely, take  $\tilde{x} = \tilde{q}_i$ , column  $i$  of  $\mathbf{Q}$ , which is not  $\tilde{0}$  because it has norm 1. Then  $\tilde{y} = \mathbf{Q}^\top \tilde{q}_i$  is column  $i$  of  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ , so  $y_i = 1$  and every other entry of  $\tilde{y}$  is 0, and the display gives  $\tilde{q}_i^\top \mathbf{A} \tilde{q}_i = \lambda_i$ . So if  $\mathbf{A}$  is positive semidefinite,  $\lambda_i \geq 0$ , and if  $\mathbf{A}$  is positive definite,  $\lambda_i > 0$ .  $\square$

**Example 0.94** (Reading definiteness off the eigenvalues).

- $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  has eigenvalues 3 and 1 (Example 0.87), both positive, so it is positive definite.
- $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  has eigenvalues 2, for the eigenvector  $(1, 1)$ , and 0, for the eigenvector  $(1, -1)$ , so it is positive semidefinite but not positive definite, as Example 0.93 found directly.

**Theorem 0.63** (A positive definite matrix has a positive definite inverse). *Let  $\mathbf{A}$  be a  $p \times p$  positive definite matrix (Definition 0.64) with real entries, with eigendecomposition  $\mathbf{A} = \mathbf{Q} \mathbf{Q}^\top$  (Definition 0.61) and eigenvalues  $\lambda_1, \dots, \lambda_p$ . Then  $\mathbf{A}$  is invertible (Definition 0.50),*

$$\mathbf{A}^{-1} = \mathbf{Q}^{-1} \mathbf{Q}^\top, \quad {}^{-1} = \begin{bmatrix} 1/\lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1/\lambda_p \end{bmatrix},$$

and  $\mathbf{A}^{-1}$  is positive definite.

### Proof

*Proof.* By Theorem 0.62, every  $\lambda_i > 0$ , so  ${}^{-1}$  is well defined, and multiplying the two diagonal matrices entry by entry gives  ${}^{-1} = {}^{-1} = \mathbf{I}_p$ . Write  $\mathbf{B} = \mathbf{Q}^{-1} \mathbf{Q}^\top$ . Then:

$$\begin{aligned}
\mathbf{AB} &= \mathbf{QQ}^\top \mathbf{Q}^{-1} \mathbf{Q}^\top && \text{(substitute)} \\
&= \mathbf{Q} \mathbf{I}_p^{-1} \mathbf{Q}^\top && (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p) \\
&= \mathbf{Q}^{-1} \mathbf{Q}^\top && \text{(identity matrix)} \\
&= \mathbf{QQ}^\top && (-1 = \mathbf{I}_p) \\
&= \mathbf{I}_p && (\mathbf{Q} \text{ is orthogonal})
\end{aligned}$$

The same steps, with  $^{-1}$  and  $^\top$  swapped, give  $\mathbf{BA} = \mathbf{I}_p$ , so  $\mathbf{B} = \mathbf{A}^{-1}$  (Definition 0.49). The steps with  $\mathbf{Q}$  use Definition 0.54, whose remark records that  $\mathbf{QQ}^\top = \mathbf{I}_p$  too, and the identity step uses Theorem 0.50.

$\mathbf{A}^{-1}$  is symmetric by Corollary 0.4. For  $\tilde{x} \neq \tilde{0}$ , let  $\tilde{y} = \mathbf{Q}^\top \tilde{x}$ , which is not  $\tilde{0}$  because  $\tilde{x} = \mathbf{Q}\tilde{y}$ . As in the proof of Theorem 0.62:

$$\begin{aligned}
\tilde{x}^\top \mathbf{A}^{-1} \tilde{x} &= \tilde{y}^\top {}^{-1} \tilde{y} && \text{(substitute } \mathbf{A}^{-1} = \mathbf{Q}^{-1} \mathbf{Q}^\top \text{)} \\
&= \sum_{i=1}^p \frac{y_i^2}{\lambda_i} && (-1 \text{ is diagonal)} \\
&> 0 && \text{(each } \lambda_i > 0, \text{ and some } y_i \neq 0)
\end{aligned}$$

So  $\mathbf{A}^{-1}$  is positive definite. □

**Example 0.95** (Inverting a positive definite matrix). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ , Example 0.89

gives  $\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$  and eigenvalues 3 and 1, so:

$$\begin{aligned}
\mathbf{A}^{-1} &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1/3 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute; } \mathbf{Q} \text{ is symmetric)} \\
&= \frac{1}{2} \begin{bmatrix} 1/3 & 1 \\ 1/3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply the first two matrices)} \\
&= \frac{1}{2} \begin{bmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} && \text{(simplify)}
\end{aligned}$$

Multiplying out,  $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \mathbf{I}_2$ .

## Determinants

**Definition 0.65** (Determinant). The **determinant** of a  $p \times p$  matrix  $\mathbf{A}$  with entries  $a_{ij}$ , written  $\det(\mathbf{A})$  or  $|\mathbf{A}|$ , is the number defined recursively in  $p$ :

- For  $p = 1$ ,  $\det(\mathbf{A}) = a_{11}$ .
- For  $p \geq 2$ ,

$$\det(\mathbf{A}) = \sum_{j=1}^p (-1)^{1+j} a_{1j} \det(\mathbf{A}_{(1j)}),$$

where  $\mathbf{A}_{(1j)}$  is the  $(p-1) \times (p-1)$  matrix left after deleting row 1 and column  $j$  of  $\mathbf{A}$ .

**Example 0.96** (Determinant of a  $2 \times 2$  matrix). For  $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ , deleting row 1 and column 1 leaves  $[d]$ , and deleting row 1 and column 2 leaves  $[c]$ . So:

$$\begin{aligned} \det(\mathbf{A}) &= (-1)^{1+1} a \det([d]) + (-1)^{1+2} b \det([c]) && \text{(definition, } p = 2) \\ &= ad - bc && \text{(definition, } p = 1) \end{aligned}$$

For example,  $\det\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\right) = 2 \cdot 2 - 1 \cdot 1 = 3$ .

*Remark 0.34* (Cofactor expansion). The recursive formula in Definition 0.65 is the *cofactor expansion* along the first row. Other sources define the determinant as a sum over all orderings of the columns and derive this expansion from it; the two definitions agree (Banerjee and Roy 2014).

Each ordering picks one entry from each row, from the column that the ordering assigns to that row, and multiplies them. Its sign is  $+1$  if the ordering takes an even number of swaps of two columns to reach from  $(1, 2, \dots, p)$ , and  $-1$  if it takes an odd number.

For  $p = 2$ , the columns have two orderings. The ordering  $(1, 2)$  keeps each column in place, picks the entries  $a_{11}$  and  $a_{22}$ , and has sign  $+1$ . The ordering  $(2, 1)$  swaps the two columns, picks the entries  $a_{12}$  and  $a_{21}$ , and has sign  $-1$ . The sum  $a_{11}a_{22} - a_{12}a_{21}$  is the value  $ad - bc$  from Example 0.96.

See also <https://en.wikipedia.org/wiki/Determinant>.

**Theorem 0.64** (Determinant of a diagonal matrix). *The determinant (Definition 0.65) of a  $p \times p$  diagonal matrix (Definition 0.48) is the product of its diagonal entries:*

$$\det \left( \begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_p \end{bmatrix} \right) = \prod_{i=1}^p d_i$$

### **i** Proof

*Proof.* By induction on  $p$ . For  $p = 1$ , the matrix is  $[d_1]$ , and its determinant is  $d_1$  by definition.

For  $p \geq 2$ , suppose the result holds for  $(p-1) \times (p-1)$  diagonal matrices, and let  $\mathbf{D}$  be  $p \times p$  and diagonal. Row 1 of  $\mathbf{D}$  is  $(d_1, 0, \dots, 0)$ , so only the  $j = 1$  term of the definition can be nonzero, and deleting row 1 and column 1 of  $\mathbf{D}$  leaves the diagonal matrix with diagonal entries  $d_2, \dots, d_p$ :

$$\begin{aligned} \det(\mathbf{D}) &= (-1)^{1+1} d_1 \det(\mathbf{D}_{(11)}) + \sum_{j=2}^p (-1)^{1+j} \cdot 0 \cdot \det(\mathbf{D}_{(1j)}) && \text{(definition)} \\ &= d_1 \det(\mathbf{D}_{(11)}) && \text{(drop the zero terms)} \\ &= d_1 \prod_{i=2}^p d_i && \text{(induction hypothesis)} \\ &= \prod_{i=1}^p d_i && \text{(combine)} \end{aligned}$$

□

**Example 0.97** (Determinant of a scaled identity matrix). For a number  $c$ ,  $c\mathbf{I}_p$  is diagonal with every diagonal entry equal to  $c$ , so  $\det(c\mathbf{I}_p) = c^p$ . In particular,  $\det(\mathbf{I}_p) = 1$ .

**Theorem 0.65** (Determinant of a product). For  $p \times p$  matrices  $\mathbf{A}$  and  $\mathbf{B}$ ,

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}).$$

**Example 0.98** (Checking the product rule). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ , Example 0.96 gives  $\det(\mathbf{A}) = 2 \cdot 1 - 1 \cdot 0 = 2$  and  $\det(\mathbf{B}) = 1 \cdot 1 - 0 \cdot 3 = 1$ . The product is

$$\mathbf{AB} = \begin{bmatrix} 2 \cdot 1 + 1 \cdot 3 & 2 \cdot 0 + 1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 3 & 0 \cdot 0 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 3 & 1 \end{bmatrix},$$

with  $\det(\mathbf{AB}) = 5 \cdot 1 - 1 \cdot 3 = 2 = \det(\mathbf{A}) \det(\mathbf{B})$ .

*Remark 0.35* (An example is not a proof). Example 0.98 checks Theorem 0.65 for one pair of  $2 \times 2$  matrices, which shows the theorem holds there but does not prove it for every pair. The proof needs properties of the determinant that these notes don't develop; see Banerjee and Roy (2014).

See also <https://en.wikipedia.org/wiki/Determinant>.

**Theorem 0.66** (The determinant of a symmetric matrix is the product of its eigenvalues). *Let  $\mathbf{A}$  be a  $p \times p$  symmetric matrix with real entries, with eigendecomposition  $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$  (Definition 0.61) and eigenvalues  $\lambda_1, \dots, \lambda_p$  on the diagonal of  $\mathbf{\Lambda}$ . Then*

$$\det(\mathbf{A}) = \prod_{i=1}^p \lambda_i.$$

### Proof

*Proof.* First,  $\det(\mathbf{Q}^\top) \det(\mathbf{Q}) = 1$ :

$$\begin{aligned} \det(\mathbf{Q}^\top) \det(\mathbf{Q}) &= \det(\mathbf{Q}^\top \mathbf{Q}) && \text{(determinant of a product)} \\ &= \det(\mathbf{I}_p) && \text{(\mathbf{Q} is orthogonal)} \\ &= 1 && \text{(determinant of the identity)} \end{aligned}$$

Then:

$$\begin{aligned} \det(\mathbf{A}) &= \det(\mathbf{Q}\mathbf{Q}^\top) && \text{(substitute the eigendecomposition)} \\ &= \det(\mathbf{Q}) \det(\mathbf{\Lambda}) \det(\mathbf{Q}^\top) && \text{(determinant of a product, twice)} \\ &= \det(\mathbf{\Lambda}) \cdot \det(\mathbf{Q}^\top) \det(\mathbf{Q}) && \text{(reorder the three numbers)} \\ &= \det(\mathbf{\Lambda}) && \text{(the first display)} \\ &= \prod_{i=1}^p \lambda_i && \text{(determinant of a diagonal matrix)} \end{aligned}$$

The product steps are Theorem 0.65, the orthogonality step is Definition 0.54, and the identity and diagonal steps are Theorem 0.64 and Example 0.97.  $\square$

**Corollary 0.5** (A positive definite matrix has a positive determinant). *If  $\mathbf{A}$  is positive definite (Definition 0.64), then  $\det(\mathbf{A}) > 0$ .*

**i** Proof

*Proof.* By Theorem 0.62, every eigenvalue of  $\mathbf{A}$  is positive, so their product, which is  $\det(\mathbf{A})$  by Theorem 0.66, is positive.  $\square$

**Example 0.99** (Two ways to the same determinant).  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  has eigenvalues 3 and 1 (Example 0.87), so Theorem 0.66 gives  $\det(\mathbf{A}) = 3 \cdot 1 = 3$ , matching  $2 \cdot 2 - 1 \cdot 1 = 3$  from Example 0.96.

## Design Matrix

**Definition 0.66** (Design matrix). In a regression model with  $n$  observations and  $p$  predictors, the **design matrix** (or *model matrix*)  $\mathbf{X}$  is the  $n \times p$  matrix whose  $i$ -th row is the covariate vector  $\tilde{x}_i^\top$  for observation  $i$ :

$$\mathbf{X} = \begin{bmatrix} \tilde{x}_1^\top \\ \tilde{x}_2^\top \\ \vdots \\ \tilde{x}_n^\top \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

*Remark 0.36* (Products with the design matrix). The product  $\mathbf{X}\tilde{\beta}$  collects the values  $\tilde{x}_i^\top \tilde{\beta}$  for all  $n$  observations into a single  $n \times 1$  vector:

$$\mathbf{X}\tilde{\beta} = \begin{bmatrix} \tilde{x}_1^\top \tilde{\beta} \\ \vdots \\ \tilde{x}_n^\top \tilde{\beta} \end{bmatrix}$$

For example, take the rank-2 matrix from Example 0.9 as  $\mathbf{X}$ , with  $n = 3$  observations and  $p = 2$  columns: an intercept column and one covariate. With  $\tilde{\beta} = (1, 2)$ ,

$$\mathbf{X}\tilde{\beta} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}.$$

The matrix  $\mathbf{X}^\top \mathbf{X}$  is  $p \times p$  and symmetric, since  $(\mathbf{X}^\top \mathbf{X})^\top = \mathbf{X}^\top (\mathbf{X}^\top)^\top = \mathbf{X}^\top \mathbf{X}$  (Theorem 0.10). When  $\mathbf{X}^\top \mathbf{X}$  is invertible, it appears in the OLS estimator  $\hat{\tilde{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \tilde{y}$ .

**Theorem 0.67** ( $\mathbf{X}^\top \mathbf{X}$  is invertible when  $\mathbf{X}$  has full column rank). If  $\mathbf{X}$  is an  $n \times p$  matrix with  $\text{rank}(\mathbf{X}) = p$  (Definition 0.26), then the  $p \times p$  matrix  $\mathbf{X}^\top \mathbf{X}$  is invertible.

### i Proof

*Proof.* Let  $\tilde{c}$  be a vector of length  $p$  with  $\mathbf{X}^\top \mathbf{X} \tilde{c} = \tilde{0}$ . Then

$$\begin{aligned} 0 &= \tilde{c}^\top \mathbf{X}^\top \mathbf{X} \tilde{c} && \text{(multiply } \mathbf{X}^\top \mathbf{X} \tilde{c} = \tilde{0} \text{ on the left by } \tilde{c}^\top) \\ &= (\mathbf{X} \tilde{c})^\top (\mathbf{X} \tilde{c}) && \text{(transpose of a product)} \\ &= \|\mathbf{X} \tilde{c}\|^2 && \text{(definition of the Euclidean norm)} \end{aligned}$$

so  $\mathbf{X} \tilde{c} = \tilde{0}$ . Because  $\mathbf{X} \tilde{c} = c_1(\text{column 1}) + \dots + c_p(\text{column } p)$  and the columns of  $\mathbf{X}$  are linearly independent,  $\tilde{c} = \tilde{0}$ . So the only solution of  $\mathbf{X}^\top \mathbf{X} \tilde{c} = \tilde{0}$  is  $\tilde{c} = \tilde{0}$ , which says that the columns of the square matrix  $\mathbf{X}^\top \mathbf{X}$  are linearly independent, and a square matrix with linearly independent columns is invertible (Banerjee and Roy 2014, Corollary 5.7, p. 143).  $\square$

**Example 0.100** (Inverting  $\mathbf{X}^\top \mathbf{X}$ ). For the rank-2 matrix  $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$  from Example 0.9:

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad (\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix},$$

and multiplying the two out gives  $\mathbf{I}_2$ .

**Definition 0.67** (Hat matrix). For an  $n \times p$  design matrix  $\mathbf{X}$  (Definition 0.66) with  $\text{rank}(\mathbf{X}) = p$ , the **hat matrix** is the  $n \times n$  matrix

$$\underset{n \times n}{\mathbf{H}} \stackrel{\text{def}}{=} \underset{n \times p}{\mathbf{X}} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{p \times p} \underset{p \times n}{\mathbf{X}^\top}$$

The inverse exists by Theorem 0.67.

**Example 0.101** (The hat matrix of an intercept-only model). With  $n = 2$  observations and only an intercept,  $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  ( $2 \times 1$ , rank 1), so  $\mathbf{X}^\top \mathbf{X} = 2$  and

$$\mathbf{H} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

Then  $\mathbf{H} \tilde{y} = (\bar{y}, \bar{y})$ , where  $\bar{y} = (y_1 + y_2)/2$ : the fitted values of an intercept-only model are

the sample mean.

**Theorem 0.68** (Hat matrix is a projection matrix). *If  $\mathbf{X}$  is an  $n \times p$  design matrix with  $\text{rank}(\mathbf{X}) = p$ , then the hat matrix  $\mathbf{H}$  (Definition 0.67) is an orthogonal projection matrix (Definition 0.52).*

### **i** Proof

*Proof.* We verify symmetry and idempotency. Both use that  $\mathbf{X}^\top \mathbf{X}$  is symmetric,  $(\mathbf{X}^\top \mathbf{X})^\top = \mathbf{X}^\top (\mathbf{X}^\top)^\top = \mathbf{X}^\top \mathbf{X}$  (Theorem 0.10), so its inverse is symmetric too (Corollary 0.4).

**Symmetry:**

$$\begin{aligned}
 \mathbf{H}^\top &= (\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top)^\top && \text{(definition of } \mathbf{H} \text{)} \\
 &= (\mathbf{X}^\top)^\top \cdot ((\mathbf{X}^\top \mathbf{X})^{-1})^\top \cdot \mathbf{X}^\top && \text{(transpose of a product, twice)} \\
 &= \mathbf{X} \cdot ((\mathbf{X}^\top \mathbf{X})^{-1})^\top \cdot \mathbf{X}^\top && \text{(transposing twice changes nothing)} \\
 &= \mathbf{X} \cdot (\mathbf{X}^\top \mathbf{X})^{-1} \cdot \mathbf{X}^\top && \text{(the inverse of a symmetric matrix is symmetric)} \\
 &= \mathbf{H} && \text{(definition of } \mathbf{H} \text{)}
 \end{aligned}$$

**Idempotency:**

$$\begin{aligned}
 \mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \cdot \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of } \mathbf{H} \text{)} \\
 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(regroup; matrix multiplication is associative)} \\
 &= \mathbf{X} \mathbf{I}_p (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of the inverse)} \\
 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(identity matrix)} \\
 &= \mathbf{H} && \text{(definition of } \mathbf{H} \text{)}
 \end{aligned}$$

□

**Example 0.102** (The intercept-only hat matrix is a projection). For  $\mathbf{H} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$  from Example 0.101,  $\mathbf{H}^\top = \mathbf{H}$ , and

$$\mathbf{H}^2 = \begin{bmatrix} 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \\ 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} = \mathbf{H},$$

as Theorem 0.68 says.

*Remark 0.37* (Why it is called the hat matrix). The hat matrix gives the fitted values in linear regression:  $\hat{\tilde{y}} = \mathbf{X}\hat{\tilde{\beta}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \tilde{y} = \mathbf{H}\tilde{y}$ . Multiplying by  $\mathbf{H}$  “puts a hat” on  $\tilde{y}$ , which is where the name comes from. In Example 0.101,  $\mathbf{H}$  puts a hat on  $\tilde{y}$  by replacing each  $y_i$  with the sample mean  $\bar{y}$ .

## Orthogonal projection onto a subspace

### Source

This section is adapted from the second half of Zhou (2024d), used under the MIT License (see the license text in Section ). It connects the orthogonal projection matrices of Definition 0.52, defined by an algebraic property, to the geometric decomposition of Theorem 0.38.

**Definition 0.68** (Orthogonal projection onto a subspace). Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^p$  and  $\tilde{y} \in \mathbb{R}^p$ . By Theorem 0.38,  $\tilde{y} = \tilde{u} + \tilde{v}$  for exactly one  $\tilde{u} \in \mathcal{S}$  and  $\tilde{v} \in \mathcal{S}^\perp$ . The vector  $\tilde{u}$  is the **orthogonal projection** of  $\tilde{y}$  onto  $\mathcal{S}$ .

**Example 0.103** (Projecting onto a line in  $\mathbb{R}^3$ ). In Example 0.55,  $\tilde{y} = (3, 1, 2)$  splits as  $(2, 2, 0) + (1, -1, 2)$ , with  $(2, 2, 0) \in \mathcal{S} = \text{span}\{(1, 1, 0)\}$  and  $(1, -1, 2) \in \mathcal{S}^\perp$ . So the orthogonal projection of  $(3, 1, 2)$  onto  $\mathcal{S}$  is  $(2, 2, 0)$ . A vector already in  $\mathcal{S}$ , such as  $(5, 5, 0)$ , is its own projection, since  $(5, 5, 0) = (5, 5, 0) + \tilde{0}$  and  $\tilde{0} \in \mathcal{S}^\perp$  (Theorem 0.36, Theorem 0.11); a vector of  $\mathcal{S}^\perp$ , such as  $(1, -1, 2)$ , projects to  $\tilde{0}$ , since  $(1, -1, 2) = \tilde{0} + (1, -1, 2)$  and  $\tilde{0} \in \mathcal{S}$  (Theorem 0.11).

**Example 0.104** (A split that is not the orthogonal one).  $(3, 1, 2) = (3, 3, 0) + (0, -2, 2)$  also writes  $(3, 1, 2)$  as a vector of  $\mathcal{S} = \text{span}\{(1, 1, 0)\}$  plus a remainder, but  $(1, 1, 0) \cdot (0, -2, 2) = -2 \neq 0$ , so the remainder is not in  $\mathcal{S}^\perp$ , and  $(3, 3, 0)$  is not the orthogonal projection of  $(3, 1, 2)$  onto  $\mathcal{S}$ .

**Theorem 0.69** (The orthogonal projection is the closest point). Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^p$ , let  $\tilde{y} \in \mathbb{R}^p$ , and let  $\tilde{u}$  be the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{S}$  (Definition 0.68). Then for every  $\tilde{w} \in \mathcal{S}$ ,

$$\|\tilde{y} - \tilde{u}\| \leq \|\tilde{y} - \tilde{w}\|,$$

with equality only when  $\tilde{w} = \tilde{u}$ .

**i** Proof

*Proof.* By Definition 0.68,  $\tilde{y} - \tilde{u} \in \mathcal{S}^\perp$ . Both  $\tilde{u}$  and  $\tilde{w}$  are in  $\mathcal{S}$ , so  $\tilde{u} - \tilde{w} = \tilde{u} + (-1)\tilde{w} \in \mathcal{S}$  (Definition 0.28), and therefore  $(\tilde{y} - \tilde{u}) \perp (\tilde{u} - \tilde{w})$  (Definition 0.38). Then

$$\begin{aligned}\|\tilde{y} - \tilde{w}\|^2 &= \|(\tilde{y} - \tilde{u}) + (\tilde{u} - \tilde{w})\|^2 && \text{(add and subtract } \tilde{u}) \\ &= \|\tilde{y} - \tilde{u}\|^2 + \|\tilde{u} - \tilde{w}\|^2 && \text{(Pythagorean theorem, @thm-norm-sum-square)} \\ &\geq \|\tilde{y} - \tilde{u}\|^2, && (\|\tilde{u} - \tilde{w}\|^2 \geq 0, \text{ @thm-norm-properties})\end{aligned}$$

with equality exactly when  $\|\tilde{u} - \tilde{w}\| = 0$ , that is, when  $\tilde{w} = \tilde{u}$  (Theorem 0.41, part 1). Taking nonnegative square roots preserves the inequality.  $\square$

**Example 0.105** (No point of the line is closer). In Example 0.103,  $\tilde{y} = (3, 1, 2)$  has projection  $\tilde{u} = (2, 2, 0)$  onto  $\mathcal{S} = \text{span}\{(1, 1, 0)\}$ , at distance  $\|(1, -1, 2)\| = \sqrt{6}$ . Another point of  $\mathcal{S}$ , such as  $\tilde{w} = (3, 3, 0)$ , is farther:  $\|\tilde{y} - \tilde{w}\| = \|(0, -2, 2)\| = \sqrt{8} > \sqrt{6}$ , and the difference of squares is  $\|\tilde{u} - \tilde{w}\|^2 = \|(-1, -1, 0)\|^2 = 2 = 8 - 6$ , by the Pythagorean theorem (Theorem 0.42), since  $\tilde{y} - \tilde{u} \in \mathcal{S}^\perp$  and  $\tilde{u} - \tilde{w} \in \mathcal{S}$ .

**Theorem 0.70** (Projecting with an orthonormal basis). *Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^p$  with  $\dim(\mathcal{S}) = r \geq 1$ , and let  $\mathbf{Q}$  be the  $p \times r$  matrix whose columns  $\tilde{q}_1, \dots, \tilde{q}_r$  are an orthonormal basis of  $\mathcal{S}$  (Definition 0.40, Corollary 0.3). Then for every  $\tilde{y} \in \mathbb{R}^p$ ,  $\mathbf{Q}\mathbf{Q}^\top \tilde{y}$  is the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{S}$ , and  $\mathbf{Q}\mathbf{Q}^\top$  is an orthogonal projection matrix (Definition 0.52).*

**i** Proof

*Proof.*  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$ . Write  $q_{ki}$  for entry  $k$  of  $\tilde{q}_i$ , which is entry  $(k, i)$  of  $\mathbf{Q}$ . Then

$$\begin{aligned}(\mathbf{Q}^\top \mathbf{Q})_{ij} &= \sum_{k=1}^p (\mathbf{Q}^\top)_{ik} q_{kj} && \text{(@def-matrix-mult)} \\ &= \sum_{k=1}^p q_{ki} q_{kj} && \text{(@def-matrix-transpose)} \\ &= \tilde{q}_i \cdot \tilde{q}_j, && \text{(@def-dot-product)}\end{aligned}$$

which is 1 if  $i = j$  and 0 otherwise (Definition 0.13), so  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$  (Definition 0.46).

**The split.** Let  $\tilde{u} = \mathbf{Q}\mathbf{Q}^\top \tilde{y}$  and  $\tilde{v} = \tilde{y} - \tilde{u}$ , so  $\tilde{y} = \tilde{u} + \tilde{v}$ .  $\tilde{u} = \mathbf{Q}(\mathbf{Q}^\top \tilde{y})$  (Theorem 0.7) is in  $\mathcal{C}(\mathbf{Q})$  (Definition 0.32), which is  $\text{span}\{\tilde{q}_1, \dots, \tilde{q}_r\} = \mathcal{S}$  (Theorem 0.22). For  $\tilde{v}$ ,

$$\begin{aligned}
\mathbf{Q}^\top \tilde{v} &= \mathbf{Q}^\top (\tilde{y} - \mathbf{Q}\mathbf{Q}^\top \tilde{y}) && \text{(substitute)} \\
&= \mathbf{Q}^\top \tilde{y} - \mathbf{Q}^\top (\mathbf{Q}\mathbf{Q}^\top \tilde{y}) && \text{(@thm-matvec-linear, with coefficients 1 and } -1) \\
&= \mathbf{Q}^\top \tilde{y} - (\mathbf{Q}^\top \mathbf{Q}) \mathbf{Q}^\top \tilde{y} && \text{(@thm-matmul-assoc)} \\
&= \mathbf{Q}^\top \tilde{y} - \mathbf{I}_r \mathbf{Q}^\top \tilde{y} && (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r, \text{ from the first part of the proof)} \\
&= \mathbf{Q}^\top \tilde{y} - \mathbf{Q}^\top \tilde{y} && \text{(@thm-identity)} \\
&= \tilde{0}_r, && \text{(arithmetic)}
\end{aligned}$$

so  $\tilde{v} \in \mathcal{N}(\mathbf{Q}^\top) = \mathcal{C}(\mathbf{Q})^\perp = \mathcal{S}^\perp$  (Theorem 0.37). By the uniqueness in Theorem 0.38,  $\tilde{u}$  is the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{S}$ .

**$\mathbf{Q}\mathbf{Q}^\top$  is symmetric and idempotent.**

$$\begin{aligned}
(\mathbf{Q}\mathbf{Q}^\top)^\top &= (\mathbf{Q}^\top)^\top \mathbf{Q}^\top && \text{(@thm-transpose-product)} \\
&= \mathbf{Q}\mathbf{Q}^\top, && \text{(transposing twice, @def-matrix-transpose)}
\end{aligned}$$

and

$$\begin{aligned}
(\mathbf{Q}\mathbf{Q}^\top)(\mathbf{Q}\mathbf{Q}^\top) &= \mathbf{Q}(\mathbf{Q}^\top \mathbf{Q})\mathbf{Q}^\top && \text{(@thm-matmul-assoc)} \\
&= \mathbf{Q}\mathbf{I}_r \mathbf{Q}^\top && (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r, \text{ from the first part of the proof)} \\
&= \mathbf{Q}\mathbf{Q}^\top. && \text{(@thm-identity)}
\end{aligned}$$

□

**Example 0.106** (The projection matrix onto a line). For  $\mathcal{S} = \text{span}\{(1, 1, 0)\}$ , the single vector  $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$  is an orthonormal basis: it spans  $\mathcal{S}$ , it is nonzero and so linearly independent, and  $\|\tilde{q}_1\| = \sqrt{\frac{1}{2} + \frac{1}{2} + 0} = 1$ . With  $\mathbf{Q} = [\tilde{q}_1]$ ,

$$\mathbf{Q}\mathbf{Q}^\top = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Applied to  $\tilde{y} = (3, 1, 2)$ , it gives  $(\frac{3+1}{2}, \frac{3+1}{2}, 0) = (2, 2, 0)$ , the projection found in Example 0.103.

**Theorem 0.71** (A projection matrix projects onto its column space, and is the only one that does).

1. If  $\mathbf{P}$  is a  $p \times p$  orthogonal projection matrix (Definition 0.52), then for every  $\tilde{y} \in \mathbb{R}^p$ ,  $\mathbf{P}\tilde{y}$  is the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{C}(\mathbf{P})$  (Definition 0.68).
2. If  $\mathbf{P}_1$  and  $\mathbf{P}_2$  are  $p \times p$  matrices such that, for every  $\tilde{y}$ , both  $\mathbf{P}_1\tilde{y}$  and  $\mathbf{P}_2\tilde{y}$  are the orthogonal projection of  $\tilde{y}$  onto the same subspace  $\mathcal{S}$ , then  $\mathbf{P}_1 = \mathbf{P}_2$ .

**i** Proof

*Proof. Part 1.* Write  $\tilde{y} = \mathbf{P}\tilde{y} + (\tilde{y} - \mathbf{P}\tilde{y})$ . The first term is in  $\mathcal{C}(\mathbf{P})$  (Definition 0.32). For the second,

$$\begin{aligned} \mathbf{P}(\tilde{y} - \mathbf{P}\tilde{y}) &= \mathbf{P}\tilde{y} - \mathbf{P}(\mathbf{P}\tilde{y}) && \text{(@thm-matvec-linear, with coefficients 1 and } -1) \\ &= \mathbf{P}\tilde{y} - (\mathbf{P}\mathbf{P})\tilde{y} && \text{(@thm-matmul-assoc)} \\ &= \mathbf{P}\tilde{y} - \mathbf{P}^2\tilde{y} && \text{(@def-matrix-power)} \\ &= \mathbf{P}\tilde{y} - \mathbf{P}\tilde{y} && (\mathbf{P} \text{ is idempotent)} \\ &= \tilde{0}_p, && \text{(arithmetic)} \end{aligned}$$

so  $\tilde{y} - \mathbf{P}\tilde{y} \in \mathcal{N}(\mathbf{P})$ . Since  $\mathbf{P} = \mathbf{P}^\top$ ,  $\mathcal{N}(\mathbf{P}) = \mathcal{N}(\mathbf{P}^\top) = \mathcal{C}(\mathbf{P})^\perp$  (Theorem 0.37). By the uniqueness in Theorem 0.38,  $\mathbf{P}\tilde{y}$  is the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{C}(\mathbf{P})$ .

**Part 2.** The orthogonal projection of a vector onto  $\mathcal{S}$  is unique (Definition 0.68), so  $\mathbf{P}_1\tilde{e}_j = \mathbf{P}_2\tilde{e}_j$  for each indicator vector  $\tilde{e}_j$  (Definition 0.9).  $\mathbf{P}_1\tilde{e}_j$  is column  $j$  of  $\mathbf{P}_1$ : by Theorem 0.13 it is the combination of the columns of  $\mathbf{P}_1$  with coefficients the entries of  $\tilde{e}_j$ , which are 1 in place  $j$  and 0 elsewhere (Definition 0.9). Likewise for  $\mathbf{P}_2$ , so the two matrices have the same columns.  $\square$

**Example 0.107** (Two routes to the same projection matrix). The matrix  $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  of Example 0.78 is an orthogonal projection matrix, and its column space is  $\text{span}\{(1, 0)\}$ , the horizontal axis. By part 1,  $\mathbf{P}(v_1, v_2) = (v_1, 0)$  is the orthogonal projection onto that axis: the remainder  $(0, v_2)$  is orthogonal to  $(1, 0)$ . The vector  $\tilde{q}_1 = (1, 0)$  has length 1 and spans the axis, so it is an orthonormal basis of it, and with  $\mathbf{B} = [\tilde{q}_1]$ ,  $\mathbf{B}\mathbf{B}^\top = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \mathbf{P}$ , as part 2 and Theorem 0.70 require.

The oblique projection  $\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$  of Example 0.78 also maps every vector into the horizontal axis. It sends  $(0, 1)$  to  $(1, 0)$ , but the remainder  $(0, 1) - (1, 0) = (-1, 1)$  is not orthogonal to the axis, so  $(1, 0)$  is not the orthogonal projection of  $(0, 1)$  onto the axis, which is  $\mathbf{P}(0, 1) = (0, 0)$ .

**Theorem 0.72** (The hat matrix projects onto the column space of the design matrix). *If  $\mathbf{X}$  is an  $n \times p$  design matrix with  $\text{rank}(\mathbf{X}) = p$ , then for every  $\tilde{y} \in \mathbb{R}^n$ ,  $\mathbf{H}\tilde{y}$  (Definition 0.67) is the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{C}(\mathbf{X})$ . So  $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$  for any  $n \times p$  matrix  $\mathbf{Q}$  whose columns are an orthonormal basis of  $\mathcal{C}(\mathbf{X})$ .*

### i Proof

*Proof.*  $\mathbf{H}$  is an orthogonal projection matrix (Theorem 0.68), so  $\mathbf{H}\tilde{\mathbf{y}}$  is the orthogonal projection of  $\tilde{\mathbf{y}}$  onto  $\mathcal{C}(\mathbf{H})$  (Theorem 0.71, part 1). It remains to show  $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ .

$\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$ .  $\mathbf{H}\tilde{\mathbf{z}} = \mathbf{X}((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \tilde{\mathbf{z}})$  (Theorem 0.7), which is in  $\mathcal{C}(\mathbf{X})$  (Definition 0.32).  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{H})$ . First,

$$\begin{aligned} \mathbf{H}\mathbf{X} &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} && (\text{@def-hat-matrix}) \\ &= \mathbf{X}((\mathbf{X}^\top \mathbf{X})^{-1}(\mathbf{X}^\top \mathbf{X})) && (\text{@thm-matmul-assoc}) \\ &= \mathbf{X}\mathbf{I}_p && (\text{@def-matrix-inverse}) \\ &= \mathbf{X}. && (\text{@thm-identity}) \end{aligned}$$

So for every  $\tilde{\mathbf{b}} \in \mathbb{R}^p$ ,  $\mathbf{X}\tilde{\mathbf{b}} = (\mathbf{H}\mathbf{X})\tilde{\mathbf{b}}$ , since  $\mathbf{H}\mathbf{X} = \mathbf{X}$ , and  $(\mathbf{H}\mathbf{X})\tilde{\mathbf{b}} = \mathbf{H}(\mathbf{X}\tilde{\mathbf{b}})$  (Theorem 0.7), which is in  $\mathcal{C}(\mathbf{H})$ .

$\mathcal{C}(\mathbf{X})$  has dimension  $p \geq 1$  (Theorem 0.26), so an orthonormal basis of it has  $p$  vectors (Definition 0.31), and  $\mathbf{Q}\mathbf{Q}^\top$  also gives the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$  (Theorem 0.70, with  $n$  in place of  $p$  and  $r = p$ ). By Theorem 0.71, part 2,  $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$ .  $\square$

**Example 0.108** (The intercept-only hat matrix, from an orthonormal basis). For the intercept-only design  $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  of Example 0.101,  $\mathcal{C}(\mathbf{X}) = \text{span}\{(1, 1)\}$ , with orthonormal basis  $\tilde{\mathbf{q}}_1 = \frac{1}{\sqrt{2}}(1, 1)$ , since  $\|\tilde{\mathbf{q}}_1\| = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1$ . With  $\mathbf{Q} = [\tilde{\mathbf{q}}_1]$ ,

$$\mathbf{Q}\mathbf{Q}^\top = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix},$$

the hat matrix found in Example 0.101. The fitted values  $(\bar{y}, \bar{y})$  are the closest point to  $\tilde{\mathbf{y}}$  on the line of constant vectors (Theorem 0.69), and the residuals  $(y_1 - \bar{y}, y_2 - \bar{y})$  are orthogonal to  $(1, 1)$ :  $(1, 1) \cdot (y_1 - \bar{y}, y_2 - \bar{y}) = y_1 + y_2 - 2\bar{y} = 0$ , since  $2\bar{y} = y_1 + y_2$ .

## Generalized inverses and linear systems

### i Source

This section is adapted from Zhou (2024c), used under the MIT License (see the license text in Section ). The source defers the existence of the Moore-Penrose inverse to the singular value decomposition; the proof here builds it from a rank factorization (Theorem 0.30) instead, and the section adds the projection  $\mathbf{X}\mathbf{G}\mathbf{X}^\top$  for any generalized inverse  $\mathbf{G}$  of  $\mathbf{X}^\top \mathbf{X}$ , which the source does not cover.

**Theorem 0.73** (A scalar factor passes through a matrix product). *For an  $m \times k$  matrix  $\mathbf{A}$ , a  $k \times n$  matrix  $\mathbf{B}$  and a number  $c$ ,*

$$\mathbf{A}(c\mathbf{B}) = c(\mathbf{AB}) = (c\mathbf{A})\mathbf{B}.$$

*In particular,  $\mathbf{A}(\mathbf{B} - \mathbf{C}) = \mathbf{AB} - \mathbf{AC}$  and  $(\mathbf{B} - \mathbf{C})\mathbf{D} = \mathbf{BD} - \mathbf{CD}$  for matrices of compatible sizes, where  $\mathbf{B} - \mathbf{C} = \mathbf{B} + (-1)\mathbf{C}$  (Theorem 0.6, Definition 0.19).*

### Proof

*Proof.*  $\mathbf{A}(c\mathbf{B}) = c(\mathbf{AB})$ . For each entry  $(i, j)$ ,

$$\begin{aligned} (\mathbf{A}(c\mathbf{B}))_{ij} &= \sum_{l=1}^k a_{il} (c\mathbf{B})_{lj} && (\text{@def-matrix-mult}) \\ &= \sum_{l=1}^k a_{il} (c b_{lj}) && (\text{@def-scalar-mult}) \\ &= \sum_{l=1}^k c (a_{il} b_{lj}) && (\text{reorder the product of numbers}) \\ &= c \sum_{l=1}^k a_{il} b_{lj} && (\text{factor } c \text{ out of the sum}) \\ &= c(\mathbf{AB})_{ij} && (\text{@def-matrix-mult}) \\ &= (c(\mathbf{AB}))_{ij}. && (\text{@def-scalar-mult}) \end{aligned}$$

$(c\mathbf{A})\mathbf{B} = c(\mathbf{AB})$ . For each entry  $(i, j)$ ,

$$\begin{aligned} ((c\mathbf{A})\mathbf{B})_{ij} &= \sum_{l=1}^k (c a_{il}) b_{lj} && (\text{@def-matrix-mult, @def-scalar-mult}) \\ &= c \sum_{l=1}^k a_{il} b_{lj} && (\text{factor } c \text{ out of the sum}) \\ &= (c(\mathbf{AB}))_{ij}. && (\text{@def-matrix-mult, @def-scalar-mult}) \end{aligned}$$

### Differences.

$$\begin{aligned} \mathbf{A}(\mathbf{B} - \mathbf{C}) &= \mathbf{A}(\mathbf{B} + (-1)\mathbf{C}) && (\text{the difference as a sum}) \\ &= \mathbf{AB} + \mathbf{A}((-1)\mathbf{C}) && (\text{@thm-matmul-distrib}) \\ &= \mathbf{AB} + (-1)(\mathbf{AC}) && (\text{first part}) \\ &= \mathbf{AB} - \mathbf{AC}, && (\text{the difference as a sum}) \end{aligned}$$

and  $(\mathbf{B} - \mathbf{C})\mathbf{D} = \mathbf{BD} - \mathbf{CD}$  the same way, using the second law in Theorem 0.8 and the second part. A column vector is a matrix with one column, so all of this applies with  $\mathbf{B}$ ,  $\mathbf{C}$  or  $\mathbf{D}$  a vector.  $\square$

**Example 0.109** (Pulling a scalar out of a product). With  $\mathbf{A} = \begin{bmatrix} 1 & 2 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ :  
 $\mathbf{A} (2\mathbf{B}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 8 \end{bmatrix} = 22$ , and  $2(\mathbf{AB}) = 2(3 + 8) = 22$ .

**Definition 0.69** (Generalized inverse). A **generalized inverse** of an  $m \times n$  matrix  $\mathbf{A}$  is an  $n \times m$  matrix  $\mathbf{G}$  such that

$$\underbrace{\mathbf{A}}_{m \times n} \underbrace{\mathbf{G}}_{n \times m} \underbrace{\mathbf{A}}_{m \times n} = \mathbf{A}.$$

**Example 0.110** (A matrix with many generalized inverses). Let  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  and  $\mathbf{G} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ . Then

$$\begin{aligned} \mathbf{AGA} &= \begin{bmatrix} g_{11} & g_{12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad (\text{@def-matrix-mult, for } \mathbf{AG}) \\ &= \begin{bmatrix} g_{11} & 0 \\ 0 & 0 \end{bmatrix}, \quad (\text{@def-matrix-mult}) \end{aligned}$$

which equals  $\mathbf{A}$  exactly when  $g_{11} = 1$ . So every matrix  $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$  is a generalized inverse of  $\mathbf{A}$ : a generalized inverse need not be unique. The zero matrix is not one, since its  $(1, 1)$  entry is 0.

**Theorem 0.74** (An invertible matrix has only one generalized inverse). *If  $\mathbf{A}$  is an invertible  $p \times p$  matrix (Definition 0.50), its only generalized inverse is  $\mathbf{A}^{-1}$ .*

**i** Proof

*Proof.*  $\mathbf{AA}^{-1}\mathbf{A} = \mathbf{I}_p\mathbf{A} = \mathbf{A}$  (Definition 0.49, Theorem 0.50), so  $\mathbf{A}^{-1}$  is a generalized inverse. If  $\mathbf{G}$  is any generalized inverse, then

$$\begin{aligned}
\mathbf{G} &= \mathbf{I}_p \mathbf{G} \mathbf{I}_p && (\text{@thm-identity}) \\
&= (\mathbf{A}^{-1} \mathbf{A}) \mathbf{G} (\mathbf{A} \mathbf{A}^{-1}) && (\text{@def-matrix-inverse}) \\
&= \mathbf{A}^{-1} (\mathbf{A} \mathbf{G} \mathbf{A}) \mathbf{A}^{-1} && (\text{@thm-matmul-assoc}) \\
&= \mathbf{A}^{-1} \mathbf{A} \mathbf{A}^{-1} && (\text{@def-generalized-inverse}) \\
&= \mathbf{I}_p \mathbf{A}^{-1} && (\text{@def-matrix-inverse}) \\
&= \mathbf{A}^{-1}. && (\text{@thm-identity})
\end{aligned}$$

□

**Example 0.111** (The generalized inverse of an invertible matrix). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$  of Example 0.75, the only generalized inverse is  $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$ . The matrix of Example 0.110, by contrast, has infinitely many, so by Theorem 0.74 it is singular.

**Theorem 0.75** (A generalized inverse solves every solvable system). *Let  $\mathbf{A}$  be  $m \times n$ , let  $\mathbf{G}$  be a generalized inverse of  $\mathbf{A}$  (Definition 0.69), and let  $\tilde{\mathbf{b}} \in \mathbb{R}^m$ . If  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  has a solution, then  $\mathbf{G}\tilde{\mathbf{b}}$  is a solution.*

#### **i** Proof

*Proof.* Let  $\tilde{\mathbf{x}}_0$  be a solution, so  $\mathbf{A}\tilde{\mathbf{x}}_0 = \tilde{\mathbf{b}}$ . Then

$$\begin{aligned}
\mathbf{A}(\mathbf{G}\tilde{\mathbf{b}}) &= \mathbf{A}\mathbf{G}(\mathbf{A}\tilde{\mathbf{x}}_0) && (\text{substitute } \tilde{\mathbf{b}} = \mathbf{A}\tilde{\mathbf{x}}_0) \\
&= (\mathbf{A}\mathbf{G}\mathbf{A})\tilde{\mathbf{x}}_0 && (\text{@thm-matmul-assoc}) \\
&= \mathbf{A}\tilde{\mathbf{x}}_0 && (\text{@def-generalized-inverse}) \\
&= \tilde{\mathbf{b}}. && (\tilde{\mathbf{x}}_0 \text{ is a solution})
\end{aligned}$$

□

**Example 0.112** (Solving with a generalized inverse). With  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  and its generalized inverse  $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$  (Example 0.110), the system  $\mathbf{A}\tilde{\mathbf{x}} = (2, 0)$  has the solution  $(2, 0)$ , and  $\mathbf{G}(2, 0) = (2, 14)$  is a solution too:  $\mathbf{A}(2, 14) = (2, 0)$ . The system  $\mathbf{A}\tilde{\mathbf{x}} = (2, 1)$  has no solution, since the second entry of  $\mathbf{A}\tilde{\mathbf{x}}$  is always 0; there  $\mathbf{G}(2, 1) = (7, 14)$  gives  $\mathbf{A}(7, 14) = (7, 0) \neq (2, 1)$ .

**Definition 0.70** (Moore-Penrose inverse). A **Moore-Penrose inverse** of an  $m \times n$  matrix  $\mathbf{A}$  is an  $n \times m$  matrix  $\mathbf{G}$  satisfying all four conditions

1.  $\mathbf{AGA} = \mathbf{A}$ ;
2.  $\mathbf{GAG} = \mathbf{G}$ ;
3.  $(\mathbf{AG})^\top = \mathbf{AG}$ ;
4.  $(\mathbf{GA})^\top = \mathbf{GA}$ .

Condition 1 says  $\mathbf{G}$  is a generalized inverse (Definition 0.69).

**Example 0.113** (Checking the four conditions). For  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ , the matrix  $\mathbf{G} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  satisfies all four:  $\mathbf{AG} = \mathbf{GA} = \mathbf{A}$  and  $\mathbf{A}^2 = \mathbf{A}$ , so conditions 1 and 2 read  $\mathbf{A} = \mathbf{A}$ , and conditions 3 and 4 hold because  $\mathbf{A}$  is symmetric.

The generalized inverse  $\mathbf{G}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$  of  $\mathbf{A}$  (Example 0.110) is not a Moore-Penrose inverse:

$\mathbf{AG}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ , which is not symmetric, so condition 3 fails.

**Theorem 0.76** (A matrix has at most one Moore-Penrose inverse). If  $\mathbf{G}_1$  and  $\mathbf{G}_2$  are both Moore-Penrose inverses of an  $m \times n$  matrix  $\mathbf{A}$  (Definition 0.70), then  $\mathbf{G}_1 = \mathbf{G}_2$ .

### **i** Proof

*Proof.*  $\mathbf{AG}_1 = \mathbf{AG}_2$ .

$$\begin{aligned}
 \mathbf{AG}_1 &= (\mathbf{AG}_1)^\top && \text{(condition 3 for } \mathbf{G}_1) \\
 &= \mathbf{G}_1^\top \mathbf{A}^\top && \text{(@thm-transpose-product)} \\
 &= \mathbf{G}_1^\top (\mathbf{AG}_2 \mathbf{A})^\top && \text{(condition 1 for } \mathbf{G}_2) \\
 &= \mathbf{G}_1^\top \mathbf{A}^\top \mathbf{G}_2^\top \mathbf{A}^\top && \text{(@thm-transpose-product, twice)} \\
 &= (\mathbf{G}_1^\top \mathbf{A}^\top) (\mathbf{G}_2^\top \mathbf{A}^\top) && \text{(@thm-matmul-associ)} \\
 &= (\mathbf{AG}_1)^\top (\mathbf{AG}_2)^\top && \text{(@thm-transpose-product, twice)} \\
 &= (\mathbf{AG}_1)(\mathbf{AG}_2) && \text{(condition 3 for } \mathbf{G}_1 \text{ and } \mathbf{G}_2) \\
 &= (\mathbf{AG}_1 \mathbf{A}) \mathbf{G}_2 && \text{(@thm-matmul-associ)} \\
 &= \mathbf{AG}_2. && \text{(condition 1 for } \mathbf{G}_1)
 \end{aligned}$$

$$\mathbf{G}_1 \mathbf{A} = \mathbf{G}_2 \mathbf{A}.$$

$$\begin{aligned}
\mathbf{G}_1 \mathbf{A} &= (\mathbf{G}_1 \mathbf{A})^\top && \text{(condition 4 for } \mathbf{G}_1) \\
&= \mathbf{A}^\top \mathbf{G}_1^\top && \text{(@thm-transpose-product)} \\
&= (\mathbf{A} \mathbf{G}_2 \mathbf{A})^\top \mathbf{G}_1^\top && \text{(condition 1 for } \mathbf{G}_2) \\
&= \mathbf{A}^\top \mathbf{G}_2^\top \mathbf{A}^\top \mathbf{G}_1^\top && \text{(@thm-transpose-product, twice)} \\
&= (\mathbf{A}^\top \mathbf{G}_2^\top) (\mathbf{A}^\top \mathbf{G}_1^\top) && \text{(@thm-matmul-assoc)} \\
&= (\mathbf{G}_2 \mathbf{A})^\top (\mathbf{G}_1 \mathbf{A})^\top && \text{(@thm-transpose-product, twice)} \\
&= (\mathbf{G}_2 \mathbf{A})(\mathbf{G}_1 \mathbf{A}) && \text{(condition 4 for } \mathbf{G}_2 \text{ and } \mathbf{G}_1) \\
&= \mathbf{G}_2 (\mathbf{A} \mathbf{G}_1 \mathbf{A}) && \text{(@thm-matmul-assoc)} \\
&= \mathbf{G}_2 \mathbf{A}. && \text{(condition 1 for } \mathbf{G}_1)
\end{aligned}$$

**Conclusion.**

$$\begin{aligned}
\mathbf{G}_1 &= \mathbf{G}_1 \mathbf{A} \mathbf{G}_1 && \text{(condition 2 for } \mathbf{G}_1) \\
&= \mathbf{G}_1 (\mathbf{A} \mathbf{G}_1) && \text{(@thm-matmul-assoc)} \\
&= \mathbf{G}_1 (\mathbf{A} \mathbf{G}_2) && \text{(the first step)} \\
&= (\mathbf{G}_1 \mathbf{A}) \mathbf{G}_2 && \text{(@thm-matmul-assoc)} \\
&= (\mathbf{G}_2 \mathbf{A}) \mathbf{G}_2 && \text{(the second step)} \\
&= \mathbf{G}_2 \mathbf{A} \mathbf{G}_2 && \text{(@thm-matmul-assoc)} \\
&= \mathbf{G}_2. && \text{(condition 2 for } \mathbf{G}_2)
\end{aligned}$$

□

**Example 0.114** (Only one of the many generalized inverses qualifies). The matrix  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  has infinitely many generalized inverses  $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$  (Example 0.110), and  $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  is a Moore-Penrose inverse (Example 0.113). By Theorem 0.76 it is the only one, so every other choice of  $g_{12}, g_{21}, g_{22}$  breaks condition 2, 3 or 4. For instance,  $g_{21} = 1$  (the rest 0) gives  $\mathbf{G} \mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ , which is not symmetric.

**Theorem 0.77** (Every matrix has a Moore-Penrose inverse). *Every  $m \times n$  matrix  $\mathbf{A}$  has a Moore-Penrose inverse (Definition 0.70), written  $\mathbf{A}^+$ ; it is unique by Theorem 0.76.*

- If  $\mathbf{A} = \mathbf{0}_{m \times n}$ , then  $\mathbf{A}^+ = \mathbf{0}_{n \times m}$ .
- If  $\text{rank}(\mathbf{A}) = r \geq 1$  and  $\mathbf{A} = \mathbf{C} \mathbf{R}$  is a rank factorization (Definition 0.35), then

$$\underbrace{\mathbf{A}^+}_{n \times m} = \underbrace{\mathbf{R}^\top}_{n \times r} \underbrace{(\mathbf{R}\mathbf{R}^\top)^{-1}}_{r \times r} \underbrace{(\mathbf{C}^\top \mathbf{C})^{-1}}_{r \times r} \underbrace{\mathbf{C}^\top}_{r \times m}.$$

### i Proof

*Proof. The two cases cover every matrix.* If  $\mathbf{A} \neq \mathbf{0}_{m \times n}$ , it has a nonzero column, which is linearly independent on its own ( $c \tilde{v} = \tilde{0}$  with  $\tilde{v} \neq \tilde{0}$  forces  $c = 0$ ), so  $\text{rank}(\mathbf{A}) \geq 1$  (Definition 0.26).

**The zero matrix.** With  $\mathbf{A} = \mathbf{0}_{m \times n}$  and  $\mathbf{G} = \mathbf{0}_{n \times m}$ , every product in conditions 1 to 4 is a zero matrix, and a zero matrix equals its own transpose, so all four conditions hold.

**The two inverses exist.** Now suppose  $\text{rank}(\mathbf{A}) = r \geq 1$ . A rank factorization exists (Theorem 0.30).  $\mathbf{C}$  is  $m \times r$  and  $\mathbf{R}$  is  $r \times n$ . Both have rank  $r$ :

$$\begin{aligned} r &= \text{rank}(\mathbf{C}\mathbf{R}) \quad (\mathbf{A} = \mathbf{C}\mathbf{R}) \\ &\leq \text{rank}(\mathbf{C}) \quad (\text{@thm-rank-product}) \\ &\leq r, \quad (\text{@cor-rank-bound, } \mathbf{C} \text{ has } r \text{ columns}) \end{aligned}$$

and, using  $(\mathbf{C}\mathbf{R})^\top = \mathbf{R}^\top \mathbf{C}^\top$  (Theorem 0.10),

$$\begin{aligned} r &= \text{rank}(\mathbf{R}^\top \mathbf{C}^\top) \quad (\text{@thm-rank-transpose}) \\ &\leq \text{rank}(\mathbf{R}^\top) \quad (\text{@thm-rank-product}) \\ &\leq r. \quad (\text{@cor-rank-bound, } \mathbf{R}^\top \text{ has } r \text{ columns}) \end{aligned}$$

So  $\mathbf{C}$  and  $\mathbf{R}^\top$  have full column rank  $r$ , and  $\mathbf{C}^\top \mathbf{C}$  and  $\mathbf{R}\mathbf{R}^\top = (\mathbf{R}^\top)^\top \mathbf{R}^\top$  are invertible (Theorem 0.67). They are symmetric:  $(\mathbf{C}^\top \mathbf{C})^\top = \mathbf{C}^\top (\mathbf{C}^\top)^\top = \mathbf{C}^\top \mathbf{C}$  (Theorem 0.10, Definition 0.16), and likewise for  $\mathbf{R}\mathbf{R}^\top$ . Write  $\mathbf{S} \stackrel{\text{def}}{=} (\mathbf{R}\mathbf{R}^\top)^{-1}$  and  $\mathbf{T} \stackrel{\text{def}}{=} (\mathbf{C}^\top \mathbf{C})^{-1}$ , both symmetric (Corollary 0.4), and  $\mathbf{G} \stackrel{\text{def}}{=} \mathbf{R}^\top \mathbf{S} \mathbf{T} \mathbf{C}^\top$ .

**Two products.**

$$\begin{aligned} \mathbf{A}\mathbf{G} &= \mathbf{C}\mathbf{R}\mathbf{R}^\top \mathbf{S} \mathbf{T} \mathbf{C}^\top && (\text{substitute } \mathbf{A} \text{ and } \mathbf{G}) \\ &= \mathbf{C}(\mathbf{R}\mathbf{R}^\top) \mathbf{S} \mathbf{T} \mathbf{C}^\top && (\text{@thm-matmul-assoc}) \\ &= \mathbf{C}\mathbf{I}_r \mathbf{T} \mathbf{C}^\top && (\text{@def-matrix-inverse}) \\ &= \mathbf{C} \mathbf{T} \mathbf{C}^\top, && (\text{@thm-identity}) \\ \mathbf{G}\mathbf{A} &= \mathbf{R}^\top \mathbf{S} \mathbf{T} \mathbf{C}^\top \mathbf{C} \mathbf{R} && (\text{substitute } \mathbf{G} \text{ and } \mathbf{A}) \\ &= \mathbf{R}^\top \mathbf{S} \mathbf{T} (\mathbf{C}^\top \mathbf{C}) \mathbf{R} && (\text{@thm-matmul-assoc}) \\ &= \mathbf{R}^\top \mathbf{S} \mathbf{I}_r \mathbf{R} && (\text{@def-matrix-inverse}) \\ &= \mathbf{R}^\top \mathbf{S} \mathbf{R}. && (\text{@thm-identity}) \end{aligned}$$

**Conditions 3 and 4.**

$$\begin{aligned}
(\mathbf{AG})^\top &= (\mathbf{CTC}^\top)^\top && \text{(first product)} \\
&= (\mathbf{C}(\mathbf{TC}^\top))^\top && \text{(@thm-matmul-assoc)} \\
&= (\mathbf{TC}^\top)^\top \mathbf{C}^\top && \text{(@thm-transpose-product)} \\
&= (\mathbf{C}^\top)^\top \mathbf{T}^\top \mathbf{C}^\top && \text{(@thm-transpose-product)} \\
&= \mathbf{C} \mathbf{T}^\top \mathbf{C}^\top && \text{(@def-matrix-transpose)} \\
&= \mathbf{CTC}^\top, && \text{(\mathbf{T} is symmetric)} \\
(\mathbf{GA})^\top &= (\mathbf{R}^\top \mathbf{SR})^\top && \text{(second product)} \\
&= (\mathbf{R}^\top (\mathbf{SR}))^\top && \text{(@thm-matmul-assoc)} \\
&= (\mathbf{SR})^\top (\mathbf{R}^\top)^\top && \text{(@thm-transpose-product)} \\
&= \mathbf{R}^\top \mathbf{S}^\top (\mathbf{R}^\top)^\top && \text{(@thm-transpose-product)} \\
&= \mathbf{R}^\top \mathbf{S}^\top \mathbf{R} && \text{(@def-matrix-transpose)} \\
&= \mathbf{R}^\top \mathbf{SR}. && \text{(\mathbf{S} is symmetric)}
\end{aligned}$$

**Condition 1.**

$$\begin{aligned}
\mathbf{AGA} &= \mathbf{CTC}^\top \mathbf{CR} && \text{(first product)} \\
&= \mathbf{C} \mathbf{T} (\mathbf{C}^\top \mathbf{C}) \mathbf{R} && \text{(@thm-matmul-assoc)} \\
&= \mathbf{C} \mathbf{I}_r \mathbf{R} && \text{(@def-matrix-inverse)} \\
&= \mathbf{A}. && \text{(@thm-identity)}
\end{aligned}$$

**Condition 2.**

$$\begin{aligned}
\mathbf{GAG} &= \mathbf{R}^\top \mathbf{SR} \mathbf{R}^\top \mathbf{STC}^\top && \text{(second product)} \\
&= \mathbf{R}^\top \mathbf{S} (\mathbf{RR}^\top) \mathbf{STC}^\top && \text{(@thm-matmul-assoc)} \\
&= \mathbf{R}^\top \mathbf{I}_r \mathbf{STC}^\top && \text{(@def-matrix-inverse)} \\
&= \mathbf{G}. && \text{(@thm-identity)}
\end{aligned}$$

□

**Example 0.115** (The Moore-Penrose inverse of a rank-one matrix). For  $\mathbf{A}$  in Example 0.28, the rank factorization  $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ ,  $\mathbf{R} = [1 \ -2 \ -2]$  (Example 0.40) gives  $\mathbf{C}^\top \mathbf{C} = 1 + 9 = 10$  and  $\mathbf{RR}^\top = 1 + 4 + 4 = 9$ , so

$$\mathbf{A}^+ = \frac{1}{9 \cdot 10} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} [1 \ 3] = \frac{1}{90} \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}.$$

As a check on condition 1,  $\mathbf{A}\mathbf{A}^+ = \mathbf{C}\mathbf{T}\mathbf{C}^\top = \frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$ , and its first row times  $\mathbf{A}$  is  $\frac{1}{10}(1 \cdot (1, -2, -2) + 3 \cdot (3, -6, -6)) = (1, -2, -2)$ , the first row of  $\mathbf{A}$ ; the second row is 3 times the first, like the second row of  $\mathbf{A}$ .

**Definition 0.71** (Consistent linear system). A linear system  $\mathbf{A}\tilde{x} = \tilde{b}$ , with  $\mathbf{A}$  an  $m \times n$  matrix and  $\tilde{b} \in \mathbb{R}^m$ , is **consistent** if it has at least one solution  $\tilde{x} \in \mathbb{R}^n$ , and **inconsistent** otherwise.

**Example 0.116** (A consistent system and an inconsistent one). With  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ ,  $\mathbf{A}(x_1, x_2) = (x_1, 0)$  (Definition 0.21).

- $\mathbf{A}\tilde{x} = (2, 0)$  is consistent:  $\tilde{x} = (2, 0)$  is a solution.
- $\mathbf{A}\tilde{x} = (2, 1)$  is inconsistent: the second entry of  $\mathbf{A}\tilde{x}$  is always 0, never 1.

**Theorem 0.78** (When a linear system has a solution). Let  $\mathbf{A}$  be  $m \times n$ , let  $\tilde{b} \in \mathbb{R}^m$ , and let  $\mathbf{G}$  be any generalized inverse of  $\mathbf{A}$  (Definition 0.69). The following statements are equivalent:

1.  $\mathbf{A}\tilde{x} = \tilde{b}$  is consistent (Definition 0.71);
2.  $\tilde{b} \in \mathcal{C}(\mathbf{A})$  (Definition 0.32);
3.  $\mathbf{A}\mathbf{G}\tilde{b} = \tilde{b}$ .

### **i** Proof

*Proof.* **1 and 2 are equivalent.** Statement 1 says  $\tilde{b} = \mathbf{A}\tilde{x}$  for some  $\tilde{x}$  (Definition 0.71), and  $\mathcal{C}(\mathbf{A})$  is the set of vectors  $\mathbf{A}\tilde{x}$  (Definition 0.32), so  $\tilde{b}$  is in it exactly when  $\tilde{b} = \mathbf{A}\tilde{x}$  for some  $\tilde{x}$ .

**1 implies 3.** By Theorem 0.75,  $\mathbf{G}\tilde{b}$  is a solution, so  $\mathbf{A}(\mathbf{G}\tilde{b}) = \tilde{b}$ , which is statement 3 (Theorem 0.7).

**3 implies 1.** If  $\mathbf{A}\mathbf{G}\tilde{b} = \tilde{b}$ , then  $\tilde{x} = \mathbf{G}\tilde{b}$  is a solution, since  $\mathbf{A}(\mathbf{G}\tilde{b}) = (\mathbf{A}\mathbf{G})\tilde{b}$  (Theorem 0.7).  $\square$

**Example 0.117** (Testing consistency with a generalized inverse). For  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  and

the generalized inverse  $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$ ,  $\mathbf{A}\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 0 & 0 \end{bmatrix}$ .

- $\tilde{b} = (2, 0)$ :  $\mathbf{A}\mathbf{G}\tilde{b} = (2, 0) = \tilde{b}$ , so the system is consistent.
- $\tilde{b} = (2, 1)$ :  $\mathbf{A}\mathbf{G}\tilde{b} = (7, 0) \neq \tilde{b}$ , so it is not, as Example 0.112 found directly.

**Theorem 0.79** (All solutions of a linear system). *Let  $\mathbf{A}$  be  $m \times n$  with a generalized inverse  $\mathbf{G}$  (Definition 0.69).*

1.  $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{I}_n - \mathbf{GA})$ :  $\tilde{x}$  solves  $\mathbf{A}\tilde{x} = \tilde{0}_m$  exactly when  $\tilde{x} = (\mathbf{I}_n - \mathbf{GA})\tilde{q}$  for some  $\tilde{q} \in \mathbb{R}^n$ .
2. If  $\mathbf{A}\tilde{x} = \tilde{b}$  is consistent (Definition 0.71), then  $\tilde{x}$  is a solution exactly when

$$\tilde{x} = \mathbf{G}\tilde{b} + (\mathbf{I}_n - \mathbf{GA})\tilde{q} \quad \text{for some } \tilde{q} \in \mathbb{R}^n :$$

one particular solution plus a vector of  $\mathcal{N}(\mathbf{A})$ .

### **i** Proof

*Proof. Part 1.* First,

$$\begin{aligned} \mathbf{A}(\mathbf{I}_n - \mathbf{GA}) &= \mathbf{A}\mathbf{I}_n - \mathbf{A}(\mathbf{GA}) && \text{(@thm-scalar-matmul)} \\ &= \mathbf{A}\mathbf{I}_n - \mathbf{AGA} && \text{(@thm-matmul-assoc)} \\ &= \mathbf{A} - \mathbf{AGA} && \text{(@thm-identity)} \\ &= \mathbf{A} - \mathbf{A} && \text{(@def-generalized-inverse)} \\ &= \mathbf{0}_{m \times n}, && \text{(arithmetic)} \end{aligned}$$

so  $\mathbf{A}(\mathbf{I}_n - \mathbf{GA})\tilde{q} = \tilde{0}_m$  for every  $\tilde{q}$  (Theorem 0.7), and  $\mathcal{C}(\mathbf{I}_n - \mathbf{GA}) \subseteq \mathcal{N}(\mathbf{A})$ . Conversely, if  $\mathbf{A}\tilde{x} = \tilde{0}_m$ , then

$$\begin{aligned} (\mathbf{I}_n - \mathbf{GA})\tilde{x} &= \mathbf{I}_n\tilde{x} - (\mathbf{GA})\tilde{x} && \text{(@thm-scalar-matmul, with } \tilde{x} \text{ as the last factor)} \\ &= \tilde{x} - (\mathbf{GA})\tilde{x} && \text{(@thm-identity)} \\ &= \tilde{x} - \mathbf{G}(\mathbf{A}\tilde{x}) && \text{(@thm-matmul-assoc)} \\ &= \tilde{x} - \mathbf{G}\tilde{0}_m && (\mathbf{A}\tilde{x} = \tilde{0}_m) \\ &= \tilde{x}, && (\mathbf{G}\tilde{0}_m = \tilde{0}_n) \end{aligned}$$

so  $\tilde{x}$  is in  $\mathcal{C}(\mathbf{I}_n - \mathbf{GA})$ , with  $\tilde{q} = \tilde{x}$ .

**Part 2.**  $\mathbf{G}\tilde{b}$  is a solution (Theorem 0.75). For any  $\tilde{x}$ ,

$$\begin{aligned} \mathbf{A}(\tilde{x} - \mathbf{G}\tilde{b}) &= \mathbf{A}\tilde{x} - \mathbf{A}(\mathbf{G}\tilde{b}) && \text{(@thm-matvec-linear, with coefficients 1 and } -1) \\ &= \mathbf{A}\tilde{x} - \mathbf{AG}\tilde{b} && \text{(@thm-matmul-assoc)} \\ &= \mathbf{A}\tilde{x} - \tilde{b}, && \text{(@thm-consistency, statement 3)} \end{aligned}$$

so  $\tilde{x}$  solves  $\mathbf{A}\tilde{x} = \tilde{b}$  exactly when  $\tilde{x} - \mathbf{G}\tilde{b} \in \mathcal{N}(\mathbf{A})$ , that is, by part 1, exactly when  $\tilde{x} - \mathbf{G}\tilde{b} = (\mathbf{I}_n - \mathbf{GA})\tilde{q}$  for some  $\tilde{q}$ .  $\square$

**Example 0.118** (All solutions of a consistent system). For  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ ,  $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$  and  $\tilde{b} = (2, 0)$ :  $\mathbf{G}\tilde{b} = (2, 14)$ ,  $\mathbf{GA} = \begin{bmatrix} 1 & 0 \\ 7 & 0 \end{bmatrix}$ , and  $\mathbf{I}_2 - \mathbf{GA} = \begin{bmatrix} 0 & 0 \\ -7 & 1 \end{bmatrix}$ , so

$$\tilde{x} = (2, 14) + (0, -7q_1 + q_2), \quad q_1, q_2 \in \mathbb{R}.$$

Since  $-7q_1 + q_2$  can be any number, the solutions are exactly the vectors  $(2, t)$ ,  $t \in \mathbb{R}$ , which is what  $\mathbf{A}(x_1, x_2) = (x_1, 0) = (2, 0)$  says directly.

**Corollary 0.6** (Solvable for every right-hand side, and solvable uniquely). *Let  $\mathbf{A}$  be  $m \times n$ .*

1.  $\mathbf{A}\tilde{x} = \tilde{b}$  is consistent for every  $\tilde{b} \in \mathbb{R}^m$  exactly when  $\text{rank}(\mathbf{A}) = m$ .
2. A consistent system  $\mathbf{A}\tilde{x} = \tilde{b}$  has exactly one solution exactly when  $\text{rank}(\mathbf{A}) = n$ .

### Proof

*Proof. Part 1.* By Theorem 0.78, the system is consistent for every  $\tilde{b}$  exactly when  $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$ .  $\mathcal{C}(\mathbf{A})$  is a subspace of  $\mathbb{R}^m$  (Theorem 0.22) with dimension  $\text{rank}(\mathbf{A})$  (Theorem 0.26), and  $\dim(\mathbb{R}^m) = m$  (Example 0.22). If the dimension is  $m$ , then  $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$  (Theorem 0.21); if  $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$ , the dimension is  $m$ .

**Part 2.** By Theorem 0.79, the solutions are a particular solution plus the vectors of  $\mathcal{N}(\mathbf{A})$ , so there is exactly one solution exactly when  $\mathcal{N}(\mathbf{A}) = \{\tilde{0}_n\}$ , that is, when  $\text{nullity}(\mathbf{A}) = 0$  (a subspace of dimension 0 has the empty list as a basis, whose span is  $\{\tilde{0}\}$ , by Definition 0.31 and Definition 0.29; and  $\{\tilde{0}\}$  has dimension 0, by Example 0.22). By Theorem 0.27, that is when  $\text{rank}(\mathbf{A}) = n$ .  $\square$

**Example 0.119** (Three systems).

- The  $2 \times 3$  matrix  $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$  of Example 0.39 has rank  $2 = m$ , so every system with it is consistent, but  $2 < 3 = n$ , so the solutions are never unique.
- The  $3 \times 2$  matrix  $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$  of Example 0.100 has rank  $2 = n$ , so a consistent system has one solution, but  $2 < 3 = m$ , so some systems are inconsistent:  $(1, 0, 0) \notin \mathcal{C}(\mathbf{X})$ ,

because  $a(1, 1, 1) + b(1, 2, 3)$  has equal differences between consecutive entries, and  $(1, 0, 0)$  does not.

- An invertible  $p \times p$  matrix  $\mathbf{A}$  has rank  $p$ : if  $\mathbf{A}\tilde{x} = \tilde{0}$  then  $\tilde{x} = \mathbf{A}^{-1}\mathbf{A}\tilde{x} = \tilde{0}$ , so its nullity is 0 and its rank is  $p$  (Theorem 0.27). So every system with it is consistent (part 1, with  $p = m$ ) and has exactly one solution (part 2, with  $p = n$ ), namely  $\mathbf{A}^{-1}\tilde{b}$ , since  $\mathbf{A}(\mathbf{A}^{-1}\tilde{b}) = \tilde{b}$ .

**Theorem 0.80** (The projection onto a column space, with any generalized inverse). *Let  $\mathbf{X}$  be an  $n \times p$  matrix of any rank, and let  $\mathbf{G}$  be any generalized inverse of  $\mathbf{X}^\top \mathbf{X}$  (Definition 0.69). Then for every  $\tilde{y} \in \mathbb{R}^n$ ,  $\mathbf{X}\mathbf{G}\mathbf{X}^\top \tilde{y}$  is the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{C}(\mathbf{X})$  (Definition 0.68). So the matrix  $\mathbf{X}\mathbf{G}\mathbf{X}^\top$  is the same for every choice of  $\mathbf{G}$ , and when  $\text{rank}(\mathbf{X}) = p$  it is the hat matrix  $\mathbf{H}$  (Definition 0.67).*

### i Proof

*Proof.*  $\mathbf{X}^\top \tilde{y}$  is in  $\mathcal{C}(\mathbf{X}^\top \mathbf{X})$ .  $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ , since  $(\mathbf{X}^\top \mathbf{X})\tilde{z} = \mathbf{X}^\top (\mathbf{X}\tilde{z})$  (Theorem 0.7). Both are subspaces of  $\mathbb{R}^p$  (Theorem 0.22) with the same dimension:  $\text{rank}(\mathbf{X}^\top \mathbf{X}) = \text{rank}(\mathbf{X}^\top)$  (Theorem 0.29, Theorem 0.26). So  $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$  (Theorem 0.21), and since  $\mathbf{X}^\top \tilde{y} \in \mathcal{C}(\mathbf{X}^\top)$  (Definition 0.32),  $\mathbf{X}^\top \tilde{y} = \mathbf{X}^\top \mathbf{X}\tilde{z}$  for some  $\tilde{z} \in \mathbb{R}^p$ .

**The split.** Let  $\tilde{u} = \mathbf{X}\mathbf{G}\mathbf{X}^\top \tilde{y} = \mathbf{X}(\mathbf{G}\mathbf{X}^\top \tilde{y})$  (Theorem 0.7), which is in  $\mathcal{C}(\mathbf{X})$  (Definition 0.32). For  $\tilde{y} - \tilde{u}$ ,

$$\begin{aligned}
 \mathbf{X}^\top (\tilde{y} - \tilde{u}) &= \mathbf{X}^\top \tilde{y} - \mathbf{X}^\top \tilde{u} && \text{(@thm-matvec-linear, with coefficients 1 and -1)} \\
 &= \mathbf{X}^\top \tilde{y} - \mathbf{X}^\top (\mathbf{X}\mathbf{G}\mathbf{X}^\top \tilde{y}) && \text{(substitute } \tilde{u}) \\
 &= \mathbf{X}^\top \tilde{y} - (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \tilde{y}) && \text{(@thm-matmul-assoc)} \\
 &= \mathbf{X}^\top \tilde{y} - (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) \tilde{z} && \text{(first step)} \\
 &= \mathbf{X}^\top \tilde{y} - ((\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X})) \tilde{z} && \text{(@thm-matmul-assoc)} \\
 &= \mathbf{X}^\top \tilde{y} - (\mathbf{X}^\top \mathbf{X}) \tilde{z} && \text{(@def-generalized-inverse)} \\
 &= \mathbf{X}^\top \tilde{y} - \mathbf{X}^\top \tilde{y} && \text{(first step)} \\
 &= \tilde{0}_p, && \text{(arithmetic)}
 \end{aligned}$$

so  $\tilde{y} - \tilde{u} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$  (Theorem 0.37). By the uniqueness in Theorem 0.38,  $\tilde{u}$  is the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{C}(\mathbf{X})$ .

**Same matrix.** Every choice of  $\mathbf{G}$  gives a matrix that projects every  $\tilde{y}$  onto  $\mathcal{C}(\mathbf{X})$ , so all choices give the same matrix (Theorem 0.71, part 2). When  $\text{rank}(\mathbf{X}) = p$ ,  $(\mathbf{X}^\top \mathbf{X})^{-1}$  is one choice (Theorem 0.67, Theorem 0.74), and it gives  $\mathbf{H}$ .  $\square$

**Example 0.120** (A design matrix with a repeated column). Let  $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ , an intercept column entered twice, so  $\text{rank}(\mathbf{X}) = 1 < 2$  and  $\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$  is not invertible: its rank is  $\text{rank}(\mathbf{X}) = 1 < 2$  (Theorem 0.29), while an invertible  $2 \times 2$  matrix has rank 2 (Example 0.119). The matrix  $\mathbf{G} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{bmatrix}$  is a generalized inverse of it:

$$\begin{aligned} (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) &= \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \quad (\text{@def-matrix-mult, for the first product}) \\ &= \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}. \quad (\text{@def-matrix-mult}) \end{aligned}$$

Then

$$\begin{aligned} \mathbf{X} \mathbf{G} \mathbf{X}^\top &= \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad (\text{@def-matrix-mult, for } \mathbf{X} \mathbf{G}) \\ &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad (\text{@def-matrix-mult}) \end{aligned}$$

the hat matrix of the intercept-only model (Example 0.101). Another generalized inverse,  $\mathbf{G}' = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ , which is one because  $(\mathbf{X}^\top \mathbf{X}) \mathbf{G}' = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$  and that times  $\mathbf{X}^\top \mathbf{X}$  is again  $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ , gives  $\mathbf{X} \mathbf{G}' = \begin{bmatrix} 0 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$  and the same  $\mathbf{X} \mathbf{G}' \mathbf{X}^\top = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ , as Theorem 0.80 says it must. Both project onto  $\mathcal{C}(\mathbf{X}) = \text{span}\{(1, 1)\}$ , the column space of the intercept-only design.

## Solving linear systems and least squares

### Source

This section is adapted from Zhou (2024c) and Zhou (2024b), used under the MIT License (see the license text in Section ). The first source states that every invertible matrix has an LU factorization; that is false without reordering rows (the  $2 \times 2$  row swap  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  has none), so this section states only how an LU factorization, when one exists, solves a system. The least squares results are proved here from the closest-point theorem (Theorem 0.69) rather than by differentiating, and the QR factorization is derived from Gram-Schmidt (Theorem 0.48).

**Definition 0.72** (Triangular matrix). A square matrix  $\mathbf{U}$  is **upper triangular** if every entry below the diagonal is 0:  $u_{ij} = 0$  whenever  $i > j$ . A square matrix  $\mathbf{L}$  is **lower triangular** if every entry above the diagonal is 0:  $\ell_{ij} = 0$  whenever  $i < j$ . A lower triangular matrix whose diagonal entries are all 1 is **unit lower triangular**.

**Example 0.121** (Triangular and not).

- $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$  is upper triangular: its entries below the diagonal, in positions  $(2, 1)$ ,  $(3, 1)$  and  $(3, 2)$ , are 0.
- $\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$  is unit lower triangular.
- A diagonal matrix (Definition 0.48) is both upper and lower triangular.
- $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  is neither: its  $(2, 1)$  entry is not 0, and neither is its  $(1, 2)$  entry.

**Theorem 0.81** (Solving a triangular system by substitution). *Let  $\mathbf{U}$  be an  $n \times n$  upper triangular matrix (Definition 0.72) whose diagonal entries  $u_{11}, \dots, u_{nn}$  are all nonzero. For every  $\tilde{\mathbf{b}} \in \mathbb{R}^n$ , the system  $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  has exactly one solution, given for  $i = n, n-1, \dots, 1$  in turn by*

$$x_i = \frac{1}{u_{ii}} \left( b_i - \sum_{j=i+1}^n u_{ij} x_j \right)$$

*(back substitution). Likewise, if  $\mathbf{L}$  is lower triangular with nonzero diagonal,  $\mathbf{L}\tilde{\mathbf{y}} = \tilde{\mathbf{b}}$  has exactly one solution, given for  $i = 1, \dots, n$  by  $y_i = \frac{1}{\ell_{ii}} \left( b_i - \sum_{j=1}^{i-1} \ell_{ij} y_j \right)$  (forward substitution).*

**i** Proof

*Proof.* Row  $i$  of  $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  reads

$$\begin{aligned}
b_i &= \sum_{j=1}^n u_{ij} x_j && (\text{@def-matvec-mult}) \\
&= \sum_{j=i}^n u_{ij} x_j && (u_{ij} = 0 \text{ for } j < i) \\
&= u_{ii} x_i + \sum_{j=i+1}^n u_{ij} x_j. && (\text{split off the } j = i \text{ term})
\end{aligned}$$

This equation is equivalent to each of

$$\begin{aligned}
u_{ii} x_i &= b_i - \sum_{j=i+1}^n u_{ij} x_j && (\text{subtract the sum from both sides}) \\
x_i &= \frac{1}{u_{ii}} \left( b_i - \sum_{j=i+1}^n u_{ij} x_j \right). && (\text{divide by } u_{ii} \neq 0)
\end{aligned}$$

Row  $n$  involves only  $x_n$ , so it fixes  $x_n$ ; once  $x_{i+1}, \dots, x_n$  are fixed, row  $i$  fixes  $x_i$ . So the rows, taken from the last to the first, hold exactly when  $\tilde{x}$  is the vector the formula builds: there is one solution, and only one. For  $\mathbf{L}$ , row  $i$  reads  $\sum_{j=1}^{i-1} \ell_{ij} y_j + \ell_{ii} y_i = b_i$ , and the same argument runs from the first row to the last.  $\square$

**Example 0.122** (Back substitution on a  $3 \times 3$  system). Solve  $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix} \tilde{x} = \begin{bmatrix} 8 \\ 1 \\ 1 \end{bmatrix}$ :

1.  $x_3 = \frac{1}{-1} \cdot 1 = -1$ ;
2.  $x_2 = \frac{1}{1/2} (1 - \frac{1}{2} \cdot (-1)) = 2 \cdot \frac{3}{2} = 3$ ;
3.  $x_1 = \frac{1}{2} (8 - 1 \cdot 3 - (-1)(-1)) = \frac{1}{2} \cdot 4 = 2$ .

So  $\tilde{x} = (2, 3, -1)$ . Without a nonzero diagonal the method fails:  $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \tilde{x} = (1, 1)$  has no solution, since its second row reads  $0 = 1$ .

**Theorem 0.82** (QR factorization). *Let  $\mathbf{A}$  be an  $m \times n$  matrix with  $\text{rank}(\mathbf{A}) = n$ . Then*

$$\underbrace{\mathbf{A}}_{m \times n} = \underbrace{\mathbf{Q}}_{m \times n} \underbrace{\mathbf{R}}_{n \times n},$$

where the columns of  $\mathbf{Q}$  are orthonormal (Definition 0.13) and  $\mathbf{R}$  is upper triangular (Definition 0.72) with positive diagonal entries.

## i Proof

*Proof.* The  $n$  columns  $\tilde{a}_1, \dots, \tilde{a}_n$  of  $\mathbf{A}$  are linearly independent (Definition 0.27), so the Gram-Schmidt process (Definition 0.41) runs all  $n$  steps and gives orthonormal  $\tilde{q}_1, \dots, \tilde{q}_n$  (Theorem 0.48, parts 1 and 3); let  $\mathbf{Q}$  have these columns. At step  $i$ ,  $\tilde{q}_i \neq \tilde{0}$ , and rearranging the orthogonalize step,

$$\begin{aligned}\tilde{a}_i &= \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j && \text{(add the sum to both sides of the orthogonalize step)} \\ &= \|\tilde{q}_i\| \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j. && \text{(normalize step: } \tilde{q}_i = \|\tilde{q}_i\| \tilde{q}_i \text{)}\end{aligned}$$

Let  $\mathbf{R}$  be the  $n \times n$  matrix with  $r_{ji} = \tilde{q}_j \cdot \tilde{a}_i$  for  $j < i$ ,  $r_{ii} = \|\tilde{q}_i\| > 0$ , and  $r_{ji} = 0$  for  $j > i$ ; it is upper triangular with positive diagonal. Column  $i$  of  $\mathbf{QR}$  is  $\mathbf{Q}$  times column  $i$  of  $\mathbf{R}$  (Definition 0.20), and

$$\begin{aligned}\mathbf{Q}(r_{1i}, \dots, r_{ni}) &= \sum_{j=1}^n r_{ji} \tilde{q}_j && \text{(@thm-matvec-columns)} \\ &= \sum_{j=1}^i r_{ji} \tilde{q}_j && (r_{ji} = 0 \text{ for } j > i) \\ &= r_{ii} \tilde{q}_i + \sum_{j=1}^{i-1} r_{ji} \tilde{q}_j && \text{(split off the } j = i \text{ term)} \\ &= \|\tilde{q}_i\| \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j && \text{(the entries of } \mathbf{R} \text{)} \\ &= \tilde{a}_i. && \text{(the display)}\end{aligned}$$

So  $\mathbf{QR} = \mathbf{A}$ . □

**Example 0.123** (A QR factorization from Gram-Schmidt). For  $\mathbf{A}$  with columns  $\tilde{a}_1 = (1, 1, 0)$ ,  $\tilde{a}_2 = (1, 0, 1)$ ,  $\tilde{a}_3 = (0, 1, 1)$ , Example 0.69 and Example 0.71 found  $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ ,  $\tilde{q}_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$ ,  $\tilde{q}_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)$ , with  $\|\tilde{q}_1\| = \sqrt{2}$ ,  $\|\tilde{q}_2\| = \sqrt{3/2}$ ,  $\|\tilde{q}_3\| = 2/\sqrt{3}$ ,  $\tilde{q}_1 \cdot \tilde{a}_2 = \tilde{q}_1 \cdot \tilde{a}_3 = \frac{1}{\sqrt{2}}$  and  $\tilde{q}_2 \cdot \tilde{a}_3 = \frac{1}{\sqrt{6}}$ . So

$$\mathbf{R} = \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}.$$

As a check on the second column,  $\frac{1}{\sqrt{2}} \tilde{q}_1 + \sqrt{3/2} \tilde{q}_2 = \frac{1}{2}(1, 1, 0) + \frac{1}{2}(1, -1, 2) = (1, 0, 1) =$

$\tilde{a}_2$ .

**Definition 0.73** (Least squares solution). Let  $\mathbf{A}$  be  $m \times n$  and  $\tilde{\mathbf{b}} \in \mathbb{R}^m$ . A **least squares solution** of  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  is a vector  $\hat{\tilde{\mathbf{x}}} \in \mathbb{R}^n$  such that

$$\|\tilde{\mathbf{b}} - \mathbf{A}\hat{\tilde{\mathbf{x}}}\| \leq \|\tilde{\mathbf{b}} - \mathbf{A}\tilde{\mathbf{x}}\| \quad \text{for every } \tilde{\mathbf{x}} \in \mathbb{R}^n.$$

The equations  $\mathbf{A}^\top \mathbf{A}\tilde{\mathbf{x}} = \mathbf{A}^\top \tilde{\mathbf{b}}$  are the **normal equations**.

**Example 0.124** (Fitting a constant). Let  $\mathbf{A} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  and  $\tilde{\mathbf{b}} = (1, 3)$ . The system  $x(1, 1) = (1, 3)$  is inconsistent (Definition 0.71): its first entry needs  $x = 1$  and its second needs  $x = 3$ . For any number  $x$ ,  $\|\tilde{\mathbf{b}} - \mathbf{A}x\|^2 = (1 - x)^2 + (3 - x)^2 = 2(x - 2)^2 + 2$ , which is smallest at  $x = 2$ , so  $\hat{x} = 2$  is the least squares solution. The value  $x = 1$  is not one: it gives  $0 + 4 = 4 > 2$ . The normal equations read  $2x = 4$ , and their solution is  $x = 2$  too.

**Theorem 0.83** (Least squares solutions solve the normal equations). Let  $\mathbf{A}$  be  $m \times n$  and  $\tilde{\mathbf{b}} \in \mathbb{R}^m$ .

1.  $\hat{\tilde{\mathbf{x}}}$  is a least squares solution (Definition 0.73) exactly when it solves the normal equations  $\mathbf{A}^\top \mathbf{A}\hat{\tilde{\mathbf{x}}} = \mathbf{A}^\top \tilde{\mathbf{b}}$ .
2. The normal equations are consistent (Definition 0.71).
3. Every least squares solution gives the same fitted vector  $\mathbf{A}\hat{\tilde{\mathbf{x}}}$ : the orthogonal projection of  $\tilde{\mathbf{b}}$  onto  $\mathcal{C}(\mathbf{A})$  (Definition 0.68).
4. The least squares solution is unique exactly when  $\text{rank}(\mathbf{A}) = n$ , and then  $\hat{\tilde{\mathbf{x}}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \tilde{\mathbf{b}}$ .

### **i** Proof

*Proof.* Let  $\tilde{\mathbf{u}}$  be the orthogonal projection of  $\tilde{\mathbf{b}}$  onto  $\mathcal{C}(\mathbf{A})$ . The vectors  $\mathbf{A}\tilde{\mathbf{x}}$ , as  $\tilde{\mathbf{x}}$  ranges over  $\mathbb{R}^n$ , are exactly the points of  $\mathcal{C}(\mathbf{A})$  (Definition 0.32). **Least squares solutions are the solutions of  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{u}}$ .** If  $\mathbf{A}\hat{\tilde{\mathbf{x}}} = \tilde{\mathbf{u}}$ , then for every  $\tilde{\mathbf{x}}$ ,  $\mathbf{A}\tilde{\mathbf{x}} \in \mathcal{C}(\mathbf{A})$ , so  $\|\tilde{\mathbf{b}} - \mathbf{A}\hat{\tilde{\mathbf{x}}}\| = \|\tilde{\mathbf{b}} - \tilde{\mathbf{u}}\| \leq \|\tilde{\mathbf{b}} - \mathbf{A}\tilde{\mathbf{x}}\|$  (Theorem 0.69), and  $\hat{\tilde{\mathbf{x}}}$  is a least squares solution. Conversely, let  $\hat{\tilde{\mathbf{x}}}$  be a least squares solution.  $\tilde{\mathbf{u}} \in \mathcal{C}(\mathbf{A})$ , so  $\tilde{\mathbf{u}} = \mathbf{A}\tilde{\mathbf{x}}_0$  for some  $\tilde{\mathbf{x}}_0$  (Definition 0.32), and

$$\begin{aligned}
\|\tilde{b} - \mathbf{A}\hat{x}\| &\leq \|\tilde{b} - \mathbf{A}\tilde{x}_0\| \quad (\text{@def-least-squares}) \\
&= \|\tilde{b} - \tilde{u}\| \quad (\tilde{u} = \mathbf{A}\tilde{x}_0) \\
&\leq \|\tilde{b} - \mathbf{A}\hat{x}\|, \quad (\text{@thm-closest-point, with } \tilde{u} = \mathbf{A}\hat{x} \in \mathcal{C}(\mathbf{A}))
\end{aligned}$$

so the last inequality is an equality, and the equality case of Theorem 0.69 gives  $\mathbf{A}\hat{x} = \tilde{u}$ . **Parts 1 and 3.**  $\mathbf{A}\hat{x} = \tilde{u}$  holds exactly when  $\tilde{b} - \mathbf{A}\hat{x} \in \mathcal{C}(\mathbf{A})^\perp$ : if  $\mathbf{A}\hat{x} = \tilde{u}$ , then  $\tilde{b} - \tilde{u} \in \mathcal{C}(\mathbf{A})^\perp$  (Definition 0.68); conversely, if  $\tilde{b} - \mathbf{A}\hat{x} \in \mathcal{C}(\mathbf{A})^\perp$ , then  $\tilde{b} = \mathbf{A}\hat{x} + (\tilde{b} - \mathbf{A}\hat{x})$  is a split into  $\mathcal{C}(\mathbf{A})$  and  $\mathcal{C}(\mathbf{A})^\perp$ , so  $\mathbf{A}\hat{x} = \tilde{u}$  by the uniqueness in Theorem 0.38. And  $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$  (Theorem 0.37), where

$$\begin{aligned}
\mathbf{A}^\top (\tilde{b} - \mathbf{A}\hat{x}) &= \mathbf{A}^\top \tilde{b} - \mathbf{A}^\top (\mathbf{A}\hat{x}) \quad (\text{@thm-scalar-matmul}) \\
&= \mathbf{A}^\top \tilde{b} - \mathbf{A}^\top \mathbf{A}\hat{x}, \quad (\text{@thm-matmul-assoc})
\end{aligned}$$

which is  $\tilde{0}_n$  exactly when the normal equations hold.

**Part 2.** The vector  $\tilde{x}_0$  above satisfies  $\mathbf{A}\tilde{x}_0 = \tilde{u}$ , so it is a least squares solution, and it solves the normal equations by part 1.

**Part 4.** The least squares solutions are exactly the solutions of  $\mathbf{A}\tilde{x} = \tilde{u}$ , a consistent system, so there is exactly one when  $\text{rank}(\mathbf{A}) = n$  and more than one otherwise (Corollary 0.6). When  $\text{rank}(\mathbf{A}) = n$ ,  $\mathbf{A}^\top \mathbf{A}$  is invertible (Theorem 0.67), and with  $\mathbf{M} \stackrel{\text{def}}{=} \mathbf{A}^\top \mathbf{A}$ ,

$$\begin{aligned}
\hat{x} &= \mathbf{I}_n \hat{x} \quad (\text{@thm-identity}) \\
&= (\mathbf{M}^{-1} \mathbf{M}) \hat{x} \quad (\text{@def-matrix-inverse}) \\
&= \mathbf{M}^{-1} (\mathbf{M}\hat{x}) \quad (\text{@thm-matmul-assoc}) \\
&= \mathbf{M}^{-1} \mathbf{A}^\top \tilde{b}. \quad (\text{the normal equations})
\end{aligned}$$

□

**Example 0.125** (A least squares line). Fit  $y = x_1 + x_2 t$  to the points  $(t, y) = (1, 1), (2, 2), (3, 2)$ :  $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$  (the  $\mathbf{X}$  of Example 0.100) and  $\tilde{b} = (1, 2, 2)$ .  $\mathbf{A}$  has

rank 2 (Example 0.9), so the solution is unique (part 4).  $\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$  has inverse  $\frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$  (Example 0.100), and  $\mathbf{A}^\top \tilde{b} = (1 + 2 + 2, 1 + 4 + 6) = (5, 11)$ , so

$$\hat{x} = \frac{1}{6} (14 \cdot 5 - 6 \cdot 11, -6 \cdot 5 + 3 \cdot 11) = \frac{1}{6} (4, 3) = \left(\frac{2}{3}, \frac{1}{2}\right).$$

The fitted values are  $\mathbf{A}\hat{x} = \left(\frac{7}{6}, \frac{5}{3}, \frac{13}{6}\right)$ , and the residuals  $\tilde{b} - \mathbf{A}\hat{x} = \left(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}\right)$  are

orthogonal to both columns of  $\mathbf{A}$ :  $-\frac{1}{6} + \frac{1}{3} - \frac{1}{6} = 0$  and  $-\frac{1}{6} + \frac{2}{3} - \frac{1}{2} = 0$ .

**Example 0.126** (Many least squares solutions, one fitted vector). With the rank-1 matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix} \text{ (Example 0.9) and } \tilde{\mathbf{b}} = (1, 2, 2), \mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix} \text{ and } \mathbf{A}^\top \tilde{\mathbf{b}} = (5, 10), \text{ so both}$$

normal equations say  $3x_1 + 6x_2 = 5$ , that is,  $x_1 + 2x_2 = \frac{5}{3}$ . There are infinitely many least squares solutions, such as  $(\frac{5}{3}, 0)$  and  $(0, \frac{5}{6})$ , as part 4 predicts for  $\text{rank}(\mathbf{A}) = 1 < 2$ ; but every one gives the same fitted vector  $\mathbf{A}\hat{\mathbf{x}} = (x_1 + 2x_2)(1, 1, 1) = \frac{5}{3}(1, 1, 1)$ , as part 3 says.

**Theorem 0.84** (Solving least squares by QR). *Let  $\mathbf{A}$  be  $m \times n$  with  $\text{rank}(\mathbf{A}) = n$ , with QR factorization  $\mathbf{A} = \mathbf{Q}\mathbf{R}$  (Theorem 0.82), and let  $\tilde{\mathbf{b}} \in \mathbb{R}^m$ . The least squares solution (Definition 0.73) is the unique solution of*

$$\underbrace{\mathbf{R}}_{n \times n} \hat{\mathbf{x}} = \underbrace{\mathbf{Q}^\top}_{n \times m} \tilde{\mathbf{b}},$$

*which back substitution computes (Theorem 0.81). When  $m = n$ , it is the unique solution of  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ .*

### **i** Proof

*Proof.*  $\mathbf{R}$  is upper triangular with positive diagonal, so  $\mathbf{R}\tilde{\mathbf{x}} = \mathbf{Q}^\top \tilde{\mathbf{b}}$  has exactly one solution  $\hat{\mathbf{x}}$  (Theorem 0.81).  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n$ , as in the proof of Theorem 0.70. Then

$$\begin{aligned} \mathbf{A}^\top \mathbf{A} \hat{\mathbf{x}} &= (\mathbf{Q}\mathbf{R})^\top (\mathbf{Q}\mathbf{R}) \hat{\mathbf{x}} && \text{(substitute } \mathbf{A} = \mathbf{Q}\mathbf{R}) \\ &= \mathbf{R}^\top \mathbf{Q}^\top (\mathbf{Q}\mathbf{R}) \hat{\mathbf{x}} && \text{(@thm-transpose-product)} \\ &= \mathbf{R}^\top (\mathbf{Q}^\top \mathbf{Q}) (\mathbf{R}\hat{\mathbf{x}}) && \text{(@thm-matmul-assoc)} \\ &= \mathbf{R}^\top \mathbf{I}_n (\mathbf{R}\hat{\mathbf{x}}) && (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n) \\ &= \mathbf{R}^\top (\mathbf{R}\hat{\mathbf{x}}) && \text{(@thm-identity)} \\ &= \mathbf{R}^\top (\mathbf{Q}^\top \tilde{\mathbf{b}}) && (\mathbf{R}\hat{\mathbf{x}} = \mathbf{Q}^\top \tilde{\mathbf{b}}) \\ &= (\mathbf{R}^\top \mathbf{Q}^\top) \tilde{\mathbf{b}} && \text{(@thm-matmul-assoc)} \\ &= (\mathbf{Q}\mathbf{R})^\top \tilde{\mathbf{b}} && \text{(@thm-transpose-product)} \\ &= \mathbf{A}^\top \tilde{\mathbf{b}}, && (\mathbf{A} = \mathbf{Q}\mathbf{R}) \end{aligned}$$

so  $\hat{\mathbf{x}}$  solves the normal equations and is the least squares solution (Theorem 0.83), which

is unique because  $\text{rank}(\mathbf{A}) = n$ . When  $m = n$ ,  $\text{rank}(\mathbf{A}) = n = m$ , so  $\mathbf{A}\tilde{x} = \tilde{b}$  is consistent for every  $\tilde{b}$  (Corollary 0.6, part 1), so its solution leaves residual  $\tilde{0}$ , is a least squares solution, and is therefore  $\hat{\tilde{x}}$ .  $\square$

**Example 0.127** (The least squares line by QR). For  $\mathbf{A}$  and  $\tilde{b}$  of Example 0.125, Gram-Schmidt gives  $\tilde{q}_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$ ; then  $\tilde{q}_1 \cdot (1, 2, 3) = \frac{6}{\sqrt{3}} = 2\sqrt{3}$ ,  $\tilde{q}_2 = (1, 2, 3) - 2(1, 1, 1) = (-1, 0, 1)$  with norm  $\sqrt{2}$ , and  $\tilde{q}_2 = \frac{1}{\sqrt{2}}(-1, 0, 1)$ . So

$$\mathbf{R} = \begin{bmatrix} \sqrt{3} & 2\sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}, \quad \mathbf{Q}^\top \tilde{b} = \left( \frac{1+2+2}{\sqrt{3}}, \frac{-1+0+2}{\sqrt{2}} \right) = \left( \frac{5}{\sqrt{3}}, \frac{1}{\sqrt{2}} \right).$$

Back substitution gives  $x_2 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$  and  $x_1 = \frac{1}{\sqrt{3}} \left( \frac{5}{\sqrt{3}} - 2\sqrt{3} \cdot \frac{1}{2} \right) = \frac{5}{3} - 1 = \frac{2}{3}$ , the solution found in Example 0.125.

**Definition 0.74** (LU factorization). An **LU factorization** of an  $n \times n$  matrix  $\mathbf{A}$  is a product  $\mathbf{A} = \mathbf{LU}$  with  $\mathbf{L}$  unit lower triangular and  $\mathbf{U}$  upper triangular (Definition 0.72).

**Example 0.128** (An LU factorization of a  $3 \times 3$  matrix). Let  $\mathbf{A} = \begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$ ,

$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ -1 & 4 & 1 \end{bmatrix}$  (unit lower triangular) and  $\mathbf{U} = \begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$  (upper triangular). By

Definition 0.20, entry  $(i, j)$  of  $\mathbf{LU}$  is  $\sum_k \ell_{ik} u_{kj}$ , so row  $i$  of  $\mathbf{LU}$  is  $\sum_k \ell_{ik}$  times row  $k$  of  $\mathbf{U}$ . Row 1 is  $1 \cdot (2, 1, -1) = (2, 1, -1)$ ; row 2 is  $-\frac{3}{2}(2, 1, -1) + (0, \frac{1}{2}, \frac{1}{2}) = (-3, -1, 2)$ , and row 3 is  $-(2, 1, -1) + 4(0, \frac{1}{2}, \frac{1}{2}) + (0, 0, -1) = (-2, 1, 2)$ . These are the rows of  $\mathbf{A}$ , so  $\mathbf{LU} = \mathbf{A}$  is an LU factorization. The below-diagonal entries of  $\mathbf{L}$  are the negatives of the multiples of earlier rows that Gaussian elimination adds to reduce  $\mathbf{A}$  to  $\mathbf{U}$ .

**Example 0.129** (An invertible matrix with no LU factorization).  $\mathbf{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  is invertible:

$\mathbf{P}^2 = \mathbf{I}_2$ , so  $\mathbf{P}^{-1} = \mathbf{P}$  (Definition 0.49); but it has no LU factorization. If  $\mathbf{P} = \mathbf{LU}$ , the  $(1, 1)$  entry gives  $0 = 1 \cdot u_{11}$ , so  $u_{11} = 0$ , and then the  $(2, 1)$  entry gives  $1 = \ell_{21} u_{11} = 0$ , which is impossible. Swapping the two rows first removes the obstacle: the swapped matrix is  $\mathbf{I}_2 = \mathbf{I}_2 \mathbf{I}_2$ .

**Theorem 0.85** (Solving a system with an LU factorization). *If  $\mathbf{A} = \mathbf{LU}$  is an LU factorization (Definition 0.74) and the diagonal entries of  $\mathbf{U}$  are all nonzero, then for every  $\tilde{\mathbf{b}} \in \mathbb{R}^n$  the system  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  has exactly one solution: solve  $\mathbf{L}\tilde{\mathbf{y}} = \tilde{\mathbf{b}}$  by forward substitution, then  $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{y}}$  by back substitution (Theorem 0.81).*

#### Proof

*Proof.*  $\mathbf{L}$  has diagonal entries 1 and  $\mathbf{U}$  has nonzero ones, so each of the two triangular systems has exactly one solution (Theorem 0.81). If  $\tilde{\mathbf{y}}$  and  $\tilde{\mathbf{x}}$  are those solutions, then  $\mathbf{A}\tilde{\mathbf{x}} = \mathbf{L}(\mathbf{U}\tilde{\mathbf{x}}) = \mathbf{L}\tilde{\mathbf{y}} = \tilde{\mathbf{b}}$  (Theorem 0.7), so  $\tilde{\mathbf{x}}$  is a solution. Conversely, if  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ , then  $\tilde{\mathbf{y}} \stackrel{\text{def}}{=} \mathbf{U}\tilde{\mathbf{x}}$  satisfies  $\mathbf{L}\tilde{\mathbf{y}} = \mathbf{L}(\mathbf{U}\tilde{\mathbf{x}}) = (\mathbf{LU})\tilde{\mathbf{x}} = \mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  (Theorem 0.7), so  $\tilde{\mathbf{y}}$  is the unique solution of that system, and  $\tilde{\mathbf{x}}$  is then the unique solution of  $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{y}}$ .  $\square$

**Example 0.130** (Two triangular solves). With  $\mathbf{L}$  and  $\mathbf{U}$  of Example 0.128, solve  $\mathbf{A}\tilde{\mathbf{x}} = (8, -11, -3)$ . Forward substitution on  $\mathbf{L}\tilde{\mathbf{y}} = (8, -11, -3)$  gives  $y_1 = 8$ ,  $y_2 = -11 + \frac{3}{2} \cdot 8 = 1$ ,  $y_3 = -3 + 8 - 4 \cdot 1 = 1$ . Back substitution on  $\mathbf{U}\tilde{\mathbf{x}} = (8, 1, 1)$  is Example 0.122, which gives  $\tilde{\mathbf{x}} = (2, 3, -1)$ . As a check,  $\mathbf{A}(2, 3, -1) = (4+3+1, -6-3-2, -4+3-2) = (8, -11, -3)$ .

## Similarity and diagonalization

#### Source

This section is adapted from Zhou (2024a), used under the MIT License (see the license text in Section ). These notes take eigenvalues to be real numbers (Definition 0.60), so the source's complex eigenvalues are left out, and eigenvalues are characterized through null spaces rather than through the characteristic polynomial. Only the source's material on similarity, diagonalization and the basic eigenvalue properties is adapted; its characteristic polynomial and algebraic multiplicity, its trace and determinant identities, and its section on symmetric matrices (covered by Theorem 0.59) are left out, and its result that eigenvalues of orthogonal matrices have modulus 1 appears here in the real form  $\pm 1$ . The source's two-vector argument that eigenvectors for distinct eigenvalues are independent is extended here to any number of eigenvectors.

**Theorem 0.86** (A square matrix is invertible exactly when it has full rank). *An  $n \times n$  matrix  $\mathbf{A}$  is invertible (Definition 0.50) exactly when  $\text{rank}(\mathbf{A}) = n$ , that is, exactly when  $\mathcal{N}(\mathbf{A}) = \{\tilde{\mathbf{0}}_n\}$ .*

## i Proof

*Proof.* **Invertible implies rank  $n$ .** If  $\mathbf{A}\tilde{x} = \tilde{0}_n$ , then

$$\begin{aligned}
 \tilde{x} &= \mathbf{I}_n \tilde{x} && (\text{@thm-identity}) \\
 &= (\mathbf{A}^{-1} \mathbf{A}) \tilde{x} && (\text{@def-matrix-inverse}) \\
 &= \mathbf{A}^{-1} (\mathbf{A} \tilde{x}) && (\text{@thm-matmul-assoc}) \\
 &= \mathbf{A}^{-1} \tilde{0}_n && (\mathbf{A}\tilde{x} = \tilde{0}_n) \\
 &= \tilde{0}_n, && (\text{@def-matvec-mult})
 \end{aligned}$$

so  $\mathcal{N}(\mathbf{A}) = \{\tilde{0}_n\}$ , the nullity is 0, and  $\text{rank}(\mathbf{A}) = n$  (Theorem 0.27).

**Rank  $n$  implies invertible.** Every system  $\mathbf{A}\tilde{y} = \tilde{b}$  has exactly one solution (Corollary 0.6, both parts, with  $m = n$ ). Let  $\tilde{y}_j$  solve  $\mathbf{A}\tilde{y}_j = \tilde{e}_j$  (Definition 0.9), and let  $\mathbf{B}$  have columns  $\tilde{y}_1, \dots, \tilde{y}_n$ . Column  $j$  of  $\mathbf{AB}$  is  $\mathbf{A}\tilde{y}_j = \tilde{e}_j$  (Definition 0.20), so  $\mathbf{AB} = \mathbf{I}_n$ . For the other order,

$$\begin{aligned}
 \mathbf{A}(\mathbf{BA} - \mathbf{I}_n) &= \mathbf{ABA} - \mathbf{AI}_n && (\text{@thm-scalar-matmul}, \text{@thm-matmul-assoc}) \\
 &= \mathbf{I}_n \mathbf{A} - \mathbf{AI}_n && (\mathbf{AB} = \mathbf{I}_n) \\
 &= \mathbf{A} - \mathbf{A} && (\text{@thm-identity}) \\
 &= \mathbf{0}_{n \times n}, && (\text{arithmetic})
 \end{aligned}$$

so every column of  $\mathbf{BA} - \mathbf{I}_n$  is in  $\mathcal{N}(\mathbf{A}) = \{\tilde{0}_n\}$  (Definition 0.20), and  $\mathbf{BA} = \mathbf{I}_n$ . So  $\mathbf{B}$  satisfies Definition 0.50. The two conditions in the statement are equivalent by Theorem 0.27: the nullity is 0 exactly when  $\mathcal{N}(\mathbf{A}) = \{\tilde{0}_n\}$ , because a subspace of dimension 0 has the empty list as a basis, whose span is  $\{\tilde{0}\}$  (Definition 0.31, Definition 0.29), and  $\{\tilde{0}\}$  has dimension 0 (Example 0.22).  $\square$

**Example 0.131** (Rank decides invertibility).

- $\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$  of Example 0.75 has rank 2:  $c_1(2, 0) + c_2(1, 1) = (2c_1 + c_2, c_2)$  is  $\tilde{0}$  only if  $c_2 = 0$  and then  $c_1 = 0$ . So it is invertible, as that example found by exhibiting the inverse.
- $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  has rank less than 2:  $(1, -1)$  is a nonzero vector in its null space. So it is singular, as Example 0.75 found.

**Definition 0.75** (Eigenspace and geometric multiplicity). Let  $\lambda$  be an eigenvalue of a  $p \times p$  matrix  $\mathbf{A}$  (Definition 0.60). The **eigenspace** of  $\lambda$  is  $\mathcal{E}_\lambda \stackrel{\text{def}}{=} \mathcal{N}(\mathbf{A} - \lambda \mathbf{I}_p)$ . It is a

subspace of  $\mathbb{R}^p$  (Theorem 0.23), and its dimension (Definition 0.31) is the **geometric multiplicity** of  $\lambda$ .

**Example 0.132** (Eigenspaces of two matrices).

- For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  of Example 0.87,  $\mathbf{A} - 3\mathbf{I}_2 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$  sends  $\tilde{v}$  to  $(v_2 - v_1)(1, -1)$ , so  $\mathcal{E}_3 = \text{span}\{(1, 1)\}$ , with geometric multiplicity 1.
- For  $\mathbf{J} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ ,  $\mathbf{J}\tilde{v} = (v_2, 0)$ .  $\mathbf{J}(1, 0) = (0, 0) = 0(1, 0)$ , so 0 is an eigenvalue. If  $\mathbf{J}\tilde{v} = \lambda\tilde{v}$  with  $\lambda \neq 0$ , the second entry gives  $\lambda v_2 = 0$ , so  $v_2 = 0$ , and then the first gives  $\lambda v_1 = 0$ , so  $\tilde{v} = \tilde{0}$ . So 0 is the only eigenvalue, and  $\mathcal{E}_0 = \mathcal{N}(\mathbf{J}) = \{(t, 0) : t \in \mathbb{R}\}$ , with geometric multiplicity 1.

**Example 0.133** (A number that is not an eigenvalue has no eigenspace). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  and  $\lambda = 2$ ,  $\mathbf{A} - 2\mathbf{I}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  sends  $\tilde{v}$  to  $(v_2, v_1)$ , which is  $\tilde{0}$  only for  $\tilde{v} = \tilde{0}$ , so  $\mathcal{N}(\mathbf{A} - 2\mathbf{I}_2) = \{\tilde{0}\}$ . And 2 is not an eigenvalue (Definition 0.60):  $\mathbf{A}\tilde{v} = 2\tilde{v}$  reads  $(2v_1 + v_2, v_1 + 2v_2) = (2v_1, 2v_2)$ , so  $v_2 = 0$  and  $v_1 = 0$ . So Definition 0.75 does not apply to 2.

**Theorem 0.87** (Eigenvalues are where  $\mathbf{A} - \lambda\mathbf{I}$  is singular). *Let  $\mathbf{A}$  be a  $p \times p$  matrix and  $\lambda$  a real number. Then  $\lambda$  is an eigenvalue of  $\mathbf{A}$  (Definition 0.60) exactly when  $\mathbf{A} - \lambda\mathbf{I}_p$  is singular (Definition 0.50). In particular,  $\mathbf{A}$  is singular exactly when 0 is one of its eigenvalues; and each eigenspace  $\mathcal{E}_\lambda$  (Definition 0.75) is a subspace of  $\mathbb{R}^p$  consisting of  $\tilde{0}$  and the eigenvectors for  $\lambda$ .*

### **i** Proof

*Proof.* For any  $\tilde{v} \in \mathbb{R}^p$ ,

$$\begin{aligned} (\mathbf{A} - \lambda\mathbf{I}_p)\tilde{v} &= \mathbf{A}\tilde{v} - (\lambda\mathbf{I}_p)\tilde{v} \quad (\text{@thm-scalar-matmul, the difference law}) \\ &= \mathbf{A}\tilde{v} - \lambda(\mathbf{I}_p\tilde{v}) \quad (\text{@thm-scalar-matmul, the scalar factor}) \\ &= \mathbf{A}\tilde{v} - \lambda\tilde{v}, \quad (\text{@thm-identity}) \end{aligned}$$

so  $\mathbf{A}\tilde{v} = \lambda\tilde{v}$  exactly when  $\tilde{v} \in \mathcal{N}(\mathbf{A} - \lambda\mathbf{I}_p)$ ; the nonzero such  $\tilde{v}$  are the eigenvectors for  $\lambda$ . So  $\lambda$  is an eigenvalue exactly when that null space contains a nonzero vector, which by Theorem 0.86 is exactly when  $\mathbf{A} - \lambda\mathbf{I}_p$  is singular. With  $\lambda = 0$ ,  $\mathbf{A} - 0\mathbf{I}_p = \mathbf{A}$ . An eigenspace is a null space, so it is a subspace (Theorem 0.23).  $\square$

**Example 0.134** (Eigenvalues and singular matrices). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ ,  $\mathbf{A} - 3\mathbf{I}_2 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$  is singular (its columns are negatives of each other, so its rank is 1), matching the eigenvalue 3 of Example 0.87.  $\mathbf{A}$  itself has rank 2, since  $c_1(2, 1) + c_2(1, 2) = (2c_1 + c_2, c_1 + 2c_2)$  is  $\tilde{0}$  only if  $c_1 = c_2 = 0$  (subtract twice the second entry from the first:  $-3c_2 = 0$ , and then the second entry gives  $c_1 = 0$ ); so it is invertible (Theorem 0.86), and 0 is not an eigenvalue. The singular matrix  $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  has eigenvalue 0, with eigenvector  $(1, -1)$ .

**Theorem 0.88** (Eigenvalues of shifts and powers). *Let  $\tilde{v}$  be an eigenvector of a  $p \times p$  matrix  $\mathbf{A}$  for the eigenvalue  $\lambda$  (Definition 0.60). Then*

1.  $\tilde{v}$  is an eigenvector of  $\mathbf{A} + s\mathbf{I}_p$  for  $\lambda + s$ , for every number  $s$ ;
2.  $\tilde{v}$  is an eigenvector of  $\mathbf{A}^k$  (Definition 0.45) for  $\lambda^k$ , for every positive integer  $k$ .

### **i** Proof

*Proof.* **Part 1.**

$$\begin{aligned}
 (\mathbf{A} + s\mathbf{I}_p)\tilde{v} &= \mathbf{A}\tilde{v} + (s\mathbf{I}_p)\tilde{v} && \text{(@thm-matmul-distrib)} \\
 &= \mathbf{A}\tilde{v} + s(\mathbf{I}_p\tilde{v}) && \text{(@thm-scalar-matmul)} \\
 &= \lambda\tilde{v} + s(\mathbf{I}_p\tilde{v}) && \text{(eigenvector)} \\
 &= \lambda\tilde{v} + s\tilde{v} && \text{(@thm-identity)} \\
 &= (\lambda + s)\tilde{v}. && \text{(add entrywise)}
 \end{aligned}$$

**Part 2, by induction on  $k$ .** For  $k = 1$  it is the assumption. If  $\mathbf{A}^{k-1}\tilde{v} = \lambda^{k-1}\tilde{v}$ , then

$$\begin{aligned}
 \mathbf{A}^k\tilde{v} &= \mathbf{A}(\mathbf{A}^{k-1}\tilde{v}) && \text{(@def-matrix-power, @thm-matmul-assoc)} \\
 &= \mathbf{A}(\lambda^{k-1}\tilde{v}) && \text{(induction hypothesis)} \\
 &= \lambda^{k-1}(\mathbf{A}\tilde{v}) && \text{(@thm-scalar-matmul)} \\
 &= \lambda^{k-1}(\lambda\tilde{v}) && \text{(eigenvector)} \\
 &= \lambda^k\tilde{v}. && \text{(multiply the numbers)}
 \end{aligned}$$

In both parts  $\tilde{v} \neq \tilde{0}$ , so  $\tilde{v}$  is an eigenvector. □

**Example 0.135** (Shifting and squaring). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  with eigenvector  $(1, 1)$  for 3

(Example 0.87):

- $\mathbf{A} - 2\mathbf{I}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  sends  $(1, 1)$  to  $(1, 1) = (3 - 2)(1, 1)$ ;
- $\mathbf{A}^2 = \begin{bmatrix} 2 \cdot 2 + 1 \cdot 1 & 2 \cdot 1 + 1 \cdot 2 \\ 1 \cdot 2 + 2 \cdot 1 & 1 \cdot 1 + 2 \cdot 2 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$  (Definition 0.20) sends  $(1, 1)$  to  $(9, 9) = 3^2(1, 1)$ .

**Theorem 0.89** (The eigenvalues of an upper triangular matrix are its diagonal entries). *The eigenvalues (Definition 0.60) of an upper triangular  $p \times p$  matrix  $\mathbf{U}$  (Definition 0.72) are exactly its diagonal entries  $u_{11}, \dots, u_{pp}$ .*

### **i** Proof

*Proof.*  $\mathbf{U} - \lambda\mathbf{I}_p$  is upper triangular, with diagonal entries  $u_{ii} - \lambda$  (Definition 0.46, Definition 0.18).

**If  $\lambda$  is not a diagonal entry**, every  $u_{ii} - \lambda \neq 0$ , so  $(\mathbf{U} - \lambda\mathbf{I}_p)\tilde{v} = \tilde{0}$  has exactly one solution,  $\tilde{v} = \tilde{0}$  (Theorem 0.81), and  $\lambda$  is not an eigenvalue (Theorem 0.87).

**If  $\lambda$  is a diagonal entry**, let  $k$  be the smallest index with  $u_{kk} = \lambda$ , and write  $\mathbf{M} = \mathbf{U} - \lambda\mathbf{I}_p$ , so  $m_{kk} = 0$  and  $m_{ii} \neq 0$  for  $i < k$ . Look for  $\tilde{v}$  with  $v_k = 1$  and  $v_j = 0$  for  $j > k$ . Entry  $i$  of  $\mathbf{M}\tilde{v}$  is

$$\begin{aligned} \sum_{j=1}^p m_{ij} v_j &= \sum_{j=i}^p m_{ij} v_j \quad (m_{ij} = 0 \text{ for } j < i) \\ &= \sum_{j=i}^k m_{ij} v_j, \quad (v_j = 0 \text{ for } j > k) \end{aligned}$$

where the last sum is empty, and so 0, when  $i > k$ . So:

- for  $i > k$  entry  $i$  is 0;
- for  $i = k$  it is  $m_{kk} v_k = 0$ ;
- for  $i < k$  it is 0 exactly when  $v_i = -\frac{1}{m_{ii}} \sum_{j=i+1}^k m_{ij} v_j$ , and these equations fix  $v_{k-1}, \dots, v_1$  in turn, as in Theorem 0.81.

So  $\mathbf{M}\tilde{v} = \tilde{0}$  with  $v_k = 1$ , hence  $\tilde{v} \neq \tilde{0}$ , and  $\lambda$  is an eigenvalue. □

**Example 0.136** (Eigenvalues read off the diagonal).  $\mathbf{U} = \begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$  of Example 0.121

has eigenvalues  $2, \frac{1}{2}$  and  $-1$ . For  $\lambda = \frac{1}{2}$  (so  $k = 2$ ), take  $v_2 = 1, v_3 = 0$ ; row 1 of  $(\mathbf{U} - \frac{1}{2} \mathbf{I}_3) \tilde{v}$  is  $\frac{3}{2} v_1 + 1 = 0$ , so  $v_1 = -\frac{2}{3}$ . Check:  $\mathbf{U}(-\frac{2}{3}, 1, 0) = (-\frac{4}{3} + 1, \frac{1}{2}, 0) = \frac{1}{2}(-\frac{2}{3}, 1, 0)$ . The matrix  $\mathbf{J}$  of Example 0.132 is upper triangular with both diagonal entries 0, and 0 is its only eigenvalue.

**Theorem 0.90** (Eigenvalues of idempotent and orthogonal matrices).

1. Every eigenvalue of an idempotent matrix (Definition 0.51) is 0 or 1.
2. Every eigenvalue of an orthogonal matrix (Definition 0.54) is 1 or  $-1$ .

### Proof

*Proof.* Let  $\tilde{v} \neq \tilde{0}$  with  $\mathbf{A}\tilde{v} = \lambda\tilde{v}$ .

**Part 1.** If  $\mathbf{A}^2 = \mathbf{A}$ , then

$$\begin{aligned} \lambda\tilde{v} &= \mathbf{A}\tilde{v} && \text{(eigenvector)} \\ &= \mathbf{A}^2\tilde{v} && \text{(idempotent)} \\ &= \lambda^2\tilde{v}, && \text{(@thm-eigen-shift-power, part 2)} \end{aligned}$$

and subtracting  $\lambda\tilde{v}$  from both sides gives  $(\lambda^2 - \lambda)\tilde{v} = \tilde{0}$ . Some entry of  $\tilde{v}$  is nonzero, so  $\lambda^2 - \lambda = \lambda(\lambda - 1) = 0$ , and a product of two numbers is 0 only when one of them is:  $\lambda = 0$  or  $\lambda = 1$ .

**Part 2.** If  $\mathbf{A}$  is orthogonal, then

$$\begin{aligned} \|\tilde{v}\| &= \|\mathbf{A}\tilde{v}\| && \text{(@thm-orthogonal-norm)} \\ &= \|\lambda\tilde{v}\| && \text{(eigenvector)} \\ &= |\lambda| \|\tilde{v}\|, && \text{(@thm-norm-properties, part 2)} \end{aligned}$$

and  $\|\tilde{v}\| > 0$  (Theorem 0.41, part 1), so dividing by  $\|\tilde{v}\|$  gives  $|\lambda| = 1$ ; since  $\lambda$  is real (Definition 0.60),  $\lambda = 1$  or  $\lambda = -1$ .  $\square$

**Example 0.137** (Projections and reflections).

- The idempotent matrix  $\begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$  of Example 0.102 sends  $(1, 1)$  to  $(1, 1)$  and  $(1, -1)$  to  $(0, 0)$ : eigenvalues 1 and 0.
- The matrix  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ , which swaps the two entries, is orthogonal: its columns  $(0, 1)$  and  $(1, 0)$  are orthonormal (Remark 0.20). It sends  $(1, 1)$  to  $(1, 1)$  and  $(1, -1)$  to  $(-1, 1)$ : eigenvalues 1 and  $-1$ .
- The rotation  $\mathbf{Q}$  of Example 0.79 has no real eigenvalue. By part 2 the only candidates

are  $\pm 1$ .  $\mathbf{Q}\tilde{v} = \tilde{v}$  reads  $-0.4v_1 - 0.8v_2 = 0$  and  $0.8v_1 - 0.4v_2 = 0$ , so  $v_1 = -2v_2$  and  $v_2 = 2v_1 = -4v_2$ , forcing  $\tilde{v} = 0$ ;  $\mathbf{Q}\tilde{v} = -\tilde{v}$  reads  $1.6v_1 - 0.8v_2 = 0$  and  $0.8v_1 + 1.6v_2 = 0$ , so  $v_2 = 2v_1$  and  $0.8v_1 + 3.2v_1 = 0$ , forcing  $\tilde{v} = \tilde{0}$  again.

**Definition 0.76** (Similar matrices). Two  $p \times p$  matrices  $\mathbf{A}$  and  $\mathbf{B}$  are **similar** if  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  for some invertible  $p \times p$  matrix  $\mathbf{P}$  (Definition 0.50).

*Remark 0.38* (Similarity read both ways). If  $\mathbf{P}$  is invertible, then  $\mathbf{P}^{-1}$  is invertible with inverse  $\mathbf{P}$ :  $\mathbf{P}^{-1}\mathbf{P} = \mathbf{P}\mathbf{P}^{-1} = \mathbf{I}_p$  (Definition 0.49) is the condition of Definition 0.50 for  $\mathbf{P}^{-1}$  with  $\mathbf{P}$  as the other factor, and that factor is unique (Remark 0.18). If  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ , then

$$\begin{aligned} \mathbf{P}\mathbf{B}\mathbf{P}^{-1} &= \mathbf{P}(\mathbf{P}^{-1}\mathbf{A}\mathbf{P})\mathbf{P}^{-1} && \text{(substitute } \mathbf{B} \text{)} \\ &= (\mathbf{P}\mathbf{P}^{-1})\mathbf{A}(\mathbf{P}\mathbf{P}^{-1}) && \text{(@thm-matmul-assoc)} \\ &= \mathbf{I}_p\mathbf{A}\mathbf{I}_p && \text{(@def-matrix-inverse)} \\ &= \mathbf{A}, && \text{(@thm-identity)} \end{aligned}$$

and the same steps with  $\mathbf{P}$  and  $\mathbf{P}^{-1}$  exchanged turn  $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1}$  back into  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ . So  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  exactly when  $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1}$ , and similarity goes both ways:  $\mathbf{A}$  is similar to  $\mathbf{B}$  through  $\mathbf{P}^{-1}$ , whose inverse is  $\mathbf{P}$ .

**Example 0.138** (A matrix similar to a diagonal one). Let  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  and  $\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ . Then

$$\begin{aligned} \mathbf{P}^2 &= \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot (-1) \\ 1 \cdot 1 + (-1) \cdot 1 & 1 \cdot 1 + (-1)(-1) \end{bmatrix} && \text{(@def-matrix-mult)} \\ &= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} && \text{(arithmetic)} \\ &= 2\mathbf{I}_2, && \text{(@def-scalar-mult, @def-identity-matrix)} \end{aligned}$$

so  $\mathbf{P}(\frac{1}{2}\mathbf{P}) = \frac{1}{2}\mathbf{P}^2 = \mathbf{I}_2$  and likewise  $(\frac{1}{2}\mathbf{P})\mathbf{P} = \mathbf{I}_2$  (Theorem 0.73):  $\mathbf{P}^{-1} = \frac{1}{2}\mathbf{P}$  (Definition 0.49). Then

$$\begin{aligned}
\mathbf{P}^{-1}\mathbf{A}\mathbf{P} &= (\tfrac{1}{2}\mathbf{P})\mathbf{A}\mathbf{P} && \text{(substitute } \mathbf{P}^{-1}\text{)} \\
&= \tfrac{1}{2}(\mathbf{P}\mathbf{A}\mathbf{P}) && \text{(@thm-scalar-matmul)} \\
&= \tfrac{1}{2}\mathbf{P}(\mathbf{A}\mathbf{P}) && \text{(@thm-matmul-assoc)} \\
&= \tfrac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 3 & -1 \end{bmatrix} && \text{(@def-matrix-mult, for } \mathbf{A}\mathbf{P}\text{)} \\
&= \tfrac{1}{2}\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix} && \text{(@def-matrix-mult)} \\
&= \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, && \text{(@def-scalar-mult)}
\end{aligned}$$

so  $\mathbf{A}$  is similar to  $\text{diag}(3, 1)$ . At the other extreme, the identity matrix is similar only to itself:  $\mathbf{P}^{-1}\mathbf{I}_p\mathbf{P} = \mathbf{P}^{-1}\mathbf{P} = \mathbf{I}_p$  for every invertible  $\mathbf{P}$  (Theorem 0.50, Definition 0.49).

**Theorem 0.91** (Similar matrices have the same eigenvalues). *If  $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  (Definition 0.76), then  $\tilde{v}$  is an eigenvector of  $\mathbf{A}$  for  $\lambda$  exactly when  $\mathbf{P}^{-1}\tilde{v}$  is an eigenvector of  $\mathbf{B}$  for  $\lambda$ . So  $\mathbf{A}$  and  $\mathbf{B}$  have the same eigenvalues.*

### **i** Proof

*Proof. From A to B.* Let  $\mathbf{A}\tilde{v} = \lambda\tilde{v}$  with  $\tilde{v} \neq \tilde{0}$ , and  $\tilde{w} = \mathbf{P}^{-1}\tilde{v}$ . Then

$$\begin{aligned}
\mathbf{B}\tilde{w} &= (\mathbf{P}^{-1}\mathbf{A}\mathbf{P})(\mathbf{P}^{-1}\tilde{v}) && \text{(substitute } \mathbf{B} \text{ and } \tilde{w}\text{)} \\
&= \mathbf{P}^{-1}\mathbf{A}(\mathbf{P}\mathbf{P}^{-1})\tilde{v} && \text{(@thm-matmul-assoc)} \\
&= \mathbf{P}^{-1}\mathbf{A}\mathbf{I}_p\tilde{v} && \text{(@def-matrix-inverse)} \\
&= \mathbf{P}^{-1}\mathbf{A}\tilde{v} && \text{(@thm-identity)} \\
&= \mathbf{P}^{-1}(\mathbf{A}\tilde{v}) && \text{(@thm-matmul-assoc)} \\
&= \mathbf{P}^{-1}(\lambda\tilde{v}) && \text{(eigenvector)} \\
&= \lambda\tilde{w}. && \text{(@thm-scalar-matmul)}
\end{aligned}$$

And  $\tilde{w} \neq \tilde{0}$ : if  $\tilde{w} = \tilde{0}$ , then  $\tilde{v} = \mathbf{I}_p\tilde{v} = \mathbf{P}(\mathbf{P}^{-1}\tilde{v}) = \mathbf{P}\tilde{0} = \tilde{0}$  (Theorem 0.50, Definition 0.49, Theorem 0.7, Definition 0.21), which is false.

**From B to A.** By Remark 0.38,  $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1} = (\mathbf{P}^{-1})^{-1}\mathbf{B}\mathbf{P}^{-1}$ , so the first part, with  $\mathbf{P}^{-1}$  in place of  $\mathbf{P}$  and the roles of  $\mathbf{A}$  and  $\mathbf{B}$  exchanged, shows that if  $\tilde{w} = \mathbf{P}^{-1}\tilde{v}$  is an eigenvector of  $\mathbf{B}$  for  $\lambda$ , then  $\mathbf{P}\tilde{w}$  is an eigenvector of  $\mathbf{A}$  for  $\lambda$ ; and  $\mathbf{P}\tilde{w} = \mathbf{P}\mathbf{P}^{-1}\tilde{v} = \tilde{v}$  (Theorem 0.7, Definition 0.49, Theorem 0.50).  $\square$

**Example 0.139** (Eigenvectors carried across a similarity). In Example 0.138,  $\mathbf{B} = \text{diag}(3, 1)$  has eigenvectors  $(1, 0)$  for 3 and  $(0, 1)$  for 1:  $\text{diag}(3, 1)(1, 0) = (3, 0)$  and  $\text{diag}(3, 1)(0, 1) = (0, 1)$ .  $\mathbf{P}(1, 0) = (1, 1)$  and  $\mathbf{P}(0, 1) = (1, -1)$  are the eigenvectors of  $\mathbf{A}$  for 3 and 1 found in Example 0.87.

**Theorem 0.92** (Eigenvectors for distinct eigenvalues are linearly independent). *If  $\tilde{v}_1, \dots, \tilde{v}_k$  are eigenvectors of a  $p \times p$  matrix  $\mathbf{A}$  for eigenvalues  $\lambda_1, \dots, \lambda_k$  that are all different, then  $\tilde{v}_1, \dots, \tilde{v}_k$  are linearly independent (Definition 0.25).*

### **i** Proof

*Proof.* By induction on  $k$ . For  $k = 1$ ,  $\tilde{v}_1 \neq \tilde{0}$  is linearly independent on its own. Suppose the result holds for  $k - 1$  eigenvectors, and  $\sum_{i=1}^k c_i \tilde{v}_i = \tilde{0}$ . Then

$$\begin{aligned} \tilde{0} &= (\mathbf{A} - \lambda_k \mathbf{I}_p) \sum_{i=1}^k c_i \tilde{v}_i \quad (\text{multiply the supposed equation by } \mathbf{A} - \lambda_k \mathbf{I}_p) \\ &= \sum_{i=1}^k c_i (\mathbf{A} - \lambda_k \mathbf{I}_p) \tilde{v}_i \quad (\text{@thm-matvec-linear}) \\ &= \sum_{i=1}^k c_i (\lambda_i - \lambda_k) \tilde{v}_i \quad (\text{@thm-eigen-shift-power, part 1, with } s = -\lambda_k) \\ &= \sum_{i=1}^{k-1} c_i (\lambda_i - \lambda_k) \tilde{v}_i. \quad (\text{the } i = k \text{ term is } \tilde{0}) \end{aligned}$$

By the induction hypothesis  $\tilde{v}_1, \dots, \tilde{v}_{k-1}$  are linearly independent, so every  $c_i (\lambda_i - \lambda_k) = 0$  for  $i < k$ , and since  $\lambda_i \neq \lambda_k$ , every  $c_i = 0$  for  $i < k$ . The supposed equation then reads  $c_k \tilde{v}_k = \tilde{0}$  with  $\tilde{v}_k \neq \tilde{0}$ , so  $c_k = 0$  too.  $\square$

**Example 0.140** (Independent eigenvectors). The eigenvectors  $(1, 1)$  for 3 and  $(1, -1)$  for 1 of Example 0.87 are linearly independent, as Theorem 0.92 says they must be:  $c_1(1, 1) + c_2(1, -1) = (c_1 + c_2, c_1 - c_2)$  is  $\tilde{0}$  only if  $c_1 = c_2 = 0$ . Two eigenvectors for the same eigenvalue need not be independent:  $(1, 1)$  and  $(2, 2)$  are both eigenvectors for 3.

**Definition 0.77** (Diagonalizable matrix). A  $p \times p$  matrix  $\mathbf{A}$  is **diagonalizable** if  $\mathbf{A} = \mathbf{X}\mathbf{X}^{-1}$  for some invertible  $p \times p$  matrix  $\mathbf{X}$  and diagonal  $p \times p$  matrix (Definition 0.48); by Remark 0.38, that is the same as  $\mathbf{A} = \mathbf{X}^{-1}\mathbf{A}\mathbf{X}$ , so a diagonalizable matrix is one similar to a diagonal matrix (Definition 0.76).

**Example 0.141** (Diagonalizable and not).

- $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  is diagonalizable: with  $\mathbf{P}$  as in Example 0.138,  $\text{diag}(3, 1) = \mathbf{P}^{-1} \mathbf{A} \mathbf{P}$ .
- $\mathbf{J} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$  is not. Suppose  $\mathbf{J} = \mathbf{X} \mathbf{X}^{-1}$ , so  $\mathbf{J} = \mathbf{X}^{-1} \mathbf{J} \mathbf{X}$  (Remark 0.38). The eigenvalues of  $\mathbf{J}$  are its diagonal entries (Theorem 0.89, since a diagonal matrix is upper triangular) and are the eigenvalues of  $\mathbf{J}$  (Theorem 0.91), of which 0 is the only one (Example 0.132). So  $\mathbf{J} = \mathbf{0}_{2 \times 2}$ , and then  $\mathbf{J} = \mathbf{X} \mathbf{0}_{2 \times 2} \mathbf{X}^{-1} = \mathbf{0}_{2 \times 2}$  (Definition 0.20), which it is not.

**Theorem 0.93** (Diagonalizable means a basis of eigenvectors). *A  $p \times p$  matrix  $\mathbf{A}$  is diagonalizable (Definition 0.77) exactly when it has  $p$  linearly independent eigenvectors. Then  $\mathbf{A} = \mathbf{X} \mathbf{X}^{-1}$  with the eigenvectors as the columns of  $\mathbf{X}$  and their eigenvalues, in the same order, on the diagonal of  $\mathbf{X}^{-1} \mathbf{A} \mathbf{X}$ .*

### **i** Proof

*Proof.* Let  $\mathbf{X}$  have columns  $\tilde{x}_1, \dots, \tilde{x}_p$  and  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_p)$ . Column  $j$  of  $\mathbf{A} \mathbf{X}$  is  $\mathbf{A} \tilde{x}_j$ , and column  $j$  of  $\mathbf{X} \mathbf{D}$  is  $\mathbf{X}$  times column  $j$  of  $\mathbf{D}$  (Definition 0.20), which is  $\mathbf{X}(\lambda_j \tilde{e}_j) = \lambda_j (\mathbf{X} \tilde{e}_j) = \lambda_j \tilde{x}_j$  (Theorem 0.73, Theorem 0.13), so

$$\mathbf{A} \mathbf{X} = \mathbf{X} \mathbf{D} \iff \mathbf{A} \tilde{x}_j = \lambda_j \tilde{x}_j \text{ for every } j. \quad (6)$$

Also,  $\mathbf{X}$  has rank  $p$  exactly when its  $p$  columns are linearly independent (Definition 0.26), which by Theorem 0.86 is exactly when  $\mathbf{X}$  is invertible.

**Eigenvectors give a diagonalization.** If  $\tilde{x}_1, \dots, \tilde{x}_p$  are linearly independent eigenvectors, then  $\mathbf{X}$  is invertible and Equation 6 gives  $\mathbf{A} \mathbf{X} = \mathbf{X} \mathbf{D}$ , so

$$\begin{aligned} \mathbf{A} &= \mathbf{A} \mathbf{I}_p && (\text{@thm-identity}) \\ &= \mathbf{A} (\mathbf{X} \mathbf{X}^{-1}) && (\text{@def-matrix-inverse}) \\ &= (\mathbf{A} \mathbf{X}) \mathbf{X}^{-1} && (\text{@thm-matmul-assoc}) \\ &= \mathbf{X} \mathbf{D} \mathbf{X}^{-1}. && (\mathbf{A} \mathbf{X} = \mathbf{X} \mathbf{D}) \end{aligned}$$

**A diagonalization gives eigenvectors.** If  $\mathbf{A} = \mathbf{X} \mathbf{D} \mathbf{X}^{-1}$ , then

$$\begin{aligned} \mathbf{A} \mathbf{X} &= (\mathbf{X} \mathbf{D} \mathbf{X}^{-1}) \mathbf{X} && (\text{substitute } \mathbf{A}) \\ &= \mathbf{X} (\mathbf{D} \mathbf{X}^{-1} \mathbf{X}) && (\text{@thm-matmul-assoc}) \\ &= \mathbf{X} \mathbf{I}_p && (\text{@def-matrix-inverse}) \\ &= \mathbf{X}, && (\text{@thm-identity}) \end{aligned}$$

so each column satisfies  $\mathbf{A}\tilde{x}_j = \lambda_j\tilde{x}_j$  (Equation 6).  $\mathbf{X}$  is invertible, so its columns are linearly independent, and in particular nonzero; they are  $p$  linearly independent eigenvectors.  $\square$

**Example 0.142** (Building the diagonalization from eigenvectors). The eigenvectors  $(1, 1)$  and  $(1, -1)$  of  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  are linearly independent (Example 0.140), so with  $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$  and  $\mathbf{D} = \text{diag}(3, 1)$ ,  $\mathbf{A} = \mathbf{X}\mathbf{X}^{-1}\mathbf{D}$ ; by Remark 0.38 this is the similarity of Example 0.138, with  $\mathbf{X} = \mathbf{P}$ . Because  $\mathbf{A}$  is symmetric, Theorem 0.59 even gives such a factorization with orthonormal eigenvectors (Definition 0.61): dividing each column of  $\mathbf{X}$  by its length  $\sqrt{2}$  gives one. By contrast, every eigenvector of  $\mathbf{J}$  in Example 0.132 has the form  $(t, 0)$ , so any two of them are multiples of each other, and  $\mathbf{J}$  has no two linearly independent eigenvectors: another way to see that it is not diagonalizable.

**Corollary 0.7** (Distinct eigenvalues make a matrix diagonalizable). *A  $p \times p$  matrix with  $p$  different eigenvalues is diagonalizable.*

#### **i** Proof

*Proof.* Choose one eigenvector for each eigenvalue. These  $p$  eigenvectors are linearly independent (Theorem 0.92), so the matrix is diagonalizable (Theorem 0.93).  $\square$

**Example 0.143** (A triangular matrix with distinct diagonal entries).  $\mathbf{U}$  of Example 0.136 has the three different eigenvalues 2,  $\frac{1}{2}$  and  $-1$  (Theorem 0.89), so it is diagonalizable, even though it is not symmetric, so Theorem 0.59 does not apply to it. The converse fails:  $\mathbf{I}_2$  is diagonal, hence diagonalizable ( $\mathbf{I}_2 = \mathbf{I}_2 \mathbf{I}_2 \mathbf{I}_2^{-1}$ ), but has only the eigenvalue 1.

**Theorem 0.94** (Powers of a diagonalizable matrix). *If  $\mathbf{A} = \mathbf{X}\mathbf{X}^{-1}\mathbf{D}$  (Definition 0.77) with  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_p)$ , then for every positive integer  $k$ ,  $\mathbf{A}^k = \mathbf{X}\mathbf{D}^k\mathbf{X}^{-1}$  and  $\mathbf{D}^k = \text{diag}(\lambda_1^k, \dots, \lambda_p^k)$ .*

#### **i** Proof

*Proof. Powers of .* If  $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$  and  $\mathbf{E} = \text{diag}(e_1, \dots, e_p)$ , with entries  $d_{il}$  and  $e_{lj}$  (so  $d_{ii} = d_i$  and  $e_{jj} = e_j$ ), entry  $(i, j)$  of  $\mathbf{DE}$  is  $\sum_l d_{il} e_{lj}$  (Definition 0.20), where  $d_{il} = 0$  unless  $l = i$  and  $e_{lj} = 0$  unless  $l = j$ . For  $i \neq j$  no term survives, so the entry is 0; for  $i = j$  only  $l = i$  survives, giving  $d_i e_i$ . So  $\mathbf{DE} = \text{diag}(d_1 e_1, \dots, d_p e_p)$ , and induction on

$k$  gives  $\mathbf{A}^k = \text{diag}(\lambda_1^k, \dots, \lambda_p^k)$ .

**Powers of  $\mathbf{A}$ , by induction on  $k$ .**  $k = 1$  is the assumption. If it holds for  $k - 1$ ,

$$\begin{aligned}
\mathbf{A}^k &= \mathbf{A}^{k-1} \mathbf{A} && (\text{@def-matrix-power, @thm-matmul-assoc}) \\
&= \mathbf{X}^{k-1} \mathbf{X}^{-1} \mathbf{A} && (\text{induction hypothesis}) \\
&= \mathbf{X}^{k-1} \mathbf{X}^{-1} \mathbf{X} \mathbf{X}^{-1} && (\text{the assumption}) \\
&= \mathbf{X}^{k-1} (\mathbf{X}^{-1} \mathbf{X}) \mathbf{X}^{-1} && (\text{@thm-matmul-assoc}) \\
&= \mathbf{X}^{k-1} \mathbf{I}_p \mathbf{X}^{-1} && (\text{@def-matrix-inverse}) \\
&= \mathbf{X}^{k-1} \mathbf{X}^{-1} && (\text{@thm-identity}) \\
&= \mathbf{X}^k \mathbf{X}^{-1}. && (\text{@def-matrix-power})
\end{aligned}$$

□

**Example 0.144** (A closed form for the powers). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ , with  $\mathbf{X}$  as in Example 0.142 and  $\mathbf{X}^{-1} = \frac{1}{2} \mathbf{X}$  (Example 0.138),

$$\begin{aligned}
\mathbf{A}^k &= \mathbf{X} \text{diag}(3^k, 1) \mathbf{X}^{-1} && (\text{@thm-diagonalizable-powers}) \\
&= \mathbf{X} \text{diag}(3^k, 1) \left(\frac{1}{2} \mathbf{X}\right) && (\text{substitute } \mathbf{X}^{-1}) \\
&= \frac{1}{2} \mathbf{X} \text{diag}(3^k, 1) \mathbf{X} && (\text{@thm-scalar-matmul}) \\
&= \frac{1}{2} \begin{bmatrix} 3^k & 1 \\ 3^k & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && (\text{@def-matrix-mult, for } \mathbf{X} \text{diag}(3^k, 1)) \\
&= \frac{1}{2} \begin{bmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{bmatrix}. && (\text{@def-matrix-mult})
\end{aligned}$$

For  $k = 2$  this gives  $\frac{1}{2} \begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ , which matches  $\mathbf{A}^2$  in Example 0.135.

## Gram matrices, Schur complements and Cholesky

### i Source

This section is adapted from Zhou (2024f), used under the MIT License (see the license text in Section ). The source's eigenvalue and quadratic-form characterizations of definiteness are already on this page (Theorem 0.62). These parts of the source are left out:

- its covariance test, which needs probability
- its material on minimization
- its positive semidefinite version of the Schur complement test

- its pivot test
- its Hadamard-product “Schur lemma”
- its notes on the LU factorization and computational cost

The source leaves the Schur complement test unproved; it is proved here from a block expansion of the quadratic form, and the Cholesky factorization is derived from it.

**Theorem 0.95** (Definite matrices are Gram matrices). *Let  $\mathbf{A}$  be a  $p \times p$  matrix.*

1.  $\mathbf{A}$  is positive semidefinite (Definition 0.63) exactly when  $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$  for some matrix  $\mathbf{B}$  with  $p$  columns.
2.  $\mathbf{A}$  is positive definite (Definition 0.64) exactly when  $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$  for some matrix  $\mathbf{B}$  with  $p$  linearly independent columns.

### **i** Proof

*Proof.* If  $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$ .  $\mathbf{A}$  is symmetric, since  $(\mathbf{B}^\top \mathbf{B})^\top = \mathbf{B}^\top (\mathbf{B}^\top)^\top = \mathbf{B}^\top \mathbf{B}$  (Theorem 0.10, Definition 0.16). For any  $\tilde{x}$ ,

$$\begin{aligned} \tilde{x}^\top \mathbf{A} \tilde{x} &= \tilde{x}^\top \mathbf{B}^\top \mathbf{B} \tilde{x} && \text{(substitute } \mathbf{A}) \\ &= (\mathbf{B} \tilde{x})^\top (\mathbf{B} \tilde{x}) && \text{(@thm-transpose-product, @thm-matmul-assoc)} \\ &= \|\mathbf{B} \tilde{x}\|^2 && \text{(@eq-l2-norm, squared)} \\ &\geq 0. && \text{(@thm-norm-properties)} \end{aligned}$$

If the columns of  $\mathbf{B}$  are linearly independent, then  $\mathcal{N}(\mathbf{B}) = \{\tilde{0}\}$  ( $\mathbf{B} \tilde{x}$  is the combination of the columns with coefficients  $x_j$ , Theorem 0.13, and only the zero combination is  $\tilde{0}$ , Definition 0.25), so for  $\tilde{x} \neq \tilde{0}$ ,  $\mathbf{B} \tilde{x} \neq \tilde{0}$  and  $\|\mathbf{B} \tilde{x}\|^2 > 0$  (Theorem 0.41, part 1).

**Only if.** Let  $\mathbf{A}$  be positive semidefinite. It is symmetric, so  $\mathbf{A} = \mathbf{Q} \mathbf{Q}^\top$  (Theorem 0.59), and every  $\lambda_i \geq 0$  (Theorem 0.62). Let  $^{1/2} \stackrel{\text{def}}{=} \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_p})$ , which is symmetric, with  $^{1/2} {}^{1/2} = \mathbf{I}$  (the product of diagonal matrices multiplies their diagonals, as in the proof of Theorem 0.94), and let  $\mathbf{B} \stackrel{\text{def}}{=} {}^{1/2} \mathbf{Q}^\top$ . Then

$$\begin{aligned} \mathbf{B}^\top \mathbf{B} &= (\mathbf{Q}^\top)^\top ({}^{1/2})^\top {}^{1/2} \mathbf{Q}^\top && \text{(@thm-transpose-product)} \\ &= \mathbf{Q} {}^{1/2} {}^{1/2} \mathbf{Q}^\top && \text{(@def-matrix-transpose; } {}^{1/2} \text{ is symmetric)} \\ &= \mathbf{Q} \mathbf{Q}^\top && ({}^{1/2} {}^{1/2} = \mathbf{I}) \\ &= \mathbf{A}. && \text{(@thm-spectral)} \end{aligned}$$

If  $\mathbf{A}$  is positive definite, it is positive semidefinite (Remark 0.33), so the same  $\mathbf{B}$  gives  $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$ ; and every  $\lambda_i > 0$  (Theorem 0.62), so  $^{1/2}$  is diagonal with nonzero diagonal,

and invertible (its inverse is  $\text{diag}(1/\sqrt{\lambda_1}, \dots, 1/\sqrt{\lambda_p})$ );  $\mathbf{Q}^\top$  is invertible with inverse  $\mathbf{Q}$  (Remark 0.20). So  $\mathbf{B}$  is invertible (Theorem 0.51), and its  $p$  columns are linearly independent (Theorem 0.86, Definition 0.26).  $\square$

**Example 0.145** (Gram factorizations).

- With  $\mathbf{R} = \begin{bmatrix} 1 & 1 \end{bmatrix}$  ( $1 \times 2$ ),  $\mathbf{R}^\top \mathbf{R} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ , the positive semidefinite matrix of Example 0.92; the two columns of  $\mathbf{R}$ , 1 and 1, are dependent, so part 2 does not apply to  $\mathbf{R}$ ;  $\mathbf{R}^\top \mathbf{R}$  is in fact not positive definite (Example 0.93).
- With  $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ , whose columns are independent (Example 0.9),  $\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$  (Example 0.100) is positive definite.
- $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$  of Example 0.93 is symmetric but not positive semidefinite, so by part 1 it is not  $\mathbf{B}^\top \mathbf{B}$  for any  $\mathbf{B}$ .

**Theorem 0.96** (Operations that preserve definiteness).

1. If  $\mathbf{C}$  is a  $p \times p$  positive definite matrix (Definition 0.64) and  $\mathbf{A}$  is a  $p \times k$  matrix with  $k$  linearly independent columns, then the  $k \times k$  matrix  $\mathbf{A}^\top \mathbf{C} \mathbf{A}$  is positive definite.
2. If  $\mathbf{A}_1$  and  $\mathbf{A}_2$  are positive definite  $p \times p$  matrices and  $\alpha_1, \alpha_2 > 0$ , then  $\alpha_1 \mathbf{A}_1 + \alpha_2 \mathbf{A}_2$  is positive definite.

### **i** Proof

*Proof. Part 1.*  $\mathbf{A}^\top \mathbf{C} \mathbf{A}$  is symmetric:  $(\mathbf{A}^\top \mathbf{C} \mathbf{A})^\top = \mathbf{A}^\top \mathbf{C}^\top (\mathbf{A}^\top)^\top = \mathbf{A}^\top \mathbf{C} \mathbf{A}$  (Theorem 0.10, twice;  $\mathbf{C}$  symmetric; Definition 0.16). For  $\tilde{x} \neq \tilde{0}_k$ ,  $\mathbf{A}\tilde{x} \neq \tilde{0}_p$ , since the columns of  $\mathbf{A}$  are independent (Theorem 0.13, Definition 0.25), and

$$\begin{aligned} \tilde{x}^\top (\mathbf{A}^\top \mathbf{C} \mathbf{A}) \tilde{x} &= (\mathbf{A}\tilde{x})^\top \mathbf{C} (\mathbf{A}\tilde{x}) \quad (\text{@thm-transpose-product, @thm-matmul-assoc}) \\ &> 0. \quad (\mathbf{C} \text{ positive definite, } \mathbf{A}\tilde{x} \neq \tilde{0}) \end{aligned}$$

**Part 2.**  $\alpha_1 \mathbf{A}_1 + \alpha_2 \mathbf{A}_2$  is symmetric: its transpose is  $\alpha_1 \mathbf{A}_1^\top + \alpha_2 \mathbf{A}_2^\top = \alpha_1 \mathbf{A}_1 + \alpha_2 \mathbf{A}_2$  (Theorem 0.9, Definition 0.19, Definition 0.16). For  $\tilde{x} \neq \tilde{0}$ ,

$$\begin{aligned}
\tilde{x}^\top (\alpha_1 \mathbf{A}_1 + \alpha_2 \mathbf{A}_2) \tilde{x} &= \tilde{x}^\top (\alpha_1 \mathbf{A}_1) \tilde{x} + \tilde{x}^\top (\alpha_2 \mathbf{A}_2) \tilde{x} && \text{(@thm-matmul-distrib, twice)} \\
&= \alpha_1 \tilde{x}^\top \mathbf{A}_1 \tilde{x} + \alpha_2 \tilde{x}^\top \mathbf{A}_2 \tilde{x} && \text{(@thm-scalar-matmul)} \\
&> 0. && \text{(each term is positive)}
\end{aligned}$$

□

**Example 0.146** (Building positive definite matrices).

- With  $\mathbf{C} = \mathbf{I}_3$  and  $\mathbf{X}$  of Example 0.145, part 1 gives again that  $\mathbf{X}^\top \mathbf{X}$  is positive definite.
- With  $\alpha_1 = 2$ ,  $\mathbf{A}_1 = \mathbf{I}_2$  (Example 0.93),  $\alpha_2 = 1$  and  $\mathbf{A}_2 = \mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$  (Example 0.145), part 2 gives that  $2\mathbf{I}_2 + \mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 5 & 6 \\ 6 & 16 \end{bmatrix}$  is positive definite.
- An aside beyond part 2:  $\mathbf{I}_2 + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  is positive definite plus positive semidefinite; its quadratic form is  $x_1^2 + x_2^2 + (x_1 + x_2)^2 > 0$  for  $\tilde{x} \neq \tilde{0}$ , so it is positive definite, in line with its eigenvalues 3 and 1 (Example 0.87).
- Independence is needed in part 1: with  $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ , whose columns are equal,  $\mathbf{A}^\top \mathbf{I}_2 \mathbf{A} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ , and  $(1, -1)$  gives quadratic form 0.

**Theorem 0.97** (Leading blocks of a positive definite matrix are positive definite). *Let  $\mathbf{X}$  be a  $p \times p$  positive definite matrix (Definition 0.64), and for  $1 \leq k \leq p$  let  $\mathbf{X}_k$  be its top-left  $k \times k$  block, with entries  $x_{ij}$  for  $i, j \leq k$ . Then  $\mathbf{X}_k$  is positive definite. Also, every diagonal entry of a positive definite matrix is positive.*

### **i** Proof

*Proof.*  $\mathbf{X}_k$  is symmetric, because  $\mathbf{X}$  is. For  $\tilde{u} \in \mathbb{R}^k$  with  $\tilde{u} \neq \tilde{0}$ , let  $\tilde{y} = (u_1, \dots, u_k, 0, \dots, 0) \in \mathbb{R}^p$ , which is nonzero, with entries  $y_i = u_i$  for  $i \leq k$  and  $y_i = 0$  for  $i > k$ . Then

$$\begin{aligned}
\tilde{u}^\top \mathbf{X}_k \tilde{u} &= \sum_{i=1}^k \sum_{j=1}^k u_i x_{ij} u_j \quad (\text{@def-matvec-mult, @def-dot-product}) \\
&= \sum_{i=1}^k \sum_{j=1}^k y_i x_{ij} y_j \quad (y_i = u_i \text{ for } i \leq k) \\
&= \sum_{i=1}^p \sum_{j=1}^p y_i x_{ij} y_j \quad (\text{each added term has a factor } y_i = 0 \text{ or } y_j = 0) \\
&= \tilde{y}^\top \mathbf{X} \tilde{y} \quad (\text{@def-matvec-mult, @def-dot-product}) \\
&> 0. \quad (\mathbf{X} \text{ positive definite, } \tilde{y} \neq \tilde{0})
\end{aligned}$$

For the diagonal entry  $x_{ii}$ , take  $\tilde{e}_i$  (Definition 0.9), which is nonzero:

$$\begin{aligned}
\tilde{e}_i^\top \mathbf{X} \tilde{e}_i &= \sum_{r=1}^p \sum_{s=1}^p (\tilde{e}_i)_r x_{rs} (\tilde{e}_i)_s \quad (\text{@def-matvec-mult, @def-dot-product}) \\
&= x_{ii} \quad (\text{only the term } r = s = i \text{ is nonzero}) \\
&> 0. \quad (\mathbf{X} \text{ positive definite})
\end{aligned}$$

□

**Example 0.147** (A quick necessary test).  $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  is positive definite (Example 0.146), and its diagonal entries 2, 2 and its  $1 \times 1$  leading block  $[2]$  are positive. The test can only rule matrices out:  $-\mathbf{I}_2$  has negative diagonal entries, so it is not positive definite, even though  $\det(-\mathbf{I}_2) = (-1)(-1) - 0 = 1 > 0$  (Definition 0.65); a positive determinant alone does not make a matrix positive definite. And positive diagonal entries are not enough:  $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$  of Example 0.93 has them but is not positive definite.

**Theorem 0.98** (The quadratic form of a block matrix). *Let  $k, m \geq 1$ , let  $\mathbf{A}$  be a symmetric  $k \times k$  matrix,  $\mathbf{B}$  a  $k \times m$  matrix and  $\mathbf{C}$  a symmetric  $m \times m$  matrix, and let  $\mathbf{X}$  be the  $(k+m) \times (k+m)$  matrix with blocks*

$$\mathbf{X} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{bmatrix},$$

*that is,  $x_{ij} = a_{ij}$ ,  $x_{i,k+j} = b_{ij}$ ,  $x_{k+i,j} = b_{ji}$  and  $x_{k+i,k+j} = c_{ij}$  for indices in range.  $\mathbf{X}$  is symmetric (Definition 0.47):  $x_{k+i,j} = b_{ji} = x_{j,k+i}$ , and  $\mathbf{A}$  and  $\mathbf{C}$  are symmetric. Split  $\tilde{v} \in \mathbb{R}^{k+m}$  as  $\tilde{v} = (\tilde{u}, \tilde{w})$ , with  $\tilde{u} \in \mathbb{R}^k$  its first  $k$  entries and  $\tilde{w} \in \mathbb{R}^m$  the rest. Then*

$$\tilde{v}^\top \mathbf{X} \tilde{v} = \tilde{u}^\top \mathbf{A} \tilde{u} + 2 \tilde{u}^\top \mathbf{B} \tilde{w} + \tilde{w}^\top \mathbf{C} \tilde{w}, \quad (7)$$

and if  $\mathbf{A}$  is invertible, with  $\mathbf{S} \stackrel{\text{def}}{=} \mathbf{C} - \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B}$  and  $\tilde{z} \stackrel{\text{def}}{=} \tilde{u} + \mathbf{A}^{-1} \mathbf{B} \tilde{w}$ ,

$$\tilde{v}^\top \mathbf{X} \tilde{v} = \tilde{z}^\top \mathbf{A} \tilde{z} + \tilde{w}^\top \mathbf{S} \tilde{w}.$$

### **i** Proof

*Proof.*  **$\mathbf{X} \tilde{v}$  in blocks.** For  $i \leq k$ ,

$$\begin{aligned} (\mathbf{X} \tilde{v})_i &= \sum_{j=1}^{k+m} x_{ij} v_j && (\text{@def-matvec-mult}) \\ &= \sum_{j=1}^k x_{ij} v_j + \sum_{j=1}^m x_{i,k+j} v_{k+j} && (\text{split the sum at } j = k) \\ &= \sum_{j=1}^k a_{ij} u_j + \sum_{j=1}^m b_{ij} w_j && (\text{the blocks of } \mathbf{X} \text{ and of } \tilde{v}) \\ &= (\mathbf{A} \tilde{u})_i + (\mathbf{B} \tilde{w})_i, && (\text{@def-matvec-mult}) \end{aligned}$$

and in the same way  $(\mathbf{X} \tilde{v})_{k+i} = (\mathbf{B}^\top \tilde{u})_i + (\mathbf{C} \tilde{w})_i$  for  $i \leq m$ .

**The expansion.**

$$\begin{aligned} \tilde{v}^\top \mathbf{X} \tilde{v} &= \sum_{i=1}^k u_i (\mathbf{X} \tilde{v})_i + \sum_{i=1}^m w_i (\mathbf{X} \tilde{v})_{k+i} && (\text{@def-dot-product, split the sum at } k) \\ &= \tilde{u} \cdot (\mathbf{A} \tilde{u} + \mathbf{B} \tilde{w}) + \tilde{w} \cdot (\mathbf{B}^\top \tilde{u} + \mathbf{C} \tilde{w}) && (\text{the blocks of } \mathbf{X} \tilde{v}) \\ &= \tilde{u}^\top \mathbf{A} \tilde{u} + \tilde{u}^\top \mathbf{B} \tilde{w} + \tilde{w}^\top \mathbf{B}^\top \tilde{u} + \tilde{w}^\top \mathbf{C} \tilde{w} && (\text{@thm-dot-linear}) \\ &= \tilde{u}^\top \mathbf{A} \tilde{u} + \tilde{u}^\top \mathbf{B} \tilde{w} + \tilde{u}^\top \mathbf{B} \tilde{w} + \tilde{w}^\top \mathbf{C} \tilde{w} && (\tilde{w}^\top \mathbf{B}^\top \tilde{u} = (\tilde{w}^\top \mathbf{B}^\top \tilde{u})^\top = \tilde{u}^\top \mathbf{B} \tilde{w}; 1 \times 1, \text{@def-matrix-transpose,}) \\ &= \tilde{u}^\top \mathbf{A} \tilde{u} + 2 \tilde{u}^\top \mathbf{B} \tilde{w} + \tilde{w}^\top \mathbf{C} \tilde{w}. && (\text{combine like terms}) \end{aligned}$$

**Completing the square.** Write  $\tilde{d} \stackrel{\text{def}}{=} \mathbf{A}^{-1} \mathbf{B} \tilde{w}$ , so  $\tilde{z} = \tilde{u} + \tilde{d}$  and  $\mathbf{A} \tilde{d} = \mathbf{B} \tilde{w}$  (Theorem 0.7, Definition 0.49, Theorem 0.50);  $\mathbf{A}^{-1}$  is symmetric (Corollary 0.4). Then

$$\begin{aligned}
\tilde{z}^\top \mathbf{A} \tilde{z} &= (\tilde{u} + \tilde{d}) \cdot \mathbf{A} (\tilde{u} + \tilde{d}) && \text{(substitute } \tilde{z} = \tilde{u} + \tilde{d} \text{)} \\
&= (\tilde{u} + \tilde{d}) \cdot (\mathbf{A} \tilde{u} + \mathbf{A} \tilde{d}) && \text{(@thm-matmul-distrib)} \\
&= \tilde{u}^\top \mathbf{A} \tilde{u} + \tilde{u}^\top \mathbf{A} \tilde{d} + \tilde{d}^\top \mathbf{A} \tilde{u} + \tilde{d}^\top \mathbf{A} \tilde{d} && \text{(@thm-dot-linear, both slots)} \\
&= \tilde{u}^\top \mathbf{A} \tilde{u} + \tilde{u}^\top \mathbf{A} \tilde{d} + \tilde{u}^\top \mathbf{A} \tilde{d} + \tilde{d}^\top \mathbf{A} \tilde{d} && \text{(as in the expansion, with } \mathbf{A}^\top = \mathbf{A} \text{)} \\
&= \tilde{u}^\top \mathbf{A} \tilde{u} + 2 \tilde{u}^\top \mathbf{A} \tilde{d} + \tilde{d}^\top \mathbf{A} \tilde{d} && \text{(combine like terms)} \\
&= \tilde{u}^\top \mathbf{A} \tilde{u} + 2 \tilde{u}^\top \mathbf{B} \tilde{w} + \tilde{d}^\top \mathbf{B} \tilde{w} && (\mathbf{A} \tilde{d} = \mathbf{B} \tilde{w}, \text{ twice)} \\
&= \tilde{u}^\top \mathbf{A} \tilde{u} + 2 \tilde{u}^\top \mathbf{B} \tilde{w} + \tilde{w}^\top \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} \tilde{w}, && (\tilde{d}^\top = \tilde{w}^\top \mathbf{B}^\top \mathbf{A}^{-1}, \text{ @thm-transpose-product, @cor-inverse-symm})
\end{aligned}$$

and

$$\begin{aligned}
\tilde{w}^\top \mathbf{S} \tilde{w} &= \tilde{w}^\top (\mathbf{C} - \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B}) \tilde{w} && \text{(substitute } \mathbf{S} \text{)} \\
&= \tilde{w}^\top (\mathbf{C} \tilde{w} - \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} \tilde{w}) && \text{(@thm-scalar-matmul, difference rule on the right)} \\
&= \tilde{w}^\top \mathbf{C} \tilde{w} - \tilde{w}^\top \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} \tilde{w}. && \text{(@thm-scalar-matmul, difference rule on the left)}
\end{aligned}$$

Adding the two displays, the  $\tilde{w}^\top \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B} \tilde{w}$  terms cancel, leaving  $\tilde{u}^\top \mathbf{A} \tilde{u} + 2 \tilde{u}^\top \mathbf{B} \tilde{w} + \tilde{w}^\top \mathbf{C} \tilde{w}$ , which is  $\tilde{v}^\top \mathbf{X} \tilde{v}$  by Equation 7.  $\square$

**Example 0.148** (The  $2 \times 2$  case). For  $\mathbf{X} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$  ( $k = m = 1$ ) and  $a \neq 0$ , the theorem reads

$$au^2 + 2buw + cw^2 = a \left(u + \frac{b}{a} w\right)^2 + \left(c - \frac{b^2}{a}\right) w^2,$$

with  $\tilde{v} = (u, w)$ : the familiar completing of the square. With  $a = c = 5$  and  $b = 4$ :  $5(u + 0.8w)^2 + 1.8w^2 = 5(u^2 + 1.6uw + 0.64w^2) + 1.8w^2 = 5u^2 + 8uw + 3.2w^2 + 1.8w^2 = 5u^2 + 8uw + 5w^2$ .

**Definition 0.78** (Schur complement). For a block matrix  $\mathbf{X} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{bmatrix}$  as in Theorem 0.98 with  $\mathbf{A}$  invertible, the **Schur complement** of  $\mathbf{A}$  in  $\mathbf{X}$  is the  $m \times m$  matrix  $\mathbf{S} = \mathbf{C} - \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B}$  defined in Theorem 0.98.

**Example 0.149** (Schur complements of  $2 \times 2$  matrices).

- For  $\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$  with  $\mathbf{A} = [5]$ :  $\mathbf{S} = 5 - 4 \cdot \frac{1}{5} \cdot 4 = \frac{25}{5} - \frac{16}{5} = \frac{9}{5}$ , that is,  $\mathbf{S} = [\frac{9}{5}]$ .

- For  $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$  with  $\mathbf{A} = [1]$ :  $\mathbf{S} = [1 - 2 \cdot 1 \cdot 2] = [-3]$ .
- For  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  the top-left block  $[0]$  is not invertible, so there is no Schur complement of it.

**Theorem 0.99** (Schur complement test). *Let  $\mathbf{X} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{bmatrix}$  be as in Theorem 0.98. Then  $\mathbf{X}$  is positive definite (Definition 0.64) exactly when  $\mathbf{A}$  is positive definite and its Schur complement  $\mathbf{S}$  (Definition 0.78) is positive definite.*

### i Proof

*Proof. If.*  $\mathbf{A}$  is positive definite, so invertible (Theorem 0.63), and  $\mathbf{S}$  is defined.  $\mathbf{X}$  is symmetric. Let  $\tilde{v} = (\tilde{u}, \tilde{w}) \neq \tilde{0}$  and  $\tilde{z} = \tilde{u} + \mathbf{A}^{-1}\mathbf{B}\tilde{w}$ . By Theorem 0.98,  $\tilde{v}^\top \mathbf{X} \tilde{v} = \tilde{z}^\top \mathbf{A} \tilde{z} + \tilde{w}^\top \mathbf{S} \tilde{w}$ , and both terms are at least 0. If  $\tilde{w} \neq \tilde{0}$ , the second term is positive. If  $\tilde{w} = \tilde{0}$ , then  $\tilde{u} \neq \tilde{0}$  and  $\tilde{z} = \tilde{u}$ , so the first term is positive. Either way  $\tilde{v}^\top \mathbf{X} \tilde{v} > 0$ .

*Only if.*  $\mathbf{A}$  is the top-left  $k \times k$  block of  $\mathbf{X}$ , so it is positive definite (Theorem 0.97), and invertible (Theorem 0.63).  $\mathbf{S}$  is symmetric:  $(\mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B})^\top = \mathbf{B}^\top (\mathbf{A}^{-1})^\top \mathbf{B} = \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B}$  (Theorem 0.10, Definition 0.16 for  $(\mathbf{B}^\top)^\top = \mathbf{B}$ , Corollary 0.4), and  $\mathbf{C}$  is symmetric, so  $\mathbf{S} = \mathbf{C} - \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B}$  is symmetric (the transpose of a difference is the difference of the transposes, entry by entry, Definition 0.16). For  $\tilde{w} \neq \tilde{0}$ , take  $\tilde{u} = -\mathbf{A}^{-1} \mathbf{B} \tilde{w}$ , so  $\tilde{z} = \tilde{0}$  and  $\tilde{v} = (\tilde{u}, \tilde{w}) \neq \tilde{0}$ ; then  $\tilde{w}^\top \mathbf{S} \tilde{w} = \tilde{v}^\top \mathbf{X} \tilde{v} > 0$  (Theorem 0.98).  $\square$

**Example 0.150** (Testing  $2 \times 2$  matrices). A symmetric  $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$  is positive definite exactly when  $a > 0$  and  $c - \frac{b^2}{a} > 0$ , that is,  $a > 0$  and  $ac > b^2$  (a  $1 \times 1$  matrix  $[s]$  is positive definite exactly when  $s > 0$ , since  $x^\top s x = s x^2$ ).

- $\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ :  $5 > 0$  and  $\frac{9}{5} > 0$  (Example 0.149), so it is positive definite.
- $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ :  $1 > 0$  but  $\mathbf{S} = -3$ , so it is not, as Example 0.93 found directly.

**Theorem 0.100** (Cholesky factorization). *A  $p \times p$  matrix  $\mathbf{A}$  is positive definite (Definition 0.64) exactly when  $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$  for some lower triangular  $\mathbf{L}$  (Definition 0.72) with positive diagonal entries. For a positive definite  $\mathbf{A}$  that  $\mathbf{L}$  is unique; it is the **Cholesky factor** of  $\mathbf{A}$ .*

**i** Proof

*Proof.* **If.**  $\mathbf{L}^\top$  is upper triangular with positive diagonal (Definition 0.16), so  $\mathbf{L}^\top \tilde{x} = \tilde{0}$  only for  $\tilde{x} = \tilde{0}$  (Theorem 0.81), and the columns of  $\mathbf{L}^\top$  are linearly independent (Theorem 0.13, Definition 0.25). So  $\mathbf{A} = (\mathbf{L}^\top)^\top \mathbf{L}^\top$  (Definition 0.16) is positive definite (Theorem 0.95, part 2).

**Only if, with uniqueness, by induction on  $p$ .** For  $p = 1$ ,  $\mathbf{A} = [a]$  with  $a > 0$  (Theorem 0.97), and  $[\ell][\ell] = [a]$  with  $\ell > 0$  holds exactly for  $\ell = \sqrt{a}$ . For  $p > 1$ , assume every  $(p-1) \times (p-1)$  positive definite matrix has exactly one such factor, and split off the first row and column.  $\mathbf{A}$  is symmetric, so its first row is the transpose of its first column:

$$\mathbf{A} = \begin{bmatrix} a_{11} & \tilde{b}^\top \\ \tilde{b} & \mathbf{A}_{22} \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} \ell_{11} & \tilde{0}_{1 \times (p-1)} \\ \tilde{g} & \mathbf{L}_{22} \end{bmatrix}, \quad \mathbf{L}^\top = \begin{bmatrix} \ell_{11} & \tilde{g}^\top \\ \tilde{0}_{(p-1) \times 1} & \mathbf{L}_{22}^\top \end{bmatrix},$$

with  $\tilde{b}, \tilde{g} \in \mathbb{R}^{p-1}$ , where  $b_i$  and  $g_i$  are entries  $(i+1, 1)$  of  $\mathbf{A}$  and  $\mathbf{L}$ , and  $(\mathbf{A}_{22})_{ij}$  and  $(\mathbf{L}_{22})_{ij}$  are their entries  $(i+1, j+1)$ ;  $\mathbf{L}^\top$  has this form by Definition 0.16. A  $p \times p$  matrix  $\mathbf{L}$  is lower triangular with positive diagonal exactly when  $\ell_{11} > 0$ , its first row is otherwise zero, and  $\mathbf{L}_{22}$  is lower triangular with positive diagonal;  $\tilde{g}$  can be anything. So choosing  $\mathbf{L}$  means choosing  $\ell_{11} > 0$ ,  $\tilde{g}$  and such an  $\mathbf{L}_{22}$ .

Entry  $(i, j)$  of  $\mathbf{L}\mathbf{L}^\top$  is  $\sum_{r=1}^p \ell_{ir} \ell_{jr}$ , with  $\ell_{ir}$  entry  $(i, r)$  of  $\mathbf{L}$  (Definition 0.20, Definition 0.16). Row 1 of  $\mathbf{L}$  is  $(\ell_{11}, 0, \dots, 0)$ , and row  $i+1$  is  $(g_i, (\mathbf{L}_{22})_{i1}, \dots, (\mathbf{L}_{22})_{i,p-1})$ . So, for  $i, j = 1, \dots, p-1$ :

$$\begin{aligned} (\mathbf{L}\mathbf{L}^\top)_{11} &= \ell_{11} \ell_{11} + \sum_{r=2}^p 0 \cdot 0 = \ell_{11}^2, & (\text{rows 1 and 1}) \\ (\mathbf{L}\mathbf{L}^\top)_{i+1,1} &= g_i \ell_{11} + \sum_{r=2}^p (\mathbf{L}_{22})_{i,r-1} \cdot 0 = \ell_{11} g_i, & (\text{rows } i+1 \text{ and } 1) \\ (\mathbf{L}\mathbf{L}^\top)_{i+1,j+1} &= g_i g_j + \sum_{r=2}^p (\mathbf{L}_{22})_{i,r-1} (\mathbf{L}_{22})_{j,r-1} & (\text{rows } i+1 \text{ and } j+1) \\ &= (\tilde{g}\tilde{g}^\top)_{ij} + (\mathbf{L}_{22}\mathbf{L}_{22}^\top)_{ij}. & (@\text{def-matrix-mult, with } s = r-1) \end{aligned}$$

Entry  $(1, j+1)$  is  $\ell_{11} g_j$  by the same computation with the roles swapped, the transpose of the  $(2, 1)$  block; it matches  $\tilde{b}^\top$  exactly when the  $(2, 1)$  block matches  $\tilde{b}$ , which is where the symmetry of  $\mathbf{A}$  is used. So  $\mathbf{L}\mathbf{L}^\top = \mathbf{A}$  says exactly

$$\ell_{11}^2 = a_{11}, \quad \ell_{11} \tilde{g} = \tilde{b}, \quad \tilde{g}\tilde{g}^\top + \mathbf{L}_{22}\mathbf{L}_{22}^\top = \mathbf{A}_{22}.$$

$a_{11} > 0$  (Theorem 0.97), so the first equation with  $\ell_{11} > 0$  forces  $\ell_{11} = \sqrt{a_{11}}$ ; the second then forces  $\tilde{g} = \frac{1}{\ell_{11}} \tilde{b}$ ; and the third becomes

$$\begin{aligned}
\mathbf{L}_{22}\mathbf{L}_{22}^\top &= \mathbf{A}_{22} - \tilde{g}\tilde{g}^\top && \text{(subtract } \tilde{g}\tilde{g}^\top) \\
&= \mathbf{A}_{22} - \left(\frac{1}{\ell_{11}}\tilde{b}\right)\left(\frac{1}{\ell_{11}}\tilde{b}\right)^\top && \text{(substitute } \tilde{g}) \\
&= \mathbf{A}_{22} - \frac{1}{\ell_{11}^2}\tilde{b}\tilde{b}^\top && \text{(@def-matrix-transpose, @thm-scalar-matmul)} \\
&= \mathbf{A}_{22} - \frac{1}{a_{11}}\tilde{b}\tilde{b}^\top. && (\ell_{11}^2 = a_{11})
\end{aligned}$$

Here  $[a_{11}]$  is invertible with inverse  $[1/a_{11}]$ , so with  $\mathbf{B} = \tilde{b}^\top$  in Definition 0.78,  $\mathbf{B}^\top [a_{11}]^{-1} \mathbf{B} = \tilde{b} \frac{1}{a_{11}} \tilde{b}^\top = \frac{1}{a_{11}} \tilde{b}\tilde{b}^\top$  (Theorem 0.73), and the right side is the Schur complement of  $[a_{11}]$  in  $\mathbf{A}$ . It is positive definite by Theorem 0.99 (with  $k = 1$ ;  $\mathbf{A}_{22}$  is symmetric because  $\mathbf{A}$  is). By the induction hypothesis it has exactly one factor  $\mathbf{L}_{22}$  of the required kind. So  $\mathbf{L}$  exists and is unique.  $\square$

**Example 0.151** (Two Cholesky factors).

- $\mathbf{A} = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix}$ :  $\ell_{11} = \sqrt{4} = 2$ ,  $\ell_{21} = 2/2 = 1$ , and  $\ell_{22}^2 = 5 - 1^2 = 4$ , so  $\ell_{22} = 2$ :  
 $\mathbf{L} = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$ , and  $\mathbf{L}\mathbf{L}^\top = \begin{bmatrix} 2 \cdot 2 & 2 \cdot 1 \\ 1 \cdot 2 & 1 \cdot 1 + 2 \cdot 2 \end{bmatrix} = \mathbf{A}$ .
- $\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ :  $\ell_{11} = \sqrt{5}$ ,  $\ell_{21} = 4/\sqrt{5}$ , and  $\ell_{22}^2 = 5 - \frac{16}{5} = \frac{9}{5}$ , the Schur complement of Example 0.149, so  $\ell_{22} = \frac{3}{\sqrt{5}}$ .
- For  $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$  the method breaks down:  $\ell_{22}^2 = 1 - 4 = -3$  has no real solution, matching the failure of the Schur complement test (Example 0.150).

## Further reading

- Strang (2023) is a widely used first course in linear algebra. It covers vectors, elimination, vector spaces, orthogonality, determinants, eigenvalues, and the singular value decomposition.
- Axler (2024) is a proof-based treatment centered on vector spaces and linear maps. A free electronic version is available from the publisher.
- Fieller (2016)
- Banerjee and Roy (2014)
- Searle and Khuri (2017)

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